

Sparse graph moments: closed tree walks

(OEIS A094149, the moments M_{2k} of $G(N, 1/N)$)

Part I: fixed collision orders and smooth inverse centers

Part II: the full sparse graph moment equivalent

Part III: dominant vertices in sparse graph moments

Part IV: larger tree walk lower bounds

Part V: a shifted Bell lower bound

Part VI: general mean degree, optimal Poisson laws, and a growing-degree transition

Five research reports of one external research session

(bundle Reports 214, 213, 212, 210 and 209), bound into one ProveIt report

Part VI: research manuscript prepared with OpenAI assistance for the ProveIt project

Parts I–V: 4 October 2026 Part VI: 8 October 2026 merged: 7 and 9 October 2026

Abstract

Five manuscripts on one object, bound into one report: the number M_{2k} of normalized closed walks of length $2k$ on trees, with time zero distinguished, which is the $2k$ -th moment of the limiting expected spectral measure of the unscaled adjacency matrix of $G(N, 1/N)$ and the sequence OEIS A094149. They form one chain, printed newest first. Part I (Report 214) proves, for every fixed collision order $p \geq 1$, $M_{2k}/(2B_{k+1}) = 1 + \sum_{j \leq p} A_j(k) + O_p(\log^{10p+6} k/k^{p+1})$ with explicit finite Bell transforms $A_j(k) \sim W(k)^{3j}/(2^j j! k^j)$, so that the first relative correction is $W(k)^3/(2k)$, and gives smooth inverse centres with an existential two-ceiling enclosure. Part II (Report 213) proves $M_{2k} \sim 2B_{k+1}$ and a superpolynomial bound for the walks without a dominant vertex. Part III (Report 212) proves that the walks having a vertex with at least $k - S_k$ departures number $(2 + o(1))B_{k+1}$ and reduces the full equivalent to a middle-return tail estimate, which Part II proves; it also gives a rigorous scalar Poisson-neighbourhood correction $\frac{1}{2}W(k)^3/k$, the coefficient Part I proves for the moments, and two counterexamples to untruncated averaging. Part IV (Report 210) counts two disjoint families exactly, $G_k = 2B_{k+1}[1 + (1 + o(1))W(k)^2/k]$ and $J_k \sim W(k)^2 B_k$, and Part V (Report 209) proves $M_{2k} \geq 2(B_{k+1} - B_k) - F_{2k}$ with equality exactly for $k \leq 3$. The proofs that Reports 212–214 repeat are printed once. The write proves that the two families of Part IV already carry the first correction, $(M_{2k} - G_k - J_k)/(2B_{k+1}) = o(W(k)^3/k)$, quotes the OEIS entry, compares the inverses with the repository's transseries volume, and gathers the open questions. The report is unrefereed, and nothing in it is formalized.

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Guide to this report

The five Parts are five manuscripts of the session bundle of Reports 1–243 (arrival commit [60f54ea06](#)), placed in this directory by commit [f79c9bef1](#) (batch 112). All are dated 4 October 2026 and all study one object: M_{2k} (written a_k in Parts I and II), the number of normalized closed walks of half-length k whose traversed edges form a tree, with time zero distinguished; for $k \geq 1$ this is OEIS A094149 and the $2k$ -th moment of the limiting expected spectral measure of the unscaled adjacency matrix of $G(N, 1/N)$ (k fixed before $N \rightarrow \infty$).

Part	Report	Delivered title	Labels	Sections here
I	214 (base)	Fixed collision orders and smooth inverse centers for A094149	sgm:co:	1–13, A, B
II	213	The full sparse graph moment equivalent	sgm:eq:	14–27, A
III	212	Dominant vertices in sparse graph moments	sgm:dv:	28–38, A
IV	210	Larger tree walk lower bounds for sparse graph moments	sgm:lw:	39–48
V	209	A shifted Bell lower bound for sparse graph moments	sgm:sb:	49–55
VI	—	Sparse graph moments: sharp corrections, optimal Poisson laws, and a growing-degree transition (added 9 October 2026)	sgm:md:	57–64

The chain. The five manuscripts are successive steps of one research session, numbered in the order written; this report prints them newest first, so that the strongest statement comes first.

- Part V (Report 209) proves the exact lower bound $M_{2k} \geq 2(B_{k+1} - B_k) - F_{2k}$ (Theorem [50.1](#)), with equality exactly for $k \leq 3$, from a height-two family and its one-step rotation, and hence $\liminf M_{2k}/B_{k+1} \geq 2$ (Corollary [51.2](#)).
- Part IV (Report 210) counts two disjoint rotation-closed families exactly, G_k (all walks whose repeated edges share a vertex) and J_k (a recoverable hub and one repeated secondary edge), with $G_k = 2B_{k+1}[1 + (1 + o(1))W(k)^2/k]$ and $J_k \sim W(k)^2 B_k$ (Theorems [39.1](#) and [39.2](#)); it shows that Part V’s family lies inside G_k .
- Part III (Report 212) proves that the walks with a vertex of at least $k - S_k$ departures number $(2 + o(1))B_{k+1}$ for every window $\log k = o(S_k)$, $S_k = o(k^{1/3})$ (Theorem [28.1](#)), and reduces $M_{2k} \sim 2B_{k+1}$ to a weighted middle-return tail estimate ([162](#)), which it does not prove. It also proves a scalar Poisson-neighbourhood correction $\frac{1}{2}W(k)^3/k$ (Theorem [35.1](#)) and gives two counterexamples to untruncated averaging.
- Part II (Report 213) proves $M_{2k} \sim 2B_{k+1}$ and that both the complement of the dominant class and the weighted middle-return tail are $o(k^{-p}B_{k+1})$ for every fixed p (Theorem [14.1](#)); so it proves Part III’s missing estimate. Its Theorem [14.2](#) is Part III’s Theorem [28.1](#). It adds a two-ceiling inverse of radius $o(1/\log x)$.
- Part I (Report 214) refines Part II to every fixed collision order, $a_k/(2B_{k+1}) = 1 + \sum_{j \leq p} A_j(k) + O_p(\log^{10p+6} k/k^{p+1})$ with $A_j(k) \sim W(k)^{3j}/(2^j j! k^j)$ (Theorem [1.1](#)), so the first relative correction is $W(k)^3/(2k)$ — the scale Part III’s question 5 names and the coefficient of its scalar theorem — and sharpens the inverse to radius $\log^{10p+5} x/x^{p+1}$ (Theorem [11.1](#)). Its Appendix B restates Part II’s complement proof.

The write adds that Part IV’s families already carry the first correction: $(M_{2k} - G_k - J_k)/(2B_{k+1}) = o(W(k)^3/k)$ (Remark 46.1). Every earlier statement that a later Part supersedes is printed as delivered, with a dated note: Part III’s unproved premise, Part II’s open first correction, Part IV’s and Part V’s sentences that the equivalent does not follow from them, and the question lists. No claim of any Part was found to be wrong.

Arrangement and duplication. The base of the merge is Report 214: its `Report214.tex` was staged as this `article.tex`; it is the most general manuscript, and its Appendix B restates the complement proof of Report 213 in full. Reports 212, 213 and 214 repeat whole proof blocks, and each is printed here once, at its first place in the order of the Parts: the elementary Bell estimates (Report 213’s Appendix A word for word, and the lemma of Report 212’s Appendix A in other words) and the complement proof (Report 213’s Sections 6–9 and the first part of Section 10, word for word up to the named differences) are printed in Part I’s Appendices A and B; the dominant-window proof (Report 212’s Sections 2–5, word for word Report 213’s Sections 2–5) is printed in Part II. The omitted blocks keep their section headings, with notes that map every numbered object they contained to its printed counterpart; references to those objects in the rest of their Part were pointed to the printed counterparts, so they show the printed number. Proofs that are the same argument in different words and different statements are kept: Part II’s Sections 14.1–14.2 (restated in Part I’s Appendix B.1), Part I’s Proposition 2.1 (Part II’s Proposition 15.1 and Lemma 15.2 combined), Part III’s Section 28.1, Part IV’s Lemma 43.1 and Part V’s Lemma 51.1 (Bell-ratio lemmas with different additional statements), and the Bell convolution $\sum_s \binom{k-1}{s} B_{k-s} = B_{k+1} - B_k$, proved in Parts II, IV and V.

Numbering. Sections are numbered continuously, statements within sections, and equations and tables globally, as delivered. Part I keeps all its numbers, including Tables 1–2 and Appendices A–B. Report 213’s Section k is Section $k + 13$ here, Report 212’s is $k + 27$, Report 210’s is $k + 38$ and Report 209’s is $k + 48$; their equations (k) are $(k + 89)$, $(k + 155)$, $(k + 194)$ and $(k + 227)$; their tables are Tables 3, 4–5, 6 and 7–8. The equations that are not printed again keep their places in the count. The appendices of Parts II and III keep the letter A. The write’s final Section 56 gathers the open questions. Labels carry `sgm:co:`, `sgm:eq:`, `sgm:dv:`, `sgm:lw:` and `sgm:sb:`; the write’s labels carry the Part prefix, or `sgm:` for this Guide, Section 56 and its items (`sgm:q:`). The bibliography is merged at the end: the five Parts use the same keys for the same works, and every detail of every variant is kept in the entry.

[write] **Part VI, added 9 October 2026 (second write).** Since its second write the report has six Parts. Parts I–V, the Guide and Section 56 are the report as merged on 7 October 2026 and independently checked; the abstract above and the Guide’s text describe them alone, and nothing of them changed except this note, the Part VI row of the table above, dated notes after Remark 46.1 and at the end of Section 56, and the title, the running head and the PDF fields. Part VI (Sections 57–64, labels `sgm:md:`) is the manuscript *Sparse graph moments: sharp corrections, optimal Poisson laws, and a growing-degree transition* (8 October 2026, “Research manuscript prepared with OpenAI assistance for the ProveIt project”). It studies the moments $m_k(c)$ of $G(N, c/N)$ for a general mean degree c ; $m_k(1) = M_{2k}$, and $m_k(c)$ is Part V’s $\mu_{2k}(c)$. It re-derives Part I’s main theorem, with weights, uniformly for c in a compact subset of $(0, \infty)$ (Theorem 60.6, normalized by $2H_k(c) = 2\mathbb{E}(X + c)^k$, $X \sim \text{Poisson}(c)$, which is $2B_{k+1}$ at $c = 1$), and so answers item 5 of Section 56; it evaluates $\log(m_k(c)/2H_k(c)) = cw^2(w + 2 + 2c)/(2k) + cw^5/(6k^2) + O(w^4/k^2)$, $w = W(k/c)$ (Corollary 60.7), which answers item 3 in part; it proves a signed Poisson–Charlier law for the deficit $k - \max_v d_w(v)$ with sharp and optimal total-variation constants (Theorems 61.1 and 61.2); and it proves a growing-degree transition $m_k(c)/2H_k(c) \sim e^{c^2 w^2/k}$ when $cw/\sqrt{k}$ stays in a compact subset of $(0, \infty)$ (Theorem 62.1). That transition is a new direction: item 4 concerns a growing *collision order*, not a growing mean degree, and is not answered. Section 57 records its provenance, what it answers here, the checks, its non-claims and its letters ($T_k(c) = \mathbb{E}X^k$ is a Touchard polynomial there, not Part I’s T_k ; D_k is the deficit, not the two-hub count; $Q(k, s)$ a Catalan quotient, not B_{k+1}).

[write] **Provenance, added 7 October 2026, batch 112.** This report was built from five manuscripts, all dated 4 October 2026, arrived in 60f54ea06 and placed in f79c9bef1. Each archive wraps a directory `Report2NN/`. *Part I:* Report 214, archive `Report214-reproducibility.zip` (615,513 bytes, 21 files); text `Report214.tex` (1051 lines, 25 pages as delivered). *Part II:* Report 213, `Report213-reproducibility.zip` (579,496 bytes, 20 files); `Report213.tex` (831 lines, 20 pages). *Part III:* Report 212, `Report212-reproducibility.zip` (555,569 bytes, 36 files); `Report212.tex` (578 lines, 16 pages). *Part IV:* Report 210, `Report210-reproducibility.zip` (431,307 bytes, 9 files); `Report210.tex` (438 lines, 12 pages). *Part V:* Report 209, `Report209-reproducibility.zip` (415,540 bytes, 12 files); `Report209.tex` (440 lines, 13 pages). Every title block reads “Report2NN” under the title, with an empty author line and an empty PDF author field; no manuscript names a person, tool or addressee, carries a “prepared for private review” line, an e-mail address or personal data, or records a ProveIt commit (so no pin is recorded). Each Part’s programs are shipped under `code/` and its data, generated tables, requirements and recorded outputs under `data/`, with the prefixes `214-orders-`, `213-equiv-`, `212-hubs-`, `210-walks-` and `209-shifted-`; the source ledgers (`SOURCES.txt`, and for Reports 213–214 `SOURCE_FILES.txt`) and the code guides are shipped at the top level under the same prefixes. Report 214’s delivered `README.txt` was staged as the report `README` and is replaced by it. Not shipped, and retrievable from the arrival commit: the five delivered PDFs, the texts and delivered `READMEs` of Reports 209, 210, 212 and 213 (their texts are Parts V, IV, III and II of this file), and the three pure checksum manifests `MANIFEST.json` of Reports 212, 213 and 214 (35, 19 and 20 entries), all verified at the write. The seven generated tables that the delivered texts read with `\input` are printed inline (the shipped files under `data/` are byte-identical to the delivered ones). *Where the merge had to choose:* the base and the order (newest first); which copy of each repeated proof block to print (the first in the order of the Parts, see the Guide); keeping the proofs that differ in wording or statement; one merged bibliography; the one paragraph style that all five deliveries share (no indent, 4 pt paragraph skip). The write prefixed the 357 delivered labels (129, 95, 61, 39 and 33 for Reports 214, 213, 212, 210 and 209; the last two counts include the labels of their generated tables) and updated the 268 references to them in the printed text (217 `\eqref`, 51 `\ref`); 57 of the labels belong to the blocks that are not printed again (33 of Report 213, 24 of Report 212), and the 8 references to them from the printed text point to the printed counterparts. It labelled one unlabelled section of Report 210, and added this Guide, the Part headers, Remarks 1.2, 11.4 and 46.1, the dated notes and Section 56. The added remarks are the last statements of their sections and the added displays are unnumbered, so every delivered number that is printed is unchanged up to the stated shifts (label tables of builds of the five delivered texts and of this one compared).

The sources, as the write read them (7 October 2026). OEIS A094149 in the internal format (Remark 1.2); the transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex` (p0:def:model, p0:def:three-inverses, p0:thm:staircase, p0:thm:lambert-core, plt:def:lw-monomial-log-datum, plt:thm:lw-template, q2:sec:bell); the neighbouring report `a064856-stirling-catalan-transforms`. Not read by the write: the cited literature; the Parts’ accounts of what they inspected are reported, not confirmed.

What was checked at intake. The batch-112 dossier (7 October 2026) read the five manuscripts, found no demonstrably wrong claim, and reran the delivered suites on copies: their outputs reproduce. It did not rerun Report 212’s C++/GMP regeneration to $k = 256$ or the PDF and ZIP builders. *What the write checked.* It read the five manuscripts in full and found no error. It compared the repeated blocks line by line (the differences are named in the notes where the blocks are omitted). With its own code (Python 3.14.4, exact integers and rationals) it recomputed the rows $F_{k,m}$ from the first-edge recurrence for $k \leq 128$, and enumerated all normalized walks for $k \leq 7$ by brute force (12,850 walks; every row agrees); the thirteen terms of A094149; the boundary identities (150); for Part I, $d_1 = 0$, d_2 and d_3 of (25) for $q \leq 19$ (hence P_2 , P_3 of (37)), and every entry of Table 2 from (5), (39) and (27); for Part II, Table 3; for Part III, the maximum and the decrease of $M_{2k}/(2B_{k+1})$ and Tables 4 and 5 for $k \leq 128$ (the note at the end of Section 34.1), including $U_{k,S}$ through the two-hub formula (116); for Part IV, Table 6 from (197) and (198), $P_6 = 15$ and $C_6 = 3$; for Part V, Table 7 and the equality range $k \leq 3$ of (234) for $k \leq 128$. It reran, on fresh copies, Report 214’s and Report 213’s `code/finite_checks.py` (all five and four outputs equal to the shipped results), Report 210’s `verify_families.py` and Report 209’s two verifiers (both pass). It checked the three checksum manifests, and that every staged file is byte-identical to its delivery. Finite computations are observations, not certificates.

Relation to the repository. Before batch 112 the only report of the repository that treated A094149 was **a064856-stirling-catalan-transforms**, whose Part I proves $B_k \leq M_{2k} \leq a_k$ for the Stirling transform a_k of the Catalan numbers (from Bauer–Golinelli) and whose item **stc:q:moments** names the relative asymptotics of M_{2k} as open and these five manuscripts as unchecked leads. Part II answers that item, and its conditional remark there ($M_{2k}/a_k \rightarrow 0$ if $M_{2k} \sim 2B_{k+1}$) becomes unconditional; a reciprocal note there is proposed separately. Neighbours by method, in the same directory: **a277364-bell-asymptotics** and **a088714-bell-scale-growth** (Bell-number saddles). No Lean or Rocq file treats these sequences, no statement of this report is formalized, and its place in the collection confers no formal status.

[write] What is not claimed, collected from the five Parts. *Part I.* Every order fixed before $k \rightarrow \infty$; no effective constant or onset; the two-ceiling enclosure is existential, not a certified finite-threshold procedure, and does not identify a single ceiling; no growing-order expansion, no exponential resummation, no evaluation of the lower logarithmic suborders; the finite table is not an onset estimate; the complement theorem is inherited from Report 213, and the Bauer–Golinelli comparison is credited as its external input; the first-edge recurrence is Bauer–Golinelli’s; bounded source comparison, no historical priority, the further directions not claimed to be globally open; same-toolchain byte identity only. *Part II.* The superpolynomial strength applies to the two tails, not to $M_{2k}/(2B_{k+1}) - 1$; no first correction, no all-order expansion, no effective onset; the inverse enclosure is existential; the Bauer–Golinelli comparison is credited; Hiesmayr–McKenzie is context, not used; Janson–Jonsson–Stefánsson and Janson only at abstract level; no worldwide priority or open-problem claim. *Part III.* The full equivalent is not proved (the premise of (162) is not); the finite tables and the scalar model are not evidence for it; the scalar Theorem 35.1 does not identify M_{2k} with its expectation; the spectral deductions are the report’s own heuristic, and the two counterexamples concern untruncated averaging, not the cited theorem; the recurrence is Bauer–Golinelli’s; no monotonicity beyond $k = 256$; no global priority or open-status claim. *Part IV.* The families are lower bounds; nothing in it implies $M_{2k} \sim 2B_{k+1}$ or an inverse theorem; Catalan-decoration and kernel methods are credited to Hainzl–de Panafieu, the recurrence and bi-star decomposition to Bauer–Golinelli; no priority claim. *Part V.* Only a lower bound and its one-sided consequence; the equivalence $L_k \sim 2B_{k+1}$ concerns the selected union; the fixed- ℓ star theorem is Bauer–Golinelli’s, and the quantified extension does not reach the Bell edge scale; the overlap product specializes the published bi-star decomposition; no historical-priority claim, no inverse theorem. *The write adds:* its checks are finite exact computations; its transseries remark claims no novelty for any inversion; its Remark 46.1 is a consequence of two theorems of the Parts and says nothing about finite k .

Reading conventions (letters the Parts use differently, with the tempting false readings; no symbol of any source was renamed).

Symbol	Meanings, by Part
a_k, M_{2k}	The moments: $a_k = M_{2k}$ in Parts I–II, M_{2k} in Parts III–V. Part II’s urn also has initial populations $a_b = b $ (Section 16); Part V’s mass ratios use a_m , Part I’s Appendix A a_y .
F	Parts I–III: $F_{k,m}$, walks with m root departures. Part V: F_n , the Fibonacci numbers, in F_{2k} . Part I, Section 10: $F(x)$ a Poisson expectation. <i>False reading:</i> Part V’s F_{2k} is not a row of $F_{k,m}$.
C, Cat	Catalan numbers: Cat_ℓ in Parts I–IV, C_ℓ in Part V. Part II: $C_{k,S}$ the complement count. Part IV: C_n , a partition count, “not a Catalan number”. Elsewhere C, C_p, C' constants.
$B, \mathcal{B}$	B_n Bell numbers throughout ($B_0 = 1$). $\mathcal{B}(x)$ the Dobinski interpolation (Parts I–II). In Part III’s tables B is B_{k+1} .
A	Part I: $A_j(k)$ the transforms; A the largest block; $\mathcal{A}_t(x)$ in Appendix B. Part II: $A_{n,s} = \binom{n+s-1}{s} B_n$; A the largest block; its $A_t(x)$ is Part I’s $\mathcal{A}_t(x)$. Parts I–III: A_N the adjacency matrix. Part V: $A_{k,\ell}$ family counts and $\mathcal{A}_k$ the height-two family. <i>False reading:</i> $A_j(k)$ is not $A_{n,s}$ or $A_{k,\ell}$.
E	Parts I–III: $E_{t,q}$ cut child walks. Part IV: E_k an auxiliary weighted sum, “not the cardinality of a claimed rotation family”. $\mathbb{E}$ expectation.

Symbol	Meanings, by Part
G, J, H	Part I: $G_j(n, s)$; $H = \sum h_i$; $H(m, s)$ the entropy in Appendix B (also Part II); $J(s) = s \log(s/\log k)$ in Appendix B. Part IV: G_k, J_k, H_k and the families $\mathcal{G}_k, \mathcal{J}_k, \mathcal{H}_k$. Part V: H the adjacency matrix of a tree (Proposition 53.1).
T	Part I: T_k in (5), $A_1(k) = T_k/B_{k+1}$; T_b urn counts. Part II: $T_t(c)$ Touchard polynomials; T a cutoff. Part IV: $T_k = \sum_n d(k, n)B_n$ and $t_k(m)$. Part V: $T_n(\alpha)$ Touchard. <i>False reading</i> : Part I's T_k and Part IV's T_k are different numbers.
R	Part I: $R(q, t) = E_{t,q}/\binom{q+t-1}{t}$. Parts II–III: $R_{k,S}$ the weighted middle-return sum. Parts IV–V: the rotation operator R .
S	Parts I–III: the window S, S_k . Part IV: $S_k = \mathbb{E}[1_{\{X \geq 3\}} \lambda_X^k]$. Part V: $S(k, \ell)$ Stirling numbers of the second kind (Parts I–II write $\left\{ \begin{smallmatrix} k \\ \ell \end{smallmatrix} \right\}$). <i>False reading</i> : Part IV's S_k is not a window.
L	Part I: $L = \lceil \log^2 k \rceil$; the target of the inverse; $L_p(T)$; a real cutoff in Appendix B. Parts II–III: an integer hub threshold; a real cutoff. Part V: $L_k, L_{k,\ell}$ the lower bounds.
P, Q	Part I: $P_h(z)$ Bell polynomials; $Q_{u,H}(\ell)$. Parts II–IV: $Q_k = B_{k+1}$. Part IV: P_n a partition count. Part V: $P_k(\alpha)$ the overlap polynomial.
D, U	Parts I–III: D_k the two-hub count; $\mathcal{U}_{k,S}, U_{k,S}$ the dominant class; D_0 (Part I), D (Part II) the envelope constant. Part IV: $D(k, n)$ rooted counts and U_k their sum, “not J_k ”.
N	The graph size of $G(N, 1/N)$; the index of Bell lemmas; the threshold $N(L)$ (Part I), $N(y)$ (Part II); $N_I(w), N_S(w)$ vertex counts (Part II).
δ, η	Part I: $\delta_p(x)$ the inverse radius; η_0 . Part II: $\delta(x), \eta(x)$. Part III: $\delta_k = r^3/k$. Part IV: $\delta_k = k/x^2 \sim W(k)^2/k, \eta_k = k^2/x^3 \sim W(k)^3/k$. <i>False reading</i> : Part III's δ_k is Part IV's η_k up to $1 + o(1)$, not its δ_k .
r, x, ρ	$r = W(k)$ in Parts III–IV; $r = W(N)$ in Part I's Appendix A; r the number of marked blocks in Part I; r the root in Part V. $x = N/W(N)$ the Bell saddle (Appendix A, Part IV), $x_k = (k+1)/W(k+1)$; x the real argument and $x_p(L)$ the centres in Part I. $\rho = W(N)$.
Z, X	$X \sim \text{Poisson}(1)$ throughout. Part I: Z the deficit law (Lemma 3.2), Z_j the urn weight, $Z = X + b$ in Section 9. Part II: Z_X a binomial variable, Z_j the urn weight. Part V: X_k the overlap set.
d, h, q	$d_w(v)$ departure counts, “not graph degrees” (Parts I–III; Part V's tree degrees d_v are graph degrees); $d_h(q)$ in Part I; $d(k, n)$ in Part IV. h : branch lengths h_i (Part I), the cutoff $\lfloor k/(64 \log k) \rfloor$ (Appendix B), $h(k, n)$ (Part IV), mass ratios h_m (Part V). q : a block size or edge multiplicity; q_ℓ in Part V.
λ, μ	Part III: eigenvalues λ . Part IV: $\lambda_m = m^2/(m-1)$. Part V: $\lambda = (3 + \sqrt{5})/2$; $\mu_{2k}(\alpha)$ the weighted moments. Part I: μ_N the tilted Bell law.

Part I

Fixed collision orders and smooth inverse centers

Bundle Report 214, archive `Report214-reproducibility.zip`, delivered as `Report214.tex` (1051 lines, 25 pages as delivered), the base of the merge. Delivered title block “Fixed collision orders and smooth inverse centers for A094149 / Report214 / 4 October 2026” (empty author line). Printed as delivered, with `[write]` remarks and notes; its section, statement, equation and table numbers are those of the delivery, its labels carry `sgm:co:`, and its two generated tables are printed inline. Its Appendices A and B are the only print in this report of the Bell estimates and of the complement proof that Reports 212 and 213 repeat. “Report213” in this Part is Part II.

Abstract of this Part (as delivered). For the normalized time-rooted closed tree walks counted by A094149, write $a_k = M_{2k}$ and let B_k be the Bell numbers. We construct explicit finite Bell transforms $A_j(k)$ such that, for every fixed $p \geq 1$,

$$\frac{a_k}{2B_{k+1}} = 1 + \sum_{j=1}^p A_j(k) + O_p\left(\frac{\log^{10p+6} k}{k^{p+1}}\right), \quad A_j(k) \sim \frac{W(k)^{3j}}{2^j j! k^j}.$$

The coefficients use finitely many child-return rows, through half-length $j + 1$. A marked Pólya urn argument controls arbitrary depth, repeated branches, and signed Newton coefficients. The first relative correction is $W(k)^3/(2k)$; after subtracting its exact Bell transform, the next leading correction is $W(k)^6/(8k^2)$. A finite Poisson interpolation gives smooth inverse centers with an existential two-ceiling error of order $\log^{10p+5} x/x^{p+1}$. The superpolynomial complement theorem from Report213 is stated precisely and its required proof is recalled in an appendix, including its credited Bauer–Golinelli input. Every order is fixed before the index tends to infinity. No effective onset, growing-order expansion, exponential resummation, or historical priority claim is made.

1 The sequence and the result

A normalized closed tree walk of length $2k$ is a sequence $w = (v_0, v_1, \dots, v_{2k-1}, v_0)$ whose traversed unordered pairs form a tree and whose consecutive vertices differ. Vertex names are assigned in first-appearance order. Two walks are identified only by a vertex bijection preserving time order; time zero is distinguished. Rotation can therefore give a different normalized walk. Write $\mathcal{W}_k$ for this set and $a_k = M_{2k} = |\mathcal{W}_k|$, with $a_0 = 1$. The positive-order terms are OEIS A094149 [17].

Let $F_{k,n}$ count walks whose initial vertex has n departures. Set $F_{0,0} = 1$ and impossible entries to zero. Then $a_k = \sum_{n=1}^k F_{k,n}$ for $k \geq 1$. If edge e is traversed m_e times in each direction, the departure count at v is

$$d_w(v) = \sum_{e \ni v} m_e, \quad \sum_e m_e = k, \quad \sum_v d_w(v) = 2k. \quad (1)$$

These are walk departure counts, not graph degrees. The Bell numbers have $B_0 = 1$. All logarithms are natural and $W(x) > 0$ satisfies $W(x)e^{W(x)} = x$.

For an integer $h \geq 0$, write $(x)_h = x(x-1)\cdots(x-h+1)$ and $x^{\bar{h}} = x(x+1)\cdots(x+h-1)$; empty products equal one. On nonnegative integers $(s)_H = 0$ for $H > s$. Any term with such a vanishing factor is set to zero before other potentially invalid indices are evaluated.

Theorem 1.1. *The explicitly defined transforms $A_j(k)$ in (27) are independent of the chosen truncation order. For every fixed integer $p \geq 1$,*

$$\frac{a_k}{2B_{k+1}} = 1 + \sum_{j=1}^p A_j(k) + O_p\left(\frac{\log^{10p+6} k}{k^{p+1}}\right). \quad (2)$$

For each fixed $j \geq 1$,

$$A_j(k) \sim \frac{W(k)^{3j}}{2^j j! k^j}. \quad (3)$$

In particular, for each fixed $p \geq 1$,

$$\frac{a_k}{2B_{k+1}} - 1 - \sum_{j=1}^{p-1} A_j(k) \sim \frac{W(k)^{3p}}{2^p p! k^p}. \quad (4)$$

The lower transforms in (4) must be retained exactly. Replacing them by their individual leading scales in (3) is not justified at the smaller residual scale. Constants and onsets depend on the fixed order; none is certified numerically here.

The first transform has the concrete form

$$A_1(k) = \frac{T_k}{B_{k+1}}, \quad T_k = \sum_{s=2}^{k-1} \binom{k}{s} \binom{s}{2} \frac{(k-s-1)B_{k-s-1} + 4B_{k-s}}{k-s+1}. \quad (5)$$

Thus

$$\frac{a_k}{2B_{k+1}} = 1 + \left(\frac{1}{2} + o(1)\right) \frac{W(k)^3}{k}, \quad (6)$$

$$\frac{a_k}{2B_{k+1}} - 1 - \frac{T_k}{B_{k+1}} = \left(\frac{1}{8} + o(1)\right) \frac{W(k)^6}{k^2}. \quad (7)$$

The second assertion does not follow merely by subtracting $W(k)^3/(2k)$.

Dependencies and organization. The new argument refines the full equivalent $a_k \sim 2B_{k+1}$ and the superpolynomial complement theorem proved in Report213 [8]. Section 3 isolates exactly the inherited reduction. Appendix B recalls a complete proof of the complement estimate needed here, so that the required dependency can be read without the companion report. Its external counting input is the coefficientwise comparison of Bauer–Golinelli [1]; that theorem is not attributed to this article. The branch construction, weighted collision estimates, coefficient formulas, and leading scales are proved below. Smooth interpolation and the inverse consequences are treated separately, without differentiating any integer-index error term.

Remark 1.2 (The OEIS entry; added by the write, 7 October 2026). OEIS A094149 [17], read on 7 October 2026 at revision #10 (13 July 2025), was submitted by Alexey Spiridonov on 4 May 2004. Its name reads “The $2k$ -th moments of the random graph $G(n, 1/n)$ (odd moments are zero). The number of walks of length $2k$ on all bushes (rooted plane trees) that start and end at the root and visit new vertices from left-to-right (but may return).” It lists thirteen terms, $a_1, \dots, a_{13} = 1, 3, 12, 57, \dots, 1368977132$ (its b-file is synthesized from them), and its formula line reads “Asymptotically between A_k (the k -th Bell number, A000110) and $\text{choose}(2k, k) \cdot A_k$.”

The write recomputed a_k for $1 \leq k \leq 128$ from the recurrence (58) with its own code: the thirteen listed terms agree, and a separate brute-force enumeration of normalized walks for $k \leq 7$ (12,850

walks) agrees with every row $F_{k,m}$, $1 \leq m \leq k \leq 7$. The formula line holds as a pair of bounds for every $k \geq 1$: summing the Bauer–Golinelli comparison (65) over ℓ and using $\text{Cat}_\ell \leq \text{Cat}_k \leq \binom{2k}{k}$ gives $B_k \leq a_k \leq \text{Cat}_k B_k \leq \binom{2k}{k} B_k$. Neither bound is sharp: by Part II’s Theorem 14.1, $a_k \sim 2B_{k+1}$, so $a_k/B_k \sim 2k/W(k) \rightarrow \infty$ by (60), while $a_k/(\binom{2k}{k} B_k) \rightarrow 0$ because $\binom{2k}{k} \geq 2^k$. The entry records no asymptotic formula and no conjecture, so the five Parts settle no OEIS conjecture, and none claims to. No OEIS edit is part of this work.

Correction (independent check of the write, 7 October 2026). The quotation above is one sentence of the formula line, not the whole line. The full line reads “See [link:1] for a complex recurrence relationship. Asymptotically between A_k (the k -th Bell number, A000110) and $\text{choose}(2k, k) * A_k$. (see [ref:1]).” Here [link:1] is the entry’s one link, Khorunzhy’s preprint [2], and [ref:1] its one reference, Spiridonov’s thesis “in preparation” (cf. [3]). That sentence is the entry’s only asymptotic statement. So “no asymptotic formula” above means no asymptotic equivalent: the entry states the bounds discussed above and nothing sharper. The conclusion is unchanged.

2 Exact branch decomposition and urn representation

For $t \geq 0$ and $q \geq 1$ define

$$E_{t,q} = \sum_{\ell=0}^t F_{t,\ell} \binom{\ell+q-1}{\ell}, \quad R(q,t) = \frac{E_{t,q}}{\binom{q+t-1}{t}}. \quad (8)$$

A child-rooted walk with ℓ excursions is divided into q ordered, possibly empty closed segments in $\binom{\ell+q-1}{\ell}$ ways. Cuts occur only at returns to the child root. Consequently $E_{t,q}$ counts all such divided child walks, including arbitrary repeated edges and depth.

Proposition 2.1. *For $n \geq 1$, $s \geq 0$,*

$$F_{n+s,n} = \sum_{\pi \in \text{Part}[n]} \sum_{\substack{t_b \geq 0 \\ \sum_b t_b = s}} \prod_{\substack{(b \in \pi) \\ b \in \pi}} E_{t_b, |b|}. \quad (9)$$

Moreover,

$$\binom{q+t-1}{t} \leq E_{t,q} \leq a_t \binom{q+t-1}{t}, \quad a_t \leq \text{Cat}_t t! \leq 4^{t-1} t! \quad (t \geq 1). \quad (10)$$

Proof. Partition the n root-excursion positions according to their first child. A block of size q specifies q visits to one branch. Concatenating the interiors gives a child walk of half-length t_b divided into q closed segments. Different branches have disjoint vertex sets, since the traversed graph is a tree. Conversely, interleave these segments in the block positions and assign global first-appearance labels. This reconstruction is unique even when visits to different children interleave; no labeling factorial is missing.

The lower bound takes the one child walk with t distinct fresh leaf excursions; the upper bound uses $\ell \leq t$ in (8). To bound a_t , encode a walk by its root-distance Dyck path and its outward child choices. There are Cat_t paths. At the i th outward step at most i choices are available, including a newly discovered child; inward steps are forced. The encoding is injective, so there are at most $\text{Cat}_t t!$ walks. Finally $\text{Cat}_1 = 1$ and $\text{Cat}_{t+1} / \text{Cat}_t < 4$. $\square$

Choose a uniform partition π of $[n]$ with block sizes q_b , and a Pólya urn with initial populations q_b . After s draws let T_b be the numbers of selections. Its exact occupancy law is

$$\mathbb{P}(T_b = t_b \mid \pi) = \frac{\prod_b \binom{q_b + t_b - 1}{t_b}}{\binom{n + s - 1}{s}}, \quad \sum_b t_b = s.$$

Dividing (9) by its fresh normalization gives

$$\frac{F_{n+s,n}}{\binom{n+s-1}{s} B_n} = \mathbb{E}_\pi \mathbb{E}_{\text{urn}} \prod_b R(q_b, T_b). \quad (11)$$

Put $w_0 = w_1 = 1$ and $w_t = 4^{t-1}t!$ for $t \geq 1$. Then

$$1 \leq R(q, t) \leq w_t, \quad R(q, 0) = R(q, 1) = 1. \quad (12)$$

The return refinement matters: $E_{t,q}$ is determined by the whole row $(F_{t,\ell})_{\ell=0}^t$, not by the scalar total a_t alone.

3 Reduction to a logarithmic deficit window

For the rest of the expansion proof take

$$S = \lceil \log^2 k \rceil, \quad n = k - s, \quad 0 \leq s \leq S.$$

Let $\mathcal{U}_{k,S}$ be the walks with some vertex having at least $k - S$ departures. The inherited complement theorem is

$$|\mathcal{W}_k \setminus \mathcal{U}_{k,S}| = o(k^{-D} B_{k+1}) \quad \text{for every fixed } D > 0. \quad (13)$$

Report213 proves this more generally whenever $S/\log k \rightarrow \infty$ and $S = o(k^{1/3})$. Appendix B supplies the proof needed at the present choice of S ; the stronger weighted middle-return theorem is not required here.

Lemma 3.1. *For every fixed $D > 0$,*

$$\frac{a_k}{2B_{k+1}} = \frac{1}{B_{k+1}} \sum_{s=0}^S \frac{k}{k-s} F_{k,k-s} + o(k^{-D}). \quad (14)$$

Proof. For any set I of departure counts, exact weighted rotation gives

$$\sum_{m \in I} \frac{2k}{m} F_{k,m} = \sum_{w \in \mathcal{W}_k} |\{v : d_w(v) \in I\}|. \quad (15)$$

Indeed a vertex of count m appears as root at exactly m of the $2k$ indexed rotations. If a rotation orbit has d distinct normalized words, each occurs $2k/d$ times among indexed rotations. Dividing by that multiplicity proves the identity even for periodic walks and rotations inducing relabelings.

Two vertices with departure count above $k/2$ must be adjacent: otherwise their incident-edge sets are disjoint and their counts sum to at most k . There are at most two, because a tree has no triangle. Let D_k count walks with two qualifying vertices. Then (15) gives

$$\sum_{s=0}^S \frac{2k}{k-s} F_{k,k-s} = |\mathcal{U}_{k,S}| + D_k.$$

The edge joining the two hubs has multiplicity $q \geq k - 2S$, since their departure counts sum to at most $k + q$. Rotate to a crossing of that edge and remove its traversals. The two sides are q -segmented child walks of total half-length at most $2S$. By (10), $E_{t,q} \leq (4k)^t$ when $t + q \leq k$, including $t = 0$. Overcounting all origins therefore gives

$$D_k \leq 2k(2S + 1)^2(4k)^{2S} = \exp(O(\log^3 k)). \quad (16)$$

Since $\log B_{k+1} \sim k \log k$, this and (13) are smaller than $k^{-D} B_{k+1}$ for every fixed D . Rearrangement proves the lemma. $\square$

The cancellation

$$\frac{k}{k-s} \binom{k-1}{s} = \binom{k}{s} \quad (17)$$

is kept exact at every order. Replacing the rotation weight by a rough $1 + O(S/k)$ estimate would hide part of the correction.

Lemma 3.2 (Fresh deficit law). *The numbers $p_k(s) = \binom{k}{s} B_{k-s}/B_{k+1}$, $0 \leq s \leq k$, are probabilities. If Z has this law, then for every fixed integer $h \geq 0$,*

$$\mathbb{E}(Z)_h = (k)_h \frac{B_{k+1-h}}{B_{k+1}} \sim W(k)^h. \quad (18)$$

For every fixed $D > 0$ and $r \geq 0$,

$$\sum_{s>S} s^r p_k(s) = o(k^{-D}). \quad (19)$$

Proof. The Bell recurrence normalizes p_k . Equivalently, start with $X \sim \text{Poisson}(1)$, tilt its law by $(X+1)^k/B_{k+1}$, and conditionally take $Z \sim \text{Bin}(k, 1/(X+1))$. Expanding gives the stated masses and factorial moments. Under this tilt $X+1$ has probabilities proportional to $y^{k+1}/y!$. The Bell concentration lemma in Appendix A puts $X \asymp k/W(k)$ except with exponentially small probability. On $X \geq (k+1)/(2W(k+1))$, the conditional binomial mean is $O(\log k)$; Chernoff bounds the probability of $Z > S$ by $(C \log k/S)^S$. Both errors are smaller than every negative power of k . As $s \leq k$, fixed power weights are absorbed by strengthening that power first. Fixed-shift Bell ratios prove (18). $\square$

The endpoint $s = k$, which would give $n = 0$, has mass $1/B_{k+1}$. Thus the exact sum restricted to $s \leq k-1$ equals $1 - 1/B_{k+1}$; its missing endpoint is superpolynomially small.

4 A weighted marked collision estimate

Fix a partition and write $A = \max_b q_b$. A draw is a collision if its color has already appeared among earlier draws. After j draws define

$$C_j = j - |\{b : T_b > 0\}|, \quad Z_j = \prod_b w_{T_b}.$$

Thus $C_s = \sum_b (T_b - 1)_+$ counts the collision excess.

Lemma 4.1. *Conditional on the partition, for every integer $d \geq 1$,*

$$\mathbb{E}_{\text{urn}}[Z_s u^{C_s}] \preceq \prod_{j=0}^{s-1} (1 + b_j u), \quad b_j = \frac{8(Aj + j^2)}{n}, \quad (20)$$

$$\mathbb{E}_{\text{urn}}[Z_s; C_s \geq d] \leq \frac{\Lambda^d}{d!} e^\Lambda, \quad \Lambda = \sum_{j < s} b_j \leq \frac{4s^2(A + s)}{n}. \quad (21)$$

Here $\preceq$ is coefficientwise comparison of polynomials in u .

Proof. A new color has weight multiplier one. A previously drawn color with count $t \geq 1$ has multiplier $w_{t+1}/w_t = 4(t+1)$. Conditional on any history of length j , the total probability-weighted multiplier of collision transitions is at most

$$\sum_{b: T_b > 0} \frac{q_b + T_b}{n + j} 4(T_b + 1) \leq \frac{8Aj + 4j^2 + 4j}{n} \leq b_j.$$

For $j = 0$ this is zero. For $j \geq 1$ use at most j occupied colors, $\sum T_b = j$, and $\sum T_b^2 \leq j^2$. The total new-color multiplier is at most one. Each weighted history therefore extends by a factor $\alpha + \beta u$ with $0 \leq \alpha \leq 1$, $0 \leq \beta \leq b_j$. Induction proves the coefficientwise bound. The coefficient of u^r in the product is at most $\Lambda^r/r!$; summing over $r \geq d$ gives the second assertion. $\square$

An additional conditional estimate will control atypical partitions:

$$\mathbb{E}_{\text{urn}} Z_s \leq e^{\tau A + \gamma}, \quad \tau = \frac{7s(s-1)}{2n}, \quad \gamma = \frac{7s(s-1)(2s-1)}{6n}. \quad (22)$$

Indeed the one-step multiplier is $1 + \sum_{T_b > 0} (q_b + T_b)(4T_b + 3)/(n + j)$, at most $1 + 7(Aj + j^2)/(n + j)$ because $4t + 3 \leq 7t$. Iterate and use $1 + u \leq e^u$.

Lemma 4.2 (Tilted partition tail). *Put $L = \lceil \log^2 k \rceil$. Uniformly for $n = k - s$, $s \leq S$, and τ from (22),*

$$\mathbb{E}_\pi[e^{\tau A}; A > L] = o(k^{-D}) \quad \text{for every fixed } D > 0. \quad (23)$$

Proof. The expected number of blocks of size q is $\binom{n}{q} B_{n-q}/B_n$. Since a block realizes the maximum,

$$\mathbb{E}[e^{\tau A}; A > L] \leq \sum_{q > L} e^{\tau q} \binom{n}{q} \frac{B_{n-q}}{B_n}.$$

Here $\tau = O(\log^4 k/k) = o(1)$. For $q \leq n/2$, adjacent Bell-ratio bounds make a summand at most $(Ce^\tau \log n/q)^q$. Its sum beyond L is bounded by $\exp[-cL \log(L/\log n)]$: beyond that point the ratio of successive upper bounds is eventually below $1/2$. For $q > n/2$, retain the first $\lfloor n/2 \rfloor$ adjacent ratios and use Bell monotonicity for the rest. This part is at most

$$(1 + e^\tau)^n (C \log n/n)^{\lfloor n/2 \rfloor} = \exp[-\Omega(n \log n)].$$

Both bounds imply the claim uniformly. In particular an untilted rare-event estimate has not been substituted for the required weighted tail. $\square$

5 Finite coefficients from Newton differences

For $h \geq 0$ set

$$c_h(q) = \sum_{t=0}^h (-1)^{h-t} \binom{h}{t} R(q, t), \quad (24)$$

$$d_h(q) = \frac{q^h}{h!} c_h(q) = \sum_{t=0}^h (-1)^{h-t} E_{t,q} \frac{(q+t)^{\overline{h-t}}}{(h-t)!}. \quad (25)$$

Then $c_0 = d_0 = 1$ and $c_1 = d_1 = 0$. The second expression in (25) shows that d_h is a polynomial in q of degree at most h . For $h \geq 1$ it has zero constant term: $E_{t,q}$ for $t \geq 1$ is divisible by q , and the $t = 0$ term also vanishes at $q = 0$.

For $j \geq 1$, $n \geq 1$, $s \geq 0$ define

$$G_j(n, s) = \sum_{r=1}^j \frac{1}{r!} \sum_{\substack{h_1, \dots, h_r \geq 2 \\ \sum_i (h_i - 1) = j}} \frac{(s)_H}{n^H} \mathbb{E}_\pi \sum_{\substack{b_1, \dots, b_r \in \pi \\ \text{ordered and distinct}}} \prod_{i=1}^r d_{h_i}(q_{b_i}), \quad H = \sum_i h_i = j + r. \quad (26)$$

An empty ordered-block sum is zero. These sums are finite; $H \leq 2j$ and $h_i \leq j + 1$. The correction transform is

$$A_j(k) = \frac{1}{B_{k+1}} \sum_{s=0}^{k-1} \binom{k}{s} B_{k-s} G_j(k-s, s). \quad (27)$$

Only child-return rows through half-length $j + 1$ are needed. No unknown large-index scalar moment appears in the coefficient construction.

Proposition 5.1. *For a fixed integer $p \geq 1$, define on an urn history*

$$L_p(T) = 1 + \sum_{j=1}^p \sum_{r=1}^j \frac{1}{r!} \sum_{\substack{h_i \geq 2 \\ \sum_i (h_i - 1) = j}} \sum_{\substack{b_1, \dots, b_r \in \pi \\ \text{ordered and distinct}}} \prod_{i=1}^r c_{h_i}(q_{b_i}) \binom{T_{b_i}}{h_i}. \quad (28)$$

Then $\prod_b R(q_b, T_b) = L_p(T)$ whenever $C_s \leq p$. For all histories,

$$|L_p(T)| \leq K_p(1 + s)^{2p}, \quad (29)$$

for a constant depending only on p . Also

$$\mathbb{E}_\pi \mathbb{E}_{\text{urn}} L_p = 1 + \sum_{j=1}^p G_j(n, s). \quad (30)$$

Proof. Finite binomial inversion gives $R(q, t) = \sum_{h=0}^t c_h(q) \binom{t}{h}$. Multiply these identities over colors. Every nonunit term has $h \geq 2$ since $c_1 = 0$. A nonzero term with marked colors b_i has $h_i \leq T_{b_i}$, and hence $\sum_i (h_i - 1) \leq C_s$. It follows that (28) omits no term when $C_s \leq p$. The $r!$ denominator removes the ordering of marked blocks without assuming their h_i are different.

By (12), $|c_h(q)| \leq \sum_{t=0}^h \binom{h}{t} w_t$, uniformly in q . Moreover

$$\sum_{\text{distinct ordered } b_i} \prod_i \binom{T_{b_i}}{h_i} \leq \frac{(s)_H}{\prod_i h_i!}.$$

The left side selects disjoint groups of draw positions of specified distinct colors; dropping those color restrictions gives the right side. There are only finitely many fixed configurations, with $H \leq 2p$, proving (29).

Finally the urn factorial-moment identity is

$$\mathbb{E}_{\text{urn}} \prod_i \binom{T_{b_i}}{h_i} = \frac{(s)_H}{n^H} \prod_i \frac{q_{b_i}^{\overline{h_i}}}{h_i!}$$

for distinct colors. Choose H distinct draw indices and use exchangeability to prove it. Substitution into (28) gives (30) exactly. $\square$

6 The remainder at every fixed collision order

On the event $C_s \leq p$, Proposition 5.1 has no error. On its complement, $\prod R \leq Z_s$, $Z_s \geq 1$, and (29) give

$$\left| \mathbb{E}_{\text{urn}} \prod_b R(q_b, T_b) - \mathbb{E}_{\text{urn}} L_p \right| \leq K'_p (1+s)^{2p} \frac{\Lambda^{p+1}}{(p+1)!} e^\Lambda. \quad (31)$$

Thus every longer branch and every arrangement of repeated, nested, or separate collisions is paid for in a weighted remainder; no positivity of the Newton coefficients is assumed.

On $A \leq L$ and $s \leq S$, (21) has $\Lambda = O(\log^6 k/k)$. Equation (31) is consequently

$$O_p \left(\log^{4p} k \left(\frac{\log^6 k}{k} \right)^{p+1} \right) = O_p \left(\frac{\log^{10p+6} k}{k^{p+1}} \right).$$

On $A > L$ use the pre-averaged bound $\prod R \leq Z_s$, (22), and (29). Lemma 4.2 bounds their contribution by

$$e^\gamma \mathbb{E}[e^{\tau A}; A > L] + K_p (1+S)^{2p} \mathbb{P}(A > L) = o(k^{-D})$$

for every fixed D . Therefore (11) and (30) imply the uniform expansion

$$\frac{F_{k,k-s}}{\binom{k-1}{s} B_{k-s}} = 1 + \sum_{j=1}^p G_j(k-s, s) + O_p \left(\frac{\log^{10p+6} k}{k^{p+1}} \right), \quad 0 \leq s \leq S. \quad (32)$$

Each homogeneous collision part of (28) obeys the same polynomial history bound, so $|G_j(n, s)| \leq K_j (1+s)^{2j}$ for every $n \geq 1$. Lemma 3.2 therefore extends the G_j sums from $s \leq S$ to $s \leq k-1$ at superpolynomial cost. Apply Lemma 3.1, use the exact cancellation (17), and sum the uniform error against probabilities $p_k(s)$. The fresh sum is $1 - 1/B_{k+1}$. This proves (2).

One can already check that the coefficients have successive polynomial orders:

$$A_j(k) = O_j(\log^{6j} k/k^j). \quad (33)$$

Indeed $|d_h(q)| \leq K_h q^h$. For a configuration with r marked blocks and $H-r=j$, on $A \leq L$,

$$\sum_{\text{distinct ordered } b_i} \prod_i q_{b_i}^{h_i} \leq A^j \left(\sum_b q_b \right)^r = A^j n^r.$$

The urn prefactor in (26) then gives $O_j(S^{2j} L^j / n^j)$. The large-block and deficit tails are again superpolynomial. The more precise leading constants require the finite Bell reduction next.

7 Bell polynomial evaluation and the first two transforms

Expand $d_h(q) = \sum_{u=1}^h \alpha_{h,u}(q)_u$. Since $\mathbb{E}(X)_u = z^u$ for $X \sim \text{Poisson}(z)$, define

$$P_h(z) = \sum_{u=1}^h \alpha_{h,u} z^u, \quad D_h(z) = \sum_{q \geq 1} d_h(q) \frac{z^q}{q!} = e^z P_h(z). \quad (34)$$

The coefficients are explicitly $\alpha_{h,u} = \Delta^u d_h(0)/u!$. Thus computing P_h uses finite rational arithmetic. The labeled-partition exponential formula gives

$$\mathbb{E}_\pi \sum_{\text{distinct ordered } b_i} \prod_i d_{h_i}(q_{b_i}) = \frac{n!}{B_n} [z^n] \exp(e^z - 1 + rz) \prod_i P_{h_i}(z). \quad (35)$$

For $X \sim \text{Poisson}(1)$ and $N \geq 0$,

$$N! [z^N] \exp(e^z - 1 + rz) = \mathbb{E}(X + r)^N = \mathbb{E}[(X)_r X^N] = \sum_{v=0}^r \mathfrak{s}(r, v) B_{N+v}, \quad (36)$$

where $\mathfrak{s}(r, v)$ are signed Stirling numbers of the first kind. The middle equality is the exact Poisson factorial shift. The $r = 0$ version is B_N . Consequently each $G_j(n, s)$ has a fixed finite rational expression in Bell numbers at fixed shifts of n . The transforms $A_j(k)$ are then the finite convolutions in (27). Thus every coefficient is explicitly computable by finite rational arithmetic, with no unresolved infinite coefficient extraction.

The first rows give

$$\begin{aligned} d_2(q) &= \frac{q(q+3)}{2}, & d_3(q) &= \frac{q(q+4)(q+5)}{6}, \\ P_2(z) &= \frac{z^2 + 4z}{2}, & P_3(z) &= \frac{z^3 + 12z^2 + 30z}{6}. \end{aligned} \quad (37)$$

For $j = 1$, (26) has one marked block and $h_1 = 2$. Counting pairs in a uniform partition gives

$$\mathbb{E}_\pi \sum_b q_b(q_b + 3) = n(n-1) \frac{B_{n-1}}{B_n} + 4n.$$

Hence

$$G_1(n, s) = \binom{s}{2} \phi_n, \quad \phi_n = \frac{(n-1)B_{n-1}/B_n + 4}{n+1}. \quad (38)$$

This proves (5), including its empty-sum convention.

For $j = 2$ the only configurations are one $h = 3$ block or two $h = 2$ blocks. Writing out (35)–(36),

$$\begin{aligned} G_2(n, s) &= \frac{\binom{s}{3}}{6n^3 B_n} [(n)_3 B_{n-2} + 12(n)_2 B_{n-1} + 30n B_n] \\ &\quad + \frac{\binom{s}{4}}{8n^4 B_n} [(n)_4 (B_{n-2} - B_{n-3}) \\ &\quad + 8(n)_3 (B_{n-1} - B_{n-2}) + 16(n)_2 (B_n - B_{n-1})]. \end{aligned} \quad (39)$$

If $n < d$, a term carrying $(n)_d$ is zero before its Bell subscripts are evaluated. Thus (39) is valid even at $n = 1, 2$. Inserting it into (27) supplies an explicit second transform with remainder $O(\log^{26} k/k^3)$.

8 The leading scale of every fixed transform

Fix j and consider one configuration in (26) with r marked blocks and $H = j + r$. The polynomial $\prod_i P_{h_i}$ has degree at most H . A monomial z^u contributes to its partition average

$$\frac{(n)_u}{B_n} \sum_{v=0}^r \mathfrak{s}(r, v) B_{n-u+v}. \quad (40)$$

Uniformly for $n = k - s$, $s \leq S$, the fixed-shift Bell ratios are

$$\frac{B_{n-u+v}}{B_n} \sim \left(\frac{W(n)}{n} \right)^{u-v}.$$

With $w = W(k)$, each fixed monomial in (40) is $O(n^r w^{H-r}) = O(n^r w^j)$. To see this also when $u < v$, its individual order is $n^v w^{u-v}$, and the ratio to $n^r w^j$ is $(w/n)^{r-v} w^{u-H}$, which is bounded. Multiplying by the urn prefactor and averaging over the fresh deficit law, whose H th factorial moment is asymptotic to w^H , gives

$$O(w^{H+j}/k^j) = O(w^{2j+r}/k^j). \quad (41)$$

All configurations with $r < j$ are therefore smaller than w^{3j}/k^j .

The only configuration with $r = j$ has every $h_i = 2$. By (37), its polynomial has leading monomial $z^{2j}/2^j$. Its partition average is

$$\frac{(n)_{2j} B_{n-j}}{2^j B_n} (1 + o(1)) \sim \frac{n^j w^j}{2^j}.$$

A one-degree polynomial loss costs a factor $1/w$; a one-unit signed-Stirling shift loss costs w/n . Include the $1/j!$ factor in (26), the urn factor $(s)_{2j}/n^{2j}$, and $\mathbb{E}(Z)_{2j} \sim w^{2j}$. All tails are controlled as before. The result is

$$A_j(k) \sim \frac{w^{3j}}{2^j j! k^j},$$

proving (3). Finally the remainder in (2) at order p is $o(w^{3p}/k^p)$, which proves (4) and completes Theorem 1.1.

For the explicit second transform, the one- $h = 3$ contribution is asymptotic to $w^5/(6k^2)$ and the two- $h = 2$ contribution to $w^6/(8k^2)$. This explains the leading coefficient $1/8$ in (7). It does not evaluate all logarithmic suborders of the full coefficient at scale k^{-2} .

9 A finite smooth interpolation of the corrections

An integer-index asymptotic does not by itself specify a useful real interpolation, and its remainder cannot simply be differentiated. We instead construct one from finitely many explicit Poisson expectations.

For one ordered branch list $h_1, \dots, h_b \geq 2$ put

$$j = \sum_i (h_i - 1), \quad H = j + b, \quad \prod_i P_{h_i}(z) = \sum_{u=b}^H p_u z^u.$$

The unnormalized contribution of its monomial z^u to $B_{k+1}A_j(k)$, before multiplying by $p_u/b!$, is

$$\sum_{n=1}^k \binom{k}{n} \frac{(k-n)_H (n)_u}{n^{\overline{H}}} \mathbb{E}(X+b)^{n-u}, \quad X \sim \text{Poisson}(1). \quad (42)$$

Terms with $n < u$ or $k-n < H$ vanish before their exponents are evaluated. Put $m = n - u$ and $v = u + H - 1$. Exact factorial cancellation gives

$$\binom{k}{n} \frac{(k-n)_H (n)_u}{n^{\overline{H}}} = k \binom{k-1}{m+v} (m+1)^{\overline{u-1}}. \quad (43)$$

For $Z = X + b \geq 1$ define the monic polynomial

$$Q_{u,H}(\ell) = (\ell - v + 1)^{\overline{u-1}} = \sum_{t=0}^{u-1} q_{u,H,t}(\ell) t.$$

The coefficients in its finite falling-factorial expansion are again obtained by finite differences. Define, for real $x > 0$,

$$\begin{aligned} \mathcal{I}_{b,H,u}(x) &= x \sum_{t=0}^{u-1} q_{u,H,t}(x-1) \mathbb{E} \left[Z^{t-v} (Z+1)^{x-1-t} \right] \\ &\quad - x \sum_{\ell=0}^{v-1} \binom{x-1}{\ell} Q_{u,H}(\ell) \mathbb{E} Z^{\ell-v}. \end{aligned} \quad (44)$$

Generalized binomials mean falling-factorial polynomials divided by $\ell!$. Because $Z \geq 1$, the negative powers are well defined. All Poisson series and their real derivatives converge locally uniformly: factorial decay dominates every fixed power of Z and $\log(Z+1)$.

Proposition 9.1. *The smooth functions*

$$\mathcal{T}_j(x) = \sum_{b=1}^j \frac{1}{b!} \sum_{\substack{h_1, \dots, h_b \geq 2 \\ \sum_i (h_i - 1) = j}} \sum_{u=b}^H p_u \mathcal{I}_{b,H,u}(x) \quad (45)$$

satisfy $\mathcal{T}_j(k) = B_{k+1}A_j(k)$ at every positive integer k .

Proof. By (43), (42) equals

$$k \mathbb{E} \left[Z^{-v} \sum_{\ell=v}^{k-1} \binom{k-1}{\ell} Q_{u,H}(\ell) Z^{\ell} \right].$$

For every nonnegative integer t , the differentiated binomial identity gives

$$\sum_{\ell=0}^{k-1} \binom{k-1}{\ell} (\ell)_t Z^{\ell} = (k-1)_t Z^t (1+Z)^{k-1-t}.$$

If $t > k-1$, both sides vanish; the right side is zero times a finite quantity. Expand Q in this basis and subtract the finitely many terms with $\ell < v$. This is exactly (44) at $x = k$. The argument also covers $k < v+1$, because binomial coefficients with $\ell > k-1$ vanish and the remaining full and boundary sums cancel. Summing configurations proves the assertion. $\square$

This construction requires only finite child-return rows through half-length $j+1$ and explicitly specified Poisson expectations. It contains neither the unknown full sequence a_k at large indices nor an arbitrary interpolation of its errors.

10 Real concentration and derivative bounds

For real $N > 0$ define

$$\mathcal{B}(N) = e^{-1} \sum_{y \geq 1} \frac{y^N}{y!}, \quad \mu_N(y) = \frac{e^{-1} y^N}{y! \mathcal{B}(N)}, \quad y_* = \frac{N}{W(N)}.$$

At positive integers $\mathcal{B}(n) = B_n$; no equality at $n = 0$ is asserted. The mass-ratio proof in Appendix A works for real N without change, giving exponential concentration outside every fixed relative neighborhood of y_* . For each fixed $d > 0$ it follows that

$$\mathbb{E}_{\mu_N} Y^{-d} = O(y_*^{-d}), \quad \mathbb{E}_{\mu_N} [Y^{-d} \log Y] = O(y_*^{-d} \log N), \quad (46)$$

$$\frac{\mathcal{B}'(N)}{\mathcal{B}(N)} = \mathbb{E}_{\mu_N} \log Y \sim \log N. \quad (47)$$

For the negative moments, split at $Y = y_*/2$: below it the bounded summands have exponentially small probability; above it their bounds are at most constant multiples of the stated scales. Relative concentration gives the lower logarithmic estimate, and Jensen with $\mathbb{E}Y = O(y_*)$ gives the upper one. This controls derivatives directly, without differentiating an asymptotic equivalent.

For a full-sum expectation in (44), write

$$F(x) = \mathbb{E}[Z^{t-v}(Z+1)^{x-1-t}], \quad N = x+1.$$

Reindex $y = Z+1 = X+b+1$ and put

$$K_t(Y) = (Y)_{b+1}(Y-1)^{t-v}Y^{-t-2}.$$

The exact Poisson reindexing gives the pair of identities

$$\frac{F(x)}{\mathcal{B}(x+1)} = \mathbb{E}_{\mu_{x+1}}[K_t(Y); Y \geq b+1], \quad (48)$$

$$\frac{F'(x)}{\mathcal{B}(x+1)} = \mathbb{E}_{\mu_{x+1}}[K_t(Y) \log Y; Y \geq b+1]. \quad (49)$$

The second line is the raw derivative F' divided by $\mathcal{B}$, not the derivative of $F/\mathcal{B}$.

Since $t \leq u-1$, $v = u+H-1$, and $H = j+b$, the kernel is nonnegative and bounded by $CY^{-(u+j)}$ on $Y \geq b+1 \geq 2$. Equations (46)–(49) bound $F/\mathcal{B}$ and $F'/\mathcal{B}$ by

$$O((W(x)/x)^{u+j}), \quad O((W(x)/x)^{u+j} \log x),$$

respectively. The prefactor $x(x-1)_t$ is $O(x^{t+1})$, and its derivative is $O(x^t)$. Since $t+1 \leq u \leq H \leq 2j$, every full-sum term, and hence their finite signed sum, satisfies

$$\mathcal{T}_j(x) = O_j \left(\mathcal{B}(x+1) \frac{W(x)^{3j}}{x^j} \right), \quad (50)$$

$$\mathcal{T}'_j(x) = O_j \left(\mathcal{B}(x+1) \frac{W(x)^{3j} \log x}{x^j} \right). \quad (51)$$

The boundary subtraction in (44) is a fixed-degree polynomial in x with finite constant coefficients; it and its derivative are negligible on these scales.

For each fixed $p \geq 0$ put

$$g_p(x) = 2 \left[\mathcal{B}(x+1) + \sum_{j=1}^p \mathcal{T}_j(x) \right], \quad (52)$$

with empty sum when $p = 0$. Then

$$\frac{g_p(x)}{2\mathcal{B}(x+1)} = 1 + O_p(W(x)^3/x), \quad \frac{g'_p(x)}{g_p(x)} \sim \log x. \quad (53)$$

In particular g_p is positive, strictly increasing, and unbounded on some eventual interval. For sufficiently large L , define $x_p(L)$ to be the unique solution there of $g_p(x_p(L)) = L$.

11 Two ceilings and displacement of the centers

Define the discrete threshold

$$N(L) = \min\{k \geq 1 : a_k \geq L\}.$$

Appending a fresh root-leaf excursion injects walks of half-length k into walks of half-length $k+1$. Deleting the last two steps recovers the source. The repeated single-edge walk is outside the image for $k \geq 1$, so a_k is strictly increasing in that range.

Theorem 11.1. *For every fixed integer $p \geq 1$ there is a constant $C_p > 0$ such that, for all sufficiently large L , with $x = x_p(L)$,*

$$\lceil x - \delta_p(x) \rceil \leq N(L) \leq \lceil x + \delta_p(x) \rceil, \quad \delta_p(x) = C_p \frac{\log^{10p+5} x}{x^{p+1}}. \quad (54)$$

The two ceilings are eventually equal or adjacent. The constant and onset are existential.

Proof. Theorem 1.1 and Proposition 9.1 imply

$$\log \frac{a_k}{g_p(k)} = O_p(\epsilon_p(k)), \quad \epsilon_p(x) = \frac{\log^{10p+6} x}{x^{p+1}}.$$

This uses $g_p(k)/(2\mathcal{B}_{k+1}) \rightarrow 1$ before taking the logarithm. Since ϵ_p is eventually decreasing and $\epsilon_p(x/2) = O_p(\epsilon_p(x))$, all integer log errors with $k \geq x/2$ are bounded by $C'_p \epsilon_p(x)$.

On $[x-\delta, x+\delta]$, (53) gives $(\log g_p)' \geq \frac{1}{2} \log x$ eventually. Choose C_p large enough that $\delta_p(x) \log x/2 > 2C'_p \epsilon_p(x)$. Integration shows

$$g_p(x-\delta) < L e^{-2C'_p \epsilon_p(x)}, \quad g_p(x+\delta) > L e^{2C'_p \epsilon_p(x)}.$$

Eventual monotonicity extends these comparisons to all integers on the corresponding sides with $k \geq x/2$. After the integer log errors are included, $a_k < L$ for $k \leq x-\delta$ and $a_k > L$ for $k \geq x+\delta$ in that range. For $k < x/2$, monotonicity of a_k , the established leading equivalent, and the Bell derivative estimate show $a_k < L$ as well, since $B_{\lfloor x/2 \rfloor + 1}/\mathcal{B}(x+1) \rightarrow 0$. Taking ceilings gives (54); at an exact lower integer boundary the argument is slightly stronger than the displayed conservative lower bound. Finally $2\delta_p(x) < 1$ eventually. $\square$

Even at high fixed order, (54) is an asymptotic enclosure, not a certified finite-threshold numerical procedure. It does not identify a single ceiling at exact sequence thresholds.

Proposition 11.2 (Real leading scales). *For each fixed $j \geq 1$, as real $x \rightarrow \infty$,*

$$\frac{\mathcal{T}_j(x)}{\mathcal{B}(x+1)} \sim \frac{W(x)^{3j}}{2^j j! x^j}. \quad (55)$$

Proof. For fixed indices in (48), $K_t(Y) = Y^{-(u+j)}(1 + O(1/Y))$ as $Y \rightarrow \infty$. The real Bell concentration and negative-moment bound imply

$$\mathbb{E}_{\mu_{x+1}}[K_t(Y); Y \geq b+1] \sim ((x+1)/W(x+1))^{-(u+j)}.$$

For completeness, below half the saddle the kernel is bounded and the probability exponentially small; above it the normalized kernel is uniformly bounded and converges to one in tilted probability. Thus the leading constant is one.

The monic polynomial $Q_{u,H}$ has highest falling-basis coefficient $q_{u,H,u-1} = 1$. Its highest- t full-sum term in (44), divided by $\mathcal{B}(x+1)$, is therefore asymptotic to $W(x)^{u+j}/x^j$. Every lower- t prefactor has smaller polynomial degree and loses at least $1/x$ with the same kernel exponent. The boundary polynomial is negligible. In (45), configurations with $b < j$ have $u \leq j+b \leq 2j-1$, so are $O(W(x)^{3j-1}/x^j)$. The unique all-pair configuration $b = j$, $h_i = 2$, has highest monomial $z^{2j}/2^j$ and ordering factor $1/j!$. Its leading coefficient is positive and cannot cancel. This proves (55) directly for real arguments. $\square$

Corollary 11.3. *For every fixed $p \geq 1$,*

$$x_p(L) - x_0(L) \sim -\frac{W(x_0)^3}{2x_0 \log x_0} \sim -\frac{W(x_0)^2}{2x_0}, \quad x_0 = x_0(L). \quad (56)$$

For successive fixed-order centers, with $y = x_{p-1}(L)$,

$$x_p(L) - x_{p-1}(L) \sim -\frac{W(y)^{3p}}{2^p p! y^p \log y}. \quad (57)$$

Proof. At the uncorrected center, Proposition 11.2 gives

$$\log \frac{g_p(x_0)}{L} \sim \frac{W(x_0)^3}{2x_0} =: \eta_0 > 0.$$

All fixed higher corrections are negligible relative to the first. The logarithmic derivative of g_p is between $\frac{1}{2} \log x_0$ and $2 \log x_0$ in a unit neighborhood for large L . Monotonicity first brackets x_p within $O(\eta_0/\log x_0)$ to the left of x_0 . Integrating the sharper equivalent (53) over this shrinking interval gives $(x_0 - x_p) \log x_0 \sim \eta_0$, proving (56). For (57), use $g_{p-1}(y) = L$ and $g_p(y) = L + 2\mathcal{T}_p(y)$, so

$$\log \frac{g_p(y)}{L} \sim \frac{W(y)^{3p}}{2^p p! y^p}.$$

The same bracketing and integration applies. This retains the full previous corrected center and does not subtract unrelated estimates for the integer inverse. $\square$

The negative sign concerns smooth centers at sufficiently large arguments. It is not a finite- k lower-bound claim for a walk subclass. Neither displacement formula narrows the two-ceiling uncertainty to a single rounding rule.

Remark 11.4 (The inverses and the transseries volume; added by the write, 7 October 2026). Compared statement by statement with the repository’s transseries volume [18]; its symbols are not used here.

1. Theorem 11.1, and Part II’s Proposition 25.1, enclose the integer staircase $N(L)$ (the volume’s N_* of `p0:def:three-inverses(1)`; a_k is strictly increasing for $k \geq 1$) between two ceilings around the inverse of a smooth comparison function, g_p here and $g = 2\mathcal{B}(x+1) = g_0$ in Part II. These functions are not interpolations of a_k ($g_p(k) \neq a_k$ in general), so the exact rounding of `p0:thm:staircase(1)` does not apply to them, and the Parts do not use it: both brackets are proved directly at the integers, from the integer-index error, the monotonicity of a_k and a lower bound for $(\log g_p)'$. They are analogues of `p0:thm:staircase(2)`, not instances. As in item (2), they do not decide $N(L)$ when the two ceilings differ, which both Parts say.
2. The centres $x_p(L)$, and Part II’s $x(y)$, are implicit inverses of explicit smooth functions, defined by monotonicity; no Part prints a formal inverse series. The Bell scale lies outside the volume’s exponential–power class `p0:def:model`, since $\log B_N \sim N \log N$ grows faster than aN^α for every $a > 0$ and $\alpha \leq 1$. The centres and the displacements of Corollary 11.3 are therefore not shown to be instances of `p0:thm:lambert-core`, `p0:thm:lambert-centered` or `plt:thm:lw-template`; the write did not examine whether a formal inverse expansion of g_p exists that is one.
3. The volume’s chapter on the Bell numbers (`q2:sec:bell`) gives the all-orders forward saddle expansion of B_n at $r = W(n)$ (`q2:thm:bell`) and names $e^{-r} = r/n$ as its transmonomial (`q2:rem:bell-scale`). The Parts do not use that expansion: they need only fixed-shift Bell ratios and the concentration Lemma A.1. Their correction scale is built from the same quantities: with $r = W(k)$, $W(k)^{3j}/k^j = r^{2j}(e^{-r})^j$, because $re^r = k$.

12 Exact finite checks and reproduction

The child-return rows are computable from the exact first-edge recurrence

$$F_{k,m} = \sum_{q=1}^m \binom{m-1}{q-1} \sum_{a+b=k-q} E_{a,q} F_{b,m-q}. \quad (58)$$

The child used by the first excursion is used q times. Choose its other positions among the remaining $m-1$ excursions, remove its cut interior walk, and retain the walk formed by all other root excursions. All recursive half-lengths are smaller than k . This is the Bauer–Golinelli departure-refined recurrence after summing out the distinct-edge count [1, Section 5.4]; it is recalled for reproducibility, not claimed as a new recurrence.

Table 1: Small exact child-return rows with zeros in impossible positions

k	$F_{k,1}$	$F_{k,2}$	$F_{k,3}$	$F_{k,4}$	$F_{k,5}$
1	1	0	0	0	0
2	1	2	0	0	0
3	3	4	5	0	0
4	12	15	15	15	0
5	57	68	66	60	52

Table 2: Finite correction diagnostics, rounded from exact rational values

k	$a_k/(2B_{k+1})$	$A_1(k)$	$A_2(k)$	Residual
2	0.300000	0.000000	0.000000	-0.700000
4	0.548077	0.807692	0.384615	-1.644231
8	1.639263	1.089895	1.003547	-1.454178
12	3.217246	0.976492	0.808301	0.432453
16	4.411279	0.893527	0.675445	1.842307
24	4.327739	0.779740	0.507240	2.040759
32	3.222888	0.702235	0.405270	1.115383

The last column of Table 2 is the exact rational residual $a_k/(2B_{k+1}) - 1 - A_1(k) - A_2(k)$, displayed in decimal form. These small indices need not be in the asymptotic range. In particular a sign, ratio, or apparent fit in this table is not an onset estimate.

The public code uses only the Python standard library for exact finite arithmetic. It freshly generates rows through $k = 32$ and independently enumerates normalized walks through $k = 7$, by following a discovered neighbor or discovering the next vertex. That enumeration does not use the recurrence, partitions, or stored moments to generate walks. The checks compare all enumerated root-return rows, reconstruct finite differences and Bell polynomials, test Newton exactness on low-collision histories, compare urn factorial moments, and check the explicit G_1, G_2 reductions and finite interpolation identities. The small scope and exact check counts are recorded in the regenerated data; no large arrays are shipped.

The release includes editable T_EX, this PDF, compact exact results, source links, and a reproduction driver. Explicit exception guards remain active under `python3 -O`. The finite harness runs normal and optimized modes, compares all output bytes, and exercises deliberate mathematical and validation failures. The outer driver verifies the source manifest, regenerates the finite outputs, rebuilds the PDF to stable bytes using a private format and installed font maps, and verifies actual archive members. With the original archive supplied, it compares the entire rebuilt ZIP byte for byte. These are same-toolchain reproducibility claims, not claims that arbitrary T_EX installations produce identical PDF bytes. No private reports, reviews, third-party full texts, or compiled binaries are included.

13 Sources and further directions

Bauer–Golinelli [1] supply the original sparse-moment tree-walk framework, the departure-refined first-edge recurrence, and the coefficientwise Stirling–Catalan comparison used in the inherited complement theorem. Report213 [8] establishes the full relative equivalent and superpolynomial complement reduction refined here. This article includes the needed complement argument for accessibility and preserves the distinction between that inherited theorem and the collision expansion.

The same-count literature also includes Khorunzhy–Vengerovsky [2], whose equations (6a,b) give a departure-refined recurrence and whose remarks give rough moment bounds. Bauer–Golinelli’s equation (13) gives $I(k, \ell) \sim 2 \left\{ \begin{smallmatrix} k \\ \ell \end{smallmatrix} \right\}$ for fixed $\ell \geq 2$. That fixed- ℓ statement cannot be summed

through the growing Bell saddle without a uniform error estimate. Spiridonov’s thesis [3, Section 3.3] records Bell–Catalan bounds and a historical conjecture of a Catalan–Stirling upper bound; the latter is already implied by Bauer–Golinelli’s coefficientwise theorem and is not a current open

input.

Hainzl–de Panafieu [4] organize tree-walk generating functions at fixed excess and obtain large-mean-degree expansions of fixed moments. Their inspected Theorems 3 and 4 use different limiting regimes. Hainzl [5] proves asymptotics for hypergraph Catalan numbers, counting tours that cross every edge exactly 2κ times with κ fixed while tree size tends to infinity. Her Theorem 1 and Proposition 1 identify stars or star-like trees and quantify a defect expansion. This is directly relevant methodological context for dominant-hub and defect arguments. Here edge multiplicities are unrestricted and summed over at fixed total walk half-length; neither the objects nor the asymptotic parameter agrees with the fixed- κ tour problem.

Condensation results for fixed-size weighted planar trees and allocations, including the factorial-weight partition-function equivalents in Janson–Jonsson–Stefánsson [6] and Janson [7], are related background rather than identities for unrestricted time-rooted walk counts.

The source record is a bounded comparison of specific inspected statements, not a worldwide literature classification. Full-text gaps include Khorunzhy–Shcherbina–Vengerovsky (2004), Rodgers–Bray (1988), and the 2026 journal version of Hainzl–de Panafieu; the ledger gives their links and distinguishes them from inspected preprints. No claim is made that these asymptotics were historically unknown or that the further directions below are globally open. The reproducibility file `SOURCES.txt` lists the public links and the precise inspection scope.

Concrete next directions. First, reducing the deliberately generous logarithmic remainder powers would improve practical inverse radii, but requires sharper weighted collision and partition estimates. Second, effective constants for the complement and local errors would turn the existential two-ceiling enclosure into a certified finite-threshold procedure. Third, computing lower logarithmic suborders of each fixed Bell transform would give an evaluated W -coefficient expansion; exact lower transforms must still be retained until their errors are below the target scale. Fourth, a uniform analysis for $p = p(k)$ would require quantitative control of coefficient growth and collision tails; the factor $1/j!$ in the fixed- j leading term alone does not justify exponential resummation. Fifth, fixed general mean degree introduces weighted Bell or Touchard structures and calls for its own uniform proof. These are specific continuations of the present method, not conclusions already established by it.

[write] **Added 7 October 2026.** Report 214 is the last manuscript of the chain, so none of these five directions is answered by another Part. They are items 1–5 of Section 56, which gathers the open questions of all five Parts. “Report213” in this Part is Part II of this report.

A Elementary Bell estimates

[write] **Printed once, 7 October 2026.** Appendices A and B of this Part are the only print in this report of three blocks that Reports 212, 213 and 214 repeat: this Appendix A (the elementary Bell estimates; Report 213’s Appendix A word for word, and, for its Lemma A.1, the first part of Report 212’s Appendix A in other words), and Appendix B.2–B.6 (Report 213’s Sections 6–9 and the first part of its Section 10, word for word up to the differences named in Part II). Parts II and III keep the headings of those sections, with notes that map their numbers to the ones printed here. Appendix B.1 restates in other words inputs that Part II prints in its Sections 14.1–14.2 (the fixed-moment identification, Hölder’s inequality, the Bauer–Golinelli comparison and the Touchard envelope); both texts are kept, because Part II’s statement section depends on its own.

This appendix establishes every Bell asymptotic needed above. Dobinski's identity follows from exponential generating functions:

$$e^{-1} \sum_{y \geq 0} \frac{e^{zy}}{y!} = \exp(e^z - 1) = \sum_{N \geq 0} B_N \frac{z^N}{N!}, \quad B_N = e^{-1} \sum_{y \geq 0} \frac{y^N}{y!}.$$

For $N \geq 1$, put $\rho = W(N)$, $x = N/\rho = e^\rho$, and $a_y = y^N/y!$ for $y \geq 1$. Let Y_N have probabilities proportional to a_y .

Lemma A.1. *For every fixed $\varepsilon \in (0, 1)$, there is $c_\varepsilon > 0$ such that, for all sufficiently large N ,*

$$\mathbb{P}(|Y_N/x - 1| > \varepsilon) \leq e^{-c_\varepsilon N}. \quad (59)$$

For every fixed positive integer p , $\mathbb{E}(Y_N/x)^p \rightarrow 1$. In particular,

$$\frac{B_{N+1}}{B_N} \sim \frac{N}{W(N)}, \quad \frac{B_{N-1}}{B_N} = O(\log N/N). \quad (60)$$

Proof. The consecutive mass ratio

$$\frac{a_{y+1}}{a_y} = \frac{(1 + 1/y)^N}{y + 1}$$

is decreasing in y . Uniformly for $y = tx + O(1)$, with t in a fixed compact subset of $(0, \infty)$,

$$\log \frac{a_{y+1}}{a_y} = \rho(1/t - 1) - \log t + o(1), \quad (61)$$

since $N/y^2 = O(\rho^2/N) = o(1)$. On the interval between $(1 - \varepsilon)x$ and $(1 - \varepsilon/2)x$, these logarithms are at least a positive constant times ρ . Multiplying a positive constant times x successive ratios produces a factor $\exp(cN)$. Every earlier term is bounded by the exterior endpoint mass, by ratio monotonicity, and there are $O(x)$ such terms. Thus the lower tail is exponentially negligible relative to an interior mass.

Across the interval between $(1 + \varepsilon/2)x$ and $(1 + \varepsilon)x$, the log ratios are at most a negative constant times ρ . This loses a factor $\exp(cN)$. Beyond that interval monotonicity bounds the remaining tail by a geometric series of ratio at most $e^{-c'\rho}$. This proves (59) after changing constants.

The upper-tail argument still works when the summands are multiplied by a fixed power $(y/x)^p$: successive polynomial factors have ratio $(1 + 1/y)^p$, uniformly tending to one there, while the mass ratio tends to zero. On the central interval, $(Y_N/x)^p$ is between $(1 - \varepsilon)^p$ and $(1 + \varepsilon)^p$. Letting $\varepsilon \downarrow 0$ proves convergence of moments. Finally $\mathbb{E}Y_N = B_{N+1}/B_N$, proving (60); apply it at adjacent indices and use $W(N) \sim \log N$ for the second bound. $\square$

The shifted law used in Lemma 3.2 is exact. Under weight $(X + 1)^k$ with $X \sim \text{Poisson}(1)$, put $y = X + 1$ to obtain

$$\frac{(X + 1)^k}{X!} = \frac{y^{k+1}}{y!}.$$

Thus $X + 1$ has the law of Y_{k+1} . In particular, with $x_k = (k + 1)/W(k + 1)$, $X/x_k \rightarrow 1$, and for large k , $X < x_k/2$ implies $X + 1 < 3x_k/4$. This gives the exponential low-tail estimate used in the fresh-term binomial mixture.

Lemma A.2. *For integers $N \geq 1$ and $r = W(N)$,*

$$N!r^{-N} \exp(e^r - 1) \leq B_N \exp(O(\log(N+1))). \quad (62)$$

Furthermore,

$$\log B_N = N \log N - N \log \log N - N + o(N), \quad B_N^{-1/N} \leq C \frac{\log(N+1)}{N} \quad (63)$$

for one fixed C and all $N \geq 1$.

Proof. Put $y = e^r = N/r$ and let q be a nearest positive integer to y . In the single Dobinski term $e^{-1}q^N/q!$, use Stirling's formula. The real exponent

$$f(u) = N \log u - u \log u + u$$

has $f'(y) = 0$ and $f''(u) = -N/u^2 - 1/u$. Rounding changes f by $O((r+1)/y)$. Consequently

$$\log B_N \geq N \log N - N \log r - N + y - O(\log N).$$

Stirling applied to $N!r^{-N}e^{y-1}$ gives the same main expression with $O(\log N)$ error, proving (62).

For the logarithmic equivalent, (60) gives $\log(B_{j+1}/B_j) = \log j - \log \log j + o(1)$. Sum over j , treating finitely many small indices separately. Integral comparison gives $\sum_{j < N} \log j = N \log N - N + O(\log N)$ and $\sum_{2 \leq j < N} \log \log j = N \log \log N + o(N)$. The Cesàro mean of the $o(1)$ error is $o(1)$. This proves the first formula in (63); exponentiation and enlargement of C give the second. $\square$

B The inherited superpolynomial complement theorem

This appendix recalls the part of Report213 [8] used in (13). Its purpose is to make the dependency independently readable. The coefficientwise counting theorem below is credited to Bauer–Golinelli; the complement proof is inherited from the preceding report, not a consequence of the new collision expansion.

Theorem B.1 (Complement estimate). *For an integer window $S = S_k$ satisfying*

$$S_k / \log k \rightarrow \infty, \quad S_k = o(k^{1/3}), \quad (64)$$

let $\mathcal{U}_{k,S} = \{w \in \mathcal{W}_k : \max_v d_w(v) \geq k - S\}$. Then, for every fixed $D > 0$,

$$|\mathcal{W}_k \setminus \mathcal{U}_{k,S}| = o(k^{-D} B_{k+1}).$$

The proof uses a low-maximum estimate, an intermediate root-return estimate, and a near-hub shoulder estimate. The root-distance encoding and weighted rotation have already been proved in the main text.

B.1 The counting input and the moment envelope

Let $I(t, \ell)$ count walks with ℓ distinct edges. Bauer–Golinelli [1, Section 5.5, equation (9)] prove the coefficientwise comparison

$$\left\{ \begin{matrix} t \\ \ell \end{matrix} \right\} \leq I(t, \ell) \leq \text{Cat}_\ell \left\{ \begin{matrix} t \\ \ell \end{matrix} \right\}. \quad (65)$$

This is the essential external counting theorem. It is stronger than the scalar Bell–Catalan bounds. The star-rooted subclass independently gives $a_t \geq B_t$.

To get a rough upper envelope, define the Touchard polynomial $T_t(c) = \sum_{\ell} \left\{ \begin{smallmatrix} t \\ \ell \end{smallmatrix} \right\} c^\ell$, with exponential generating function $\exp(c(e^z - 1))$. Since $\text{Cat}_\ell \leq 4^\ell$, positivity and a positive-radius coefficient bound at $r = W(t)$ give

$$a_t \leq T_t(4) \leq t! r^{-t} \exp(4(e^r - 1)).$$

The single-saddle bound in Appendix A compares this with B_t , so for one fixed constant D_0 and all integers $t \geq 1$,

$$a_t \leq B_t \exp\left(\frac{D_0 t}{\log(t+1)}\right). \quad (66)$$

Indeed the logarithmic ratio is at most $3t/W(t) + O(\log t)$; enlarge D_0 over finitely many small indices.

We also need an exact moment interpolation inequality

$$a_t \leq a_s^{t/s} \quad (s \geq 1, 0 \leq t \leq s). \quad (67)$$

For clarity, it does not require interchanging a graph-size limit with a growing moment. Let A_N be the adjacency matrix of $G(N, 1/N)$. At each fixed t , the normalized trace expansion has finitely many connected patterns. A pattern with v vertices and e edges contributes $(N)_v N^{-e-1}$. Since $e \geq v - 1$, exactly the tree patterns survive, each with limit one. Thus

$$\lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E} \text{tr}(A_N^{2t}) = a_t.$$

This is the fixed-moment identification in Bauer–Golinelli [1, Section 5.3]. Push the expected empirical spectral measure under $\lambda \mapsto \lambda^2$ and apply Hölder, then take the fixed-index limits. This proves (67) for every integer pair; it can subsequently be used at indices depending on k .

B.2 A finite positive coefficient majorant

The exact branch decomposition (9) and its upper cut bound give, for $m \geq 1, s \geq 0$,

$$F_{m+s,m} \leq \sum_{\pi \in \text{Part}[m]} \sum_{\sum_b t_b = s} \prod_{b \in \pi} a_{t_b} \binom{|b| + t_b - 1}{t_b}.$$

For a fixed target s , truncate each block's interior half-length to $0 \leq t \leq s$ *before* evaluating any series. The labeled-partition exponential formula and coefficient positivity imply, for every $x, z > 0$,

$$F_{m+s,m} \leq m! [x^m z^s] \exp\left(e^x - 1 + \sum_{t=1}^s a_t z^t \mathcal{A}_t(x)\right) \quad (68)$$

$$\leq m! x^{-m} z^{-s} \exp\left(e^x - 1 + \sum_{t=1}^s a_t z^t \mathcal{A}_t(x)\right), \quad (69)$$

where, for $t \geq 1$,

$$\mathcal{A}_t(x) = \sum_{q \geq 1} \frac{x^q}{q!} \binom{q+t-1}{t} = e^x \sum_{j=1}^t \binom{t-1}{j-1} \frac{x^j}{j!}. \quad (70)$$

In the coefficient expression x, z are formal variables; in the next line they are positive evaluation points. The exponent contains only finitely many moment indices. Terms $t > s$ cannot affect the desired coefficient, so this is a finite-majorant argument, not an assumption that an ordinary moment generating function converges.

The identity in (70) follows from

$$\binom{q+t-1}{t} = \sum_{j=1}^t \binom{t-1}{j-1} \binom{q}{j}$$

and $\sum_{q \geq j} \binom{q}{j} x^q / q! = e^x x^j / j!$. Two bounds will handle different ranges of t :

$$\mathcal{A}_t(x) \leq e^x x(1+x)^{t-1}, \quad \mathcal{A}_t(x) \leq e^x \exp(2\sqrt{tx}). \quad (71)$$

For the first, discard $1/j!$ in the finite sum. For the second, use $\binom{t-1}{j-1} \leq \binom{t}{j} \leq t^j / j!$ and extend to $j \geq 0$. The inequality

$$\sum_{j \geq 0} \frac{u^j}{(j!)^2} \leq e^{2\sqrt{u}}$$

follows by comparing to the even powers on the right: $\binom{2j}{j} \leq 4^j$. Both estimates hold for every $t \geq 1$ and $x > 0$.

B.3 Bell entropy with cancellation preserved

Define

$$H(m, s) = (m + s) \log(m + s) - m \log m - s \log s.$$

This is positive when $m, s > 0$. In particular,

$$H(m, s) \geq \min(m, s) \log 2, \quad H(m, s) \geq \frac{2ms}{m+s}. \quad (72)$$

The second follows from $-\log u \geq 1 - u$ applied to $u = m/(m + s)$ and $1 - u$. More precisely $H(m, s) \geq p \log(k/p)$ for $p = \min(m, s)$ and $k = m + s$.

Lemma B.2. *Uniformly over integers $m + s = k$ with $\min(m, s) \geq k^{0.49}$,*

$$\log \frac{B_m B_s}{B_k} = -(1 + o(1)) H(m, s). \quad (73)$$

The notation means that the ratio of the left side to $-H(m, s)$ tends uniformly to one.

Proof. Appendix A proves the adjacent ratio and hence

$$\log \frac{B_{j+1}}{B_j} = \log j - \log \log j + o(1), \quad \log B_p = p \log p - p \log \log p - p + o(p).$$

Put $p = \min(m, s)$ and $q = k - p \geq k/2$. The adjacent-ratio errors are uniformly $o(1)$ on $j \geq q$, so their sum over $q \leq j < k$ is $o(p)$. Also

$$\begin{aligned} \sum_{j=q}^{k-1} \log j &= k \log k - q \log q - p + O(\log k), \\ \sum_{j=q}^{k-1} \log \log j &= p \log \log k + O\left(\frac{p^2}{k \log k}\right). \end{aligned}$$

The first error is $o(p)$ under $p \geq k^{0.49}$; so is the second. Subtracting this *short sum of adjacent ratios* gives

$$\log \frac{B_p B_q}{B_k} = -H(m, s) + p \log \frac{\log k}{\log p} + o(p).$$

Finally

$$0 \leq \log \frac{\log k}{\log p} \leq \frac{\log(k/p)}{\log p}, \quad H(m, s) \geq p \log(k/p) \geq p \log 2.$$

Thus the remaining terms are $o(H)$ uniformly. Subtracting three separate coarse $o(k)$ Bell estimates would not preserve the required cancellation near the small-part endpoint; the adjacent-ratio argument does. $\square$

B.4 The intermediate return range

Set

$$h = \left\lfloor \frac{k}{64 \log k} \right\rfloor, \quad L = \frac{1}{20} \sqrt{k \log k}. \quad (74)$$

The letter L here is a real cutoff, distinct from the largest-block cutoff in the main text. Eventually $h > L > k^{0.49}$.

Proposition B.3. *Uniformly for integers $m + s = k$, $m \geq h$, and $s \geq L$, for all sufficiently large k ,*

$$\frac{F_{k,m}}{B_{k+1}} \leq \exp\{-H(m, s)/2\}. \quad (75)$$

Proof. In (69) choose

$$x = W(m), \quad \epsilon = \frac{m}{8k}, \quad \eta = 1 - \epsilon, \quad z = \eta a_s^{-1/s}, \quad T = \left\lceil \frac{64 \log k}{\epsilon^2} \right\rceil. \quad (76)$$

The star subclass gives $a_s \geq B_s$, and Appendix A gives

$$z \leq C \frac{\log(s+1)}{s}, \quad x \leq \log k, \quad \frac{c}{\log k} \leq \epsilon \leq \frac{1}{8}.$$

Uniformly in the stated range,

$$\frac{\epsilon^2 s}{(\log k)^3} \rightarrow \infty, \quad T < s, \quad T(1+x)z = o(1). \quad (77)$$

Indeed if $m \geq k/2$, then $\epsilon^2 s \geq s/256 \geq L/256$; if $m < k/2$, then $s \geq k/2$ and $\epsilon^2 s \geq m^2/(128k) \geq c'k/\log^2 k$. Both dominate $\log^3 k$. The last conclusion follows from $T(1+x)z \leq C \log^3 k/(\epsilon^2 s)$, with the harmless ceiling absorbed.

Large interior half-lengths. For $T < t \leq s$, (67) yields $a_t z^t \leq \eta^t \leq e^{-\epsilon t}$. Since $T \geq 64x/\epsilon^2$, one has $2\sqrt{tx} \leq \epsilon t/2$. The second bound in (71) now gives

$$\sum_{t>T}^s a_t z^t \mathcal{A}_t(x) \leq e^x \sum_{t>T} e^{-\epsilon t/2} = O(k^{-10}). \quad (78)$$

For an explicit comparison of scales, $e^x \leq k$, $(1 - e^{-\epsilon/2})^{-1} = O(\log k)$, and $\epsilon T/2 \geq 32 \log k/\epsilon \geq 256 \log k$.

Small interior half-lengths. Use the rough walk bound (10), the first bound in (71), and $xe^x = m$:

$$\begin{aligned} \sum_{t=1}^T a_t z^t \mathcal{A}_t(x) &\leq mz \sum_{t=1}^T 4^{t-1} t! [(1+x)z]^{t-1} \\ &= mz \{1 + O((1+x)z)\} = mz(1 + o(1)). \end{aligned} \quad (79)$$

The first term of the inner sum is one, the second is $8(1+x)z$, and each successive ratio is $4(t+1)(1+x)z = o(1)$ uniformly by (77). The exact value $a_1 = 1$ prevents any loss in its leading term.

Combining (69), (62), and the two splits gives

$$\log \frac{F_{k,m}}{B_{k+1}} \leq \log \frac{B_m a_s}{B_k} - s \log \eta + O(\log k) + (1 + o(1))mz. \quad (80)$$

Replacing B_{k+1} by the smaller B_k only weakens the upper bound. We compare every remaining term to $H = H(m, s)$.

First, (66) and Lemma B.2 show

$$\log \frac{B_m a_s}{B_k} \leq -(1 - o(1))H + \frac{D_0 s}{\log(s+1)}.$$

The positive error is $o(H)$. If $s \leq k/2$, use $H \geq s \log 2$ and $s \rightarrow \infty$. If $s > k/2$, then $h \leq m < k/2$ and

$$H \geq m \log(k/m) \geq c \frac{k \log \log k}{\log k}, \quad \frac{s}{\log(s+1)} = O(k/\log k).$$

The lower bound follows by the endpoints and the sole interior maximum of $m \log(k/m)$ on $[h, k/2]$.

Second, since $\epsilon \leq 1/8$,

$$-s \log \eta \leq \frac{s\epsilon}{1-\epsilon} \leq \frac{ms}{7k} \leq \frac{H}{14}. \quad (81)$$

Third, $mz = o(H)$. For $s > k/2$, divide $mz \leq Cm \log(s+1)/s$ by $H \geq m \log 2$. For $L \leq s \leq k/2$, divide by $H \geq s \log(k/s)$ to get

$$\frac{mz}{H} \leq \frac{Ck \log(s+1)}{s^2 \log(k/s)}.$$

If $s \leq k^{3/4}$ this is at most $Ck/s^2 = O(1/\log k)$; if $s \geq k^{3/4}$ it is $O(\log k/\sqrt{k})$. Both tend uniformly to zero.

Finally, $\min(m, s) \geq L > k^{0.49}$ eventually and $H \geq L \log 2$, so $\log k = o(H)$. Equation (80) is therefore at most $-(13/14 - o(1))H$, proving the weaker convenient bound (75). $\square$

There are at most k entries in this range and their rotation weights are at most $2k/h = O(\log k)$. Since $L/\log k \rightarrow \infty$, their weighted sum is $o(k^{-p} B_{k+1})$ for every fixed $p > 0$.

B.5 The high root shoulder with a growing tilt

The main-text collision argument treated a small largest-block tilt. The shoulder between that window and L requires a separate estimate that permits the tilt to grow.

Lemma B.4. *For the largest block A of a uniform partition of $[n]$, there is a fixed C such that, uniformly for $0 \leq \tau \leq 0.02 \log n$ and all sufficiently large n ,*

$$\log \mathbb{E} e^{\tau A} \leq C e^{\tau} \log n. \quad (82)$$

Proof. The expected number of j -blocks is $\binom{n}{j} B_{n-j} / B_n$. For $j \leq n/2$, adjacent Bell ratios on $[n/2, n]$ give

$$\binom{n}{j} \frac{B_{n-j}}{B_n} \leq \frac{(C \log n)^j}{j!}.$$

Since $e^{\tau A}$ is bounded by one plus the sum of $e^{\tau|b|}$ over the blocks, summing these terms costs at most $\exp(C e^{\tau} \log n)$. For $j > n/2$, retain the first $\lfloor n/2 \rfloor$ adjacent ratios and use Bell monotonicity for the remainder. The total is at most

$$(1 + e^{\tau})^n \left(\frac{C \log n}{n} \right)^{\lfloor n/2 \rfloor} \leq \exp(-cn \log n).$$

Indeed its logarithm is at most $(0.02 - 1/2 + o(1))n \log n$. Taking a logarithm of the whole upper bound and enlarging C proves the assertion uniformly. $\square$

Proposition B.5. *For any $S = S_k$ satisfying (64), uniformly over integers $S \leq s \leq L$,*

$$\frac{F_{k,k-s}}{B_{k+1}} \leq \exp \left\{ -\frac{1}{2} s \log(s / \log k) \right\} \quad (83)$$

for all sufficiently large k .

Proof. Put $n = k - s$ and $\tau = 7s(s-1)/(2n)$. The conditional urn estimate (22) holds for arbitrary s , without a small-tilt assumption. Therefore

$$F_{k,k-s} \leq \binom{k-1}{s} B_{k-s} \exp \left\{ \frac{7s(s-1)(2s-1)}{6n} \right\} \mathbb{E} e^{\tau A}. \quad (84)$$

Here $n \geq k/2$, and

$$\tau \leq \frac{7s^2}{k} \leq 0.0175 \log k < 0.02 \log n$$

eventually. Lemma B.4 applies. The other urn exponent is at most $(14/3)s^3/k$.

Adjacent Bell ratios on $[k-s, k+1]$ and $\binom{k-1}{s} \leq (ek/s)^s$ give fixed constants C, C' with

$$\binom{k-1}{s} \frac{B_{k-s}}{B_{k+1}} \leq \frac{C \log k}{k} \left(\frac{C' \log k}{s} \right)^s. \quad (85)$$

Set $J(s) = s \log(s / \log k)$. Uniformly for $S \leq s \leq L$, one has $J(s) / \log k \rightarrow \infty$ and $O(s) = o(J(s))$, since $S / \log k \rightarrow \infty$. The cubic error obeys

$$\frac{(14/3)s^3/k}{J(s)} \leq 0.03 \quad (86)$$

for all sufficiently large k . To check the constant uniformly, $s^2 / \log(s / \log k)$ is increasing once $\log(s / \log k) > 1/2$, so the maximum occurs at $s = L$. Its limit there is

$$\frac{28}{3} \left(\frac{1}{20} \right)^2 = \frac{7}{300} < 0.03.$$

The largest-block cost is also negligible: $Ce^\tau \log n = o(J(s))$ uniformly. For $s \leq k^{0.1}$, $\tau = o(1)$, so it is $O(\log k) = o(J)$. For $s > k^{0.1}$, $e^\tau \leq k^{0.02}$ while $J(s) \geq ck^{0.1} \log k$. Taking logarithms of (84) and using (85)–(86) gives $-J + 0.03J + o(J)$, which proves (83). $\square$

The weighted sum over this shoulder is at most

$$4kB_{k+1} \exp\{-\tfrac{1}{2}S \log(S/\log k)\}.$$

It is $o(k^{-p}B_{k+1})$ for every fixed $p > 0$, because $S \log(S/\log k)/\log k \rightarrow \infty$. All walks in the shoulder are covered, including unrestricted depth and repeated secondary edges.

B.6 The maximum departure split

Fix any admissible S . A walk outside $\mathcal{U}_{k,S}$ either has maximum departure count at most h , or has a vertex with count strictly between h and $k - S$.

In the first class, every vertex has at most h used incident edges. The Dyck-path discovery encoding from Proposition 2.1 has at most h choices at each outward step. In detail, when fewer than h children have been discovered, existing choices plus the next fresh child give at most h choices. If h children already exist, a further fresh child is forbidden in this class. There is no hidden $h + 1$ factor. The number of walks is at most

$$\text{Cat}_k h^k \leq (4h)^k.$$

Appendix A gives $\log B_k = k \log k - k \log \log k - k + o(k)$. With the cutoff (74),

$$\frac{(4h)^k}{B_{k+1}} \leq \exp\{-(\log 16 - 1 + o(1))k\}. \quad (87)$$

This is exponentially negligible since $\log 16 > 1$.

By the exact rotation identity (15), the second class is covered by

$$\sum_{h < m < k-S} \frac{2k}{m} F_{k,m}. \quad (88)$$

Write $s = k - m > S$. If $s \geq L$, Proposition B.3 applies. If $s < L$, Proposition B.5 applies, since $S < L$ eventually. The two ranges leave no gap. Their weighted sums are smaller than every fixed negative power of k on the B_{k+1} scale. Together with (87), this proves

$$|\mathcal{W}_k \setminus \mathcal{U}_{k,S}| = o(k^{-p}B_{k+1}) \quad \text{for every fixed } p > 0. \quad (89)$$

This proves Theorem B.1. In particular the window $S = \lceil \log^2 k \rceil$ used in the main text satisfies its hypotheses. The result is superpolynomial on the B_{k+1} scale and is independent of all collision-order refinements.

Part II

The full sparse graph moment equivalent

Bundle Report 213, archive `Report213-reproducibility.zip`, delivered as `Report213.tex` (831 lines, 20 pages). Delivered title block “The full sparse graph moment equivalent / Report213 / 4 October 2026” (empty author line). Printed as delivered, with **[write]** notes, except that its Sections 6–9, the first part of its Section 10 and its Appendix A are not printed again: they are word for word in Part I’s Appendices (a note under each heading maps their numbers, and references to them point to Part I’s copies). Report 213’s Section k is Section $k + 13$ here, its equation (k) is equation $(k + 89)$ (the equations not printed again keep their places in the count), its Table 1 is Table 3, and its appendix keeps the letter A; statements move with their sections. Its labels carry `sgm:eq::`; its table is printed inline. This Part’s own text is self-contained apart from those blocks.

Abstract of this Part (as delivered). For the normalized closed tree walks counted by OEIS A094149, with time zero distinguished, we prove the relative equivalent $M_{2k} \sim 2B_{k+1}$. For any integer window S_k with $S_k / \log k \rightarrow \infty$ and $S_k = o(k^{1/3})$, the complement of the family having a vertex with at least $k - S_k$ departures is $o(k^{-p}B_{k+1})$ for every fixed $p > 0$. The same superpolynomial bound holds for the weighted middle-return sum. The proof includes the arbitrary-depth dominant-window argument, a finite positive-coefficient majorant, a cancellation-preserving Bell entropy estimate, a growing-tilt Pólya-urn shoulder bound, and exact rotation coverage. Its essential external counting input is the coefficientwise comparison of Bauer and Golinelli. An elementary Touchard bound supplies the needed envelope. A Bell-interpolation inverse has an existential two-ceiling enclosure of radius $o(1/\log x)$. The full relative equivalent has only an $o(1)$ error here; no first correction, all-order expansion, effective onset, or priority claim is made.

14 Statement and normalization

A *normalized closed tree walk* is a sequence

$$w = (v_0, v_1, \dots, v_{2k-1}, v_0)$$

whose traversed unordered pairs form a tree and whose consecutive vertices differ. Two sequences are identified only by a vertex bijection preserving their time order. Labels $0, 1, \dots$ assigned by first appearance select a unique representative. Time zero is distinguished: rotation generally gives a different normalized walk. Let $\mathcal{W}_k$ be this set and $a_k = M_{2k} = |\mathcal{W}_k|$, with $a_0 = 1$. For $k \geq 1$ these are the positive-order terms of OEIS A094149 [17].

If an edge e is traversed m_e times in each direction, the number of departures from a vertex v is

$$d_w(v) = \sum_{e \ni v} m_e, \quad \sum_e m_e = k, \quad \sum_v d_w(v) = 2k. \quad (90)$$

These are walk counts, not graph degrees. Write $F_{k,m}$ for the number of walks with m departures from their initial vertex, set $F_{0,0} = 1$, and set impossible entries to zero. Then $a_k = \sum_{m=1}^k F_{k,m}$ for $k \geq 1$. Let B_j denote the Bell numbers, with $B_0 = 1$.

For an integer $S \geq 0$ define

$$\mathcal{U}_{k,S} = \{w \in \mathcal{W}_k : \max_v d_w(v) \geq k - S\}, \quad U_{k,S} = |\mathcal{U}_{k,S}|, \quad C_{k,S} = a_k - U_{k,S},$$

and the weighted middle-return sum

$$R_{k,S} = \sum_{S < m < k-S} \frac{2k}{m} F_{k,m}. \quad (91)$$

The dominant family allows arbitrary tree depth, off-hub repeated edges, and nesting. Its only restriction is a total edge-multiplicity budget of S away from at least one vertex.

Theorem 14.1. *Let $S = S_k$ be any integer sequence satisfying*

$$S_k / \log k \longrightarrow \infty, \quad S_k = o(k^{1/3}). \quad (92)$$

Then for every fixed $p > 0$,

$$C_{k,S_k} = o(k^{-p} B_{k+1}), \quad R_{k,S_k} = o(k^{-p} B_{k+1}). \quad (93)$$

Moreover,

$$M_{2k} \sim 2B_{k+1}. \quad (94)$$

All logarithms are natural. Uniform asymptotic estimates below are for sufficiently large k ; their thresholds and implicit constants are not made effective. Admissible examples include $\lfloor k^{1/4} \rfloor$ and $\lceil \log^2 k \rceil$. Slow finite onset is possible. The strength of (93) applies to the two specified tails, *not* to $M_{2k}/(2B_{k+1}) - 1$.

The proof is self-contained apart from the explicitly stated Bauer–Golinelli coefficientwise counting theorem. Every needed dominant-window argument is included; no earlier report is required. Section 25 gives the inverse consequence and states its limits. The finite checks in Section 26 verify exact formulas and encodings, not asymptotic thresholds.

[write] **Added 7 October 2026.** Within this report: Theorem 14.1 proves, in superpolynomial form and for every admissible window, the premise $R_{k,S_k} = o(B_{k+1})$ of Part III’s sufficient implication (162); the weighted middle-return sum (91) is Part III’s (161). Theorem 14.2 is Part III’s Theorem 28.1, the same statement (Report 212 writes the window condition $S_k = o(k^{1/3})$ as $S_k^3/k \rightarrow 0$). Part I (Report 214) refines (94) to every fixed collision order. The proof of Theorem 14.2 (Sections 15–18) is printed here once; Part III’s copy is not printed again. Report 213’s Sections 6–9, the first part of its Section 10 and its Appendix A are printed once, in Part I’s Appendices A and B; the sentence above that the proof “is self-contained” is true of Report 213 as delivered, and in this report those blocks are read in Part I.

14.1 Fixed sparse graph moments and Hölder

Let A_N be the unscaled adjacency matrix of $G(N, 1/N)$. For every fixed integer $t \geq 0$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E} \operatorname{tr}(A_N^{2t}) = a_t. \quad (95)$$

For $t = 0$ both sides equal one. At fixed positive t , expand the trace into normalized patterns. A connected pattern with v vertices and e distinct edges contributes $(N)_v N^{-e-1}$. There are finitely many patterns. Since $e \geq v - 1$, precisely tree patterns survive, each with limit one. This is the tree-walk moment identification of Bauer–Golinelli [1, Section 5.3], with its elementary argument recalled here.

For finite N , the expected empirical spectral measure is a probability measure. Push it forward under $\lambda \mapsto \lambda^2$ to a probability measure on $[0, \infty)$. Hölder’s inequality for that measure, followed by the fixed-moment limits (95), gives

$$a_t \leq a_s^{t/s} \quad (s \geq 1, 0 \leq t \leq s). \quad (96)$$

This exact inequality is now available for every integer pair. Applying it later to indices depending on k does not interchange $N \rightarrow \infty$ with a growing-moment limit. No infinite-dimensional spectral construction or high-probability extreme-eigenvalue theorem is needed.

14.2 The external counting theorem and an elementary envelope

Let $I(t, \ell)$ count these normalized walks using exactly ℓ distinct edges, let $\left\{ \begin{smallmatrix} t \\ \ell \end{smallmatrix} \right\}$ be a Stirling number of the second kind, and let $\text{Cat}_\ell = (\ell + 1)^{-1} \binom{2\ell}{\ell}$. The essential external input is Bauer–Golinelli [1, Section 5.5, equation (9)]:

$$\left\{ \begin{smallmatrix} t \\ \ell \end{smallmatrix} \right\} \leq I(t, \ell) \leq \text{Cat}_\ell \left\{ \begin{smallmatrix} t \\ \ell \end{smallmatrix} \right\}. \quad (97)$$

The coefficientwise comparison is a credited prior theorem. Its sum provides more than a fixed- ℓ star asymptotic. Only the upper comparison is essential below; the elementary star-rooted subclass also gives $a_t \geq B_t$.

Define the Touchard polynomial $T_t(c) = \sum_\ell \left\{ \begin{smallmatrix} t \\ \ell \end{smallmatrix} \right\} c^\ell$. Its exponential generating function is

$$\sum_{t \geq 0} T_t(c) \frac{z^t}{t!} = \exp(c(e^z - 1)).$$

Since $\text{Cat}_\ell \leq 4^\ell$, positivity and (97) give, at $r = W(t)$,

$$a_t \leq T_t(4) \leq t! r^{-t} \exp(4(e^r - 1)).$$

Here $W(t)e^{W(t)} = t$, with $W(t) > 0$. The single-Dobinski-term estimate proved in Appendix A says

$$t! r^{-t} \exp(e^r - 1) \leq B_t \exp(O(\log(t + 1))). \quad (98)$$

Consequently, for one fixed $D > 0$ and all integers $t \geq 1$,

$$a_t \leq B_t \exp\left(\frac{Dt}{\log(t + 1)}\right). \quad (99)$$

Indeed the logarithmic ratio is at most $3t/W(t) + O(\log t)$; use $W(t) \sim \log t$ and enlarge D for finitely many indices. No sharper Stirling–Catalan envelope expansion is needed.

Theorem 14.2 (Dominant window). *Under (92),*

$$F_{k,k-s} = (1 + o(1)) \binom{k-1}{s} B_{k-s} \quad \text{uniformly for } 0 \leq s \leq S_k, \quad (100)$$

$$\sum_{s=0}^{S_k} F_{k,k-s} \sim B_{k+1}, \quad (101)$$

$$U_{k,S_k} \sim 2B_{k+1}. \quad (102)$$

The next four sections prove this auxiliary theorem for full trees. The tail argument then completes Theorem 14.1.

15 Exact decomposition into child branches

For $t \geq 0$ and $q \geq 1$ put

$$E_{t,q} = \sum_{j=0}^t F_{t,j} \binom{j+q-1}{q-1}. \quad (103)$$

A child-rooted closed walk with j root excursions can be divided into q ordered, possibly empty closed segments in $\binom{j+q-1}{q-1}$ ways: distribute its j consecutive excursions among q slots. Cuts must be at returns to the child root; repeated cuts and empty end slots are allowed. Thus $E_{t,q}$ counts all child walks of half-length t with such a division. In particular $E_{0,q} = 1$.

Proposition 15.1. *For $n \geq 1$ and $s \geq 0$,*

$$F_{n+s,n} = \sum_{\pi \in \text{Part}[n]} \sum_{\substack{t_b \geq 0 \\ \sum_{b \in \pi} t_b = s}} \prod_{b \in \pi} E_{t_b, |b|}. \quad (104)$$

Moreover, for all $t \geq 0, q \geq 1$,

$$\binom{t+q-1}{t} \leq E_{t,q} \leq M_{2t} \binom{t+q-1}{t}. \quad (105)$$

Proof. Cut a root-based walk at its n returns to the root. Two excursion positions belong to the same block of π precisely when they use the same child. A block b of size q therefore specifies q visits to one branch. Concatenate their interior child walks; they have total half-length t_b and are divided into q closed segments. Distinct branches have disjoint vertex sets because the underlying graph is a tree. Their interiors use total half-length s .

Conversely, choose the partition, one divided child walk for each block, and interleave the segments according to the block positions. Only then assign global first-appearance labels. Even if visits to different children interleave, local labels in disjoint branches reconstruct the global labels uniquely. There is no additional labeling factorial. This proves (104).

For the upper bound in (105), use $j \leq t$ in (103) and sum the $F_{t,j}$. For the lower bound, take the single child walk consisting of t distinct fresh leaf excursions. It has t root excursions and hence the displayed number of cuts. The argument also covers $t = 0$. $\square$

Lemma 15.2. *For $t \geq 1$,*

$$M_{2t} \leq \text{Cat}_t \, t! \leq 4^{t-1} t!, \quad \text{Cat}_t = \frac{1}{t+1} \binom{2t}{t}. \quad (106)$$

Proof. The distance to the initial vertex is a Dyck path, with Cat_t choices. At its j th outward step, a normalized walk can choose an already discovered child or the next new child. At most $j-1$ outward discoveries have occurred, so there are at most j choices. Indexing children by discovery order gives an injective encoding by the distance path and these choices. Inward steps are forced. Thus the product of the outward choice bounds is $t!$. Finally $\text{Cat}_1 = 1$ and $\text{Cat}_{t+1} / \text{Cat}_t < 4$. $\square$

16 A Pólya urn bound for arbitrary child trees

Set

$$A_{n,s} = \binom{n+s-1}{s} B_n, \quad w_0 = w_1 = 1, \quad w_t = 4^{t-1} t! \quad (t \geq 1).$$

Choose π uniformly from $\text{Part}[n]$, and conditionally run a Pólya urn whose colors are its blocks and whose initial color populations are $a_b = |b|$. At each draw, choose a color proportionally to its current population and add one unit of that color. If T_b records its number of draws after s steps, then

$$\mathbb{P}(T_b = t_b \text{ for every } b \mid \pi) = \frac{\prod_b \binom{a_b+t_b-1}{t_b}}{\binom{n+s-1}{s}} \left(\sum_b t_b = s \right). \quad (107)$$

This follows either by multiplying draw probabilities or by the rising-factorial multinomial identity. Proposition 15.1 and Lemma 15.2 give

$$1 \leq \frac{F_{n+s,n}}{A_{n,s}} \leq \mathbb{E}_\pi \mathbb{E}_{\text{urn}} \prod_b w_{T_b}. \quad (108)$$

The lower normalization follows by summing the numerator of (107); it is the same for every partition.

Let $A = \max_b a_b$, and after j draws write $Z_j = \prod_b w_{T_b}$. Adding a first draw to an unoccupied color multiplies its weight by one. Adding to an occupied color with $t \geq 1$ draws multiplies it by $4(t+1)$. Consequently

$$\begin{aligned} \frac{\mathbb{E}[Z_{j+1} \mid \text{history}]}{Z_j} &= 1 + \sum_{b:T_b > 0} \frac{(a_b + T_b)(4T_b + 3)}{n + j} \\ &\leq 1 + \frac{7(Aj + j^2)}{n + j}. \end{aligned}$$

Here $4t + 3 \leq 7t$, $\sum_b a_b T_b \leq Aj$, and $\sum_b T_b^2 \leq j^2$. Once the partition is fixed the last bound is deterministic. Iterating conditional expectations and using $1 + u \leq e^u$ yields

$$\mathbb{E}_{\text{urn}} Z_s \leq \exp \left\{ \frac{7As(s-1)}{2n} + \frac{7s(s-1)(2s-1)}{6n} \right\}. \quad (109)$$

This estimate is the step that controls unrestricted child trees. Their rough factorial cost is paid only when the urn repeatedly selects the same branch.

Lemma 16.1. *Let A be the largest block of a uniform random partition of $[n]$. If $0 \leq \tau_n \leq 1$ eventually and $\tau_n \log^2 n \rightarrow 0$, then*

$$\mathbb{E} e^{\tau_n A} = 1 + o(1).$$

The conclusion is uniform over any range of τ bounded by such a τ_n .

Proof. The elementary Bell saddle lemma in Section A implies $B_{r-1}/B_r = O(\log r/r)$. The expected number of blocks of size j is exactly $\binom{n}{j} B_{n-j}/B_n$. Put $L = \lceil \log^2 n \rceil$. On $A < L$, the integrand is at most $e^{\tau_n L} = 1 + o(1)$. On $A \geq L$, one of the blocks has size A , so

$$\mathbb{E}[e^{\tau A}; A \geq L] \leq \sum_{j=L}^n e^{\tau j} \binom{n}{j} \frac{B_{n-j}}{B_n}. \quad (110)$$

For $j \leq n/2$, multiply adjacent Bell ratio bounds, with all their indices in $[n/2, n]$. One fixed constant C gives

$$\binom{n}{j} \frac{B_{n-j}}{B_n} \leq \left(\frac{Ce \log n}{j} \right)^j.$$

Since $\tau \leq 1$, this part of (110) is at most $\sum_{j \geq L} (C'e \log n/j)^j = o(1)$. For $j > n/2$, keep the first $\lfloor n/2 \rfloor$ adjacent ratios and use the monotonicity of Bell numbers for the rest. The remaining sum is bounded by

$$(1+e)^n \left(\frac{C \log n}{n} \right)^{\lfloor n/2 \rfloor} = o(1).$$

The constants are independent of the admissible τ . The lower bound $\mathbb{E}e^{\tau A} \geq 1$ completes the proof. $\square$

Apply (109) with $n = k - s$ and $0 \leq s \leq S_k$. Its partition-independent exponent is $O(S_k^3/k) = o(1)$, while the coefficient of A obeys

$$0 \leq \tau = \frac{7s(s-1)}{2n} = O(S_k^2/k), \quad \tau \log^2 n = o(1).$$

The last implication follows from $S_k = o(k^{1/3})$. Lemma 16.1 and (108) therefore prove (100) uniformly over the entire stated window. For the illustrative quarter-power choice, the two error scales are $O(k^{-1/4})$ and $O(k^{-1/2})$ respectively; the proof is not restricted to that choice.

17 The fresh terms concentrate in the window

The normalization in (100) satisfies the exact Bell convolution

$$\sum_{s=0}^{k-1} \binom{k-1}{s} B_{k-s} = B_{k+1} - B_k. \quad (111)$$

For example, this follows by subtracting the Bell recurrences at indices k and $k-1$ and applying Pascal's identity. We need the sum truncated at S_k .

Let $X \sim \text{Poisson}(1)$, let $Q_k = B_{k+1}$, and set

$$x_k = \frac{k+1}{W(k+1)}.$$

Dobinski's identity and its shift give

$$B_n = \mathbb{E}X^n, \quad Q_k = \mathbb{E}(X+1)^k. \quad (112)$$

Conditionally on X , take $Z_X \sim \text{Bin}(k-1, 1/(X+1))$. Direct multiplication of the binomial probability proves

$$\binom{k-1}{s} B_{k-s} = \mathbb{E} \left[X(X+1)^{k-1} \mathbb{P}(Z_X = s \mid X) \right] \quad (0 \leq s \leq k-1). \quad (113)$$

At $X = 0$ both sides of the integrand identity are interpreted by their polynomial value, which is zero since $k-s \geq 1$.

Under the probability law tilted by $(X + 1)^k/Q_k$, Section A shows that $X/x_k \rightarrow 1$ and that $\mathbb{P}(X < x_k/2) \leq e^{-ck}$ for some fixed $c > 0$ and all sufficiently large k . On $X \geq x_k/2$,

$$\mathbb{E}[Z_X | X] = \frac{k-1}{X+1} = O(\log k).$$

Because $S_k/\log k \rightarrow \infty$, the binomial Chernoff bound

$$\mathbb{P}(Z_X > S_k | X) \leq \left(\frac{e\mathbb{E}[Z_X | X]}{S_k} \right)^{S_k}$$

is $o(1)$ uniformly on this half-line. In (113), the remaining factor relative to the outer tilt is $X/(X+1) \leq 1$. Thus

$$\sum_{s>S_k}^{k-1} \binom{k-1}{s} B_{k-s} = o(Q_k).$$

Also $B_k = o(B_{k+1})$ by the adjacent Bell ratio. Combining this fact with (111) and the uniform estimate (100) proves (101).

18 Rotation and the exact two hub correction

For any set $I \subseteq \{1, \dots, k\}$, let $N_I(w)$ be the number of vertices with departure count in I . The following identity is valid without any assumption of a unique hub:

$$\sum_{m \in I} \frac{2k}{m} F_{k,m} = \sum_{w \in \mathcal{W}_k} N_I(w). \quad (114)$$

In a rotation orbit with d distinct normalized words, every word appears $2k/d$ times among the indexed rotations. A vertex of count m is the root at exactly m indices. The indexed weighted sum is therefore $2kN_I(w)$, and division by $2k/d$ gives $dN_I(w)$. This proves (114), including periodic walks and rotations that induce vertex relabelings.

Fix an integer threshold $L > k/2$, and let $U_{k,L}^*$ count walks with at least one vertex having at least L departures. Let $D_{k,L}$ count those with two such vertices. Two vertices whose visit counts exceed $k/2$ must be adjacent: otherwise their incident-edge sets are disjoint and their visit counts sum to at most k . Three qualifying vertices would therefore be pairwise adjacent, which is impossible in a tree. There are at most two.

Proposition 18.1. *For every integer $k \geq 1$ and $k/2 < L \leq k$,*

$$V_{k,L} := \sum_{m=L}^k \frac{2k}{m} F_{k,m} = U_{k,L}^* + D_{k,L}, \quad (115)$$

$$D_{k,L} = \sum_{q=1}^k \frac{k}{q} \sum_{a+b=k-q} \hat{E}_{a,q}(L) \hat{E}_{b,q}(L), \quad (116)$$

$$\hat{E}_{a,q}(L) = \sum_{j=0}^a \mathbf{1}_{\{q+j \geq L\}} F_{a,j} \binom{j+q-1}{q-1}. \quad (117)$$

In particular $U_{k,L}^ = V_{k,L} - D_{k,L}$ is an exact finite formula, despite its rational weights.*

Proof. The cyclic group of $2k$ indexed time rotations acts on normalized words. Suppose an orbit has d distinct words; each word occurs $2k/d$ times among all indexed rotations. If a representative has h qualifying vertices, the sum of root weights over all indexed rotations is

$$\sum_{v \text{ qualifying}} d_w(v) \frac{2k}{d_w(v)} = 2kh.$$

Dividing by $2k/d$, this orbit contributes dh to $V_{k,L}$. Since h is either zero, one, or two, this proves (115). No free action or absence of periodicity is assumed.

For a walk with two qualifying vertices, their joining edge is unique. Let its directional multiplicity be q . Start the walk by crossing this edge and remove all of its traversals. On each side remain q ordered closed segments, including possibly empty segments, with respective total half-lengths a, b and $a + b = k - q$. If a concatenated side walk has j root departures, its endpoint in the original walk has $q + j$. The condition that both endpoints qualify is therefore precisely (117). Rejoining the alternating crossings reconstructs one normalized word beginning with this edge, so the inner product in (116) counts these representatives exactly.

In an orbit of d distinct words, exactly $2q$ of the $2k$ indexed rotations start by crossing that central edge, in either direction. There are thus dq/k distinct starting words counted by the product. Multiplication by k/q makes their total contribution d . This reasoning remains valid when a rotation interchanges the endpoints. In particular there is no extra orientation factor. Summing proves (116). $\square$

Now put $L = k - S_k$; this exceeds $k/2$ eventually. Write $D_k = D_{k,k-S_k}$. Equation (101) and $S_k = o(k)$ give

$$V_{k,k-S_k} = \sum_{s=0}^{S_k} \frac{2k}{k-s} F_{k,k-s} \sim 2B_{k+1}. \quad (118)$$

It remains to show $D_k = o(B_{k+1})$.

If the qualifying endpoints are u, v and the joining edge has multiplicity q , (90) gives $d_w(u) + d_w(v) \leq k + q$. Hence $q \geq k - 2S_k$. Rotating to one crossing and then allowing every possible time origin, even with overcounting, yields

$$D_k \leq 2k \sum_{r=0}^{2S_k} \sum_{a+b=r} E_{a,k-r} E_{b,k-r}.$$

For $a \geq 1$, $a \leq r$, and $q = k - r \geq 1$, (105)–(106) give

$$E_{a,q} \leq 4^{a-1} a! \binom{q+a-1}{a} \leq (4k)^a;$$

the same last bound holds at $a = 0$. Therefore

$$D_k \leq 2k(2S_k + 1)^2 (4k)^{2S_k} = \exp(O(S_k \log k)) = o(B_{k+1}). \quad (119)$$

For the final comparison, the single Dobinski summand at $j = \lfloor \sqrt{k} \rfloor$ already gives $\log B_{k+1} \geq (1/2 + o(1))k \log k$; its factorial cost is only $O(\sqrt{k} \log k)$. Equations (115), (118), and (119) prove (102), completing Theorem 14.2.

19 A finite positive coefficient majorant

[write] **Printed once, 7 October 2026.** Report 213’s Section 6 is printed once, as Appendix B.2 of Part I: Report 214’s Appendix B restates it word for word, apart from writing $\mathcal{A}_t$ for Report 213’s A_t . Its numbered objects, each with the number Report 213 gave it (shifted as in this Part) and the number of its printed counterpart in Part I: (120) $\mapsto$ (68); (121) $\mapsto$ (69); (122) $\mapsto$ (70); (123) $\mapsto$ (71). References to them elsewhere in this Part point to the printed counterparts.

20 Bell entropy with cancellation preserved

[write] **Printed once, 7 October 2026.** Report 213’s Section 7 is printed once, as Appendix B.3 of Part I (word for word). Its numbered objects, each with the number Report 213 gave it (shifted as in this Part) and the number of its printed counterpart in Part I: (124) $\mapsto$ (72); Lemma 20.1 $\mapsto$ Lemma B.2; (125) $\mapsto$ (73). References to them elsewhere in this Part point to the printed counterparts.

21 The intermediate return range

[write] **Printed once, 7 October 2026.** Report 213’s Section 8 is printed once, as Appendix B.4 of Part I. The two texts differ in four places only: $\mathcal{A}_t$ for A_t ; D_0 for the envelope constant D of (99); the rough walk bound and the single-term Bell estimate are cited there as (10) and (62) instead of Report 213’s (106) and (98) (the same inequalities); and the letter L is said there to differ from Part I’s largest-block cutoff, whereas Report 213 says it differs from the temporary hub threshold of Section 18. Its numbered objects, each with the number Report 213 gave it (shifted as in this Part) and the number of its printed counterpart in Part I: (126) $\mapsto$ (74); Proposition 21.1 $\mapsto$ Proposition B.3; (127) $\mapsto$ (75); (128) $\mapsto$ (76); (129) $\mapsto$ (77); (130) $\mapsto$ (78); (131) $\mapsto$ (79); (132) $\mapsto$ (80); (133) $\mapsto$ (81). References to them elsewhere in this Part point to the printed counterparts.

22 The high root shoulder with a growing tilt

[write] **Printed once, 7 October 2026.** Report 213’s Section 9 is printed once, as Appendix B.5 of Part I; the texts differ only in the first sentence, where Report 213 says “The dominant-window proof treated a small largest-block tilt” and Report 214 “The main-text collision argument treated a small largest-block tilt”. Its numbered objects, each with the number Report 213 gave it (shifted as in this Part) and the number of its printed counterpart in Part I: Lemma 22.1 $\mapsto$ Lemma B.4; (134) $\mapsto$ (82); Proposition 22.2 $\mapsto$ Proposition B.5; (135) $\mapsto$ (83); (136) $\mapsto$ (84); (137) $\mapsto$ (85); (138) $\mapsto$ (86). References to them elsewhere in this Part point to the printed counterparts.

23 The maximum departure split and the full equivalent

[write] **Printed once, 7 October 2026.** The first part of Report 213’s Section 10, the maximum departure split, is printed once, as Appendix B.6 of Part I, where the Dyck-path encoding is cited from Proposition 2.1 instead of Report 213’s Lemma 15.2 and the conclusion (89) is written $|\mathcal{W}_k \setminus \mathcal{U}_{k,S}| = o(k^{-p}B_{k+1})$ instead of $C_{k,S} = o(k^{-p}B_{k+1})$ (the same quantity). It proves Report 213’s complement estimate $C_{k,S} = o(k^{-p}B_{k+1})$ for every fixed $p > 0$ and every admissible window. Its numbered objects, each with the number Report 213 gave it (shifted as in this Part) and the number of its printed counterpart in Part I: (139) $\mapsto$ (87); (140) $\mapsto$ (88);

(141) $\mapsto$ (89). References to them elsewhere in this Part point to the printed counterparts. Report 213's text resumes after it.

The dominant-window theorem now implies

$$a_k = U_{k,S} + C_{k,S} \sim 2B_{k+1}.$$

This establishes the full relative equivalent without assuming any middle-return tail. The next section proves that original weighted tail separately. The distinction matters: a walk with a dominant hub may still contribute to the middle-return sum when rooted elsewhere.

24 Recovery of the weighted middle return tail

By (114),

$$R_{k,S} = \sum_{w \in \mathcal{W}_k} N_S(w), \quad N_S(w) = |\{v : S < d_w(v) < k - S\}|. \quad (142)$$

There are at most $k + 1$ vertices, so $0 \leq N_S(w) \leq k + 1 \leq 2k$ for $k \geq 1$.

Introduce a smaller admissible window

$$S_0 = \lfloor \sqrt{S \log k} \rfloor.$$

Then $S_0 / \log k \rightarrow \infty$, $S_0 = o(S)$, $S_0 = o(k^{1/3})$, and $S_0^3/k \leq (S \log k)^{3/2}/k = o(1)$. Let $\mathcal{U}_0 = \mathcal{U}_{k,S_0}$. Equation (89), applied with S_0 , gives

$$|\mathcal{W}_k \setminus \mathcal{U}_0| = o(k^{-p} B_{k+1}) \quad \text{for every fixed } p > 0. \quad (143)$$

Consider $w \in \mathcal{U}_0$ with $N_S(w) > 0$, and choose a qualifying hub u . Put $s = k - d_w(u) \leq S_0$. Every vertex *distinct from u and not adjacent to u* has all its incident edges outside the hub's incident-edge set, and thus has departure count at most s . An adjacent vertex has departure count at most $q + s$, where q is the multiplicity of its edge to u . The hub itself is not middle, because $k - S_0 > k - S$. Consequently some adjacent middle vertex forces a hub edge of multiplicity

$$q > S - S_0 \geq S/2 \quad (144)$$

eventually. In the root partition at u , this edge corresponds to a block of size q .

Let $F_{k,k-s}^{\text{bad}}$ count hub-rooted walks of deficit $s \leq S_0$ whose root partition has a block of size at least $S/2$. Apply the urn domination *conditional on the partition*, before restricting the partition event. Equation (109) gives, uniformly for $s \leq S_0$,

$$F_{k,k-s}^{\text{bad}} \leq \binom{k-1}{s} B_{k-s} \exp(O(S_0^3/k)) \mathbb{E}[e^{\tau A}; A \geq S/2], \quad \tau = O(S_0^2/k) = o(1), \quad (145)$$

where A is the largest block in a uniform partition of $[n]$, $n = k - s$. There is no conditioning on a previously approximated, untilted law.

The block union bound gives

$$\begin{aligned} \mathbb{E}[e^{\tau A}; A \geq S/2] &\leq \sum_{q \geq \lceil S/2 \rceil}^n e^{\tau q} \binom{n}{q} \frac{B_{n-q}}{B_n} \\ &\leq \exp\{-cS \log(S/\log k)\} \end{aligned} \quad (146)$$

for one fixed $c > 0$ and all sufficiently large k , uniformly in $s \leq S_0$. For $q \leq n/2$, the summands are bounded by $(C \log k/q)^q$ because $\tau = o(1)$. Starting at $q \geq S/2$, the base tends to zero. Bound it by $2C \log k/S$ and sum the resulting geometric series to obtain the displayed exponent after reducing c . For $q > n/2$, the same retained-adjacent-ratio estimate as in Lemma B.4 is $\exp(-\Omega(k \log k))$, which is smaller. Since $S/\log k \rightarrow \infty$, (146) is $o(k^{-p})$ for every fixed $p > 0$.

Sum (145) over $0 \leq s \leq S_0$. The nonnegative fresh factors are bounded by the exact full sum (111); the other urn exponent is $o(1)$. Thus

$$\sum_{s=0}^{S_0} F_{k,k-s}^{\text{bad}} = o(k^{-p} B_{k+1}) \quad \text{for every fixed } p > 0.$$

Rotate these bad hub-rooted walks with weights $2k/(k-s) \leq 4$ eventually. The same orbit argument as (114), now marking a hub with the stated partition property, covers every walk in $\mathcal{U}_0$ having a middle vertex, by (144). It remains valid when two hubs qualify. Therefore

$$|\{w \in \mathcal{U}_0 : N_S(w) > 0\}| = o(k^{-p} B_{k+1}) \quad \text{for every fixed } p > 0.$$

Combine this with (143) and multiply by $N_S(w) \leq 2k$ in (142). Because the count estimates hold for every fixed exponent, the extra factor $2k$ is absorbed by using a larger exponent first. We obtain

$$R_{k,S} = o(k^{-p} B_{k+1}) \quad \text{for every fixed } p > 0,$$

completing Theorem 14.1.

25 A scoped inverse consequence

For real $x > 0$ define the Bell interpolation and its scaled shift

$$\mathcal{B}(x) = e^{-1} \sum_{j \geq 1} \frac{j^x}{j!}, \quad g(x) = 2\mathcal{B}(x+1).$$

The series and all its derivatives converge on every compact real interval, by factorial domination. At positive integer arguments $\mathcal{B}(n) = B_n$. Omitting $j = 0$ is appropriate here; no assertion $\mathcal{B}(0) = B_0$ is made. The function g is positive, strictly increasing, and unbounded, so its inverse $x(y) = g^{-1}(y)$ exists for all sufficiently large y .

Define the discrete threshold

$$N(y) = \min\{k \geq 1 : a_k \geq y\}, \quad y > 1.$$

The sequence is strictly increasing for $k \geq 1$. To see this, append a fresh leaf excursion at the root of any normalized walk of half-length k . Deleting the final two steps recovers it, so the map is injective. For $k \geq 1$ the repeated single-edge walk of half-length $k+1$ is outside the image, as its final excursion visits an already used vertex. Thus $a_{k+1} > a_k$, while $a_0 = a_1 = 1$.

Proposition 25.1. *There exists a positive function $\delta(x) = o(1/\log x)$ such that, for all sufficiently large y ,*

$$\lceil x(y) - \delta(x(y)) \rceil \leq N(y) \leq \lceil x(y) + \delta(x(y)) \rceil. \quad (147)$$

The two ceilings are eventually equal or adjacent. The radius and onset are existential, with no explicit numerical error certificate asserted.

Proof. The function $f(x) = \log \mathcal{B}(x)$ is convex as a log-sum-exponential. For integers n and $n \leq x \leq n+1$, convexity bounds its derivative by the outer unit secants:

$$\log B_n - \log B_{n-1} \leq f'(x) \leq \log B_{n+2} - \log B_{n+1}.$$

Both bounds are asymptotic to $\log n$, by (60). Hence

$$(\log g)'(x) \sim \log x. \quad (148)$$

This uses convexity, not differentiation of an asymptotic saddle remainder.

The full equivalent gives $e_k = \log(a_k/g(k)) = o(1)$. For large x , put

$$\eta(x) = \sup_{\substack{k \in \mathbb{Z} \\ k \geq x/2}} |e_k| + \frac{1}{x}, \quad \delta(x) = \frac{4\eta(x)}{\log x}.$$

Then $\eta(x) > 0$, $\eta(x) \rightarrow 0$, and $\delta = o(1/\log x)$. Uniformly on $[x-1, x+1]$, (148) bounds $(\log g)'$ below by $\frac{1}{2} \log x$ for all sufficiently large x . Integrating over intervals of length $\delta(x) < 1$ gives

$$g(x+\delta) \geq g(x)e^{2\eta}, \quad g(x-\delta) \leq g(x)e^{-2\eta}.$$

For every integer k with $x/2 \leq k \leq x-\delta$, monotonicity of g yields $a_k \leq g(k)e^\eta < g(x)$. For integers $k \geq x+\delta$, it yields $a_k \geq g(k)e^{-\eta} > g(x)$.

The integers below $x/2$ also have $a_k < g(x)$ eventually. It is enough by monotonicity to use $k = \lfloor x/2 \rfloor$; the leading equivalent at that integer, together with integration of (148), gives $a_k/g(x) \rightarrow 0$. Taking $x = x(y)$ proves the conservative lower ceiling and the upper ceiling in (147). Since $2\delta < 1$ eventually, their difference is at most one. $\square$

The definition of η uses the unknown tail error in the leading equivalent. Thus (147) is an asymptotic enclosure, not an effective inversion algorithm. A uniform single-ceiling formula at exact sequence thresholds does not follow from a relative $o(1)$ equivalent alone. No expansion for the full moments or their inverse beyond this statement is established here.

[write] **Added 7 October 2026.** Part I's Theorem 11.1 sharpens this enclosure for every fixed $p \geq 1$, to radius $C_p \log^{10p+5} x/x^{p+1}$ around corrected centres $x_p(L)$; at $p=0$ its function g_0 of (52) is the present g , so $x_0(L) = x(L)$, and Corollary 11.3 measures how far the corrected centres move from it. Remark 11.4 compares both enclosures with the repository's transseries volume.

26 Exact finite checks and reproduction

The first-edge decomposition gives the exact recurrence

$$F_{k,m} = \sum_{q=1}^m \binom{m-1}{q-1} \sum_{a+b=k-q} E_{a,q} F_{b,m-q}. \quad (149)$$

The child visited by the first excursion is visited q times. Its first position is fixed and its remaining positions can be chosen in $\binom{m-1}{q-1}$ ways. Its interior is a cut child walk of half-length a ; removing those excursions leaves a root walk of half-length b and $m-q$ departures. All recursive half-lengths are smaller than k . This is the Bauer–Golinelli first-edge recurrence after summing out the distinct-edge count [1, Section 5.4], not a new recurrence claimed here.

Three useful exact boundary checks are

$$F_{k,1} = a_{k-1}, \quad F_{k,k} = B_k, \quad F_{k,k-1} = (k-1)B_{k-1} \quad (k \geq 2 \text{ for the last}). \quad (150)$$

They respectively remove the outer edge, partition the root excursions with no interior steps, and place a single interior leaf excursion.

Table 3: Selected exact finite moments with rounded relative ratios

k	a_k	B_{k+1}	$a_k/(2B_{k+1})$
1	1	2	0.2500000000
2	3	5	0.3000000000
4	57	52	0.5480769231
8	69331	21147	1.6392632525
12	177877932	27644437	3.2172464210
16	731080093581	82864869804	4.4112788405
24	40149216663514343623	4638590332229999353	4.3277390099
32	10504009843061009466883830711	1629595892846007606764728147	3.2228879225

Table 3 is generated from exact integers. The ratio need not be close to one at these modest indices: it rises substantially before later decreasing. These values are checks of normalization and computation; they do not estimate the theorem’s onset or prove its limit.

The public standard-library Python code freshly generates the recurrence through $k = 32$ and independently enumerates canonical tree walks through $k = 8$. The enumeration chooses a discovered neighbor or the next fresh vertex directly, without using F , E , Bell numbers, or the recurrence to generate walks. Its discovery codes are checked for injectivity. All 82 181 positive-length walks through $k = 8$ are included. The checks compare every root-return row and verify the maximum-departure code bound, exact weighted rotation, and the two-hub formula.

Separate finite checks reconstruct branch cuts by weak compositions and sum directly over set partitions. Rational coefficient extraction from the exponential majorant is compared with the partition calculation. Further checks cover (70), the coefficientwise root majorant, integer-power forms of (96), the rough walk bound, the fresh Bell convolution, and the boundary identities. This independent small enumeration is distinct from repeated execution of the recurrence at larger k .

Correctness guards are explicit exceptions, not Python assertions. They remain active under `python3 -O`. The reproduction harness runs both modes, compares all deterministic output bytes, and deliberately exercises corrupted inputs and injected mathematical failures. The accompanying `README.txt` states exact commands and the scope of each test.

The release contains this PDF, editable $\text{\TeX}$, source links, compact exact data and checks, and a deterministic reproduction driver. No large extended arrays, compiled binaries, third-party full texts, or private review records are included. The driver regenerates the finite files and rebuilds the PDF in isolated directories. It checks each regenerated member and the actual whole ZIP bytes. PDF and whole-package byte identity are scoped to the same $\text{\TeX}$, font, Python, and compression toolchain recorded with the release; mathematical integer outputs need no third-party Python package. Finite checks and successful reproduction do not substitute for the asymptotic proofs.

27 Source credit and bounded literature scope

The essential external theorem is exactly (97), from Bauer–Golinelli [1, Section 5.5, equation (9)]. Their Sections 5.3 and 5.4 also supply the fixed-moment framework and departure-refined recurrence

recalled here. Their Section 5.6 discusses fixed-tree-size star asymptotics, a distinct regime from the growing-index sum treated in this article. The elementary Touchard envelope, Bell estimates, and remaining arguments above are written out so that no asymptotic expansion from an earlier report is required.

A bounded later-source check inspected the definitions and main theorem statements of Hainzl–de Panafieu [4]. Their Theorem 3 organizes fixed-excess tree-walk generating functions, and Theorem 4 treats fixed moments as mean degree grows. Those inspected statements do not give the present fixed-mean-degree, high-moment relative equivalent. A 2026 journal record was located, but only its abstract and introduction were retrieved in the underlying check; no full-version identity or exhaustive nonoverlap is assumed.

Hiesmayr–McKenzie [9] give sharp high-probability extreme-eigenvalue expansions for sparse finite graphs, including constant mean degree. Their Theorem 1.2 and local expansion in Lemma 4.10 operate on their specified good events and neighborhoods. They are relevant context, but they are not used in this proof: no conversion of such a high-probability finite-graph assertion into an annealed high-moment equivalent is presumed. The graph-size limit in (95) is taken at fixed moment order.

Condensation papers of Janson–Jonsson–Stefánsson [6] and Janson’s survey [7] were located at abstract level as related background. No theorem from those works is used or represented as fully audited here. The precise public links and inspection scope are recorded in `SOURCES.txt`. The bounded checks did not locate a statement of the same relative equivalent in the inspected material. This is not a claim of worldwide priority, absence from all literature, or global open-problem status.

Further directions. The present theorem leaves concrete next targets: a matching first correction for the full moments; a justified expansion to every fixed order; quantitative error constants and onset bounds; an effective inverse enclosure; and the analogous high-moment problem at fixed general mean degree. These are directions for further analysis of this argument, not assertions that the questions are globally open. The superpolynomial tail bounds alone do not settle the first two, because the remaining approximation error lies within the dominant family.

[write] **Added 7 October 2026.** The first two directions are answered within this report, by Part I (Report 214, the same session’s next manuscript): the first relative correction is $(\frac{1}{2} + o(1))W(k)^3/k$ by (6), and Theorem 1.1 gives an expansion to every fixed collision order with explicit finite Bell transforms. The statement above that the present theorem leaves them open is kept as delivered; it is true of Report 213. Quantitative constants and onsets, an effective inverse enclosure, and fixed general mean degree remain open: items 1 and 5 of Section 56.

A Elementary Bell estimates

[write] **Printed once, 7 October 2026.** Report 213’s Appendix A is printed once, as Appendix A of Part I (word for word, except that Part I’s text says the shifted law is used in its Lemma 3.2, where Report 213’s says Section 17; both uses are covered by it). Its numbered objects, each with the number Report 213 gave it (shifted as in this Part) and the number of its printed counterpart in Part I: Lemma A.1 $\mapsto$ Lemma A.1; (151) $\mapsto$ (59); (152) $\mapsto$ (60); (153) $\mapsto$ (61); Lemma A.2 $\mapsto$ Lemma A.2; (154) $\mapsto$ (62); (155) $\mapsto$ (63). References to them elsewhere in this Part point to the printed counterparts.

Part III

Dominant vertices in sparse graph moments

Bundle Report 212, archive `Report212-reproducibility.zip`, delivered as `Report212.tex` (578 lines, 16 pages). Delivered title block “Dominant vertices in sparse graph moments / Report212 / 4 October 2026” (empty author line). Printed as delivered, with `[write]` notes, except that its Sections 2–5 (word for word Part II’s Sections 15–18) and the Bell lemma of its Appendix A (printed in Part I) are not printed again; a note under each heading maps their numbers, and references to them point to the printed copies. Report 212’s Section k is Section $k + 27$ here, its equation (k) is equation $(k + 155)$, its Tables 1–2 are Tables 4–5, and its appendix keeps the letter A; statements move with their sections. Its labels carry `sgm:dv::`; its two tables are printed inline. “Report209” and “Report210” in this Part are Parts V and IV.

Abstract of this Part (as delivered). Let M_{2k} count normalized closed tree walks of length $2k$, with time zero distinguished, and let B_j be the Bell numbers. We prove that for every integer sequence S_k satisfying $\log k = o(S_k)$ and $S_k = o(k^{1/3})$, the entire class having a vertex with at least $k - S_k$ departures has size $(2 + o(1))B_{k+1}$. The class permits arbitrary depth, repeated edges, and nesting. A partition decomposition and a Pólya urn control its rooted count uniformly; an exact weighted rotation identity includes the two-hub correction, even for periodic walks. We reduce the full equivalent $M_{2k} \sim 2B_{k+1}$ to an explicit sufficient middle-return tail estimate, which is not proved here. Separately, a truncated Poisson-neighborhood model has a rigorous variance correction, while two counterexamples show why its untruncated average is invalid. Exact finite data and reproducible checks support the formulas without establishing the missing tail. Published recurrences and relevant spectral theorems are credited, with their distinct hypotheses and order of limits made explicit.

28 The broad dominant vertex theorem

A *normalized tree walk* is a sequence

$$w = (v_0, v_1, \dots, v_{2k-1}, v_0)$$

whose traversed unordered pairs form a tree and whose consecutive vertices differ. Two sequences are identified only by a vertex bijection preserving their order. Labeling vertices $0, 1, \dots$ by first appearance selects a unique representative. Rotating the time origin generally gives a different normalized walk. Write $\mathcal{W}_k$ for these walks and $M_{2k} = |\mathcal{W}_k|$, with $M_0 = 1$. For $k \geq 1$, these are the positive-order terms of OEIS A094149 [17].

For v in the underlying tree let $d_w(v)$ be the number of departures from v during one period. This is a *walk visit count*, not the graph degree. Every edge e is traversed equally often in each direction; call that multiplicity $m_e \geq 1$. Then

$$d_w(v) = \sum_{e \ni v} m_e, \quad \sum_e m_e = k, \quad \sum_v d_w(v) = 2k. \quad (156)$$

Let $F_{k,m}$ count walks whose initial vertex has m departures. Set $F_{0,0} = 1$ and all impossible entries to zero. Thus $M_{2k} = \sum_{m=1}^k F_{k,m}$ for $k \geq 1$.

For an integer $S \geq 0$ define

$$\mathcal{U}_{k,S} = \{w \in \mathcal{W}_k : \max_v d_w(v) \geq k - S\}, \quad U_{k,S} = |\mathcal{U}_{k,S}|.$$

Equivalently, some vertex has total edge multiplicity at most S away from it. There is no separate bound on depth, branching, repeated-edge pattern, or the nesting of excursions.

Theorem 28.1. *Let (S_k) be any integer sequence with*

$$\frac{S_k}{\log k} \rightarrow \infty, \quad \frac{S_k^3}{k} \rightarrow 0. \quad (157)$$

Then, as $k \rightarrow \infty$,

$$F_{k,k-s} = (1 + o(1)) \binom{k-1}{s} B_{k-s} \quad \text{uniformly for } 0 \leq s \leq S_k, \quad (158)$$

$$\sum_{s=0}^{S_k} F_{k,k-s} \sim B_{k+1}, \quad (159)$$

$$U_{k,S_k} \sim 2B_{k+1}. \quad (160)$$

Here B_n counts partitions of $[n]$, with $B_0 = 1$.

Examples are $S_k = \lfloor k^{1/4} \rfloor$ and $S_k = \lceil (\log k)^2 \rceil$. Both satisfy the conditions only in the asymptotic sense needed by the theorem. In particular, $\lceil (\log k)^2 \rceil$ can be a substantial fraction of k at small k .

Boundary of the result. The theorem counts a broad class exactly to leading order. It does not show that the complement is negligible. Define the weighted middle-return tail

$$R_{k,S} = \sum_{S < m < k-S} \frac{2k}{m} F_{k,m}. \quad (161)$$

We will prove the sufficient implication

$$R_{k,S_k} = o(B_{k+1}) \implies M_{2k} \sim 2B_{k+1}. \quad (162)$$

The premise in (162) remains unproved in this report. The exact recurrence in Section 34 turns it into a concrete scalar-array problem. The numerical tables and the separate scalar model do not supply this uniform tail estimate.

[write] **Added 7 October 2026.** The premise of (162) is proved in Part II: Theorem 14.1 gives $R_{k,S_k} = o(k^{-p} B_{k+1})$ for every fixed $p > 0$ and every window satisfying (157), so $M_{2k} \sim 2B_{k+1}$ by (94); Part II reaches the equivalent by bounding the complement of the dominant class directly and proves the weighted tail separately. The statement above that the premise “remains unproved” is kept as delivered; it is true of Report 212, and Part II, the same session’s next manuscript, removes it. Theorem 28.1 is Part II’s Theorem 14.2; its proof, Report 212’s Sections 2–5 (Sections 29–32 here), is word for word Part II’s Sections 15–18 and is printed there only. “Report209” and “Report210” in this Part are Parts V and IV of this report.

28.1 The sparse adjacency interpretation

Let A_N be the unscaled adjacency matrix of $G(N, 1/N)$. For each fixed k ,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E} \operatorname{tr}(A_N^{2k}) = M_{2k}. \quad (163)$$

Indeed, expansion of the trace groups closed walks by their normalized pattern. A connected pattern with v vertices and e distinct edges contributes $(N)_v / N^{e+1}$, where $(N)_v = N(N-1) \cdots (N-v+1)$.

There are only finitely many patterns at fixed k . Connectedness gives $e \geq v - 1$; exactly the tree patterns survive, each with limit one. A closed tree walk has even length. This is the standard tree-walk moment mechanism underlying Bauer–Golinelli [1].

The order of limits is important: first $N \rightarrow \infty$ with k fixed, then $k \rightarrow \infty$ in Theorem 28.1. A high-probability statement about the most extreme finite- N eigenvalues is not automatically an estimate for this annealed high-moment limit.

28.2 Relation to the preceding lower bounds

Report209 established a shifted-Bell lower family. Report210 enlarged the family to all walks whose repeated edges share one vertex, allowing arbitrary fresh tree attachments, and added a disjoint family with one repeated secondary edge. In that notation,

$$G_k = 2B_{k+1} \left[1 + (1 + o(1)) \frac{W(k)^2}{k} \right], \quad J_k \sim W(k)^2 B_k,$$

with $M_{2k} \geq G_k + J_k$. Here $W(t)e^{W(t)} = t$ and $W(t) > 0$. The present class is defined by a multiplicity budget, rather than by requiring repeated edges to share a vertex. It includes arbitrary off-hub repetitions and depth within that budget. No containment of the entire previous family in a particular finite window is asserted. The proof below is self-contained and does not require either earlier report or its files.

29 Exact decomposition into child branches

[write] **Printed once, 7 October 2026.** Report 212’s Section 2 is word for word Report 213’s Section 2, printed in Part II as Section 15. Its numbered objects, each with the number Report 212 gave it (shifted as in this Part) and the number of its printed counterpart in Part II: (164) $\mapsto$ (103); Proposition 29.1 $\mapsto$ Proposition 15.1; (165) $\mapsto$ (104); (166) $\mapsto$ (105); Lemma 29.2 $\mapsto$ Lemma 15.2; (167) $\mapsto$ (106). References to them elsewhere in this Part point to the printed counterparts.

30 A Pólya urn bound for arbitrary child trees

[write] **Printed once, 7 October 2026.** Report 212’s Section 3 is word for word Report 213’s Section 3, printed in Part II as Section 16. Its numbered objects, each with the number Report 212 gave it (shifted as in this Part) and the number of its printed counterpart in Part II: (168) $\mapsto$ (107); (169) $\mapsto$ (108); (170) $\mapsto$ (109); Lemma 30.1 $\mapsto$ Lemma 16.1; (171) $\mapsto$ (110). References to them elsewhere in this Part point to the printed counterparts.

31 The fresh terms concentrate in the window

[write] **Printed once, 7 October 2026.** Report 212’s Section 4 is word for word Report 213’s Section 4, printed in Part II as Section 17. Its numbered objects, each with the number Report 212 gave it (shifted as in this Part) and the number of its printed counterpart in Part II: (172) $\mapsto$ (111); (173) $\mapsto$ (112); (174) $\mapsto$ (113). References to them elsewhere in this Part point to the printed counterparts.

32 Rotation and the exact two hub correction

[write] **Printed once, 7 October 2026.** Report 212’s Section 5 is word for word Report 213’s Section 5, printed in Part II as Section 18, except that Report 213 opens the section with the general rotation identity (114), which Report 212 does not state, and that Report 212’s last sentence ends “completing Theorem 1.1” (Theorem 28.1 here) where Report 213’s ends “completing Theorem 14.2”. The two theorems are the same statement. Its numbered objects, each with the number Report 212 gave it (shifted as in this Part) and the number of its printed counterpart in Part II: Proposition 32.1 $\mapsto$ Proposition 18.1; (175) $\mapsto$ (115); (176) $\mapsto$ (116); (177) $\mapsto$ (117); (178) $\mapsto$ (118); (179) $\mapsto$ (119). References to them elsewhere in this Part point to the printed counterparts.

33 A precise sufficient middle return estimate

The reduction in (162) is useful because it is stated only in terms of the root-return array F , without requiring an unspecified spectral exceptional-set estimate. It is a sufficient condition, not a claimed equivalence.

Proposition 33.1. *If $1 \leq S < k/2$, then*

$$U_{k,S} \leq M_{2k} \leq U_{k,S} + (4S)^k + R_{k,S}. \quad (180)$$

Under (157), $(4S_k)^k = o(B_{k+1})$. Consequently (162) holds.

Proof. If every vertex has at most S departures, every underlying graph degree is at most S . The distance-path encoding used in Lemma 15.2 has at most S choices for each outward step, with existing children indexed by discovery order and a new child taking the next index. Hence there are at most $\text{Cat}_k S^k \leq (4S)^k$ such walks.

Every other walk outside $\mathcal{U}_{k,S}$ has some vertex with departure count strictly between S and $k - S$. The orbit-weight calculation proving (115), with this middle interval in place of the high interval, shows that $R_{k,S}$ counts every walk once per vertex in that interval. It therefore covers all these remaining walks. It can also count walks in $\mathcal{U}_{k,S}$ and can count a walk more than once; only the upper bound is needed. This proves (180).

Finally, $\log S_k \leq (1/3 + o(1)) \log k$ under (157). Compare $k \log(4S_k)$ with the single-summand Bell lower bound used after (119). The gap is of order $k \log k$, so $(4S_k)^k = o(B_{k+1})$. $\square$

A stronger, convenient sufficient target would be

$$\max_{S_k < m < k - S_k} \frac{F_{k,m}}{B_{k+1}} = o\left(\frac{1}{k \log k}\right), \quad (181)$$

since $\sum_{S_k < m < k - S_k} 2k/m = O(k \log k)$. Neither (181) nor the weaker (162) premise is established here.

There is a reason not to ask for a uniform e^{-ck} bound over this whole interval. The lower bound from (108) remains valid outside the proved window. At $m = k - S_k - 1$, with $s = S_k + 1$, it gives

$$\frac{F_{k,k-S_k-1}}{B_{k+1}} \geq \binom{k-1}{S_k+1} \frac{B_{k-S_k-1}}{B_{k+1}} \geq \exp[-O(S_k \log k)].$$

The last bound follows from the adjacent Bell asymptotic, since all the relevant indices are comparable to k . As $S_k \log k = o(k)$, a bound e^{-ck} with fixed $c > 0$ cannot hold uniformly. This does not preclude the sufficient polynomial-scale target (181).

34 The exact recurrence and finite diagnostics

The first-edge decomposition gives

$$F_{k,m} = \sum_{q=1}^m \binom{m-1}{q-1} \sum_{a+b=k-q} E_{a,q} F_{b,m-q}. \quad (182)$$

To see this, the child visited by the first excursion is visited q times in total. Its first position is forced and its other positions can be chosen in $\binom{m-1}{q-1}$ ways. Its interior is a cut child walk of half-length a counted by $E_{a,q}$. Removing these q excursions leaves a root walk of half-length b with $m - q$ root departures. The total relation is $a + b + q = k$. Conversely these data reconstruct the original normalized walk. All recursive half-lengths are smaller than k .

Equation (182) is the Bauer–Golinelli first-edge recurrence after summing over the number of distinct edges [1, Section 5]. It and the established tree-walk moment framework are prior infrastructure, not claims of novelty. The package supplies an exact C++/GMP implementation of this recurrence and separate small-row checks.

Three useful boundary identities are

$$F_{k,1} = M_{2k-2}, \quad F_{k,k} = B_k, \quad F_{k,k-1} = (k-1)B_{k-1} \quad (k \geq 2 \text{ in the last}). \quad (183)$$

The first removes the unique outer edge. The second counts root-child partitions with no interior steps. The third uses exactly one interior leaf excursion, placed in one of the $k - 1$ root excursions. They check the normalization and indexing independently of large decimal ratios.

34.1 What the two windows show

In the tables, $B = B_{k+1}$, $U = U_{k,S}$, $C = M_{2k} - U$ is the exact complement, and $R = R_{k,S}$. All counts and rational ratios are exact before display rounding. The two-hub identity is used only when $L = k - S > k/2$. At $k = 16$, the logarithm-squared choice has $S = 8$, outside that finite condition; its union and complement cells are therefore unavailable rather than extrapolated. Its middle interval is empty, so $R = 0$ there says nothing about (162).

Table 4: Exact finite diagnostics for $S = \lfloor k^{1/4} \rfloor$

k	$M_{2k}/(2B)$	S	$U/(2B)$	$(M_{2k} - U)/B$	R/B
16	4.41127884	2	0.77836576	7.26582616	34.00631125
32	3.22288792	2	0.59200767	5.26176050	42.73474055
64	1.94703094	2	0.43939211	3.01527766	49.94394360
128	1.54990993	3	0.57172639	1.95636708	55.24848671
256	1.34150872	4	0.66705528	1.34890689	67.98633086

The quarter-power window grows extremely slowly: at $k = 256$, $S = 4$. Its weighted middle tail is still large on the Bell scale in the evaluated range. The logarithm-squared window has $S = 31$ at $k = 256$, and its exact complement and weighted middle tail are much smaller. These are finite facts for two asymptotically admissible choices, not a uniform proof for either choice. In particular the latter choice need not yet satisfy a small numerical ratio S^3/k ; the theorem’s sufficient estimates can have a slow onset.

Table 5: Exact finite diagnostics for $S = \lceil (\log k)^2 \rceil$

k	$M_{2k}/(2B)$	S	$U/(2B)$	$(M_{2k} - U)/B$	R/B
16	4.41127884	8	N/A	N/A	0
32	3.22288792	13	3.01222613	0.42132359	0.60701508
64	1.94703094	18	1.94650720	0.00104748	0.00214961
128	1.54990993	24	1.54990986	1.262342×10^{-7}	3.631896×10^{-7}
256	1.34150872	31	1.34150872	1.230307×10^{-12}	4.672174×10^{-12}

The full moment ratio initially rises: within the exact range through 256, $M_{2k}/(2B_{k+1})$ attains its maximum near 4.71281008 at $k = 19$, then strictly decreases, reaching about 1.34150872 at $k = 256$. The decrease is checked by exact rational comparisons. It is consistent with the proposed equivalent and does not determine a limit. No monotonicity beyond the computed range is asserted.

[write] **Added 7 October 2026.** The write recomputed these finite values with its own implementation of (182) for $k \leq 128$ (exact integers, then rational ratios): the maximum of $M_{2k}/(2B_{k+1})$ over $k \leq 128$ is 4.71281008... at $k = 19$, the ratio decreases strictly for $19 \leq k \leq 128$, and every entry of Tables 4 and 5 at $k = 16, 32, 64$ and 128 agrees with the printed digits ($k = 256$ was not recomputed). These remain finite observations; item 12 of Section 56 records the monotonicity question.

35 A separate Poisson neighborhood calculation

The combinatorial theorem above is independent of spectral approximation. The following scalar model isolates a local high-degree mechanism and gives a sharp warning about averaging it outside its domain.

Let $X \sim \text{Poisson}(1)$ and, conditionally on $X = m$, let $Y \sim \text{Poisson}(m)$. Set

$$r = W(k), \quad x = \frac{k+1}{W(k+1)}, \quad Q_k = B_{k+1}, \quad \delta_k = \frac{r^3}{k}.$$

For a fixed real constant c , use the event

$$\mathcal{A}_k = \{X \geq x/2, |Y - X| \leq \sqrt{X} \log X\}. \quad (184)$$

For sufficiently large k , the bases below are positive on this event.

Theorem 35.1. *For every fixed $c \in \mathbb{R}$,*

$$\mathbb{E} \left[\left(X + \frac{Y}{X} + \frac{c}{X} \right)^k ; \mathcal{A}_k \right] = Q_k \left[1 + \left(\frac{1}{2} + o(1) \right) \frac{r^3}{k} \right]. \quad (185)$$

This is a scalar statement; it does not identify M_{2k} with this expectation or prove a graph-moment correction.

Proof. Fix $m \geq x/2$, write $Z = Y - m$, and put $t = k/[m(m+1)]$. On $|Z| \leq \sqrt{m} \log m$, Taylor's formula gives

$$k \log \frac{m + Y/m + c/m}{m+1} = tZ + ct + O \left(\frac{k \log^2 m}{m^3} + \frac{k}{m^4} \right). \quad (186)$$

All estimates in this proof are uniform for $m \geq x/2$; constants may depend on fixed c . The error in (186) is $o(\delta_k)$: the function $\log^2 m/m^3$ is decreasing for all sufficiently large m , and at the lower endpoint its ratio to δ_k is $O(r^2/k) = o(\delta_k)$.

The centered Poisson moment-generating function is explicit:

$$\begin{aligned}\mathbb{E}e^{tZ} &= \exp\{m(e^t - 1 - t)\} \\ &= 1 + \frac{k^2}{2m(m+1)^2} + o(\delta_k).\end{aligned}\tag{187}$$

Indeed $t = O(r^2/k)$, $mt^2 = O(\delta_k)$, $mt^3 = O(r^5/k^2) = o(\delta_k)$, and $\delta_k^2 = o(\delta_k)$.

The truncation also has to be checked under the exponential weight. Exponential tilting replaces the Poisson mean m by $me^t = m + O(k/m) = m + O(r)$. This displacement is $o(\sqrt{m} \log m)$ uniformly on the half-line. A Poisson Chernoff bound at the tilted mean therefore gives

$$\mathbb{E}[e^{tZ}; |Z| > \sqrt{m} \log m] \leq \mathbb{E}e^{tZ} \exp(-c_0 \log^2 m) = o(\delta_k)$$

for one fixed $c_0 > 0$. This is stronger than an unweighted tail bound and is the estimate needed in (187). Combining it with (186) yields the same expansion for the truncated conditional expectation after division by $(m+1)^k$.

Finally divide the outer expectation by Q_k . Under the shifted Dobinski tilt, $X/x \rightarrow 1$ and $X < x/2$ has exponentially small probability. On $m \geq x/2$, the normalized function

$$\frac{k^2/[m(m+1)^2]}{k^2/x^3}$$

is uniformly bounded and tends to one in tilted probability. Its mean therefore tends to one. Since $k^2/x^3 \sim r^3/k$, averaging (187) proves (185). $\square$

35.1 Two explicit failures of untruncated averaging

For $c = 0$, the event $X = 1$ contributes exactly

$$\mathbb{E}\left[\left(X + \frac{Y}{X}\right)^k; X = 1\right] = e^{-1}\mathbb{E}[(1 + Y_1)^k] = e^{-1}Q_k, \quad Y_1 \sim \text{Poisson}(1).\tag{188}$$

It is disjoint from $\mathcal{A}_k$ for large k . Nonnegativity and Theorem 35.1 imply

$$\liminf_{k \rightarrow \infty} \frac{\mathbb{E}[(X + Y/X)^k; X \geq 1]}{Q_k} \geq 1 + e^{-1}.$$

Thus dropping the restriction already creates an extra leading constant. In the local-neighborhood interpretation this uses a parent-hub formula when the large hub is actually a child; the required spectral mass allocation is absent.

For $c = 2$, the same event contributes $e^{-1}\mathbb{E}[(Y_1 + 3)^k]$. The Poisson shift identity and the Bell saddle lemma give

$$\mathbb{E}[(Y_1 + 3)^k] = \mathbb{E}[(Y_1)_3 Y_1^k] \sim \left(\frac{k}{W(k)}\right)^3 B_k,\tag{189}$$

where $(y)_3 = y(y-1)(y-2)$. Since $B_{k+1}/B_k \sim k/W(k)$,

$$\frac{\mathbb{E}[(Y_1 + 3)^k]}{Q_k} \sim \left(\frac{k}{W(k)}\right)^2 \rightarrow \infty.$$

An untruncated second-order local formula can therefore predict a spurious divergent multiplier even though its valid high-degree window has equivalent Q_k .

The coefficient $\frac{1}{2}r^3/k$ in (185) is consistent with the relative scale of the repeated-secondary-edge lower family in Report210. This consistency is useful evidence, but is neither an upper bound nor a second-order theorem for M_{2k} .

[write] **Added 7 October 2026.** Part I proves the corresponding statement for the moments themselves: $M_{2k}/(2B_{k+1}) = 1 + (\frac{1}{2} + o(1))W(k)^3/k$ by (6). Its proof does not use Theorem 35.1, and no Part identifies M_{2k} with the scalar expectation in (185); the equality of the two coefficients is an observation, not a consequence of either theorem.

36 What the spectral literature supplies

The bounded source check concerns unscaled adjacency moments at mean degree one. It does not establish a global priority claim or global open-problem status. The following statements are separated from the deductions of this report.

36.1 A rigorous high degree expansion

Hiesmayr–McKenzie [9, Theorem 1.2] prove that, for $\log^{-1/9} N \leq d \leq \log^{1/40} N$, with high probability each of the top $\exp(\log^{1/8} N)$ adjacency eigenvalues of $G(N, d/N)$ is associated with a vertex v and obeys

$$\lambda = \sqrt{\alpha_v + \frac{\beta_v}{\alpha_v} + \frac{d^2 + d}{\alpha_v}} + O((d^{3/2} + 1)u^{-11/6}). \quad (190)$$

Here α_v is its degree, β_v its second-sphere size, and u the extremal-degree scale. Their Section 1.4 treats the negative edge. Writing λ_{loc} for the top eigenvalue of a controlled radius- r hub neighborhood, their Lemma 4.10 gives

$$\lambda_{\text{loc}}^2 = \alpha + \frac{\beta}{\alpha} + \frac{\beta^{(1,1)} + \beta^{(2)}}{\alpha^2} - \frac{\beta^2}{\alpha^3} + O((d^2 + d)u^{-5/3}), \quad (191)$$

where $\beta^{(1,1)}$ is the third-sphere size and $\beta^{(2)}$ sums squared offspring counts of neighbors. These estimates operate on specified good neighborhoods and events; Lemma 4.2 and Section 6 supply that framework. Theorem 1.4 supplies localization. The checked preprint is the source of these claims; the journal full text was not independently inspected.

36.2 Deductions and the missing transfer

The following interpretation is our deduction. At critical mean degree, a Poisson branching neighborhood conditioned on degree a has second-sphere size distributed as $\text{Poisson}(a)$. Its typical third-sphere size is a , and the sum of squared neighbor offspring counts is about $2a$. Substitution into (191) suggests a squared energy $a + 1 + 2/a$. Allocating one positive and one negative eigenvalue to a hub would then suggest $2\mathbb{E}[(X + 1)^k] = 2B_{k+1}$. The allocation is essential; a root-only eigenvector computation does not itself produce the trace factor two.

At the Bell saddle $a \asymp k/W(k)$, an error η in squared energy induces a relative multiplier of order $\exp(W(k)\eta)$. Thus $\eta = o(1/W(k))$ would suffice locally, or an eigenvalue error $o(1/(\sqrt{a}W(k)))$. The

precision in (190) is compatible with this requirement on its controlled high-degree configurations. Local precision, however, leaves four distinct tasks:

1. Bound the *moment-weighted* trace on exceptional neighborhoods. A probability $o(1)$ alone gives no bound relative to B_{k+1} .
2. Cover the whole relevant spectral region, beyond the specified finite- N extreme ranks.
3. Justify the order of limits and any choice of $N = N(k)$ by quantitative finite-size moment comparison and tail control.
4. Allocate spectral mass to hubs without duplicating contributions from nearby hubs or discarding the remaining eigenvalues.

The counterexamples in Section 35 make the first and fourth issues concrete. They are not defects in the cited theorem, whose claims have a different scope.

36.3 Other checked sources and operator distinctions

Bauer–Golinelli [1] provide the original exact recurrence, tree-walk bounds, and fixed-tree-size asymptotics. Hainzl–de Panafieu [4] develop later tree-walk and kernel/decorating methods. These are important counting tools; an asymptotic with a fixed number of distinct edges cannot by itself settle this high-moment regime, whose important tree sizes grow.

Bhattacharya–Bhattacharya–Ganguly [10, Theorems 1.1–1.2] establish logarithmic large-deviation scales for the sparse spectral edge, including fixed mean degree. Their fixed multiplicative eigenvalue deviations do not retain the constant squared-energy shift needed to distinguish B_k from B_{k+1} at relative precision.

Addario-Berry–Lugosi–Oliveira [11] give degree-sensitive closed-walk coding and rewiring tools for uniformly labeled trees; their principal theorem compares the top eigenvalue to the square root of the maximal degree. The model and precision differ from the present unconditioned critical moment problem. The deterministic tools are possible ingredients for future bounds, not a ready-made relative equivalent.

Heydenreich–Müller–Terveer [12], in the checked July 2026 version, study simple-random-walk return probabilities on supercritical survival-conditioned trees and the lower spectral edge of the *graph Laplacian*. Their Theorems 1.1 and 1.4 do not address critical unscaled adjacency high moments. A Laplacian high-energy remark cannot be transferred to adjacency without a separate argument.

Bordenave’s survey [13] was used as a citation and scope cross-check; its Theorem 4.5 discussion points to sharper edge results rather than stating the present relative equivalent. No theorem stating or immediately proving $M_{2k} \sim 2B_{k+1}$ was located in these inspected sources. This is a bounded report of the search, not a claim that the problem is globally open.

37 Questions that would materially advance the result

1. **Can the middle-return tail be bounded directly?** The immediate target is (162), with either admissible window. A useful approach must control a weighted sum over m , not just each fixed m or a fixed number of edges. The exact recurrence and orbit weights specify the required quantity.

2. **Can the urn window be enlarged?** The rough bound $M_{2t} \leq 4^{t-1}t!$ produces the error $O(S^3/k)$. A sharper collision estimate or a more refined child-walk weight could move the current $o(k^{1/3})$ boundary. Such an improvement would strengthen the proved class without by itself resolving its complement.
3. **Can dominant hubs be selected canonically in the complement?** The exact two-hub correction works above $k/2$. A useful global coding scheme would charge complicated off-hub pieces to a selected vertex while preserving enough factorial savings. Degree-sensitive walk coding and tree rewiring are possible tools, but departure counts and graph degrees must not be conflated.
4. **Can a weighted spectral transfer be proved?** One would need a trace estimate on exceptional neighborhoods and quantitative control of the $N \rightarrow \infty$, then $k \rightarrow \infty$ limit. A high-probability local expansion alone cannot supply it.
5. **What is the first correction for the full moments?** The scalar correction and the disjoint repeated-secondary-edge lower family suggest the scale $W(k)^3/k$ relative to $2B_{k+1}$. A matching upper bound must account for all competing repeated-edge configurations. The present theorem does not provide that upper bound or its coefficient.

[write] **Status of these questions, 7 October 2026.** Question 1 is answered within this report: Part II bounds the weighted middle-return tail superpolynomially for every admissible window (Theorem 14.1). Question 5 is answered by Part I: the first relative correction of the full moments is $(\frac{1}{2} + o(1))W(k)^3/k$ by (6), the scale this question names and the coefficient of the scalar Theorem 35.1; by Remark 46.1 it is carried by Part IV's family with one repeated secondary edge. Questions 2–4 remain open; they are items 6, 7 and 8 of Section 56, where the deductions of Section 36.2 (the hub allocation and the four missing tasks) are recorded as the heuristic they are.

38 Reproduction and verification scope

The accompanying archive contains this article and editable source, an exact C++/GMP generator, standard-library Python checks, a small complete fixture, compact exact diagnostic results, and a reproducible driver. It does not contain large extended raw arrays, third-party papers, or review correspondence. Public source links and the bounded inspection scope are listed in `SOURCES.txt`.

The verification levels must be distinguished:

- Direct enumeration of normalized walks through $k = 8$ is independent of both recurrences. It checks every root-return row, the partition/cut formula, the first-edge recurrence, the weighted rotation identity, and the exact two-hub formula.
- A separate Python implementation of the first-edge recurrence reaches $k = 16$ and compares every root row, together with previously checked edge-resolved reference totals.
- A fresh C++/GMP run regenerates complete rows through $k = 256$ for the public tables. Repeating this source algorithm is an independent execution, not an independent mathematical algorithm for every high row.
- Row sums, complete index sets, boundary identities, and exact rational finite-ratio comparisons are consistency checks. Reading stored higher rows alone is not fresh generation.

Every correctness guard is an explicit runtime check, retained under Python optimization. Deliberately corrupted inputs test named guards under both normal Python and `-O`; an assertion-free positive run alone would not establish that failure detection works. The integer computations and rational formulas precede decimal formatting; floating-point approximations do not generate the counts.

The `README.txt` gives exact commands and dependencies. The default full replay generates the public data and tables and rebuilds the PDF; the quick mode has a separately stated smaller verification scope. Optional generation through $k = 512$ is available from source and is unnecessary for any displayed value or proof. The distributed fixture and compact tables are deliberately much smaller than that extended raw array. Package verification checks the actual archive inventory, each member's bytes, and the whole archive digest. PDF byte identity is scoped to the same TeX and font toolchain. None of these finite checks replaces the all- k arguments above.

A Elementary Bell saddle estimates

[write] **Printed once, 7 October 2026.** The first part of Report 212's Appendix A, the elementary Bell lemma with its proof, is the same lemma as Part I's Lemma A.1 with the same proof in slightly different words (Report 213 prints a third copy, also not printed again); it is printed once, in Part I. Its numbered objects, each with the number Report 212 gave it (shifted as in this Part) and the number of its printed counterpart in Part I: Lemma A.1 $\mapsto$ Lemma A.1; (192) $\mapsto$ (59); (193) $\mapsto$ (60); (194) $\mapsto$ (61). References to them elsewhere in this Part point to the printed counterparts. Report 212's text resumes with its last paragraph, which also justifies (189).

The shifted law used in Sections 31 and 35 is exact, not an approximation. Under weight $(X + 1)^k$ for $X \sim \text{Poisson}(1)$, substitute $y = X + 1$ to get

$$\frac{(X + 1)^k}{X!} = \frac{y^{k+1}}{y!}.$$

Thus $X + 1$ has precisely the law of Y_{k+1} . In particular $X/x_k \rightarrow 1$, and for large k , $X < x_k/2$ implies $X + 1 < 3x_k/4$, giving the exponential estimate used above. Also the fixed-moment conclusion applied to $(Y_N)_3 = Y_N^3 - 3Y_N^2 + 2Y_N$ justifies (189).

Part IV

Larger tree walk lower bounds for sparse graph moments

Bundle Report 210, archive `Report210-reproducibility.zip`, delivered as `Report210.tex` (438 lines, 12 pages). Delivered title block "Larger tree walk lower bounds for sparse graph moments / Report210 / 4 October 2026" (empty author line). Printed as delivered, with [write] notes and one remark. Report 210's Section k is Section $k + 38$ here, its equation (k) is equation $(k + 194)$ and its Table 1 is Table 6; statements move with their sections. Its text names three of its sections by number: its "Section 1", "Section 4" and "Sections 5–7" are Sections 39, 42 and 43–45 here. Its labels carry `sgm:lw::`; the write labelled its Section 8 (`sgm:lw:sec:what`); its table is printed inline. "Report209" in this Part is Part V. This Part is self-contained.

Abstract of this Part (as delivered). Let M_{2k} count closed tree walks of length $2k$ up to vertex relabeling, with the initial time distinguished. These are the positive-order terms of

OEIS A094149. We count two disjoint families exactly. The first, of size G_k , contains every walk whose repeated edges share a vertex, including arbitrary fresh tree attachments and every time rotation. The second, of size J_k , has a uniquely recoverable hub and one repeated secondary edge. We prove $M_{2k} \geq G_k + J_k$ for every $k \geq 1$, and, with $r = W(k)$,

$$G_k = 2B_{k+1} [1 + (1 + o(1))r^2/k], \quad J_k \sim r^2 B_k.$$

Consequently $M_{2k} \geq 2B_{k+1} + (1 - o(1))W(k)^2 B_k$. The additive correction is larger than that supplied by fresh trees alone, but remains $o(B_{k+1})$. The complete one-hub family has limiting ratio $G_k/B_{k+1} = 2$; this is a limitation for that structural class, not an upper bound for all moments. The proofs and standard-library exact checks are self-contained. Published tree-walk, Catalan-decoration, and kernel infrastructure is credited; no priority, full-moment asymptotic, or inverse theorem is claimed.

39 The exact lower envelope

A *normalized tree walk* is a sequence $w = (v_0, v_1, \dots, v_{2k-1}, v_0)$ whose traversed unordered pairs form a tree. Consecutive vertices are distinct. Two sequences are equivalent if one vertex bijection sends their entries to one another in order. Labeling vertices $0, 1, 2, \dots$ by actual first appearance gives a unique representative. The start remains distinguished: rotations are generally different walks.

Let $\mathcal{W}_k$ be this set and $M_{2k} = |\mathcal{W}_k|$. For $k \geq 1$ these are the terms of A094149 [17]; set $M_0 = 1$ separately. They are also the limiting expected normalized-trace moments of the unscaled adjacency matrix of $G(N, 1/N)$, with k fixed before $N \rightarrow \infty$ [1]. The asymptotics in this article subsequently let $k \rightarrow \infty$.

Deleting an edge of a tree separates it into two components, so a closed walk crosses that edge equally often in either direction. Call an edge *simple* if its total traversal count is two, and *repeated* if its count exceeds two. A repeated edge thus has total count at least four. Let B_n be the Bell number, counting partitions of $[n]$, with $B_0 = 1$. Let $W(t) > 0$ solve $W(t)e^{W(t)} = t$ for $t > 0$.

Define $\mathcal{G}_k$ by the requirement that all repeated edges have a common incident vertex, with the empty repeated-edge set allowed. Write $G_k = |\mathcal{G}_k|$. To define $\mathcal{J}_k$, first consider walks rooted at a vertex c with these properties:

1. The underlying tree has height at most two from c .
2. Exactly one edge between depths one and two is traversed four times. Every other edge between these depths is simple.
3. At least three edges incident to c are repeated.

Take the full closure of these walks under time rotation and call it $\mathcal{J}_k$, of size J_k . The vertex c is recoverable without knowing the initial time: it is the unique vertex incident to at least three repeated edges. Every other vertex is incident to at most two. The repeated secondary edge avoids c , so $\mathcal{J}_k \cap \mathcal{G}_k = \emptyset$.

Theorem 39.1. *For $n \geq 1$ put*

$$P_n = B_n - \frac{3^n + 1 + n(n-1)}{2} + n2^{n-1}, \tag{195}$$

$$C_1 = 0, \quad C_n = B_{n-1} - 3^{n-2} + (n-2)2^{n-3} \quad (n \geq 2), \tag{196}$$

where the last product is zero at $n = 2$. For every integer $k \geq 1$,

$$G_k = \sum_{n=1}^k \frac{2k}{2k-n} \binom{2k-n}{k-n} B_n - \frac{4^k - \binom{2k}{k}}{2}, \quad (197)$$

$$J_k = \sum_{n=1}^{k-2} k \binom{k-1}{n+1} [2P_n + (n-1)C_n], \quad (198)$$

$$M_{2k} \geq G_k + J_k. \quad (199)$$

Empty sums are zero. Both G_k and J_k are actual family sizes. In particular $J_k = 0$ for $k < 8$ and $J_8 = 360$.

The symbol C_n is a partition count, not a Catalan number. We write $\text{Cat}_j = \binom{2j}{j}/(j+1)$ for Catalan numbers throughout.

Theorem 39.2. As $k \rightarrow \infty$, with $r = W(k)$,

$$G_k = 2B_{k+1} \left[1 + (1 + o(1)) \frac{r^2}{k} \right], \quad (200)$$

$$J_k \sim \frac{r^3}{k} B_{k+1} \sim r^2 B_k. \quad (201)$$

Thus the exact lower envelope in (199) satisfies

$$G_k + J_k = 2B_{k+1} + (1 + o(1))W(k)^2 B_k, \quad M_{2k} \geq 2B_{k+1} + (1 - o(1))W(k)^2 B_k. \quad (202)$$

In the inequality, $o(1)$ can be chosen nonnegative. For each fixed integer $h \geq 2$,

$$\frac{G_k}{B_{k+h}} \sim 2 \left(\frac{W(k)}{k} \right)^{h-1} \longrightarrow 0. \quad (203)$$

In particular, counting any family that stays inside $\mathcal{G}_k$ more precisely cannot produce a diverging multiplier of B_{k+1} . Adding J_k improves the additive order from $W(k)B_k$ to $W(k)^2 B_k$, but still does not produce such a multiplier.

Table 6: Direct enumeration and the exact disjoint lower envelope

k	G_k	J_k	$G_k + J_k$	M_{2k}	Omitted
1	1	0	1	1	0
2	3	0	3	3	0
3	12	0	12	12	0
4	57	0	57	57	0
5	298	0	298	303	5
6	1642	0	1642	1747	105
7	9348	0	9348	10727	1379
8	54623	360	54983	69331	14348

40 Fresh trees attached to a hub

Let $\mathcal{H}_k$ consist of the walks in which every edge not incident to the initial vertex is simple. Write $H_k = |\mathcal{H}_k|$. Cutting at successive returns to the root gives n root excursions. Their first root-child steps define a partition of $[n]$: two positions lie in the same block precisely when those excursions visit the same child.

After reaching that child, each excursion traverses a fresh rooted plane tree before returning to the root. Every nonroot edge is crossed once in each direction. Such an edge cannot recur in another excursion, even if the root child does recur. The order in which its child subtrees are traversed is exactly the plane order. Therefore the Catalan series

$$C(z) = \sum_{j \geq 0} \text{Cat}_j z^j = 1 + zC(z)^2 = \frac{1 - \sqrt{1 - 4z}}{2z}$$

counts one decoration, with z marking a simple edge. Conversely, choose a root-child partition and one fresh plane tree for each excursion, concatenate their contour walks, and only then assign global first-appearance labels. This recovers each normalized walk once. The global discovery order need not itself be depth-first. No extra factorial for labels or fresh vertices is present.

With n root departures and total half-length k , this bijection gives

$$H_k = \sum_{n=1}^k h(k, n) B_n, \quad h(k, n) = [z^{k-n}] C(z)^n = \frac{n}{2k-n} \binom{2k-n}{k-n}. \quad (204)$$

For completeness, let $f(z) = zC(z)$. Then $f = z/(1-f)$, so formal Lagrange inversion gives

$$[z^k] f(z)^n = \frac{n}{k} [u^{k-n}] (1-u)^{-k} = \frac{n}{k} \binom{2k-n-1}{k-n} = h(k, n).$$

All formal series manipulations here and below are coefficientwise; no convergence of the ordinary Bell-number generating series is assumed.

A block of size b in the root partition corresponds to an edge traversed $2b$ times. Thus repeated root edges correspond to nonsingleton blocks. For partitions of $[n]$, the numbers having zero, exactly one, and at least two nonsingleton blocks are respectively

$$1, \quad 2^n - n - 1, \quad B_n - 2^n + n. \quad (205)$$

The middle count chooses the unique nonsingleton block as any subset of size at least two.

41 All time rotations without a freeness assumption

Define R by moving the initial position forward one time step and normalizing vertex names:

$$R(w) = \text{norm}(v_1, v_2, \dots, v_{2k-1}, v_0, v_1).$$

This is a permutation of $\mathcal{W}_k$ with $R^{2k} = 1$. Individual normalized words can have smaller periods, including periods shortened by a vertex permutation. The following double count incorporates those stabilizers automatically.

Lemma 41.1 (Time indexed rotation count). *Let $\mathcal{F} \subseteq \mathcal{W}_k$ be closed under rotation. Suppose each walk has an equivariantly determined set of exactly q admissible vertices. Let $\mathcal{H}$ be those walks of $\mathcal{F}$ whose initial vertex is admissible. For $u \in \mathcal{H}$, let $n(u)$ count occurrences of its initial vertex in the $2k$ time positions, omitting the repeated final entry. Then*

$$|\mathcal{F}| = \frac{2k}{q} \sum_{u \in \mathcal{H}} \frac{1}{n(u)}. \quad (206)$$

Proof. Give every pair (u, t) , with $u \in \mathcal{H}$ and $0 \leq t < 2k$, weight $1/n(u)$ and send it to $R^t u$. Its total source weight is $2k \sum_u 1/n(u)$. For a target w and time index t , there is exactly one possible source, $u = R^{-t}w$. It is in $\mathcal{H}$ when the corresponding time position of w is an admissible vertex. If an admissible vertex occurs n_v times, its time indices supply n_v separate contributions of $1/n_v$, totaling one. The target receives total weight q . Time indices are counted separately even when they give the same normalized source, so no assumption about orbit length is used. $\square$

If at least two repeated edges share an endpoint, that endpoint is unique. Apply the lemma with $q = 1$ and with that endpoint admissible. If exactly one edge is repeated, its two endpoints are the admissible vertices and $q = 2$. Their visit counts may differ; the lemma still applies. Finally, a walk with no repeated edges is the contour of a plane rooted tree and contributes Cat_k . Using (204) and (205) gives

$$\begin{aligned} G_k &= \text{Cat}_k + \sum_{n=1}^k \frac{2k}{n} h(k, n) (B_n - 2^n + n) + \sum_{n=1}^k \frac{k}{n} h(k, n) (2^n - n - 1) \\ &= \text{Cat}_k + \sum_{n=1}^k \frac{k}{n} h(k, n) (2B_n - 2^n + n - 1). \end{aligned} \quad (207)$$

This also proves that $\mathcal{G}_k$ is precisely the full rotation closure of $\mathcal{H}_k$.

To simplify, put $d(k, n) = (k/n)h(k, n)$ and $T_k = \sum_n d(k, n)B_n$. Since $z = f(1 - f)$ and $f' = 1/\sqrt{1 - 4z}$,

$$\begin{aligned} \sum_{k \geq 1} \sum_n d(k, n) 2^n z^k &= z \frac{d}{dz} [-\log(1 - 2f)] = \frac{2z}{1 - 4z}, \\ \sum_{k \geq 1} \sum_n d(k, n) z^k &= z \frac{d}{dz} [-\log(1 - f)] = \frac{1}{2} ((1 - 4z)^{-1/2} - 1). \end{aligned}$$

Also $\sum_{n \geq 1} f^n = f/(1 - f) = C - 1$, so $\sum_n d(k, n)n = k \text{Cat}_k$. Substituting these three identities into (207) and using $(k + 1) \text{Cat}_k = \binom{2k}{k}$ yields

$$G_k = 2T_k - \frac{4^k - \binom{2k}{k}}{2}, \quad (208)$$

which is (197).

41.1 Equality at small orders and explicit omitted walks

In a tree, a pairwise-intersecting collection of edges has a common vertex: if two edges meet at v , a third meeting both without containing v would form a triangle. Thus any walk outside $\mathcal{G}_k$ has

two disjoint repeated edges. Each costs at least two outward crossings, and their connecting path contains at least one further edge. Hence $k \geq 5$, proving $G_k = M_{2k}$ for $1 \leq k \leq 4$.

For every $k \geq 5$, take a path $a - b - c - d$. First bounce from a along $a - b$ and back $k - 4$ times, then follow

$$a - b - c - d - c - d - c - b - a.$$

The total length is $2k$. The outer edges are repeated, and the middle edge is simple, so the walk is outside $\mathcal{G}_k$. It has no vertex incident to three repeated edges, so it is also outside $\mathcal{J}_k$. Thus $G_k + J_k < M_{2k}$ for every $k \geq 5$, while equality holds for $1 \leq k \leq 4$.

A walk bouncing on one edge has normalized rotation period one, despite the literal alternating vertex word having period two. This simple example explains why multiplying a rooted count by $2k$ without the reciprocal visit weight would be wrong.

42 One repeated secondary edge

We now count $\mathcal{J}_k$. Start with a walk rooted at its distinguished hub c , and let n be its root-excursion count. Let $s = k - n$ be the number of outward secondary steps. These steps are *occurrences*, not necessarily distinct edges. For this family exactly one secondary edge is used twice outward, so $s \geq 2$.

Choose a partition π of $[n]$ for the root-child sequence and a weak composition $j_1 + \dots + j_n = s$, where j_i counts secondary leaf-and-back tokens in excursion i . Choose exactly one pair of tokens with the same root child and identify their leaves. All other tokens create distinct fresh leaves. This produces one secondary edge with four traversals and makes all other secondary edges simple. The walk recovers the partition, composition, and repeated pair, so this is a bijection after first-appearance normalization.

42.1 The unrestricted rooted count

Temporarily omit the condition of three repeated root edges. Pairs of tokens in the same excursion contribute, over all weak compositions,

$$\sum_{j_1 + \dots + j_n = s} \sum_i \binom{j_i}{2} = n[z^{s-2}](1 - z)^{-n-2} = n \binom{k-1}{s-2}.$$

Every root partition is allowed for these pairs, giving a factor B_n . For a fixed pair of distinct excursions $i < j$, the sum of $j_i j_j$ over compositions is $\binom{k-1}{s-2}$. Their root children must coincide; merging positions i, j leaves B_{n-1} partitions. Therefore the unrestricted rooted count at fixed n is

$$D(k, n) = \binom{k-1}{s-2} \left[nB_n + \binom{n}{2} B_{n-1} \right] \quad (s = k - n \geq 2). \quad (209)$$

Set it to zero otherwise, and define $U_k = \sum_n D(k, n)$.

The quantity U_k counts an *unrestricted rooted* family. It is not J_k , and it need not be disjoint from G_k . Nor may one simply rotate all of it with a unique-hub weight: an intrinsic unique hub is not yet guaranteed.

42.2 Removing ambiguous hubs exactly

The required root partitions have at least three nonsingleton blocks. Their number is P_n . To prove (195), a singleton block contributes z to a partition exponential generating function, and a nonsingleton block contributes $e^z - 1 - z$. Thus partitions with at most two nonsingleton blocks have exponential generating function

$$e^z \left[1 + (e^z - 1 - z) + \frac{(e^z - 1 - z)^2}{2} \right] = \frac{1}{2}e^{3z} - ze^{2z} + \frac{1}{2}(1 + z^2)e^z.$$

Multiplying its z^n coefficient by $n!$ gives $(3^n + 1 + n(n-1))/2 - n2^{n-1}$, the count subtracted from B_n .

Let C_n count valid partitions in which a specified pair belongs to the same block. Without the restriction there are B_{n-1} such partitions. If the pair's block has a additional elements, the remaining $n-2-a$ elements may have at most one nonsingleton block for the partition to be invalid. The number of partitions of h elements with at most one nonsingleton block is $2^h - h$. Hence the number to subtract is

$$\sum_{a=0}^{n-2} \binom{n-2}{a} (2^{n-2-a} - (n-2-a)) = 3^{n-2} - (n-2)2^{n-3},$$

with the stated zero-product convention at $n=2$. This proves (196). Replacing B_n, B_{n-1} in (209) gives the restricted rooted count

$$D_{\text{good}}(k, n) = \binom{k-1}{s-2} \left[nP_n + \binom{n}{2} C_n \right]. \quad (210)$$

There is now exactly one admissible hub, as proved in Section 1. Lemma 41.1 gives $J_k = \sum_n (2k/n) D_{\text{good}}(k, n)$. Since $\binom{k-1}{s-2} = \binom{k-1}{n+1}$, this is (198). The three nonsingleton blocks require $n \geq 6$, and the repeated secondary edge requires $s \geq 2$, so $k \geq 8$. At $k=8$, only $n=6$ contributes; $P_6 = 15$ and $C_6 = 3$ give $J_8 = 360$.

42.3 A useful unrestricted weighted sum

Define

$$E_k = \sum_{n=1}^{k-2} k \binom{k-1}{n+1} [2B_n + (n-1)B_{n-1}]. \quad (211)$$

This is the sum $\sum_n (2k/n) D(k, n)$. It is distinct from both U_k and J_k ; we use it as an algebraic aid, not as the cardinality of a claimed rotation family.

A root partition with at most two nonsingleton blocks injects into a ternary word: label singleton positions 0 and the two possible nonsingleton blocks 1, 2 in order of their least elements. Thus there are at most 3^n excluded partitions. For each, there are $\binom{k-1}{s}$ weak compositions and at most $\binom{s}{2} \leq k^2/2$ token pairs. Multiplying by the weight $2k/n \leq 2k$ gives

$$0 \leq E_k - J_k \leq k^3 \sum_{n=1}^k 3^n \binom{k-1}{k-n} = 3k^3 4^{k-1}. \quad (212)$$

This deliberately coarse exponential error will be negligible on the Bell scale. The exact nonnegative lower bound is always (199); the auxiliary expression $G_k + E_k - 3k^3 4^{k-1}$ can be weak or negative at small k .

43 Bell numbers and a precise Poisson saddle

This section establishes the probability estimates needed to pass from exact counts to Theorem 39.2. The set-partition exponential generating function is $\exp(e^z - 1)$. Expanding it as $e^{-1} \sum_{m \geq 0} e^{mz}/m!$ gives Dobinski's identity

$$B_j = \frac{1}{e} \sum_{m \geq 0} \frac{m^j}{m!} = \mathbb{E}[X^j], \quad X \sim \text{Poisson}(1). \quad (213)$$

At $j = 0$ the term 0^0 is interpreted as one. Shifting $m + 1 = y$ also gives

$$Q_k := \mathbb{E}[(X + 1)^k] = B_{k+1}. \quad (214)$$

Put

$$N = k + 1, \quad \rho = W(N), \quad x = N/\rho = e^\rho, \quad \delta_k = k/x^2 \sim W(k)^2/k.$$

In particular $x \rightarrow \infty$, $\rho \rightarrow \infty$, and $\delta_k \rightarrow 0$. Under the tilted probability law

$$\mathbb{P}_k(X = m) = \frac{(m + 1)^k}{e m! Q_k}, \quad m \geq 0, \quad (215)$$

the variable $Y = X + 1$ has unnormalized masses $a_y = y^N/y!$ for $y \geq 1$.

Lemma 43.1. *Under (215), $X/x \rightarrow 1$ in probability. Each fixed relative tail has probability at most $\exp(-c_\epsilon k + o(k))$ for some $c_\epsilon > 0$. In addition, for every fixed $0 < c < \log 2 - 1/2$ and all sufficiently large k ,*

$$\frac{\mathbb{E}[(X + 2)^k; X \leq x/2]}{Q_k} \leq e^{-ck}. \quad (216)$$

Consequently

$$\frac{B_{k+1}}{B_k} \sim \frac{k}{W(k)}, \quad \frac{4^k}{Q_k} = o(\delta_k). \quad (217)$$

Proof. The consecutive mass ratios

$$\frac{a_{y+1}}{a_y} = \frac{(1 + 1/y)^N}{y + 1}$$

strictly decrease in y . At $y = \lfloor tx \rfloor$, with fixed $t > 0$,

$$\log \frac{a_{y+1}}{a_y} = \rho(1/t - 1) - \log t + o(1).$$

Here $N/y^2 = O(\rho/x) = o(1)$ and $\log x = \rho$. On the interval from $(1 - \epsilon)x$ to $(1 - \epsilon/2)x$, the logarithmic ratios are at least a positive multiple of ρ for fixed $0 < \epsilon < 1$ and large k . Multiplying across an interval of length proportional to x makes the lower endpoint mass at most $e^{-c_\epsilon N + o(N)}$ times an interior mass. There are only $O(x)$ terms further to the left, each no larger than the endpoint mass. The analogous interval from $(1 + \epsilon/2)x$ to $(1 + \epsilon)x$ gives the upper tail estimate, and the subsequent decreasing ratios sum to a bounded geometric factor. This proves concentration of Y/x , and hence of X/x .

For the stronger shifted estimate, let $a = \lfloor x/2 \rfloor$, $q = \lfloor x \rfloor$, and $b_m = (m + 2)^k/m!$. Its consecutive ratios decrease and at $m = a$ have logarithm $\rho + O(1) > 0$. Thus b_m increases on $0 \leq m \leq a$

for large k . Bound the numerator of (216) by $(a+1)b_a/e$ and Q_k below by $(q+1)^k/(eq!)$. The elementary factorial estimate $\log(j!) = j \log j - j + O(\log(j+1))$ yields

$$\begin{aligned} \log \frac{(a+1)(a+2)^k q!}{a!(q+1)^k} &= k \log(1/2) + (q-a) \log x + O(x + \log k) \\ &= -(\log 2 - 1/2)k + o(k), \end{aligned}$$

because $x \log x = N$. This proves (216).

Finally $B_k/B_{k+1} = \mathbb{E}_k[1/Y]$. On $Y \geq x/2$, the quantity x/Y is bounded by two and converges in probability to one. On $Y < x/2$, its contribution is at most x times an exponentially small probability. Hence $\mathbb{E}_k[x/Y] \rightarrow 1$, proving $B_{k+1}/B_k \sim x \sim k/W(k)$. For the last assertion, fix any integer $m > 4$ in (214); its single term gives $Q_k \geq (m+1)^k/(em!)$. The resulting exponential decay of $4^k/Q_k$ is faster than the reciprocal-polynomial scale δ_k . $\square$

The elementary asymptotics $W(k+1) \sim W(k)$ and $W(k) \sim \log k$ follow from $W(k) + \log W(k) = \log k$. They justify the equivalent scales used above.

44 The complete one hub asymptotic

Replace Bell numbers in T_k by scalar powers:

$$t_k(m) = \sum_{n=1}^k \frac{k}{n} h(k, n) m^n, \quad T_k = \mathbb{E}[t_k(X)].$$

Since $\sum_{n \geq 1} m^n f^n/n = -\log(1 - mf)$, formal differentiation gives

$$\sum_{k \geq 1} t_k(m) z^k = \frac{zm}{\sqrt{1 - 4z(1 - mf(z))}}. \quad (218)$$

For integers $m \geq 3$ put $\lambda_m = m^2/(m-1) = m+1 + 1/(m-1)$. Rationalizing (218), or multiplying its denominator by its conjugate, gives

$$t_k(m) = \frac{\lambda_m^k}{2} + \frac{m-2}{2m} \sum_{j=0}^{k-1} \lambda_m^{k-j} \binom{2j}{j}. \quad (219)$$

To see the relevant cancellation explicitly, $\sum_{j \geq 0} \binom{2j}{j} \lambda_m^{-j} = (1 - 4/\lambda_m)^{-1/2} = m/(m-2)$. The infinite version of the sum in (219) therefore supplies the other half of λ_m^k . Its positive tail is bounded using $\binom{2j}{j} \leq 4^j$:

$$0 \leq \lambda_m^k - t_k(m) \leq \frac{m}{2(m-2)} 4^k \leq \frac{3}{2} 4^k. \quad (220)$$

The exceptional scalar values, read from (218), are $t_k(0) = 0$, $t_k(1) = \binom{2k}{k}/2$, and $t_k(2) = 4^k/2$. It follows uniformly after Poisson averaging that

$$T_k = S_k + O(4^k), \quad S_k = \mathbb{E}[\mathbf{1}_{\{X \geq 3\}} \lambda_X^k]. \quad (221)$$

On $X \geq x/2$, for large k ,

$$\begin{aligned} \frac{\lambda_X^k}{(X+1)^k} &= \left(1 + \frac{1}{X^2-1}\right)^k \\ &= 1 + \frac{k}{X^2} + O(\delta_k^2 + \delta_k/x^2), \end{aligned} \quad (222)$$

uniformly. Indeed the base perturbation is $O(x^{-2})$, its logarithm differs from it by $O(x^{-4})$, and the exponential argument is $O(\delta_k) = o(1)$. Consequently,

$$\begin{aligned} \mathbb{E}_k \left[\left(\frac{\lambda_X^k}{(X+1)^k} - 1 \right); X \geq x/2 \right] &= \delta_k \mathbb{E}_k[(x/X)^2; X \geq x/2] + O(\delta_k^2) \\ &= \delta_k(1 + o(1)). \end{aligned}$$

The last step uses Lemma 43.1 and the bound $(x/X)^2 \leq 4$. For $3 \leq m < x/2$, $\lambda_m \leq m+2$, so (216) bounds the remaining contribution by $e^{-ck} = o(\delta_k)$. The omitted baseline terms $m = 0, 1, 2$ are also negligible. Thus

$$\frac{S_k}{Q_k} = 1 + (1 + o(1))\delta_k. \quad (223)$$

Equations (208), (221), (223), and (217) prove (200). In particular

$$G_k - 2B_{k+1} \sim 2W(k)B_k. \quad (224)$$

Iterating (217) a fixed $h-1$ times proves (203). This last conclusion depends on the fixedness of h and concerns G_k , not M_{2k} .

45 The secondary edge asymptotic

To evaluate E_k in (211), replace B_n by m^n and B_{n-1} by m^{n-1} , and call the resulting polynomial $e_k(m)$. For $m \geq 1$ and $k \geq 2$, direct binomial summation gives

$$e_k(m) = \frac{k}{m^2} \left\{ (m+1)^{k-2} [2m^2 + (k-1)m - 2] + 2 + (k-3)m - 2(k-1)m^2 \right\}. \quad (225)$$

One way to verify it is to start with

$$\sum_{n=1}^{k-2} \binom{k-1}{n+1} m^n = \frac{(1+m)^{k-1} - 1 - (k-1)m}{m},$$

and use the derivative of this expression for the factor $n-1$. The formula is zero at $k=2$ after cancellation. Set $e_1(m) = 0$ directly and $e_k(0) = 0$ from the defining polynomial; its apparent pole in (225) should not be evaluated at zero. In the $n=1$ term, the factor $n-1$ is zero before any power m^{n-1} is considered.

Dobinski gives $E_k = \mathbb{E}[e_k(X)]$. The last three terms inside the braces, after multiplication by k/m^2 , are uniformly $O(k^2)$ for $m \geq 1$. Dividing by Q_k and using the tilted law yields

$$\frac{E_k}{Q_k} = \mathbb{E}_k[g_k(X); X \geq 1] + O(k^2/Q_k), \quad g_k(m) = \frac{k[2m^2 + (k-1)m - 2]}{m^2(m+1)^2}. \quad (226)$$

Set $\eta_k = k^2/x^3 \sim W(k)^3/k$. On $m \geq x/2$, the contribution of the numerator term $2m^2$, divided by η_k , is at most $8x/k \rightarrow 0$. The contribution of $(k-1)m$ is exactly

$$\left(1 - \frac{1}{k}\right) \left(\frac{x}{m}\right)^3 \left(\frac{m}{m+1}\right)^2.$$

It is uniformly bounded by eight and converges in tilted probability to one. The remaining term, after division by η_k , is uniformly negligible. Bounded convergence in probability therefore proves

$$\frac{\mathbb{E}_k[g_k(X); X \geq x/2]}{\eta_k} \rightarrow 1.$$

On $1 \leq m < x/2$, $|g_k(m)| = O(k^2)$, while the tilted tail is exponentially small by Lemma 43.1. This part is $o(\eta_k)$. The error $O(k^2/Q_k)$ in (226) is negligible by the same fixed-term Bell bound used previously. We have proved

$$E_k \sim \frac{W(k)^3}{k} B_{k+1} \sim W(k)^2 B_k. \quad (227)$$

Bell numbers exceed every fixed-base exponential, again by a fixed term in Dobinski's sum. Hence the error $3k^3 4^{k-1}$ in (212) is $o(W(k)^2 B_k)$, proving (201). Finally (224) is smaller than (201) by a factor asymptotic to $2/W(k)$, so addition proves (202) and completes both main theorems.

46 What the construction establishes

The complete class $\mathcal{G}_k$ permits any depth of fresh simple-edge attachments, with all normalized rotations included. Its first correction above $2B_{k+1}$ is asymptotic to $2W(k)B_k$. This fully counted class proves that those operations alone cannot yield a diverging multiplier of B_{k+1} or a bound of order B_{k+2} .

The family $\mathcal{J}_k$ leaves that class by adding a repeated edge away from the hub. Its count is asymptotic to $W(k)^2 B_k$, which is a genuinely larger additive order. However

$$\frac{W(k)^2 B_k}{B_{k+1}} \sim \frac{W(k)^3}{k} \rightarrow 0.$$

Both families together still have size asymptotic to $2B_{k+1}$. This observation gives no upper bound on the walks omitted from their union. In particular, nothing proved here implies $M_{2k} \sim 2B_{k+1}$, boundedness or divergence of M_{2k}/B_{k+1} , or an inverse characterization of a full-moment sequence.

[write] **Added 7 October 2026.** Part II proves $M_{2k} \sim 2B_{k+1}$ (Theorem 14.1), so $M_{2k}/B_{k+1} \rightarrow 2$, and Part I proves its expansion to every fixed order. The sentence above is kept as delivered; it is true of Report 210. With them, Remark 46.1 at the end of this section shows that $G_k + J_k$ already carries the first relative correction of M_{2k} .

The exact equality range of (199) is $k = 1, 2, 3, 4$. Even when J_k first becomes positive at $k = 8$, the union is far from exhaustive: it counts 54 983 of 69 331 normalized walks. Such small-order fractions are corroboration of the count, not evidence for an unproved large-order equivalence.

46.1 Relation to the preceding shifted Bell construction

The height-two rooted family with all secondary edges simple has size

$$\sum_{s=0}^{k-1} \binom{k-1}{s} B_{k-s} = B_{k+1} - B_k.$$

The first expression follows from the root-partition and weak-composition encoding in Section 4, before identifying any secondary tokens. The identity follows by subtracting the Bell recurrences $B_{j+1} = \sum_{i=0}^j \binom{j}{i} B_i$ at $j = k$ and $j = k - 1$, then using Pascal's identity. This family lies in $\mathcal{H}_k$, and its union with a one-step rotation lies in $\mathcal{G}_k$. The preceding shifted-Bell bound, treated separately in Report209, is therefore subsumed by this exact family enlargement. The proofs here do not require that report or its files.

46.2 Extensions beyond a single secondary pair

A natural next construction would allow several repeated secondary edges while keeping an intrinsically recoverable hub. The token encoding would then require compatible leaf partitions in place of one distinguished pair. Determining whether such an enlargement changes the leading B_{k+1} scale would require estimates uniform in the amount of secondary repetition. A full-moment asymptotic would also require control of the complementary walks, which no estimate in this report supplies.

[write] **Added 7 October 2026.** Part II answers the question of this subsection about the leading scale: every family of walks is contained in $\mathcal{W}_k$, and $M_{2k} \sim 2B_{k+1}$, so no enlargement changes the leading B_{k+1} scale; by Remark 46.1, none changes the first relative correction either. Exact counts of the larger families remain open (item 9 of Section 56).

Remark 46.1 (The two families carry the first correction; added by the write, 7 October 2026). With $r = W(k)$, Theorem 39.2 gives $G_k/(2B_{k+1}) = 1 + (1 + o(1))r^2/k$ and $J_k/(2B_{k+1}) = (\frac{1}{2} + o(1))r^3/k$, hence, since $r^2/k = o(r^3/k)$,

$$\frac{G_k + J_k}{2B_{k+1}} = 1 + \left(\frac{1}{2} + o(1)\right) \frac{W(k)^3}{k}.$$

Part I's (6) is the same expansion for M_{2k} . Subtracting, and using (199),

$$0 \leq \frac{M_{2k} - G_k - J_k}{2B_{k+1}} = o\left(\frac{W(k)^3}{k}\right).$$

So the walks outside $\mathcal{G}_k \cup \mathcal{J}_k$ are negligible at the first relative correction, which is carried by $\mathcal{J}_k$ alone ($J_k/(2B_{k+1}) \sim W(k)^3/(2k)$), while $G_k - 2B_{k+1} \sim 2W(k)B_k$ by (224) is of smaller order. This is a consequence of two asymptotic theorems and says nothing about finite k : at $k = 8$ the omitted walks are 14,348 of 69,331 (Table 6). The coefficient $\frac{1}{2}$ is also that of Part III's scalar Theorem 35.1. Which walks carry Part I's next correction, $(\frac{1}{8} + o(1))W(k)^6/k^2$ after the exact first transform, as in (7), is not examined.

[write] **Part VI and the scale $W(k)^2/k$, 9 October 2026.** Part VI evaluates the whole of the scale $1/k$: at $c = 1$ its Corollary 60.7 gives $M_{2k}/(2B_{k+1}) = 1 + W(k)^3/(2k) + 2W(k)^2/k + O(W(k)^6/k^2)$ (there $w = W(k/c)$ is $W(k)$, and $H_k(1) = B_{k+1}$). With $G_k/(2B_{k+1}) = 1 + (1 + o(1))W(k)^2/k$ from Theorem 39.2, this gives, by subtraction,

$$\frac{M_{2k} - G_k}{2B_{k+1}} = \frac{W(k)^3}{2k} + (1 + o(1)) \frac{W(k)^2}{k}.$$

So the walks outside $\mathcal{G}_k$ carry, besides the first correction $W(k)^3/(2k)$, also one of the two units of the coefficient of $W(k)^2/k$. How that unit splits between $\mathcal{J}_k$ (whose count is known here only to leading order, $J_k/(2B_{k+1}) \sim W(k)^3/(2k)$) and the walks outside $\mathcal{G}_k \cup \mathcal{J}_k$ is not determined; the remark's $(M_{2k} - G_k - J_k)/(2B_{k+1}) = o(W(k)^3/k)$ is neither sharpened nor contradicted. Nothing else is claimed.

47 Prior work and the limits of the source check

Bauer and Golinelli [1, §5.5, Eq. (9)] prove the Stirling–Catalan bounds for fixed tree size, and their moment recurrence and bi-star decomposition provide established exact tree-walk machinery. Their fixed-size two-star asymptotic is in §5.6, Eq. (13). Those results, including the older Bell lower bound and star contributions, are prior infrastructure rather than claims of this report.

Hainzl and de Panafieu [4] distinguish simple and excess edges and construct kernel decompositions and Catalan decorations. Their preprint Definition 6 and Lemma 10, followed by Definition 12 and Lemma 14, supply the general combinatorial setting behind fresh-tree attachments. The root-excursion construction here is a specialization of that mechanism. The purpose here is the exact Bell convolution for the complete one-hub class, stabilizer-aware time-rotation count, and the disjoint secondary-repeat correction, each with a self-contained proof. These are not presented as newly invented Catalan or kernel methods.

The source check covered the Bauer–Golinelli primary full text and the Hainzl–de Panafieu 2024 full preprint, including its printed page 15 discussion of the different asymptotic regimes. The arXiv record listed only v1 when checked. A 2026 journal version has DOI [10.1016/j.ejc.2026.104386](https://doi.org/10.1016/j.ejc.2026.104386); its full text was not independently accessed, so statements about its complete contents would be unsupported. The conference version is linked in the references. The precise formulas in this report were not located in the inspected sections, but that bounded observation is not evidence of priority. Neither the OEIS comparison nor a search failure establishes a global open-problem status.

48 Exact checks and reproducibility

The accompanying package contains this PDF, its complete TeX source, a Python standard-library verifier, generated exact data and table fragments, public source links, and a deterministic build driver. It has no network or neighboring-file dependency. The verifier performs these independent finite checks:

1. It generates every normalized closed tree walk through $k = 8$ directly, by traversing an existing adjacent edge or attaching the next unused vertex. It never uses either family formula to generate walks. Every retained walk returns to vertex 0 after $2k$ steps. Distance pruning cannot remove a completable walk.
2. It classifies $\mathcal{G}_k$, $\mathcal{H}_k$, and $\mathcal{J}_k$ geometrically, checks their formulas and disjointness, and verifies the rooted $D(k, n)$ counts. A normalized one-step rotation preserves membership in the rotation-closed classes.
3. It independently enumerates set partitions through $n = 10$, verifying P_n and C_n against their definitions. It also checks Catalan convolution coefficients and the exact scalar binomial identity (225) in finite ranges.
4. It generates exact family rows through $k = 32$ and evaluates the exact rational inequalities at selected larger orders through $k = 512$. These large-order values do not prove the limits; Sections 5–7 do.

Every correctness guard uses an explicit exception, never a Python `assert`. The driver propagates ordinary or optimized Python mode to its verifier and tests intentional guard failure in both modes.

It regenerates the integer JSON and the tables actually included by this TeX source, then compiles the PDF to byte stability with fixed metadata. Finally it writes an archive with sorted members, fixed timestamps and permissions, and no compression.

From the extracted package directory, run

```
python3 reproduce.py --out ../run-normal
python3 -O reproduce.py --out ../run-optimized
```

Each output directory must be new and outside the source directory. The source is read-only input. The driver requires generated data, tables, and PDF to match the bundled baselines exactly. Its receipt hashes every public archive member and the whole ZIP. Python checks use only the standard library; typesetting additionally needs the TeX packages listed in `requirements.txt`. PDF byte identity is scoped to the same TeX and font toolchain. A different toolchain that changes PDF bytes is reported as a mismatch, not silently accepted.

The deliberate failure option `--self-test-failure` must exit nonzero before writing output. The authoring-only `--initialize` option permits missing baselines when preparing a new version; it is not a release verification mode. These checks are finite audits of the stated mathematics and packaging. They do not replace the all- k bijections or the analytic estimates.

Part V

A shifted Bell lower bound for sparse graph moments

Bundle Report 209, archive `Report209-reproducibility.zip`, delivered as `Report209.tex` (440 lines, 13 pages). Delivered title block “A shifted Bell lower bound for sparse graph moments / Report209 / 4 October 2026” (empty author line). Printed as delivered, with `[write]` notes; its own table of contents is replaced by the report’s. Report 209’s Section k is Section $k + 48$ here, its equation (k) is equation $(k + 227)$ and its Tables 1–2 are Tables 7–8; statements move with their sections. Its labels carry `sgm:sb:`; its two tables are printed inline. Report 209 normalizes vertex names from 1, the other Parts from 0; the walks counted are the same. This Part is self-contained.

Abstract of this Part (as delivered). Let M_{2k} count normalized closed walks of length $2k$ whose underlying graph is a tree; for $k \geq 1$ these are the terms of OEIS A094149. We prove the coefficientwise lower bound

$$I(k, \ell) \geq 2 \sum_{s=0}^{\ell-1} \binom{k-1}{s} S(k-s, \ell-s) - \binom{2k-\ell}{\ell-1},$$

and its sum $M_{2k} \geq 2(B_{k+1} - B_k) - F_{2k}$. The right side counts a height-two family together with its one-step time rotation. Their exact overlap consists of restricted double-star walks and is enumerated by the even Fibonacci numbers. Equality in the summed bound holds precisely for $k = 1, 2, 3$. A self-contained adjacent-Bell ratio argument gives $M_{2k}/B_k \geq (2 - o(1))k/W(k)$, where W is the positive Lambert function. We also give the nonnegative-parameter corollary and a secondary quantified extension of the known fixed-edge star-dominance theorem. Neither a matching full-moment asymptotic nor an inverse theorem for the full moment sequence is established. Prior results are credited explicitly; no historical-priority claim is made.

49 Scope and definitions

A walk here is an ordered sequence

$$w = (v_0, v_1, \dots, v_{2k-1}, v_0)$$

with consecutive vertices distinct. Its underlying graph has the distinct visited vertices as vertices and the traversed unordered pairs as edges. We require that graph to be a tree. In particular, every edge of the underlying graph is used. Two walks are equivalent if a bijection of their vertex sets sends their entries to one another in order. A unique normalized representative labels v_0 by 1 and assigns subsequent labels in order of first appearance. The initial position remains distinguished: we do not quotient by time rotation.

Write $\mathcal{W}_k$ for these equivalence classes and $I(k, \ell)$ for the number whose tree has ℓ edges. Every used edge of a closed tree walk is traversed equally often in the two directions: deleting the edge gives a cut which must be crossed equally often each way. Thus $1 \leq \ell \leq k$ for $k \geq 1$. Set

$$M_{2k} = \sum_{\ell=1}^k I(k, \ell) \quad (k \geq 1), \quad M_0 = I(0, 0) = 1.$$

The positive-order sequence begins 1, 3, 12, 57, 303, 1747, ... and is A094149 [17]. For a nonnegative parameter α , define

$$\mu_{2k}(\alpha) = \sum_{\ell=1}^k I(k, \ell) \alpha^\ell. \quad (228)$$

These are the limiting even moments of the *unscaled* adjacency matrix of an Erdős–Rényi graph with edge probability α/N , in the expected normalized-trace sense, with k fixed before $N \rightarrow \infty$ [1, §5.3]. At $\alpha = 1$ they equal M_{2k} . Dividing an adjacency matrix by $\sqrt{\alpha}$ changes its $2k$ th moment by α^{-k} when $\alpha > 0$; that normalization is not implicit here.

Let $S(k, \ell)$ be a Stirling number of the second kind, with $S(0, 0) = 1$ and zero outside $0 \leq \ell \leq k$. Let $B_k = \sum_{\ell} S(k, \ell)$, let $C_\ell = \binom{2\ell}{\ell}/(\ell + 1)$, and let $F_0 = 0, F_1 = 1, F_{n+2} = F_{n+1} + F_n$.

49.1 Established comparisons and the present claim

Bauer and Golinelli prove [1, §5.5, Eq. (9)]

$$S(k, \ell) \leq I(k, \ell) \leq C_\ell S(k, \ell). \quad (229)$$

Their two rooted star contributions give $I(k, \ell) \geq 2S(k, \ell)$ for $\ell \geq 2$; they also prove [1, §5.6, Eq. (13)]

$$I(k, \ell) \sim 2S(k, \ell) \sim \frac{2\ell^k}{\ell!} \quad \text{for each fixed } \ell \geq 2. \quad (230)$$

In particular $M_{2k} \geq 2B_k - 1$ is already known. The lower bound proved below gives a factor growing like $k/\log k$ over B_k . Its proof is self-contained and counts an explicit subset. Its overlap product is a specialization of the published bi-star decomposition, as explained in Section 55. This report does not assert that the bound is historically new, resolve the older Bell–Catalan comparison as an open conjecture, or establish a full-moment asymptotic.

50 The coefficientwise shifted Bell bound

Theorem 50.1. *For integers $k \geq 1$ and $1 \leq \ell \leq k$, put*

$$A_{k,\ell} = \sum_{s=0}^{\ell-1} \binom{k-1}{s} S(k-s, \ell-s), \quad (231)$$

$$O_{k,\ell} = \binom{2k-\ell}{\ell-1}, \quad L_{k,\ell} = 2A_{k,\ell} - O_{k,\ell}. \quad (232)$$

Then $L_{k,\ell}$ is a positive integer counting an explicit subset of the walks counted by $I(k, \ell)$, and

$$I(k, \ell) \geq L_{k,\ell}. \quad (233)$$

Consequently,

$$M_{2k} \geq L_k := 2(B_{k+1} - B_k) - F_{2k} \quad (k \geq 1). \quad (234)$$

Equality in (234) holds exactly for $k = 1, 2, 3$ among positive integers.

50.1 An explicit height two family

Let $\mathcal{A}_k \subseteq \mathcal{W}_k$ consist of walks such that, relative to their starting root r , every vertex has depth at most two and every edge from depth one to depth two is traversed exactly twice. Edges incident to r may have any positive even traversal count. Write $\mathcal{A}_{k,\ell}$ for the subfamily with ℓ edges.

Cut a walk immediately after each return to r . This yields $n \geq 1$ root excursions. Excursion i starts $r \rightarrow c$, visits $j_i \geq 0$ leaves by two-step moves from c , and returns $c \rightarrow r$. Each depth-two leaf visited this way is fresh: using it again would use its edge more than twice. If s is the number of depth-two edges, then

$$j_1 + \cdots + j_n = s, \quad n + s = k.$$

The sequence of root children is encoded by a set partition of $[n]$: two positions are in the same block exactly when the corresponding excursions use the same child. For fixed s there are B_{k-s} such partitions and

$$\binom{n+s-1}{s} = \binom{k-1}{s}$$

weak compositions of s into n nonnegative parts.

This is a bijection. Given the partition and composition, create one root child per block. In excursion i visit its child, attach and visit j_i fresh leaves, and return to the root. Normalize all vertex labels by their actual first appearance *after* reconstructing the walk. Conversely, the walk recovers its child partition and each j_i . This global normalization is important: the discovery order of labels need not be a depth-first order. No factorial for ordering blocks or fresh leaves is missing.

If there are ℓ edges, the partition has $\ell - s$ blocks, so this bijection proves $|\mathcal{A}_{k,\ell}| = A_{k,\ell}$. Summing gives

$$|\mathcal{A}_k| = \sum_{s=0}^{k-1} \binom{k-1}{s} B_{k-s} = B_{k+1} - B_k. \quad (235)$$

For the last identity use the set-partition recurrence $B_{n+1} = \sum_{j=0}^n \binom{n}{j} B_j$, obtained by choosing the elements outside the block of a distinguished element. Subtract the recurrences for B_{k+1} and B_k ,

and apply Pascal's rule:

$$B_{k+1} - B_k = \sum_{j=1}^k \binom{k-1}{j-1} B_j.$$

Changing j to $k - s$ gives (235).

50.2 Time rotation and its inverse

Define R by moving the initial position one step forward and then normalizing:

$$R(w) = \text{norm}(v_1, v_2, \dots, v_{2k-1}, v_0, v_1).$$

Vertex relabeling commutes with this operation, so R is well-defined on equivalence classes. Moving the last nonrepeated entry to the front and normalizing gives its inverse. Hence R is a permutation of $\mathcal{W}_k$ preserving the number of edges. It need not be an involution.

For a fixed k and, if desired, a fixed ℓ , inclusion-exclusion gives

$$|\mathcal{A}_k \cup R(\mathcal{A}_k)| = 2|\mathcal{A}_k| - |\mathcal{A}_k \cap R(\mathcal{A}_k)|. \quad (236)$$

For the intersection it is convenient to count

$$X_k = \{w \in \mathcal{A}_k : R(w) \in \mathcal{A}_k\} = \mathcal{A}_k \cap R^{-1}(\mathcal{A}_k).$$

The map R is a bijection from X_k to $\mathcal{A}_k \cap R(\mathcal{A}_k)$, again preserving ℓ . This justifies the overlap computation without assuming R^2 is the identity.

50.3 The exact double star overlap

Take $w \in X_k$, with first step $r \rightarrow a$. Its tree has height at most two from both adjacent vertices r and a . Delete the edge $\{r, a\}$. A vertex on the r side at distance d from r has distance $d + 1$ from a , forcing $d \leq 1$. The same holds on the a side. Thus the tree is a double star: the central edge $\{r, a\}$, together with p leaves at r and s leaves at a .

Membership in $\mathcal{A}_k$ from r forces each a -leaf edge to be used twice. Membership after rotation to a forces each r -leaf edge to be used twice. The central edge may be traversed $2q$ times, $q \geq 1$. Therefore

$$k = q + p + s, \quad \ell = 1 + p + s. \quad (237)$$

Conversely, every double-star walk starting with $r \rightarrow a$, with all noncentral edges used exactly twice, satisfies both membership conditions. This proves necessity and sufficiency, including $p = 0$ or $s = 0$ and the single-edge case.

The sequence of excursions from r contains q central excursions and p single-use leaf excursions. The first excursion is central, so the number of their interleavings is

$$\binom{q+p-1}{p}.$$

During the q central excursions, distribute the s fresh a -leaf visits in a weak composition. This gives $\binom{q+s-1}{s}$ choices. The walk recovers both choices. Side leaves are labeled by first encounter and

introduce no extra factorial. For fixed k, ℓ , equation (237) gives $q = k - \ell + 1$ and $p + s = \ell - 1$, whence

$$\begin{aligned} |X_k \cap \mathcal{W}_{k,\ell}| &= \sum_{p+s=\ell-1} \binom{q+p-1}{p} \binom{q+s-1}{s} \\ &= \binom{2q+\ell-2}{\ell-1} = \binom{2k-\ell}{\ell-1} = O_{k,\ell}. \end{aligned} \quad (238)$$

Here $\mathcal{W}_{k,\ell}$ denotes the ℓ -edge part of $\mathcal{W}_k$. The convolution follows by extracting $x^{\ell-1}$ from $(1-x)^{-q}(1-x)^{-q}$.

The total overlap has the formal generating function

$$\begin{aligned} \sum_{k \geq 1} |X_k| z^k &= \sum_{q \geq 1} z^q \left(\sum_{p \geq 0} \binom{q+p-1}{p} z^p \right)^2 \\ &= \sum_{q \geq 1} z^q (1-z)^{-2q} = \frac{z}{1-3z+z^2}. \end{aligned} \quad (239)$$

The even Fibonacci numbers have this generating function: $F_2 = 1$, $F_4 = 3$, and $F_{2k} = 3F_{2k-2} - F_{2k-4}$. Thus $|X_k| = F_{2k}$.

Substitute (235) and (238) into (236). Since $\mathcal{A}_k \cup R(\mathcal{A}_k) \subseteq \mathcal{W}_k$, both inequalities follow. Every $\mathcal{A}_{k,\ell}$ is nonempty: a root-centered ℓ -edge star can be covered in k excursions whenever $k \geq \ell$. Hence $L_{k,\ell} \geq A_{k,\ell} > 0$.

50.4 Equality and strictness

For all applicable k the first two edge coefficients of the lower bound are

$$L_{k,1} = 1, \quad L_{k,2} = 2S(k, 2) = 2^k - 2,$$

which equal $I(k, 1)$ and $I(k, 2)$. Indeed, for two edges the underlying tree is the three-vertex path. Rooting at its center gives $S(k, 2)$ walks and rooting at an endpoint gives another $S(k, 2)$; equivalently use the two star-rooting bijections in Section 53.

On the diagonal $I(k, k) = C_k$: each edge is used exactly twice and the walk is the contour traversal of its tree, with the children ordered by first discovery. Such ordered rooted trees are counted by C_k . Also

$$A_{k,k} = 2^{k-1}, \quad O_{k,k} = k, \quad L_{k,k} = 2^k - k.$$

These identities give equality at $k = 1, 2, 3$. At $k = 4$ the lower-bound row is $(1, 14, 26, 12)$, whereas the exact row is $(1, 14, 28, 14)$, yielding $53 < 57$.

For strictness at every $k \geq 4$, start at an endpoint of a four-edge path, walk to the other endpoint and back,

$$(1, 2, 3, 4, 5, 4, 3, 2, 1),$$

and append $k - 4$ bounces $1 \rightarrow 2 \rightarrow 1$. The starting-root height is four, so the walk is outside $\mathcal{A}_k$. Its last nonrepeated vertex is 2. Inverse rotation starts at that vertex, whose maximum distance in the path is three, so the inverse-rotated walk is outside $\mathcal{A}_k$ as well. Thus the original walk is outside $R(\mathcal{A}_k)$, proving strictness and completing Theorem 50.1.

Remark 50.2 (The zero endpoint). The numerical extension at $k = 0$ is $M_0 = 1 \geq 2(B_1 - B_0) - F_0 = 0$. The family identities above apply only to $k \geq 1$: the natural empty-walk family at $k = 0$ has cardinality one. If a sequence L_k is wanted from zero onward, set $L_0 = 1$ separately.

51 The Bell ratio consequence

We include the asymptotic input with proof. Write $r = W(n) > 0$, so $re^r = n$, and $x = e^r = n/r$.

Lemma 51.1. *As $n \rightarrow \infty$,*

$$\frac{B_{n+1}}{B_n} \sim \frac{n}{W(n)}. \quad (240)$$

Moreover, B_n grows faster than every fixed-base exponential.

Proof. The exponential generating function for set partitions is $\exp(e^z - 1)$. Expanding the outer exponential and equating coefficients proves Dobinski's identity

$$B_n = e^{-1} \sum_{m \geq 0} \frac{m^n}{m!}.$$

For $n > 0$ set $a_0 = 0$ and $a_m = m^n/m!$ for $m \geq 1$. Then B_{n+1}/B_n is the mean of m under weights proportional to a_m . Their consecutive ratio is

$$h_m := \frac{a_{m+1}}{a_m} = \frac{(1 + 1/m)^n}{m + 1} \quad (m \geq 1).$$

It decreases strictly in m . For $m = \lfloor tx \rfloor$, uniformly when t stays in a compact subinterval of $(0, \infty)$,

$$\log h_m = r(1/t - 1) - \log t + o(1). \quad (241)$$

Indeed $n \log(1 + 1/m) = n/m + O(n/m^2)$, with $n/m^2 = O(r^2/n) = o(1)$; the rounding error has the same order, and $\log(m + 1) = r + \log t + o(1)$.

Fix $0 < \varepsilon < 1$. Formula (241) implies that for all sufficiently large n , throughout the lower bridge

$$(1 - \varepsilon)x \leq m \leq (1 - \varepsilon/2)x$$

one has $h_m \geq e^{c_\varepsilon r}$ for some $c_\varepsilon > 0$. Throughout the upper bridge from $(1 + \varepsilon/2)x$ to $(1 + \varepsilon)x$ one has $h_m \leq e^{-c'_\varepsilon r}$. Each bridge has a number of integer steps proportional to x . Products along the bridges therefore suppress an outer-end weight relative to its inner-end weight by $e^{-c'_\varepsilon x r} = e^{-c'_\varepsilon n}$, after adjusting positive constants for integer endpoints.

To make the tail control explicit, let m_- and b_- be integer points within bounded distance of $(1 - \varepsilon)x$ and $(1 - \varepsilon/2)x$. Monotonicity implies that the weights up to m_- increase. Consequently

$$\sum_{m \leq m_-} a_m \leq O(x) e^{-c'_\varepsilon n} a_{b_-}.$$

For integer points b_+, m_+ near $(1 + \varepsilon/2)x, (1 + \varepsilon)x$, monotonicity gives, with $\theta = e^{-c_\varepsilon r}$,

$$a_{m_+ + j} \leq e^{-c'_\varepsilon n} a_{b_+} \theta^j \quad (j \geq 0).$$

The geometric series then bounds the upper-tail mass by $e^{-c'_\varepsilon n} a_{b_+} / (1 - \theta)$ and its first moment by

$$e^{-c'_\varepsilon n} a_{b_+} \left(\frac{m_+}{1 - \theta} + \frac{\theta}{(1 - \theta)^2} \right).$$

Dividing each bound by $\sum_m a_m$, which is at least the corresponding inner-end weight, proves that the mass outside $[(1 - \varepsilon)x, (1 + \varepsilon)x]$ is $o(1)$ and its upper first moment is $o(x)$. The lower first

moment is bounded by $O(x)$ times its $o(1)$ mass. Thus the mean divided by x has lower and upper limits between $1 - \varepsilon$ and $1 + \varepsilon$. Letting $\varepsilon \downarrow 0$ proves (240).

Finally, for every fixed positive integer m , Dobinski's sum gives $B_n \geq m^n/(em!)$. For any fixed base $a > 0$, choose an integer $m > a$; then $B_n/a^n \rightarrow \infty$. $\square$

Corollary 51.2. *The full moment sequence satisfies*

$$\liminf_{k \rightarrow \infty} \frac{M_{2k}}{B_{k+1}} \geq 2, \quad \frac{M_{2k}}{B_k} \geq (2 - o(1)) \frac{k}{W(k)} \rightarrow \infty. \quad (242)$$

The explicit lower-bound sequence itself satisfies $L_k \sim 2B_{k+1}$.

Proof. Set $\lambda = (3 + \sqrt{5})/2$. The Fibonacci recurrence gives $F_{2k} = O(\lambda^k)$. The previous lemma yields $F_{2k}/B_k \rightarrow 0$ and $B_k/B_{k+1} \rightarrow 0$. Therefore

$$\frac{L_k}{B_{k+1}} = 2 - 2 \frac{B_k}{B_{k+1}} - \frac{F_{2k}}{B_{k+1}} \rightarrow 2.$$

Use $M_{2k} \geq L_k$ and (240). Since $W(k)e^{W(k)} = k$, one has $k/W(k) \rightarrow \infty$. $\square$

The equivalence $L_k \sim 2B_{k+1}$ refers only to the selected union. It does not justify $M_{2k} \sim 2B_{k+1}$. The remaining tree-walk families are nonnegative and are not controlled on that scale by the present argument.

[write] **Added 7 October 2026.** Part II proves $M_{2k} \sim 2B_{k+1}$ (Theorem 14.1). So the $\liminf$ in (242) is a limit, equal to 2, the second bound there is sharp in the form $M_{2k}/B_k \sim 2k/W(k)$, and $M_{2k} \sim L_k$. The two sentences above are kept as delivered; they are true of Report 209.

52 A weighted version

Define the Touchard polynomial $T_n(\alpha) = \sum_{j=0}^n S(n, j)\alpha^j$.

Corollary 52.1. *For $k \geq 1$ and every real $\alpha \geq 0$,*

$$\mu_{2k}(\alpha) \geq 2 \sum_{s=0}^{k-1} \binom{k-1}{s} \alpha^s T_{k-s}(\alpha) - P_k(\alpha), \quad (243)$$

where

$$P_k(\alpha) = \sum_{\ell=1}^k \binom{2k-\ell}{\ell-1} \alpha^\ell, \quad (244)$$

$$\sum_{k \geq 1} P_k(\alpha) z^k = \frac{\alpha z}{1 - (1 + 2\alpha)z + \alpha^2 z^2}. \quad (245)$$

In particular $P_1 = \alpha$, $P_2 = \alpha + 2\alpha^2$, and

$$P_k = (1 + 2\alpha)P_{k-1} - \alpha^2 P_{k-2} \quad (k \geq 3). \quad (246)$$

Proof. Multiply (233) by α^ℓ and sum. For the family term change variables to $t = \ell - s$:

$$\sum_{\ell} A_{k,\ell} \alpha^\ell = \sum_{s=0}^{k-1} \binom{k-1}{s} \alpha^s \sum_{t \geq 1} S(k-s, t) \alpha^t.$$

Here $k-s \geq 1$, so the missing $t=0$ term vanishes, giving the Touchard polynomial. The overlap parameters (237) give the formal identity

$$\begin{aligned} \sum_{k \geq 1} P_k(\alpha) z^k &= \alpha \sum_{q \geq 1} z^q \left(\sum_{p \geq 0} \binom{q+p-1}{p} (\alpha z)^p \right)^2 \\ &= \alpha \sum_{q \geq 1} z^q (1 - \alpha z)^{-2q}, \end{aligned}$$

which simplifies to (245). Its first coefficients and denominator give (246). $\square$

At $\alpha = 1$, $P_k(1) = F_{2k}$ and (243) is (234). At $\alpha = 0$, both sides are zero for positive k . For every $\alpha > 0$, equality in (243) holds exactly for $k = 1, 2, 3$: the witness for $k \geq 4$ has four edges and contributes the positive weight α^4 to the omitted family. Negative α is outside the inequality's stated domain.

53 A secondary quantified star estimate

The fixed- ℓ theorem (230) belongs to Bauer and Golinelli. The following elementary quantitative bound extends it uniformly in a restricted range. It is independent of the proof of Theorem 50.1 and does not reach the edge counts relevant to the full Bell sum.

Proposition 53.1. *For $\ell \geq 3$ and $k \geq \ell$, set*

$$q_\ell = \sqrt{1 - \frac{2}{\ell} + \frac{4}{\ell^2}}, \quad u_{k,\ell} = \ell(1 - 1/\ell)^k.$$

Whenever $u_{k,\ell} < 1$,

$$0 \leq \frac{I(k, \ell)}{2S(k, \ell)} - 1 \leq \frac{(C_\ell - 2)\ell!}{2(1 - u_{k,\ell})} q_\ell^k. \quad (247)$$

For every fixed $\varepsilon > 0$, the ratio tends to one uniformly over integers

$$\ell \rightarrow \infty, \quad k \geq (1 + \varepsilon)\ell^2 \log \ell. \quad (248)$$

For $\ell = 2$ the identity $I(k, 2) = 2S(k, 2)$ holds exactly.

Proof. Give each normalized walk a plane rooted tree by ordering the children of every vertex in their order of first discovery. This assigns each walk to one of the C_ℓ plane rooted tree shapes. Conversely, for a fixed plane rooted tree, keep only covering walks whose child-discovery order agrees with its plane order. The resulting admissible walks correspond bijectively to the assigned normalized classes. In particular their count is no larger than the number of all closed walks starting at that fixed root.

For $\ell \geq 2$ there are two plane rooted shapes of the unrooted ℓ -edge star. With the center as root, the walk is a sequence of k leaf excursions, and the partition of their positions into ℓ nonempty leaf

classes gives $S(k, \ell)$. With a leaf as root, the walk alternates between a leaf and the center. Record the k leaf positions at the even times $0, 2, \dots, 2k - 2$, with the final return repeating the initial leaf. These positions again form a partition into ℓ leaf classes, with the first block identifying the root leaf. Normalizing the remaining leaf classes by first discovery gives $S(k, \ell)$ walks. Thus the two stars contribute exactly $2S(k, \ell)$.

For a nonstar tree with $\ell \geq 3$ edges, let d_v be its vertex degrees and $e_v = d_v - 1$. The $\ell + 1$ excesses are nonnegative integers with

$$\sum_v e_v = \ell - 1, \quad \max_v e_v \leq \ell - 2.$$

Moving mass from a smaller positive excess to a larger one, until the latter reaches its cap, can only increase the sum of squares. Its maximum is therefore $(\ell - 2)^2 + 1$. It follows that

$$\sum_v d_v^2 = \sum_v e_v^2 + 2(\ell - 1) + (\ell + 1) \leq \ell^2 - \ell + 4.$$

For its adjacency matrix H , a direct count of four-step closed walks gives

$$\text{tr } H^4 = 2 \sum_v d_v^2 - 2\ell.$$

For example, use $\text{tr } H^4 = \sum_{u,v} (H_{uv}^2)^2$: the diagonal contributes $\sum_v d_v^2$, and the off-diagonal terms count ordered vertex pairs at distance two, namely $\sum_v d_v(d_v - 1)$.

The tree is bipartite, so the eigenvalues occur in opposite pairs. If ρ is the spectral radius, then $2\rho^4 \leq \text{tr } H^4$, whence

$$\rho^2 \leq \sqrt{\ell^2 - 2\ell + 4} = \ell q_\ell.$$

The spectral theorem gives $(H^{2k})_{rr} \leq \rho^{2k}$. Sum this bound over the $C_\ell - 2$ nonstar plane rooted shapes to obtain

$$I(k, \ell) - 2S(k, \ell) \leq (C_\ell - 2)\ell^k q_\ell^k. \quad (249)$$

Finally $\ell!S(k, \ell)$ counts onto maps from $[k]$ to $[\ell]$. The union bound on missing at least one target gives

$$\ell!S(k, \ell) \geq \ell^k (1 - \ell(1 - 1/\ell)^k).$$

Divide (249) by twice this positive lower bound to obtain (247).

For the uniform conclusion, $0 < q_\ell < 1$ and the right side decreases with k whenever $u_{k,\ell} < 1$. At $k_0 = \lceil (1 + \varepsilon)\ell^2 \log \ell \rceil$,

$$\log q_\ell = -1/\ell + O(1/\ell^2), \quad \log(C_\ell \ell!) = \ell \log \ell + O(\ell), \quad u_{k_0,\ell} \leq \ell e^{-k_0/\ell} \rightarrow 0.$$

Thus the logarithm of the right side is at most $-\varepsilon \ell \log \ell + O(\ell + \log \ell)$, which tends to $-\infty$. Monotonicity gives uniformity for all $k \geq k_0$. $\square$

The scale $\ell \asymp k/W(k)$ illustrates the limitation. Since $\log(k/W(k)) = W(k)$, at $\ell = k/W(k)$ one has

$$\frac{k}{\ell^2 \log \ell} = \frac{W(k)}{k} \rightarrow 0.$$

The sufficient range (248) is therefore far from this Bell-scale edge count. The more restrictive sufficient condition $k/(\ell^2 \log(\ell + 1)) \rightarrow \infty$, also sufficient for the proposition, misses it.

54 Exact checks and reproducibility

The all- k result is supplied by the bijections and arguments above. Finite computations audit the implementation, sequence identification, overlap, and coefficients; they do not replace those proofs.

54.1 An exact recurrence and an independent enumeration

For reproducibility we use the Bauer–Golinelli recurrence [1, §5.4, Eq. (8)]. Let $D_{k,\ell,m}$ count normalized walks with ℓ edges and m root excursions. Set $D_{0,0,0} = 1$ and all inadmissible entries to zero. Removing the first edge splits a nonempty walk into two rooted component walks of half-lengths a, b , edge counts i, j , and root-excursion counts u, v . If the first edge is used $2q$ times, then

$$a + b + q = k, \quad i + j + 1 = \ell, \quad q + v = m.$$

The excursions on either component are split into q possibly empty consecutive groups; the first excursion of the whole walk uses the distinguished edge. These choices give

$$D_{k,\ell,m} = \sum D_{a,i,u} D_{b,j,v} \binom{u+q-1}{q-1} \binom{v+q-1}{q-1}, \quad (250)$$

where the sum is over $q \geq 1$ and the nonnegative indices satisfying the displayed constraints. Thus $I(k, \ell) = \sum_m D_{k,\ell,m}$. The public script `verify_recurrence.py` computes all rows through $k = 16$, checks (229), the 13-term A094149 prefix recorded in the source audit, and the $k = 10$ coefficient row printed by Bauer and Golinelli.

The separate script `verify_lower_bound.py` constructs every normalized tree walk through $k = 8$ by depth-first search. At each step it may follow an existing neighbor or append a new vertex as a leaf at the current vertex; joining a nonneighbor old vertex would create a cycle. It keeps exactly closed walks of length $2k$. This enumeration does not use the partition/composition construction or recurrence (250). It checks the geometric family predicate, rotation and inverse rotation, overlap and union totals, and every edge coefficient. It also checks coefficientwise inequalities through $k = 16$ against the recurrence, the strictness witness, and the weighted overlap recurrence as exact polynomial identities.

54.2 Running the public package

The ZIP contains the TeX source and PDF, two verification scripts, a build script, documentation, generated JSON data, and the two generated table fragments. It contains no third-party full texts or private research notes. Python uses only the standard library. A TeX installation with pdfLaTeX and the packages listed in `requirements.txt` is needed to rebuild the PDF.

Extract the archive and run the following from its `Report209` directory, using two fresh output directories outside that directory:

```
python3 reproduce.py --out ../run-normal
python3 -O reproduce.py --out ../run-optimized
```

Each run independently regenerates all finite-check data and tables, compares their bytes to the bundled baselines, recompiles the PDF until its bytes stabilize, and creates a deterministic ZIP. All guards are explicit exceptions and remain active with Python's `-O`. The build also invokes both

Table 7: Exact moments and the selected union through $k = 16$

k	B_k	L_k	M_{2k}	$M_{2k} - L_k$
1	1	1	1	0
2	2	3	3	0
3	5	12	12	0
4	15	53	57	4
5	52	247	303	56
6	203	1204	1747	543
7	877	6149	10727	4578
8	4140	33027	69331	36304
9	21147	187072	467963	280891
10	115975	1118425	3280353	2161928
11	678570	7052343	23785699	16733356
12	4213597	46815312	177877932	131062620
13	27644437	326388377	1368977132	1042588755
14	190899322	2383800635	10825357094	8441556459
15	1382958545	18193535164	87852127346	69658592182
16	10480142147	144767277005	731080093581	586312816576

Table 8: The first strict case by number of edges

ℓ	$A_{4,\ell}$	$O_{4,\ell}$	$L_{4,\ell}$	$I(4, \ell)$
1	1	1	1	1
2	10	6	14	14
3	18	10	26	28
4	8	4	12	14

verification scripts with `-O` when the driver is optimized. The output receipt records hashes of every member and the whole archive. These hashes are outside the archive to avoid a circular self-hash.

Use `--self-test-failure` to confirm the driver fails intentionally before creating an output directory. Both verification scripts accept that flag too. A frozen source tree is never edited by the build. Output directories must be absent before a build; successful outputs can be retained and compared without overwriting them. The package’s deterministic timestamps and metadata support byte-identical builds in the same toolchain. Byte identity across different TeX versions or fonts is not promised.

The publication checks for this report compared the ordinary and optimized regenerated JSON, tables, PDF, every ZIP member, and the entire ZIP. The computational ranges are fixed and modest; no floating-point asymptotic fitting is used.

55 Source scope and further research

55.1 Attribution and source access

The full Bauer–Golinelli preprint was checked for the comparison (229), fixed-edge theorem (230), recurrence (250), and the bi-star decomposition in Eq. (12). In that decomposition the component factors have the form

$$S(k', \ell') \binom{k' + q - 1}{q - 1}, \quad S(k'', \ell'') \binom{k'' + q - 1}{q - 1}.$$

Specializing $k' = \ell' = s$, $k'' = \ell'' = p$ gives the exact overlap product $\binom{q+s-1}{s} \binom{q+p-1}{p}$, including empty components. Thus the overlap enumeration uses a restriction of a published decomposition. The height-two family is larger than the bi-star class: for example,

$$(1, 2, 3, 2, 1, 4, 5, 4, 1) \in \mathcal{A}_4$$

has a four-edge path as its underlying graph, of diameter four, while every bi-star has diameter at most three. The family count and rotation union provide the additional construction in this report. This distinction alone does not establish historical priority or exclude an equivalent bound elsewhere.

The author-hosted Spiridonov thesis [3], especially its Bell–Catalan discussion, was checked as background. Its older comparison conjecture is already covered by Bauer–Golinelli. The full 2024 preprint of Hainzl and de Panafieu [4] was checked; its results address fixed excess and large-mean-degree spectral expansions. Its printed page 15 discusses the bivariate asymptotics relevant to full moments. The 2026 journal bibliographic record was located, but publisher access to its full text was blocked; that journal version was not independently read. Statements about the full contents of the 2024 preprint are not silently attributed to the 2026 journal text.

Additional source checks considered Hainzl’s fixed-multiplicity hypergraph Catalan work [5], selected full-text sections of the spectral-tail approximation work of Valigi et al. [14], and the abstracts of two papers of Khorunzhiy [15, 16]. Those papers have different restrictions, parameter regimes, or levels of approximation. No exact statement of (234) or matching rigorous full-moment asymptotic was located in this bounded set of checks. This is not an exhaustive literature survey, a proof of novelty, or a global claim about what remains open as of the report date.

55.2 Questions suggested by the construction

Several concrete directions would sharpen the present analysis:

1. Enlarge the selected family by allowing repeated depth-two edges, while retaining an encoding with controllable multiplicities. Any proposed enlargement needs an injective construction or an exact multiplicity correction.
2. Study the union of several time rotations of the height-two family. Pairwise overlaps alone would not determine that larger union; higher intersections would need explicit control.
3. Obtain coefficient estimates uniform around growing edge counts of order $k/W(k)$. The secondary spectral-gap bound here is insufficient there, so a fixed- ℓ limit cannot be interchanged with the full sum on its strength.
4. Compare these constructions with bivariate generating functions for tree walks, with careful matching of adjacency normalization and limiting regime.

These are directions suggested by this proof, not assertions that their answers are absent from the literature. The established conclusion remains the all- k lower bound, its exact selected-family interpretation, and the one-sided consequence (242). No asymptotic inversion or full-sequence inverse theorem follows from these lower bounds alone.

[write] **Status of these questions, 7 October 2026.** Item 1 is pursued by Part IV, which counts every walk whose repeated edges share a vertex (a class containing the height-two family $\mathcal{A}_k$; see the end of Section 46) and a family with one repeated depth-two edge and a recoverable hub; with Part II's equivalent, no enlargement can change the leading scale $2B_{k+1}$, and by Remark 46.1 none changes the first relative correction. Items 2–4 remain open; they are part of items 9, 10 and 11 of Section 56.

56 Further questions and research

[write] **Added 7 October 2026.** This section gathers, under Vladimir's standing rule of 4 October 2026, every open question and every unproved claim of the five Parts, with its source, the argument offered and what is missing. Nothing here is claimed. No claim of any Part was found to be false. Superseded statements are kept where they stand as earlier routes, each with a dated note: Part III's unproved premise (162) (proved in Part II), Part II's open first correction and fixed-order expansion (proved in Part I), Part IV's "nothing proved here implies $M_{2k} \sim 2B_{k+1}$ " and Part V's one-sided consequence (242) (both completed by Part II), and Part III's questions 1 and 5 and Part V's item 1 (answered as stated in the notes after them).

1. **Effective constants, onsets and inverses.** *Sources:* Part I, directions 2 and its non-claims ("no effective onset"; the two-ceiling constant C_p is existential); Part II, further directions ("quantitative error constants and onset bounds; an effective inverse enclosure"). *Missing:* explicit constants for the complement estimate (Appendix B), the collision remainder and the deficit tails, which would turn Theorem 11.1 into a certified finite-threshold procedure. The finite tables (Tables 2, 3, 4, 5) show that the asymptotic regime starts late: $M_{2k}/(2B_{k+1})$ is 1.34 at $k = 256$.
2. **Smaller logarithmic powers in the remainder.** *Source:* Part I, direction 1. *Missing:* sharper weighted collision and partition estimates than Lemmas 4.1 and 4.2; the power $10p+6$ is deliberately generous.

3. **Evaluated suborders of each transform.** *Source:* Part I, direction 3, and the last sentence of Section 8, on the coefficient $\frac{1}{8}$ of (7) (“It does not evaluate all logarithmic suborders of the full coefficient at scale k^{-2} ”). *Missing:* the lower logarithmic terms of $A_j(k)$ in powers of $W(k)$; the exact lower transforms must be kept until their errors are below the target scale.
4. **Growing collision order.** *Source:* Part I, direction 4 and its non-claims (no growing-order expansion, no exponential resummation). *Missing:* quantitative control of the growth of the coefficients and of the collision tails for $p = p(k)$; the factor $1/j!$ of the leading term alone does not justify resummation.
5. **Fixed general mean degree.** *Sources:* Part I, direction 5; Part II, further directions; Part V’s weighted Corollary 52.1 is the only statement of the report for the moments $\mu_{2k}(\alpha)$ of $G(N, \alpha/N)$. *Missing:* a uniform proof for weighted Bell (Touchard) structures in place of B_{k+1} .
6. **A larger dominant window.** *Source:* Part III, question 2. *Sketch:* the rough bound $M_{2t} \leq 4^{t-1}t!$ produces the urn error $O(S^3/k)$, which limits the window of Theorem 14.2 to $S_k = o(k^{1/3})$. *Missing:* a sharper collision estimate or child-walk weight. Not needed for the equivalent any more (Part II bounds the complement directly), but it would strengthen the dominant-class theorem.
7. **Canonical hub selection in the complement.** *Source:* Part III, question 3. Part II bounds the complement by entropy and urn estimates, not by a coding of the complicated walks; a global coding that charges off-hub pieces to a selected vertex, with factorial savings, remains open as a method. Departure counts and graph degrees must not be conflated.
8. **A weighted spectral transfer.** *Source:* Part III, Section 36.2 and question 4. *Argument offered (heuristic only, as Report 212 labels it):* in a Poisson branching neighbourhood conditioned on hub degree a , Hiesmayr–McKenzie’s local expansion suggests the squared energy $a + 1 + 2/a$; allocating one positive and one negative eigenvalue to each hub suggests $2\mathbb{E}[(X+1)^k] = 2B_{k+1}$, the leading term that Part II proves combinatorially. *Missing:* the four tasks listed there — a moment-weighted trace bound on exceptional neighbourhoods, coverage of the whole relevant spectral region, the order of limits with quantitative finite-size control, and an allocation of spectral mass to hubs without duplication. Part III’s two counterexamples (Section 35.1) show that untruncated averaging fails; they refute that heuristic step, not a published theorem.
9. **Exact counts of larger explicit families.** *Sources:* Part IV, Section 46.2 (several repeated secondary edges with a recoverable hub); Part V, items 1–2 of its questions (repeated depth-two edges; unions of several time rotations, whose higher intersections need control). Their leading scale and first correction are settled by Parts I–II and Remark 46.1; *missing* are exact counts or injective encodings, and the identification of the walks that carry the second correction $W(k)^6/(8k^2)$.
10. **Edge-count coefficients near the Bell edge scale.** *Source:* Part V, item 3. *Missing:* estimates of $I(k, \ell)$ uniform for $\ell \asymp k/W(k)$; the range of Proposition 53.1 is far from it (Section 53).
11. **Bivariate generating functions.** *Sources:* Part V, item 4; all five Parts’ comparisons with Hainzl–de Panafieu [4]. *Missing:* a matching of adjacency normalization and limiting regime between their fixed-excess generating functions and the present high-moment asymptotics.

12. **Monotonicity of the ratio.** *Source:* Part III, Section 34.1: $M_{2k}/(2B_{k+1})$ attains its maximum 4.71281008... at $k = 19$ and decreases strictly through $k = 256$ (exact comparisons; the write confirmed $k \leq 128$). *Missing:* a proof that the decrease persists; it does not follow from Part I, whose first correction is positive but whose expansion has no effective onset. Report 212 asserts nothing beyond the computed range.
13. **Unresolved source comparisons.** No Part claims priority. Not read in full by any Part: the 2026 journal version of Hainzl–de Panafieu (all five Parts read the 2024 preprint only); Khorunzhy–Shcherbina–Vengerovsky (2004) and Rodgers–Bray (1988) (Part I); the journal full text of Hiesmayr–McKenzie (Part III read the preprint); Janson–Jonsson–Stefánsson and Janson’s survey (abstract level, Part II); Khorunzhiy’s two papers (abstract level) and Valigi et al. (selected sections) (Part V). The write read none of the cited literature; the Parts’ accounts of what they inspected are reported, not confirmed.

[write] **Independent check of the write, 7 October 2026.** An independent adversarial check of the write read OEIS A094149 again (revision #10, 13 July 2025) and recomputed every number the write added with its own code, under different methods where possible.

- It enumerated all normalized closed tree walks for $k \leq 9$ by brute force (467,963 walks at $k = 9$). The rows $F_{k,m}$ agree, as do the coefficientwise comparison (65), the counts G_k (by the geometric predicate) and J_k (by the defining conditions on a centre c) against (197)–(198), including $J_8 = 360$ and $J_9 = 9180$. So does $U_{k,S}$ against the two-hub formula (116) for every S with $k - S > k/2$.
- With its own implementation of (58) it computed $k \leq 256$, one step further than the write, which stopped at $k = 128$. The maximum of $M_{2k}/(2B_{k+1})$ is 4.7128100813441074388... at $k = 19$, and the decrease is strict through $k = 256$. Every entry of Tables 4 and 5, including the $k = 256$ rows, agrees. So do Tables 3, 6 and 7, and the equality set $\{1, 2, 3\}$ of (234).
- It computed $A_1(k)$ and $A_2(k)$ from the definitions (26) and (27), counting marked blocks directly by $\binom{n}{q}B_{n-q}$ and $\binom{n}{q}\binom{n-q}{q'}B_{n-q-q'}$ rather than through (35). All 28 entries of Table 2 agree, as do T_k of (5), $d_1 = 0$, d_2 and d_3 for $q < 40$, and P_2, P_3 .
- It confirmed Remark 1.2’s bounds and the strict increase of a_k for $k \leq 256$, and re-derived Remarks 11.4 and 46.1 by hand. As a finite illustration only, the ratio $((M_{2k} - G_k - J_k)/(2B_{k+1}))/((W(k)^3/k)$ is 4.46, 0.89 and 0.23 at $k = 16, 64$ and 256.

It also confirmed the provenance figures (archive sizes, file and line counts, delivered page counts), the three manifests, the byte identity of all 80 staged files, the 57 number mappings of the notes under the omitted blocks, and the label numbering (300 printed delivered labels unchanged up to the stated shifts, 324 labels in all). One correction was needed: Remark 1.2 quoted one sentence of the OEIS formula line as the whole line. It is corrected there by a dated note. No other claim of the write was found wrong. The reciprocal note that the Guide’s provenance note calls “proposed separately” has since been added to `a064856-stirling-catalan-transforms` (its item `stc:q:moments`). The check confirmed its limits $M_{2k}/B_k \rightarrow \infty$ and $M_{2k}/a_k \rightarrow 0$, and its finite figures: $B_k \leq M_{2k} \leq a_k$ for $k \leq 32$, and M_{2k}/a_k equal to 1 for $k \leq 3$, then strictly decreasing to 5.065×10^{-5} at $k = 32$.

[write] **Part VI and these questions, 9 October 2026.** Part VI (Section 57) answers part of these questions; the table of Section 57.2 gives the details, and the notes above, written before Part VI arrived, are kept. *Item 5* (fixed general mean degree): **answered** for every compact $K \subset (0, \infty)$. With $H_k(c) = \mathbb{E}(X + c)^k$, $X \sim \text{Poisson}(c)$, in place of B_{k+1} , Part VI’s Theorem 60.6 proves $m_k(c)/(2H_k(c)) = 1 + \sum_{j \leq p} A_j(k, c) + O_{K,p}(\log^{10p+6} k/k^{p+1})$, uniformly in $c \in K$, with finitely computable transforms $A_j(k, c) \sim_K c^j w^{3j}/(2^j j! k^j)$, $w = W(k/c)$; it proves its own weighted complement theorem (Theorem 60.4), and its case $c = 1$ is Part I’s Theorem 1.1 (its transforms are then Part I’s: the write checked $A_1(k, 1) = T_k/B_{k+1}$ exactly for $3 \leq k \leq 60$ and d_2, d_3, P_2, P_3 at $c = 1$). Part V’s weighted Corollary 52.1 is the same model with $\alpha = c$: its right

side is $2(H_k(c) - cH_{k-1}(c)) - P_k(c)$, by $\binom{k}{s} = \binom{k-1}{s} + \binom{k-1}{s-1}$ (checked exactly for $k \leq 30$ at $c = \frac{1}{2}, 1, 2$), so by Theorem 60.6 the lower bound is asymptotic to $\mu_{2k}(c)$ for each fixed $c > 0$, as at $c = 1$; this is the write's consequence. Effective constants and growing or vanishing c stay open (Part VI's questions). *Item 3* (evaluated suborders): **answered in part**. Corollary 60.7 evaluates the whole scale $1/k$ of the first transform, $A_1(k, c) = cw^2(w + 2 + 2c)/(2k) + O_K(w^4/k^2)$ (at $c = 1$: $W(k)^3/(2k) + 2W(k)^2/k$), and the two leading terms w^6 and w^5 of the logarithm at scale k^{-2} , $\log(m_k/2H_k) = P_1(k, c) + cw^5/(6k^2) + O_K(w^4/k^2)$; so the coefficient $\frac{1}{8}$ of Part I's (7) (Part VI's $c^2/8$ at $c = 1$) is the term $P_1^2/2$ of the exponential of the first correction, and the next term of the logarithm is $cw^5/(6k^2)$. The lower logarithmic suborders at scale k^{-2} (powers w^4 and below) and all suborders of A_j , $j \geq 3$, stay open. *Item 4* (growing collision order): **not answered**. Part VI's "growing degree" is a growing mean degree $c = c_k$ with $cw/\sqrt{k}$ in a compact set, where $m_k(c)/(2H_k(c)) = e^{c^2w^2/k}\{1 + O(k^{-1/4}\log^{13/4}k)\}$ (Theorem 62.1); the factor comes from the exact Catalan quotient $Q(k, s) = \prod_{j < s} (k + j)/(k - j)$, not from summing the A_j . It is a new direction, recorded with Part VI's own twelve questions (Section 64). *Items 6 and 7*: Part VI's Corollary 62.6 concerns the critical growing- c window only; nothing answered. *Item 1*: Part VI's Table 9 shows the same late onset at $c = \frac{1}{2}, 1, 2$. *Item 13*: Part VI read Khorunzhy–Shcherbina–Vengerovsky (2004) in full (its account, not confirmed by the write). The other items are untouched.

Part VI

Sharp corrections at every fixed mean degree, optimal Poisson laws, and a growing-degree transition

A research manuscript of 8 October 2026, archive `sparse_graph_moments_research.zip`, delivered as `sparse_graph_asymptotics/article.tex` with five section files and a bibliography file (40 pages). Delivered title block “Sparse graph moments / Sharp corrections, optimal Poisson laws, and a growing-degree transition / Research manuscript prepared with OpenAI assistance for the ProveIt project / 8 October 2026”. Printed as delivered after Section 57, which the write added, with [write] notes; the manuscript’s Section k is Section $k + 57$ here, its statements, research questions and equations $(k.m)$ are $(k+57.m)$, its Table 1 is Table 9, and its Figures 1 and 2 keep their numbers. Its labels carry the prefix `sgm:md:`. It credits Parts I–V as its “direct repository precursor” and re-derives what it uses of them, with weights.

Abstract of this Part (as delivered). Let $m_k(c)$ be the limiting expected spectral moment of order $2k$ of $G(N, c/N)$, with the graph-size limit taken first, and put $H_k(c) = \mathbb{E}(X + c)^k$ for $X \sim \text{Pois}(c)$. We develop three refinements of its high-moment asymptotics. First, uniformly for c in any compact subinterval K of $(0, \infty)$, we prove a finite-collision expansion to every fixed order and evaluate

$$\log \frac{m_k(c)}{2H_k(c)} = \frac{cw^2(w+2+2c)}{2k} + \frac{cw^5}{6k^2} + O_K\left(\frac{w^4}{k^2}\right), \quad w = W(k/c).$$

Second, for the maximum-departure deficit D_k of a moment-weighted closed tree walk, we determine a signed Poisson–Charlier expansion. It gives sharp total variation constants both before and after recentering the Poisson mean, and proves optimality among all Poisson means. Third, uniformly when $cw/\sqrt{k}$ stays in a compact subinterval of $(0, \infty)$, we prove

$$\frac{m_k(c)}{2H_k(c)} = \exp(c^2w^2/k) \left\{ 1 + O\left(k^{-1/4} \log^{13/4} k\right) \right\}.$$

An exact Catalan forest identity explains this transition; new weighted allocation and complement estimates justify it for the full moment. Proofs, exact recurrence checks, symbolic collision polynomials, and clearly delimited numerical diagnostics are included. The earlier ProveIt work at $c = 1$ is explicitly credited, and no worldwide-priority or formal-verification claim is made.

Keywords. Sparse random graphs; high spectral moments; closed tree walks; Touchard polynomials; Lambert W function; Poisson approximation; total variation; Catalan forests.

57 Provenance, scope and notation of Part VI

[Added 9 October 2026.] Part VI was added in a second write of this report, after Parts I–V had been merged (`e4199e04f`) and independently checked (`5f0331dc8`). It is the manuscript *Sparse graph moments: sharp corrections, optimal Poisson laws, and a growing-degree transition*, which starts from item 5 of Section 56 (“Fixed general mean degree”) and answers it. Everything of the manuscript is printed in this Part except its title page (recorded in the source block above), its table of contents and its reference list (merged into this report’s bibliography): its abstract and keywords above, and its seven sections as Sections 58–64, in the manuscript’s order. Its letters are kept (Section 57.5). This section and the notes marked [write] are the write’s; all other text of Sections 58–64 is the manuscript’s.

57.1 The manuscript

The archive `sparse_graph_moments_research.zip` (2,095,612 bytes; 33 files in one wrapper directory `sparse_graph_asymptotics/`, 3,896,295 bytes unpacked) arrived in commit `b28d0850b` (8 October 2026, 08:39 PDT). Its 23 supporting files were placed in this directory by commit `803f4d937` (9 October 2026), which also retired the archive, with the prefix `06-meandeg-`, the Part number; all 23 are byte-identical to the delivery (confirmed at the write). The programs, the Makefile and `build.sh` are `code/06-meandeg-*`; the data files, the two generated $\text{\TeX}$ fragments, `requirements.txt` and the records `claim_ledger.json`, `provenance.json` and `verification_report.json` are `data/06-meandeg-*`; the two figures (PDF, included here, and PNG previews) are `figures/06-meandeg-*`; the delivered `source_audit.txt` is `06-meandeg-source_audit.txt` at the top of the directory. `data/06-meandeg-exact_rows.json` (2,888,373 bytes) holds the exact rows to half-length 128 and is regenerated in seconds by `verify_exact.py` with `--order 128`. Not shipped, and retrievable from the arrival commit: the manuscript sources (`article.tex`, `sections/03_compact_moments.tex` to `sections/07_research.tex`, `references.tex`), its PDF, the delivery `README.md` and the checksum list `SHA256SUMS`.

The manuscript is `article.tex` (485 lines, its Sections 1–2), five section files (2,422 lines, its Sections 3–7) and `references.tex` (99 lines), 3,006 lines in all, dated 8 October 2026; a build of the delivered text reproduces its 40-page PDF, with no warnings. Its title page and PDF author field read “Research manuscript prepared with OpenAI assistance” (title page: “for the ProveIt project”). Its pins (its bibliography entry [25], `data/06-meandeg-provenance.json` and `06-meandeg-source_audit.txt`) are ProveIt commit `9570aaede2458830be9b3bbb072481c88f4f5a4c` (7 October 2026, 21:31 PDT), at which this report’s `article.tex` is the blob `f6a72e04cf`, Parts I–V as merged and checked and unchanged until this write, and commit `adc7f1241b42` of the `openai/math` repository, which is outside this repository and was not checked.

How the manuscript was merged. Its 158 labels carry the prefix `sgm:md:`, and its 192 references to them (144 `\eqref`, 48 `\ref`) were updated. Its five section files are printed at the places where its driver inputs them; the moment table is input from the shipped `data/06-meandeg-moment_table.tex` (unchanged; it has no label), and the two figures under their prefixed names. Its citation keys are entries of the merged bibliography, marked [Part VI], except four that Parts I–V cite already, which are printed once with Part VI’s details added and to which its citations were re-keyed: `BauerGolinelli2001` is `bg` (4 citations), `Hainzl2024` is `hdp` (2), `Khorunzhiy2016` is `khori` (2) and `OEISA094149` is `oeis` (1). Its key `ProveIt2026` is this report at the pin, that is, Parts I–V; its entry `ArratiaGoldsteinGordon1990` is listed but not cited, as delivered. Its macros `\TV`, `\normone`, `\Var`, its `definition` style, its `question` environment (“Research question”, numbered within sections) and `graphicx` were added to the preamble; its `\E`, `\Cat`, `\Wc` are those of Parts I–V. Five of its macros differ from those of Parts I–V and are redefined, as delivered, at the start of this Part and for it only: `\Poi` (“Pois”, not “Poisson”), `\Stir`, `\Part` ($\mathcal{P}$, not “Part”), `\fall` (an underlined power $x^{\underline{h}}$, which is Parts I–V’s $(x)_h$) and `\rise` (the same rising power); its `\P` is $\mathbb{P}$, Parts I–V’s `\Prob`. From this Part on, equations are numbered within sections, as delivered, while Parts I–V keep their consecutive numbers. The delivered paragraph indentation, page geometry, running heads, keyword line placement and table of contents are not reproduced; the delivered `\clearpage` commands are kept. No other text was changed. The write labelled the twelve research questions of Section 64. The numbering, against a build of the delivered text: its Sections 1–7 are Sections 58–64 here, its Theorem 3.6 is Theorem 60.6, Corollary 3.7 is Corollary 60.7, Theorems 4.1 and 4.2 are Theorems 61.1 and 61.2, Corollary 4.6 is Corollary 61.6, Theorem 5.1 is Theorem 62.1; every statement, research question, subsection and equation $k.m$ is $(k+57).m$; its Table 1 is Table 9

(Parts I–V have Tables 1–8), and its Figures 1 and 2 are unchanged (all 158 labels compared).

What the manuscript says about the repository. Its account of Parts I–V (“At the inspected snapshot it proves the $c = 1$ dominant-vertex reduction and all fixed collision orders, and explicitly lists general mean degree as a further question”) is correct; item 5 is that question. Its credit boundary is kept as delivered: Parts I–V are the “direct repository precursor”, and “Merely improving a displayed logarithmic exponent in that way is a corollary of the earlier method, not a separate new theorem at $c = 1$ ”. Its proposal to file the package as a separate report `a094149-weighted-moments-and-critical-transition/` was declined at placement: the collection keeps one report per A-number, and the manuscript re-derives this report’s main theorem, so it is this report’s Part VI. Its remark on the `openai/math` universality manuscript is context, not a premise, and was not checked.

57.2 What Part VI answers in Parts I–V

Parts I–V	After Part VI
Item 5 (fixed general mean degree)	Answered for every compact $K \subset (0, \infty)$: Theorem 60.6, all fixed collision orders uniformly in $c \in K$, with $2H_k(c)$, $H_k(c) = \mathbb{E}(X + c)^k$, in place of $2B_{k+1}$; its own weighted complement theorem (Theorem 60.4). Growing and vanishing c are Part VI’s questions.
Item 3 (evaluated suborders)	Answered in part: Corollary 60.7 evaluates the whole scale $1/k$, $A_1 = cw^2(w + 2 + 2c)/(2k) + O_K(w^4/k^2)$ (at $c = 1$: $W^3/(2k) + 2W^2/k$), and the terms w^6 (the $P_1^2/2$ behind Part I’s $\frac{1}{8}$) and w^5 (namely $cw^5/(6k^2)$ in the logarithm) at scale k^{-2} . Open: the lower suborders at k^{-2} and beyond.
Item 4 (growing collision order)	Not answered. Part VI’s growing-degree theorem (Theorem 62.1) lets the mean degree grow, $cw/\sqrt{k}$ in a compact set, not the collision order; it is a new direction.
Items 6, 7	Not answered; Corollary 62.6 concerns the critical growing- c window only.
Part I’s Theorem 1.1	Proved again , as the case $c = 1$ of Theorem 60.6, by the same method with weights (credited there as the precursor).
Part II’s equivalent and complement	Proved again at $c = 1$ by Theorem 60.4 and Theorem 60.6.
Part V’s Corollary 52.1	Same model ($\alpha = c$); its lower bound is $2(H_k(c) - cH_{k-1}(c)) - P_k(c)$, hence asymptotic to $\mu_{2k}(c)$ for each fixed c (the write’s consequence, note at the end of Section 56).
Remark 46.1	At $c = 1$, $(M_{2k} - G_k)/(2B_{k+1}) = W^3/(2k) + (1 + o(1))W^2/k$ (the write’s consequence, note after the remark); the remark itself is not sharpened.
Items 1, 2, 8–13	Untouched; Part VI’s Table 9 shows the same late onset for $c = \frac{1}{2}, 1, 2$.

Dated notes at the end of Section 56 and after Remark 46.1 say the same; the earlier notes there are kept as written.

What Part VI proves again: Part I’s main theorem and Part II’s complement estimate at $c = 1$ (with weights, for every compact K), the first-child recurrence (Bauer–Golinelli’s), and the coefficients d_2, d_3, P_2, P_3 at $c = 1$. *What is new to the report:* the uniformity in c , the evaluated logarithmic expansion, the deficit laws and the growing-degree transition.

57.3 What the write read and checked

At placement (the intake dossier, 9 October 2026; own code): a brute-force enumeration of all normalized closed tree walks for $k \leq 7$ (12,850 walks), with the polynomials $m_1, \dots, m_4$; its own implementation of the first-child recurrence at $c = \frac{1}{2}, 1, 2$ for $k \leq 48$, equal to `data/06-meandeg-exact_rows.json`, to the brute force, to the 13 terms of A094149 and to Part II’s moments; the Catalan quotient identity for $2 \leq k < 80$; and the residual of Corollary 60.7 at $k = 32, 64, 128$ (decreasing, with a constant growing with c , consistent with O_K and not decisive). No claim was found wrong, and the triage’s reading that Part VI answers item 4 was corrected.

At the write (9 October 2026; Python 3.14.4, SymPy 1.14.0, mpmath 1.3.0; own code, kept with the intake records and not shipped). Every proof of Sections 59–62 was read line by line, with no error found; rechecked by hand in particular: the exact cancellation $\frac{k}{k-s} \binom{k-1}{s} = \binom{k}{s}$; the leading pattern $h_1 = \dots = h_j = 2$ of (60.31); the bookkeeping of the logarithmic form (60.37) from (60.35), (60.39), (60.40); the two total-variation constants ($\frac{1}{2} \epsilon \mathbb{E}|Y - \lambda|$ with $\mathbb{E}|Y - \lambda| \sim \sqrt{2\lambda/\pi}$, and $\frac{1}{4} \mathbb{E}|Z^2 - 1| = \varphi(1)$); the closed form of $F(t)$, its convexity and minimum; the mean $\lambda + 2r$ and variance $\lambda + 4r$ of $Y + 2J$; $\log Q(k, s)$ to third order; and the limit $e^{\beta^2/4}$. With its own code: the first-child recurrence, integer-scaled, at $c = \frac{1}{2}, 1, 2$ for $k \leq 128$, equal to every scaled moment of `data/06-meandeg-exact_rows.json` and to the 13 terms of A094149; the rows with an indeterminate c for $k \leq 5$, giving $m_1, \dots, m_4$ and the Newton coefficients d_2, d_3, d_4 and P_2, P_3, P_4 , equal to the printed ones and to the generated `data/06-meandeg-collision_polynomials.tex`, with $[q^3]d_3 = 1/(6c^2)$, and at $c = 1$ equal to Part I’s d_2, d_3, P_2, P_3 ; $A_1(k, c)$ exactly from (60.38), equal at $c = 1$ to Part I’s T_k/B_{k+1} for $3 \leq k \leq 60$, with $(A_1 - P_1)/(w^4/k^2)$ at $k = 20, 40, 60$ equal to 0.344, 0.292, 0.271 ($c = \frac{1}{2}$), 1.29, 1.08, 1.00 ($c = 1$) and 5.65, 4.79, 4.45 ($c = 2$), bounded as (60.35) states; all eighteen entries of Table 9; the identities $\mathbb{E}C_2(Y; u)^2 = 2u^2$ and $\text{Var}(Y)_2 = 4u^3 + 2u^2$; $F(t)$ against quadrature at four points; $\mathbb{E}|Y - \lambda| = 2\lambda\pi_\lambda(\lfloor \lambda \rfloor)$ at three means; the Catalan quotient, the ratio (62.14) and the saddle (62.20) exactly; and the identity behind the Part V comparison. It did not rerun the shipped programs (the intake’s and the write’s own recurrences reproduce the shipped exact rows). *Not read by the write*: Bauer–Golinelli beyond Parts I–V’s quotations, Khorunzhy–Shcherbina–Vengerovsky, Hainzl–de Panafieu, Khorunzhiy, Pittel, Arratia–Goldstein–Gordon, Semerjian–Cugliandolo, Rodgers–Bray and the `openai/math` repository; the manuscript’s accounts of them are reported, not confirmed.

57.4 What is not claimed

From the manuscript: all growing-order results concern the limiting moments $m_k(c)$, the graph-size limit being taken first, and “none of the uniform conclusions below is asserted for a simultaneous limit of finite matrices with $k = k_N$ ”; the error bound of the growing-degree theorem is “not claimed sharp” and is “deliberately conservative”; the sharp remainder after order p does not assert uniformity for an order growing with k ; D_k is a statistic of the moment-weighted walks, “not the degree of a typical vertex of a finite graph”, and the variables of the compound approximation are auxiliary; the Bauer–Golinelli comparison is the one external enumerative input and is cited, the first-child recurrence is theirs, and Catalan enumeration, growing-parameter saddles and joint large-order analysis are not claimed new; the $c = 1$ all-orders construction is “not claimed anew”, and “Merely improving a displayed logarithmic exponent in that way is a corollary of the earlier method”; the bounded source audit “cannot certify worldwide priority”; no peer review and no proof-assistant formalization; the exact checks test finite identities and the floating-point diagnostics illustrate slow onset, and “neither supplies a proof of a limit, a uniform remainder, or historical novelty” (the

deficit and critical figures use explicitly labelled models, not the full walk law); the twelve research questions are “proposed projects”. The write adds: its checks are exact finite computations and floating evaluations; its two consequences (Part V’s bound, the scale W^2/k against Remark 46.1) follow from proved statements and say nothing about finite k ; nothing was submitted anywhere.

57.5 Reading conventions

Part VI keeps its letters. Where they clash with Parts I–V, read them as follows (the write’s own text in Part VI uses Part VI’s letters).

Symbol	Part VI	Parts I–V
$m_k(c)$, c	the weighted moment $\sum c^{e(\omega)}$; $m_k(1) = M_{2k} = a_k$; c the mean degree	a_k , M_{2k} ; Part V’s $\mu_{2k}(\alpha)$ with $\alpha = c$; c also a constant
$T_n(c)$, $H_k(c)$	Touchard $\mathbb{E}X^n$, $X \sim \text{Pois}(c)$ ($T_n(1) = B_n$); $H_k(c) = \mathbb{E}(X + c)^k$ ($H_k(1) = B_{k+1}$)	Part I’s T_k (a transform sum), T_b ; Part II’s and V’s Touchard $T_t(c)$, $T_n(\alpha)$; Part IV’s T_k ; H , $H(m, s)$
w , λ , ϵ	$w = W(k/c)$ ($= W(k)$ at $c = 1$); $\lambda = cw$; $\epsilon = w^2/k$	$r = W(k)$ (Parts III–IV); λ eigenvalues (III), λ_m (IV), $\lambda = (3 + \sqrt{5})/2$ (V)
$F_{k,n}(c)$, $E_{t,q}^c$, R_c , γ_h , d_h , P_h , G_j , $A_j(k, c)$	weighted versions; at $c = 1$ Part I’s $F_{k,n}$, $E_{t,q}$, $R(q, t)$, $c_h(q)$, $d_h(q)$, $P_h(z)$, $G_j(n, s)$, $A_j(k)$	Part V’s F_{2k} Fibonacci; Part I’s $F(x)$
$P_1(k, c)$	$cw^2(w + 2 + 2c)/(2k)$	$P_h(z)$ (Part I), $P_k(\alpha)$ (Part V), P_n (IV)
D_k , $\nu_{k,c}$, θ_0	the maximum-departure deficit and its law; the recentred mean $cw + cw^3/k$	D_k the two-hub count (I–III); D_0 , D constants
$p_{k,c}(s)$, S	the fresh law; S a variable with this law	Part I’s $p_k(s)$ and Z (Lemma 3.2); S , S_k windows
S_0 , L , h , M , $J(s)$, $\mathcal{H}$, Ent	cutoffs and entropies of the complement proofs	the same roles in Part I’s Appendix B (S , L , h , J , $H(m, s)$)
v_t	$M^{t-1}t!$ (Section 60); $c^{\bar{t}}/c^t$ (Section 62)	
$Q(k, s)$, $C(z)$, $f_q(t)$, C_2	the Catalan quotient; the Catalan series; $[z^t]C(z)^q$; the second Charlier polynomial	$Q_k = B_{k+1}$ (II–IV), $Q_{u,H}$ (I); C , C_ℓ Catalan numbers and constants
K , a , b , u , β , α	a compact $[a, b]$; $u = cw/\sqrt{k}$; $c_k = \beta\sqrt{k}/\log k$; α_* , α coefficients, $\alpha = S_1/\sqrt{V}$ -type ratios	a_k ; Part V’s α the mean degree
Z , Y , J	$Z \sim N(0, 1)$ (Section 61), Z the binomial mixture (Section 62); Y Poisson; $J \sim \text{Pois}(\lambda\epsilon/2)$, $J(\omega)$ a count	Z the deficit law, Z_j (I); J_k , $\mathcal{J}_k$ (IV)
$\mathcal{U}_{k,S}$, $\mathcal{V}_{k,S}$, $I(k, j)$	the dominant class (Part II’s $\mathcal{U}_{k,S}$); its simple-off-hub subclass; walks with j edges	$\mathcal{U}_{k,S}$, $U_{k,S}$; Bauer–Golinelli’s count
$x^{\underline{h}}$, $\{^n_j\}$, $\mathcal{P}[n]$	$x(x-1)\cdots(x-h+1)$; Stirling numbers; set partitions	$(x)_h$; Stirling numbers, also $S(n, j)$; Part an operator name

58 Introduction and principal results

High spectral moments of sparse random graphs are controlled by walks with a heavily visited vertex. Their enumeration is substantially more delicate than counting stars alone. Repeated visits can support nontrivial interiors below the vertex, and a local enumeration is useful only after all

other contributing walks have been bounded. This article develops that distinction quantitatively. It identifies the first corrections to the moment, the optimal Poisson approximation to the dominant vertex's departure deficit, and a parameter scale on which deeper but unrepeated interiors create a finite multiplicative change.

The basic quantity is

$$m_k(c) = \lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E} \operatorname{Tr} A_{N,c}^{2k}, \quad c > 0, \quad (58.1)$$

where $A_{N,c}$ is the adjacency matrix of the undirected simple graph $G(N, c/N)$. The limit in (58.1) is taken with k, c fixed. After taking it, $m_k(c)$ is an explicitly defined polynomial in c , and it makes sense to study $m_k(c_k)$ as k grows. We never interchange these limits. In particular, none of the uniform conclusions below is asserted for a simultaneous limit of finite matrices with $k = k_N$.

Put

$$T_k(c) = \mathbb{E} X^k, \quad H_k(c) = \mathbb{E}(X + c)^k, \quad X \sim \operatorname{Pois}(c), \quad w = W(k/c), \quad (58.2)$$

where $W(x) > 0$ solves $W(x)e^{W(x)} = x$. The exact shifted moment $H_k(c)$ is the useful normalization. Replacing it by a first saddle equivalent would discard more precision than some of the corrections we determine. For $c = 1$, $T_k(1) = B_k$ is a Bell number and $H_k(1) = B_{k+1}$. The sequence $m_k(1)$, starting at $k = 1$, is OEIS A094149 [17].

58.1 Three contributions

A growing-degree transition. Theorem 62.1 proves that, for fixed $0 < a < b < \infty$, uniformly when $a \leq cw/\sqrt{k} \leq b$,

$$\frac{m_k(c)}{2H_k(c)} = \exp\left(\frac{c^2 w^2}{k}\right) \left\{1 + O_{a,b}\left(k^{-1/4} \log^{13/4} k\right)\right\}. \quad (58.3)$$

Thus, for each fixed $\beta > 0$,

$$c_k = \frac{\beta\sqrt{k}}{\log k} \implies \frac{m_k(c_k)}{2H_k(c_k)} \longrightarrow e^{\beta^2/4}. \quad (58.4)$$

This is a change in the leading equivalent itself, rather than just an additional term for bounded c . The error bound is uniform and vanishing, but is not claimed sharp.

The finite multiplier has an exact combinatorial source. If the distinguished vertex has $n = k - s$ departures, then the weighted number of walks for which every edge away from that vertex is traversed once in each direction is

$$T_n(c) c^s [z^s] C(z)^n, \quad C(z) = 1 + zC(z)^2. \quad (58.5)$$

After applying the exact change-of-root weight, the relevant quotient is

$$Q(k, s) = \frac{[z^s] C(z)^{k-s}}{\binom{k-1}{s}} = \prod_{j=1}^{s-1} \frac{k+j}{k-j}, \quad 1 \leq s < k, \quad Q(k, 0) = 1. \quad (58.6)$$

Since s concentrates near cw and $\log Q(k, s) = s(s-1)/k + O(s^4/k^3)$, this quotient becomes $e^{c^2 w^2/k}$ at the critical scale. The substantive estimates in Section 62 show that repeated off-hub edges and the full complement of the near-hub class have sufficiently small total weight.

Uniform fixed-parameter corrections. For every compact $K \subset (0, \infty)$, Theorem 60.6 constructs exact finite transforms $A_j(k, c)$ such that, for each fixed $p \geq 1$,

$$\frac{m_k(c)}{2H_k(c)} = 1 + \sum_{j=1}^p A_j(k, c) + O_{K,p} \left(\frac{\log^{10p+6} k}{k^{p+1}} \right), \quad (58.7)$$

$$A_j(k, c) \sim_K \frac{c^j w^{3j}}{2^j j! k^j}. \quad (58.8)$$

Here $\sim_K$ means uniform relative convergence on K . Corollary 60.7 evaluates the initial transforms more precisely. Writing

$$P_1(k, c) = \frac{cw^2(w+2+2c)}{2k}, \quad (58.9)$$

it gives

$$\frac{m_k(c)}{2H_k(c)} = 1 + P_1(k, c) + \left(\frac{c^2}{8} + o_K(1) \right) \frac{w^6}{k^2}, \quad (58.10)$$

$$\log \frac{m_k(c)}{2H_k(c)} = P_1(k, c) + \frac{cw^5}{6k^2} + O_K \left(\frac{w^4}{k^2} \right). \quad (58.11)$$

The apparently different second terms are compatible: exponentiating the leading logarithmic correction contributes $P_1^2/2 \sim c^2 w^6/(8k^2)$. At $c = 1$, (58.11) specializes to

$$\log \frac{m_k(1)}{2B_{k+1}} = \frac{w^2(w+4)}{2k} + \frac{w^5}{6k^2} + O \left(\frac{w^4}{k^2} \right).$$

The earlier all-orders construction at $c = 1$ is not claimed anew here.

Sharp and optimal Poisson laws. Give each normalized closed tree walk ω of length $2k$ probability $c^{e(\omega)}/m_k(c)$, where $e(\omega)$ is its number of distinct edges. Let $d_\omega(v)$ count departures from vertex v and define

$$D_k = k - \max_v d_\omega(v), \quad \nu_{k,c} = \mathcal{L}(D_k). \quad (58.12)$$

This is a statistic of the moment-weighted walks, not the degree of a typical vertex of a finite graph. For fixed compact K , uniformly in $c \in K$, Theorems 61.1 and 61.2 prove

$$d_{\text{TV}}(\nu_{k,c}, \text{Pois}(cw)) \sim_K \sqrt{\frac{c}{2\pi}} \frac{w^{5/2}}{k}, \quad (58.13)$$

$$d_{\text{TV}} \left(\nu_{k,c}, \text{Pois} \left(cw + \frac{cw^3}{k} \right) \right) \sim_K \varphi(1) \frac{w^2}{k}, \quad (58.14)$$

$$\inf_{\theta \geq 0} d_{\text{TV}}(\nu_{k,c}, \text{Pois}(\theta)) \sim_K \varphi(1) \frac{w^2}{k}, \quad \varphi(1) = \frac{e^{-1/2}}{\sqrt{2\pi}}. \quad (58.15)$$

We use $d_{\text{TV}} = \frac{1}{2} \|\cdot\|_1$. A signed expansion and an explicit Gaussian absolute-error profile explain both constants and the scale of every asymptotically optimal mean. The optimization in (58.15) ranges over all nonnegative means; it is not merely a comparison between two chosen candidates. A positive approximation using an independent sum $Y + 2J$, with $Y \sim \text{Pois}(cw)$ and $J \sim \text{Pois}(cw^3/(2k))$, removes the leading quadratic error and has total variation error $o_K(w^2/k)$ (Corollary 61.6). These are auxiliary variables defining an approximation, not independent components asserted to exist inside the actual walk.

58.2 Prior work and the boundary of the novelty claim

Bauer and Golinelli developed the moment enumeration and proved the coefficientwise Stirling–Catalan comparison used below [1, Section 5.5, equation (9)]. Their fixed-edge-count equivalent does not by itself determine the sum through the moving saddle of the number of edges. Khorunzhiy, Shcherbina, and Vengerovsky established limiting moment recurrences and a resolvent equation for weighted random graphs [19]. These are foundational results, not consequences of the present work. Hainzl and de Panafieu enumerate tree walks of fixed excess by rational functions of the Catalan generating function and obtain finite-order large-mean expansions of fixed moments [4]. Khorunzhiy studies joint high moments and growing dilution in other regimes [15], and gives varying-parameter Touchard asymptotics [20]. We therefore do not claim that either Catalan enumeration, a growing-parameter saddle, or joint large-order analysis is new in itself. The particular comparison (58.3), its critical scale, and the full weighted complement proof are the contribution asserted here. The Touchard estimates needed for it are proved in the article rather than assumed under an unspecified uniformity condition.

Poisson approximation for component counts in random set partitions is also classical [21]. The additional distributional content here concerns the walk’s maximum-departure deficit after its collision bias is included: the signed quadratic term, the two sharp constants, and optimality over the whole Poisson family.

The direct repository precursor is the merged ProveIt manuscript *A094149 sparse graph moments* [25]. At the inspected snapshot it proves the $c = 1$ dominant-vertex reduction and all fixed collision orders, and explicitly lists general mean degree as a further question. Its ideas are credited and rederived with weights and uniform estimates in Section 60. One overlap is particularly important: once an expansion to every fixed order is available, the sharp equivalent of the remainder after order p follows by applying it at order $p + 1$. Merely improving a displayed logarithmic exponent in that way is a corollary of the earlier method, not a separate new theorem at $c = 1$.

The requested `openai/math` repository was inspected as additional research context [26]. Its random-regular-graph universality manuscript concerns a different ensemble and statistic. No unverified universality claim from that repository is an input to the proofs here. The archive records immutable repository revisions and the scopes of the primary sources that were actually inspected. This bounded audit supports a precise extension claim relative to those sources; it cannot certify worldwide priority. The manuscript contains mathematical proofs and independent finite checks, but has not been peer reviewed or formalized in a proof assistant.

[write] **The precursor and the sources at the pins, 9 October 2026.** “The merged ProveIt manuscript” of the preceding paragraph is Parts I–V of this report, read at the pin 9570aaede2, where its text is the same as before this write (Section 57.1); the account of it is correct. The general-mean-degree question it “explicitly lists” is item 5 of Section 56. The manuscript’s reading of Khorunzhiy–Shcherbina–Vengerovsky in full (its 06-meandeg-source_audit.txt) closes, by its own account, the access gap that item 13 records for Part I; the write did not read that paper. The `openai/math` snapshot was not checked.

58.3 Proof organization

Section 59 fixes the combinatorial model and gives an exact recurrence used by the verification code. Section 60 proves the compact-parameter expansions, including the global complement estimate. Section 61 uses that control to identify the probability law at an accuracy smaller than its optimal Poisson error. Section 62 proves a separate complement theorem in the growing-degree regime;

it does not substitute an unbounded parameter into a compact-parameter estimate. Section 63 explains what the computational artifacts check. Section 64 proposes concrete next problems and specifies the obstacles left by the present proofs.

59 Definitions, limiting moments, and exact enumeration

59.1 Normalized walks and weights

A normalized walk of length $2k$ is a word $\omega = (v_0, v_1, \dots, v_{2k})$ with $v_0 = v_{2k} = 0$, no two consecutive vertices equal, and each previously unseen vertex assigned the next unused nonnegative integer. Its traversed graph has an edge $\{v_{i-1}, v_i\}$ whenever that pair occurs. Let $\mathcal{W}_k$ consist of these walks whose traversed graph is a tree. The time-zero position is distinguished. Walks are not identified simply because one is a cyclic rotation of another; normalization is applied after any change of time origin. The empty walk is the sole member of $\mathcal{W}_0$.

If an edge e is traversed r_e times in each direction, the equality of the two counts follows by cutting the tree at that edge. Hence

$$\sum_e r_e = k, \quad d_\omega(v) = \sum_{e \ni v} r_e, \quad \sum_v d_\omega(v) = 2k, \quad d_\omega(v) \leq k. \quad (59.1)$$

Write $e(\omega)$ for the number of distinct edges, and define

$$m_k(c) = \sum_{\omega \in \mathcal{W}_k} c^{e(\omega)}, \quad F_{k,n}(c) = \sum_{\substack{\omega \in \mathcal{W}_k \\ d_\omega(0)=n}} c^{e(\omega)}. \quad (59.2)$$

In particular, $m_k(c) = \sum_{n=1}^k F_{k,n}(c)$ for $k \geq 1$, $F_{0,0}(c) = m_0(c) = 1$, and all other inadmissible entries are zero. Each m_k is a polynomial of degree k with nonnegative integer coefficients. The initial values are

$$m_1 = c, \quad m_2 = c + 2c^2, \quad m_3 = c + 6c^2 + 5c^3, \quad m_4 = c + 14c^2 + 28c^3 + 14c^4.$$

Proposition 59.1 (Identification of the limiting moment). *For every fixed $k \geq 0$ and $c > 0$, the limit in (58.1) exists and equals (59.2).*

Proof. Expand the trace into closed vertex sequences. A normalized sequence using v vertices and e distinct edges has N^v assignments of distinct graph labels and expected edge product $(c/N)^e$. After division by N , its contribution is $N^v c^e / N^{e+1}$. Its traversed graph is connected, so $v \leq e + 1$. Equality holds precisely for a tree. At fixed k there are finitely many normalized words: those with $v < e + 1$ vanish in the limit, while each tree word contributes c^e . On a tree, every used edge is traversed in both directions, so $e \leq k$. This proves the claim; the case $k = 0$ is immediate. $\square$

59.2 Partitions, factorials, and a computable recurrence

Write $\mathcal{P}[n]$ for the partitions of $\{1, \dots, n\}$ and $\{^n_j\}$ for a Stirling number of the second kind. The Touchard identity is

$$T_n(c) = \sum_{j=0}^n \left\{ \begin{matrix} n \\ j \end{matrix} \right\} c^j = \sum_{\pi \in \mathcal{P}[n]} c^{|\pi|} = e^{-c} \sum_{r \geq 0} \frac{c^r r^n}{r!}. \quad (59.3)$$

For $n = 0$, the last expression uses $0^0 = 1$. The identity follows, for example, by expanding r^n in falling factorials and using $\mathbb{E}X^j = c^j$ for a Poisson variable. We use $x^{\underline{h}} = x(x-1)\cdots(x-h+1)$ and $x^{\overline{h}} = x(x+1)\cdots(x+h-1)$, both equal to 1 at $h = 0$.

For $t \geq 0$ and $q \geq 1$ define the segmented child count

$$E_{t,q}(c) = E_{t,q}^c = \sum_{\ell=0}^t \binom{\ell+q-1}{\ell} F_{t,\ell}(c). \quad (59.4)$$

The two notations for this count are used interchangeably in the proofs. A child-rooted walk with ℓ excursions has $\binom{\ell+q-1}{\ell}$ ways to be divided into q ordered, possibly empty segments, since cuts may be inserted at its returns to the child root. In particular $E_{0,q}(c) = 1$.

Proposition 59.2 (First-child recurrence). *For $1 \leq n \leq k$,*

$$F_{k,n}(c) = c \sum_{q=1}^n \binom{n-1}{q-1} \sum_{a+b=k-q} E_{a,q}(c) F_{b,n-q}(c). \quad (59.5)$$

Together with the stated initial conditions, this is a triangular exact recurrence in k .

Proof. The first root excursion chooses a particular first child. Suppose this root-child edge is used in q excursions. Choose its other $q-1$ positions among the remaining $n-1$ root excursions. Its interior consists of q segments of combined half-length a , with weight $E_{a,q}(c)$. The other root excursions form a normalized walk of half-length b with $n-q$ root departures, weighted by $F_{b,n-q}(c)$. The selected edge has weight c , regardless of its multiplicity q , and the half-lengths satisfy $a+b+q=k$. The order of first discovery determines the labels uniquely when these objects are reassembled. This gives a bijection and proves (59.5). Since $a, b \leq k-1$, all inputs come from earlier half-lengths. $\square$

For a rational parameter $c = p/q_0$, the code stores $q_0^k F_{k,n}(p/q_0)$. These are integers. The extra factor supplied by a root edge used q times is then pq_0^{q-1} , which avoids floating-point or repeated rational arithmetic in the recurrence. This scaling does not change the combinatorial weight of a repeated edge.

59.3 The external counting bound and change of root

Let $I(k, j)$ be the number of normalized walks in $\mathcal{W}_k$ with j edges. Bauer–Golinelli’s comparison is

$$\left\{ \begin{matrix} k \\ j \end{matrix} \right\} \leq I(k, j) \leq \text{Cat}_j \left\{ \begin{matrix} k \\ j \end{matrix} \right\}, \quad \text{Cat}_j = \frac{1}{j+1} \binom{2j}{j}. \quad (59.6)$$

We cite its proof in [1, Section 5.5]; this is the one substantive external enumerative inequality used in the global estimates. In particular,

$$T_k(c) \leq m_k(c) \leq T_k(4c). \quad (59.7)$$

This rough upper bound alone is far from the precision required by the main results.

An exact averaging identity keeps track of the time-zero convention. For any relabeling- and rotation-invariant class of walks and any vertex property preserved by relabeling and rotation, let $J(\omega)$ be the number of vertices having the property. For $k \geq 1$,

$$\sum_{\omega} c^{e(\omega)} J(\omega) = \sum_{\omega} c^{e(\omega)} \frac{2k}{d_{\omega}(0)} \mathbf{1}_{\{0 \text{ has the property}\}}. \quad (59.8)$$

Indeed, average over the $2k$ possible time shifts. A vertex v occurs as the departure vertex at exactly $d_\omega(v)$ of them. Its total averaged contribution is therefore 1 if it has the property. Each fixed time shift is a bijection of normalized time-rooted words, even for periodic words, and preserves the number of distinct edges. No division by a putative full orbit size is needed. The estimates below show that the cases of zero or multiple near-maximal vertices contribute negligibly, allowing (59.8) to identify the full moment from high-return rooted rows.

60 Uniform high-moment expansions at fixed mean degree

Throughout this section $K = [a, b]$ is a fixed compact subinterval of $(0, \infty)$. Constants may depend on K , and on an explicitly fixed expansion order, but never on $c \in K$ or on the large index. We use the weighted walks and departure counts defined in Section 59. In particular $m_k(c)$ denotes the $2k$ -th spectral moment, and $F_{k,n}(c)$ its refinement by the number n of departures from the time-zero vertex. The two comparison sequences are

$$T_n(c) = \mathbb{E}X^n, \quad H_k(c) = \mathbb{E}(X + c)^k, \quad X \sim \text{Pois}(c).$$

Their exponential generating functions are $\exp(c(e^z - 1))$ and $\exp(c(e^z - 1) + cz)$, respectively. It is $H_k(c)$, rather than $T_{k+1}(c)$, that provides the normalization for general c . At $c = 1$ the Poisson shift identity gives $H_k(1) = B_{k+1}$.

60.1 A quantitative saddle estimate

We begin with estimates whose error terms are sufficiently precise to evaluate the first correction. This also supplies the uniformity required later in the counting argument.

Lemma 60.1 (Touchard and shifted Touchard saddles). *Let $w_n = W(n/c)$. Uniformly for $c \in K$,*

$$T_n(c) = \frac{n! \exp(c(e^{w_n} - 1))}{w_n^n \sqrt{2\pi n(1 + w_n)}} \left(1 + O_K\left(\frac{w_n}{n}\right)\right). \quad (60.1)$$

If $r_n > 0$ is determined by $cr_n(e^{r_n} + 1) = n$, and $\beta_n = n(1 + r_n) - cr_n^2$, then

$$H_n(c) = \frac{n! \exp(c(e^{r_n} - 1 + r_n))}{r_n^n \sqrt{2\pi \beta_n}} \left(1 + O_K\left(\frac{w_n}{n}\right)\right). \quad (60.2)$$

Moreover

$$r_n - w_n = O_K(w_n/n), \quad \frac{H_n(c)}{T_n(c)} = e^{cw_n} (1 + O_K(w_n/n)). \quad (60.3)$$

Proof. For the first assertion, Cauchy's formula on the circle of radius $w = w_n$ reduces the problem to the integral of

$$\exp Q(\theta), \quad Q(\theta) = c(e^{we^{i\theta}} - e^w) - in\theta, \quad -\pi \leq \theta \leq \pi.$$

The linear term vanishes. The quadratic term is $-\beta\theta^2/2$, where $\beta = n(1 + w)$. For every fixed $j \geq 2$, repeated application of $z d/dz$ to e^z bounds the j -th phase derivative on $|\theta| \leq 1/w$ by $C_j n(1 + w)^{j-1}$. Put $\epsilon = w/n$, $t = \sqrt{\beta}\theta$, and $U = \epsilon^{-1/12}$. On $|t| \leq U$, Taylor's theorem through degree four gives

$$Q(\theta) = -t^2/2 + i\alpha_3 t^3 + \alpha_4 t^4 + O_K(\epsilon^{3/2}|t|^5), \quad |\alpha_3| \leq C\epsilon^{1/2}, \quad |\alpha_4| \leq C\epsilon.$$

The cubic term is imaginary and odd. Its first-order contribution integrates to zero on the symmetric arc. After its cancellation, the integrated error is $O_K(\epsilon)$: expansion of the exponential is bounded by a constant times $\epsilon(t^4 + t^6)e^{-t^2/4}$, together with a smaller integrable remainder. Indeed the cubic term is at most $O(\epsilon^{1/4})$ on this arc. The central integral is therefore $\sqrt{2\pi/\beta}(1 + O_K(\epsilon))$.

For completeness, there is no unestimated portion of the contour. When $U/\sqrt{\beta} \leq |\theta| \leq 1/w$, the inequalities for $1 - \cos \theta$ and $1 - \cos(w \sin \theta)$ imply $\Re Q(\theta) \leq -\kappa \beta \theta^2$, for a fixed $\kappa > 0$. On $1/w \leq |\theta| \leq \pi$, discarding the second cosine in

$$\Re e^{we^{i\theta}} = e^{w \cos \theta} \cos(w \sin \theta)$$

gives

$$\Re Q(\theta) \leq -ce^w \{1 - e^{w(\cos \theta - 1)}\} \leq -\kappa_K n/w^2.$$

Here $1 - \cos \theta \geq 2\theta^2/\pi^2$ on the whole interval. Both outer integrals are smaller than the central integral times every fixed negative power of n . This proves (60.1).

For H_n , the additional term cz changes the saddle equation to $cr(e^r + 1) = n$. The mean value theorem at w_n gives

$$|r_n - w_n| \leq C_K \frac{w_n^2}{n(1 + w_n)} \leq C_K w_n/n.$$

The second logarithmic derivative at its saddle is $\beta_n = cr_n((1 + r_n)e^{r_n} + 1)$, the value in the statement. The same derivative estimates apply, and $cr_n(\cos \theta - 1) \leq 0$ improves the real-part estimates. Thus the preceding central and outer arc argument proves (60.2).

To compare H_n with T_n , write $g(z) = c(e^z - 1) - n \log z$. Then $g'(w_n) = 0$, and $g''(z) = O_K(n/w_n)$ throughout the interval joining r_n to w_n . Hence

$$g(r_n) + cr_n - g(w_n) = cw_n + O_K(w_n/n).$$

Also $\beta_n/[n(1 + w_n)] = 1 + O_K(w_n/n)$. Comparing the two proved saddle formulas establishes (60.3). $\square$

Lemma 60.2 (Ratios, factorial moments, and fresh tails). *For each fixed nonnegative integer h , uniformly on K ,*

$$\frac{T_{n-h}(c)}{T_n(c)} = \left(\frac{w_n}{n}\right)^h (1 + O_{K,h}(w_n/n)), \quad \frac{H_{n-h}(c)}{H_n(c)} = \left(\frac{w_n}{n}\right)^h (1 + O_{K,h}(w_n/n)). \quad (60.4)$$

In particular,

$$n \frac{T_{n-1}(c)}{T_n(c)} = w_n + O_K(w_n^2/n). \quad (60.5)$$

The numbers

$$p_{k,c}(s) = \frac{\binom{k}{s} c^s T_{k-s}(c)}{H_k(c)}, \quad 0 \leq s \leq k, \quad (60.6)$$

form a probability distribution. If S has this distribution and $w = W(k/c)$, then

$$\mathbb{E} S^h = k^h c^h \frac{H_{k-h}(c)}{H_k(c)} = (cw)^h (1 + O_{K,h}(w/k)). \quad (60.7)$$

For every fixed $D > 0$ and $j \geq 0$, writing $S_0 = \lceil \log^2 k \rceil$, one also has

$$\sum_{s > S_0} s^j p_{k,c}(s) = o_K(k^{-D}). \quad (60.8)$$

Proof. For ratios it is important to difference the explicit leading term rather than to differentiate an unspecified error. Let

$$L(n) = \log \Gamma(n+1) + c(e^{w_n} - 1) - n \log w_n - \frac{1}{2} \log(2\pi n(1+w_n)).$$

Using $w'_n = w_n/[n(1+w_n)]$ and the cancellation imposed by the saddle equation, direct differentiation gives $L'(n) = \log(n/w_n) + O_K(1/n)$. Integrating over the fixed interval $[n-h, n]$, and then adding the two $O_{K,h}(w_n/n)$ endpoint errors in (60.1), proves the first ratio. The explicit logarithm in (60.2) has derivative $\log(n/r_n) + O_K(1/n)$; use $r_n/w_n = 1 + O_K(1/n)$ to obtain the other ratio.

The binomial theorem normalizes (60.6), and extracting s^h from its finite sum proves the exact identity in (60.7). For the tail assertion, tilt the $\text{Pois}(c)$ distribution of X by $(X+c)^k/H_k(c)$. Conditionally on X , take $S \sim \text{Bin}(k, c/(X+c))$. Expansion shows that this mixture has exactly the masses (60.6). The tilted masses, up to normalization, are $c^x(x+c)^k/x!$. Their successive logarithmic ratios are

$$k \log \left(1 + \frac{1}{x+c} \right) + \log c - \log(x+1).$$

They decrease in x . At $x = tk/w + O(1)$, with t in a fixed compact subset of $(0, \infty)$, they equal $w(1/t - 1) - \log t + o_K(1)$. Multiplying the ratios across intervals of length comparable with k/w proves that each fixed relative tail outside $X \asymp k/w$ has mass at most $e^{-\kappa_K k}$. The upper remainder is bounded by a geometric series using the decreasing ratios. On the central range the conditional binomial mean is $O_K(w)$; its tail beyond S_0 is at most $(C_K w/S_0)^{S_0}$. Both estimates are superpolynomially small. Since $S \leq k$, fixed power weights are absorbed by strengthening D . $\square$

60.2 Exact weighted branches

For $q \geq 1$ and $t \geq 0$, define

$$E_{t,q}^c = \sum_{\ell=0}^t F_{t,\ell}(c) \binom{\ell+q-1}{\ell}, \quad R_c(q, t) = \frac{E_{t,q}^c}{c^t \binom{q+t-1}{t}}. \quad (60.9)$$

A child-rooted walk with ℓ excursions can be divided into q ordered, possibly empty closed segments in $\binom{\ell+q-1}{\ell}$ ways.

Proposition 60.3. *For $n \geq 1$ and $s \geq 0$,*

$$F_{n+s,n}(c) = \sum_{\pi \in \mathcal{P}[n]} c^{|\pi|} \sum_{\substack{t_b \geq 0 \\ \sum_{b \in \pi} t_b = s}} \prod_{b \in \pi} E_{t_b, |b|}^c. \quad (60.10)$$

Under the probability distribution $\mathbb{P}_c(\pi) = c^{|\pi|}/T_n(c)$, let a Pólya urn have initial populations $q_b = |b|$, and let T_b be its occupancies after s draws. Then

$$\frac{F_{n+s,n}(c)}{c^s \binom{n+s-1}{s} T_n(c)} = \mathbb{E}_{\pi, c} \mathbb{E}_{\text{urn}} \prod_{b \in \pi} R_c(q_b, T_b). \quad (60.11)$$

With $M = 4 \max(1, a^{-1})$, set $v_0 = v_1 = 1$ and $v_t = M^{t-1}t!$ for $t \geq 1$. Uniformly on K ,

$$1 \leq R_c(q, t) \leq v_t, \quad R_c(q, 0) = R_c(q, 1) = 1. \quad (60.12)$$

Proof. Partition the root's n excursion positions according to their first child. A block of size q introduces one root-child edge, with weight c , and q visits to its interior walk. Different child interiors use disjoint vertex sets. Splitting each interior at its root returns and then restoring the prescribed excursion positions reconstructs the walk uniquely. This proves (60.10), including all repeated edges and all depths. The Pólya occupancy probabilities are

$$\frac{\prod_b \binom{q_b + t_b - 1}{t_b}}{\binom{n + s - 1}{s}}, \quad \sum_b t_b = s.$$

Dividing the partition identity by its fresh normalization proves (60.11).

The lower bound in (60.12) uses an interior star walk with t distinct fresh edges, of weight c^t . The upper cut bound is $E_{t,q}^c \leq m_t(c) \binom{q+t-1}{t}$. An unweighted length- $2t$ walk has a root-distance Dyck path and, at its i -th outward step, at most i child choices, including a newly discovered child. Thus its total count is at most $\text{Cat}_t t! \leq 4^{t-1} t!$ for $t \geq 1$. Each such walk has between one and t edges, so

$$c^{-t} m_t(c) \leq 4^{t-1} t! \max(1, c^{-1})^{t-1} \leq v_t.$$

The exact one-step value is $m_1(c) = c$, which also proves $R_c(q, 1) = 1$. $\square$

60.3 Uniform suppression of walks without a dominant vertex

The following reduction is the step that permits expansions of a local branch calculation to become assertions about the complete moment. It is proved for the whole compact interval, rather than inferred from the unweighted case.

Theorem 60.4 (Compact-parameter complement and rotation). *Let $S_0 = \lceil \log^2 k \rceil$. For every fixed $D > 0$,*

$$\sum_{\substack{w \in \mathcal{W}_k \\ \max_v d_w(v) < k - S_0}} c^{e(w)} = o_K(k^{-D} T_k(c)). \quad (60.13)$$

The weighted mass of walks with two vertices having at least $k - S_0$ departures has the same bound. Consequently

$$\frac{m_k(c)}{2H_k(c)} = \frac{1}{H_k(c)} \sum_{s=0}^{S_0} \frac{k}{k-s} F_{k,k-s}(c) + o_K(k^{-D}). \quad (60.14)$$

The error statements remain valid after multiplication by any fixed power of k .

Proof. We give the three ranges of the estimate explicitly. The external combinatorial input is the coefficientwise Bauer–Golinelli comparison

$$\left\{ \begin{matrix} t \\ \ell \end{matrix} \right\} \leq I(t, \ell) \leq \text{Cat}_\ell \left\{ \begin{matrix} t \\ \ell \end{matrix} \right\},$$

where $I(t, \ell)$ counts the tree walks with ℓ edges [1]. Its positive weighted sum, followed by Lemma 60.1, gives

$$T_t(c) \leq m_t(c) \leq T_t(4c) \leq T_t(c) \exp \left(\frac{C_K t}{\log(t+1)} \right). \quad (60.15)$$

For the final inequality, evaluate the exponential generating function of $T_t(4c)$ at $W(t/c)$. Relative to $T_t(c)$, the extra exponent is $3t/W(t/c) + O_K(\log t)$, and a larger constant covers finitely many initial indices.

We shall also use the exact moment inequality

$$m_t(c) \leq m_s(c)^{t/s}, \quad 0 \leq t \leq s. \quad (60.16)$$

To justify it without a growing-index interchange, first keep s, t fixed. In the expected normalized trace of the adjacency matrix of $G(N, c/N)$, each connected walk pattern contributes its weight times $N^{v-e-1}(1 + O(N^{-1}))$. Only tree patterns survive as $N \rightarrow \infty$. Hölder applied to the expected empirical spectral measure pushed forward by $x \mapsto x^2$, followed by this fixed-index limit, proves (60.16) for every integer pair. It can subsequently be applied at the particular pair needed in an estimate.

Write $\mathcal{H}(m, s) = k \log k - m \log m - s \log s$, $k = m + s$. The adjacent ratio in Lemma 60.2 implies, uniformly when $\min(m, s) \geq k^{0.49}$,

$$\log \frac{T_m(c)T_s(c)}{T_k(c)} = -(1 + o_K(1))\mathcal{H}(m, s). \quad (60.17)$$

Here is the cancellation needed near the endpoints. Put $p = \min(m, s)$, $q = k - p$. Sum the adjacent logarithmic ratios only over $q \leq j < k$, where their errors are uniformly $o_K(1)$. Together with $\log T_p(c) = p \log p - p \log \log p - p + o_K(p)$, this gives

$$\log \frac{T_p(c)T_q(c)}{T_k(c)} = -\mathcal{H}(m, s) + p \log \frac{\log k}{\log p} + o_K(p).$$

The extra logarithm is at most $\log(k/p)/\log p$, while $\mathcal{H}(m, s) \geq p \log(k/p) \geq p \log 2$. This proves (60.17). Subtracting three coarse $o(k)$ estimates would not suffice here.

Set $B = \max(1, b)$, $C_0 = 2M$, and choose a fixed $\delta > 0$ with $\delta^2 \leq 0.005/C_0$. Our cutoffs are

$$h = \left\lfloor \frac{k}{64B \log k} \right\rfloor, \quad L = \delta \sqrt{k \log k}.$$

Eventually $h > L > k^{0.49}$.

Intermediate root departures. For $m \geq h$ and $s = k - m \geq L$, the branch identity and positivity give, for every $x, z > 0$,

$$F_{m+s,m}(c) \leq m! x^{-m} z^{-s} \exp \left(c(e^x - 1) + c \sum_{t=1}^s m_t(c) z^t \mathcal{A}_t(x) \right), \quad (60.18)$$

where

$$\mathcal{A}_t(x) = e^x \sum_{j=1}^t \binom{t-1}{j-1} \frac{x^j}{j!} \leq e^x x(1+x)^{t-1}, \quad \mathcal{A}_t(x) \leq e^x e^{2\sqrt{tx}}.$$

The expression follows by summing $x^q \binom{q+t-1}{t}/q!$ over block sizes $q \geq 1$. For the second bound use $\binom{t-1}{j-1} \leq t^j/j!$ and $\sum_j u^j/(j!)^2 \leq e^{2\sqrt{u}}$. The exponent is truncated at $t = s$ before evaluation; no convergent ordinary generating function of all moments is assumed.

In (60.18) take

$$x = W(m/c), \quad \varepsilon = \frac{m}{8k}, \quad \eta = 1 - \varepsilon, \quad z = \eta m_s(c)^{-1/s}, \quad J = \left\lceil \frac{64 \log k}{\varepsilon^2} \right\rceil.$$

Equations (60.15) and (60.1) give $z \leq C_K \log(s+1)/s$, and $\varepsilon \gg_K 1/\log k$. The two cases $m \geq k/2$ and $m < k/2$ show

$$\frac{\varepsilon^2 s}{\log^3 k} \longrightarrow \infty, \quad J < s, \quad J(1+x)z = o_K(1)$$

uniformly in the specified range. If $t > J$, Hölder yields $m_t(c)z^t \leq e^{-\varepsilon t}$. Since $2\sqrt{tx} \leq \varepsilon t/2$, the large- t part of the exponent is $O_K(k^{-10})$. For $t \leq J$, the bound $m_t(c) \leq B^t 4^{t-1} t!$ shows that successive upper-bound terms after the first have ratios $o_K(1)$. The first term is exactly cmz , because $m_1(c) = c$ and $cxe^x = m$. The small- t sum is therefore $cmz(1 + o_K(1))$.

Comparing the saddle prefactor with $T_m(c)$ gives

$$\log \frac{F_{k,m}(c)}{T_k(c)} \leq \log \frac{T_m(c)m_s(c)}{T_k(c)} - s \log \eta + O_K(\log k) + O_K(mz).$$

By (60.17) and (60.15), its first term is at most $-(1 - o_K(1))\mathcal{H} + C_K s / \log(s+1)$. The extra term is $o_K(\mathcal{H})$: for $s \leq k/2$ use $\mathcal{H} \geq s \log 2$; for $s > k/2$, use $\mathcal{H} \geq m \log(k/m) \gg_K k \log \log k / \log k$. Also

$$-s \log \eta \leq \frac{ms}{7k} \leq \frac{\mathcal{H}}{14}.$$

Finally $mz = o_K(\mathcal{H})$. For $s > k/2$ this follows from $mz \leq C_K m \log(s+1)/s$. Otherwise divide by $\mathcal{H} \geq s \log(k/s)$; on $L \leq s \leq k^{3/4}$ the resulting bound is $O_K(k/s^2) = O_K(1/\log k)$, and on $s \geq k^{3/4}$ it is $O_K(\log k/\sqrt{k})$. Thus, eventually throughout the intermediate range,

$$F_{k,m}(c)/T_k(c) \leq e^{-\mathcal{H}(m,s)/2}. \quad (60.19)$$

The high-departure shoulder. For an urn history let $A = \max_b q_b$ and $Z_s = \prod_b v_{T_b}$. An old color with current count $t \geq 1$ has weight multiplier $M(t+1)$; a new color has multiplier one. Since $M(t+1) - 1 \leq C_0 t$, successive conditional expectations give

$$\mathbb{E}_{\text{urn}} Z_s \leq e^{\tau A + \gamma}, \quad \tau \leq \frac{C_0 s^2}{2n}, \quad \gamma \leq \frac{C_0 s^3}{3n}. \quad (60.20)$$

Indeed the extra probability-weighted multiplier at step j is at most $C_0(Aj + j^2)/(n+j)$; summing it and using $1 + u \leq e^u$ proves the displayed bounds.

The weighted partition distribution satisfies

$$\mathbb{E}_{\pi,c} \# \{q\text{-blocks}\} = c \binom{n}{q} \frac{T_{n-q}(c)}{T_n(c)}.$$

Consequently, uniformly for $0 \leq \tau \leq 0.02 \log n$,

$$\log \mathbb{E}_{\pi,c} e^{\tau A} \leq C_K e^{\tau} \log n. \quad (60.21)$$

For $q \leq n/2$, the adjacent ratio bound makes the expected block count at most $C_K (C_K \log n)^q / q!$; summing the $e^{\tau q}$ -weighted terms proves the claim in this range. For $q > n/2$, retain the first $\lfloor n/2 \rfloor$ adjacent ratios. To bound the remainder, use $T_j(c) \geq a$ for $j \geq 1$ and the fact that $T_j(c)$ is nondecreasing for $j \geq 1$. The possible final value $T_0(c) = 1$ changes the estimate by at most $\max(1, a^{-1})$. This portion is bounded by a constant times

$$(1 + e^{\tau})^n \left(\frac{C_K \log n}{n} \right)^{\lfloor n/2 \rfloor} = e^{-\Omega_K(n \log n)}.$$

A maximal block realizes A , so the sum over blocks bounds $e^{\tau A}$. This proves (60.21), including its required exponential tilt.

Take $S_0 \leq s \leq L$ and $n = k - s$. Then $\tau \leq 0.005 \log k < 0.02 \log n$ eventually. Equations (60.11)–(60.21) imply

$$\frac{F_{k,k-s}(c)}{T_k(c)} \leq c^s \binom{k-1}{s} \frac{T_{k-s}(c)}{T_k(c)} e^{\gamma \mathbb{E}_{\pi,c} e^{\tau A}}.$$

The fresh factor is at most $(C_K \log k/s)^s$. Write $J(s) = s \log(s/\log k)$. The choice of δ guarantees $\gamma \leq J(s)/50$ uniformly for large k : the limiting upper bound for the endpoint ratio is $(4C_0/3)\delta^2 < 0.007$. Also $C_K e^\tau \log n = o_K(J(s))$. For $s \leq k^{0.1}$ use $\tau = o_K(1)$; for $s > k^{0.1}$, use $e^\tau \leq k^{0.005}$ and $J(s) \gg k^{0.1} \log k$. The harmless $O_K(s)$ in the fresh logarithm is $o_K(J(s))$. It follows that

$$F_{k,k-s}(c)/T_k(c) \leq e^{-J(s)/2}. \quad (60.22)$$

Low maximum, and passage from rooted counts. If every departure count is at most h , a root-distance Dyck encoding has at most h choices at each outward step. When h existing children are present, a new child is forbidden; there is no additional $h+1$ factor. Multiplying by the weight bound B^k , the total mass is at most $(4Bh)^k$. Since

$$\log T_k(c) = k \log k - k \log \log k - k + o_K(k),$$

its ratio to $T_k(c)$ is at most $\exp[-(\log 16 - 1 + o_K(1))k]$.

For any set I of departure counts, weighted indexed rotation gives the exact identity

$$\sum_{n \in I} \frac{2k}{n} F_{k,n}(c) = \sum_{w \in \mathcal{W}_k} c^{e(w)} |\{v : d_w(v) \in I\}|. \quad (60.23)$$

A vertex of departure count n appears at exactly n of the $2k$ indexed roots. If a normalized rotation orbit has fewer than $2k$ distinct words, division by their common repetition multiplicity gives the same identity; no freeness is assumed. Every walk outside the required dominant class has either low maximum or a vertex in the intermediate or shoulder ranges. The sums of (60.19) and (60.22), with at most k indices and rotation factors at most $2k/h = O_K(\log k)$, are superpolynomially small. This proves (60.13).

Two vertices whose departures exceed $k/2$ must be adjacent, because otherwise their incident-edge multiplicities are disjoint and sum to at most k . A tree has no three pairwise adjacent vertices, so there are at most two such vertices. If both departures are at least $k - S_0$, their joining edge has multiplicity at least $k - 2S_0$. Rotate to a crossing of this edge. On deleting its crossings, the two segmented interior walks have total half-length at most $2S_0$. The elementary cut bound used in Proposition 60.3 bounds their number, including origins, by $\exp(O(\log^3 k))$. Their weights contribute at most B^k . This remains superpolynomially small relative to $T_k(c)$. Use (60.23) with $I = [k - S_0, k]$, subtract this double counting, and observe $H_k(c) \geq T_k(c)$ to obtain (60.14). $\square$

60.4 The finite collision expansion

We now construct every correction by a finite exact operation. For $h \geq 0$, set

$$\gamma_h(q; c) = \sum_{t=0}^h (-1)^{h-t} \binom{h}{t} R_c(q, t), \quad (60.24)$$

$$\begin{aligned} d_h(q; c) &= \frac{q^{\bar{h}}}{h!} \gamma_h(q; c) \\ &= \sum_{t=0}^h (-1)^{h-t} c^{-t} E_{t,q}^c \frac{q + t^{\bar{h-t}}}{(h-t)!}. \end{aligned} \quad (60.25)$$

Thus $d_0 = 1$, $d_1 = 0$, and $d_h(q; c)$ is a polynomial of degree at most h in q ; for $h \geq 1$ its constant term is zero. Its coefficients belong to $\mathbb{Q}[c, c^{-1}]$.

For $j \geq 1$, $n \geq 1$, $s \geq 0$, define

$$G_j(n, s; c) = \sum_{r=1}^j \frac{1}{r!} \sum_{\substack{h_i \geq 2 \\ \sum_{i=1}^r (h_i - 1) = j}} \frac{s^H}{n^H} \mathbb{E}_{\pi, c} \sum_{\substack{b_1, \dots, b_r \in \pi \\ \text{ordered, distinct}}} \prod_{i=1}^r d_{h_i}(q_{b_i}; c), \quad H = \sum_{i=1}^r h_i = j + r. \quad (60.26)$$

The convention is that a term with $s^H = 0$ vanishes before any otherwise invalid index is evaluated. Finally set

$$A_j(k, c) = \sum_{s=0}^{k-1} p_{k,c}(s) G_j(k - s, s; c). \quad (60.27)$$

Here $H \leq 2j$ and each $h_i \leq j + 1$, so only the finitely many child rows with half-length at most $j + 1$ enter a fixed correction.

Lemma 60.5 (Uniform local expansion). *For each fixed $p \geq 1$, uniformly for $c \in K$ and $0 \leq s \leq S_0 = \lceil \log^2 k \rceil$,*

$$\frac{F_{k,k-s}(c)}{c^s \binom{k-1}{s} T_{k-s}(c)} = 1 + \sum_{j=1}^p G_j(k - s, s; c) + O_{K,p} \left(\frac{\log^{10p+6} k}{k^{p+1}} \right). \quad (60.28)$$

For all $n \geq 1$ and $s \geq 0$, $|G_j(n, s; c)| \leq C_{K,j}(1 + s)^{2j}$.

Proof. Let $C_s = s - |\{b : T_b > 0\}|$ count collision excess, and let $Z_s = \prod_b v_{T_b}$. At step j , conditional on the history, the probability-weighted multiplier of all transitions to old colors is bounded by

$$\sum_{T_b > 0} \frac{q_b + T_b}{n + j} M(T_b + 1) \leq \frac{2M(A_j + j^2)}{n} = b_j.$$

For new colors the total multiplier is at most one. Marking each old-color transition by u , induction gives a coefficientwise polynomial inequality

$$\mathbb{E}_{\text{urn}}[Z_s u^{C_s}] \preceq \prod_{j=0}^{s-1} (1 + b_j u), \quad \Lambda = \sum_{j < s} b_j \leq C_K s^2 (A + s)/n.$$

It follows that

$$\mathbb{E}_{\text{urn}}[Z_s; C_s \geq d] \leq \Lambda^d e^\Lambda / d!. \quad (60.29)$$

Binomial inversion gives $R_c(q, t) = \sum_{h=0}^t \gamma_h(q; c) \binom{t}{h}$. Expand the product over blocks and retain only terms for which $\sum (h_i - 1) \leq p$. Call the resulting polynomial $L_p(T)$. Because $\gamma_1 = 0$, every nonconstant mark has $h_i \geq 2$. A nonzero marked term satisfies $h_i \leq T_{b_i}$, so the truncation is exact whenever $C_s \leq p$. For fixed p , (60.12) bounds all involved γ_h uniformly on K . Moreover

$$\sum_{\text{ordered, distinct } b_i} \prod_i \binom{T_{b_i}}{h_i} \leq \frac{s^H}{\prod_i h_i!}.$$

Consequently $|L_p(T)| \leq C_{K,p}(1 + s)^{2p}$. Since $\prod R_c \leq Z_s$ and $Z_s \geq 1$, (60.29) yields

$$\left| \mathbb{E}_{\text{urn}} \prod_b R_c(q_b, T_b) - \mathbb{E}_{\text{urn}} L_p(T) \right| \leq C_{K,p}(1 + s)^{2p} \Lambda^{p+1} e^\Lambda.$$

The distinct-color urn factorial identity

$$\mathbb{E}_{\text{urn}} \prod_i \binom{T_{b_i}}{h_i} = \frac{s^H}{n^{\overline{H}}} \prod_i \frac{q_{b_i}^{\overline{h_i}}}{h_i!}$$

shows, exactly, that $\mathbb{E}_{\pi,c} \mathbb{E}_{\text{urn}} L_p = 1 + \sum_{j=1}^p G_j(n, s; c)$. It follows either from exchangeability of the selected draw indices or directly from the Pólya occupancy formula.

On $A, s \leq S_0$, one has $\Lambda = O_K(\log^6 k/k)$. Thus the preceding error is

$$O_{K,p} \left(\frac{\log^{10p+6} k}{k^{p+1}} \right).$$

For $A > S_0$, use (60.20) rather than an unweighted rare-event estimate. Its tilt is $\tau = O_K(\log^4 k/k)$, and the proof of (60.21), with the block sum restricted to $q > S_0$, gives

$$\mathbb{E}_{\pi,c}[e^{\tau A}; A > S_0] = o_K(k^{-D})$$

for every fixed D : the $q \leq n/2$ part is bounded by $\exp[-\kappa_K \log^2 k \log \log k]$, and the other part is $e^{-\Omega_K(n \log n)}$. Together with the polynomial bound on L_p , this proves (60.28). Applying that polynomial bound to an individual homogeneous part also gives the stated global bound on G_j . $\square$

Theorem 60.6 (All fixed collision orders). *The transforms (60.27) are independent of truncation order. For every fixed $p \geq 1$, uniformly on K ,*

$$\frac{m_k(c)}{2H_k(c)} = 1 + \sum_{j=1}^p A_j(k, c) + O_{K,p} \left(\frac{\log^{10p+6} k}{k^{p+1}} \right). \quad (60.30)$$

Every $A_j(k, c)$ has a finite rational expression involving Touchard polynomials and c , and

$$A_j(k, c) = \frac{c^j w^{3j}}{2^j j! k^j} (1 + o_K(1)), \quad w = W(k/c). \quad (60.31)$$

In particular the remainder after any fixed $p \geq 0$ exact transforms has the sharp equivalent

$$\frac{m_k(c)}{2H_k(c)} - 1 - \sum_{j=1}^p A_j(k, c) = \frac{c^{p+1} w^{3p+3}}{2^{p+1} (p+1)! k^{p+1}} (1 + o_K(1)). \quad (60.32)$$

Proof. Insert Lemma 60.5 into (60.14). The identity

$$\frac{k}{k-s} \binom{k-1}{s} = \binom{k}{s}$$

must be kept exact: it creates the probability weights $p_{k,c}(s)$. The fresh sum is $1 - c^k/H_k(c)$, whose missing endpoint is superpolynomially small. The bound on G_j and (60.8) extend each fixed correction to $s \leq k-1$. This proves (60.30).

For a finite evaluation, write

$$d_h(q; c) = \sum_{u=1}^h \alpha_{h,u}(c) q^u, \quad P_h(z; c) = \sum_{u=1}^h \alpha_{h,u}(c) z^u.$$

The exponential formula for a partition with r ordered marked blocks gives

$$\mathbb{E}_{\pi, c} \sum_{\text{ordered, distinct } b_i} \prod_i d_{h_i}(q_{b_i}; c) = \frac{n!}{T_n(c)} [z^n] c^r e^{c(e^z - 1) + rz} \prod_i P_{h_i}(z; c). \quad (60.33)$$

If $X \sim \text{Pois}(c)$, the exact Poisson shift is

$$c^r \mathbb{E}(X + r)^N = \mathbb{E} X^r X^N = \sum_{v=0}^r \mathfrak{s}(r, v) T_{N+v}(c),$$

where $\mathfrak{s}(r, v)$ are signed Stirling numbers of the first kind. Thus a monomial z^u in the finite product in (60.33) contributes

$$\frac{n^u}{T_n(c)} \sum_{v=0}^r \mathfrak{s}(r, v) T_{n-u+v}(c).$$

This proves the finite computability claim, including all normalizing powers of c .

Fix a collision pattern with r marked blocks and $H = \sum h_i = j + r$. The quantitative fixed-shift ratios show its partition average is $O_{K,j}(n^r w^j)$ on $n = k - s$, $s \leq S_0$. Multiplying by the urn factor and using $\mathbb{E} S^H = O_K(w^H)$ bounds its contribution to A_j by $O_{K,j}(w^{2j+r}/k^j)$. Only $r = j$ can attain the largest logarithmic power. That case has $h_1 = \dots = h_j = 2$. Direct calculation from the first two child rows gives

$$d_2(q; c) = \frac{q(q+1)}{2c} + q, \quad P_2(z; c) = \frac{z^2 + 2(1+c)z}{2c}. \quad (60.34)$$

Its leading monomial is $z^{2j}/(2c)^j$. Equation (60.33) and the ratio lemma therefore give partition average $n^j w^j / (2c)^j (1 + o_K(1))$. The fresh factorial moment contributes $(cw)^{2j}$; the urn denominator contributes k^{-2j} , and ordering contributes $1/j!$. This proves (60.31). The global polynomial bounds and fresh tails justify all discarded endpoints.

Finally apply (60.30) at order $p+1$. Its error is $o_K(w^{3p+3}/k^{p+1})$, so (60.31) proves (60.32). The sharp remainder is a consequence of the entire fixed-order hierarchy; it does not assert uniformity for an order growing with k . $\square$

60.5 An evaluated correction and an exponential refinement

The finite transforms preserve information at every smaller scale. The next result evaluates enough of their structure to exhibit a cancellation that an isolated leading equivalent does not reveal.

Corollary 60.7. *Uniformly on K , with $w = W(k/c)$, put*

$$P_1(k, c) = \frac{cw^2(w + 2 + 2c)}{2k}.$$

Then

$$A_1(k, c) = P_1(k, c) + O_K(w^4/k^2), \quad (60.35)$$

and

$$\frac{m_k(c)}{2H_k(c)} = 1 + P_1(k, c) + \left(\frac{c^2}{8} + o_K(1) \right) \frac{w^6}{k^2}. \quad (60.36)$$

More precisely, the logarithm has the expansion

$$\frac{m_k(c)}{2H_k(c)} = \exp \left\{ \frac{cw^2(w+2+2c)}{2k} + \frac{cw^5}{6k^2} + O_K(w^4/k^2) \right\}. \quad (60.37)$$

In particular

$$m_k(c) \sim 2H_k(c) \sim 2 \left(\frac{k}{cw} \right)^c T_k(c)$$

uniformly on the compact interval.

Proof. For a weighted random partition, $\mathbb{E} \sum_b q_b^2 = n(n-1)T_{n-1}(c)/T_n(c)$, and $\sum_b q_b = n$. Consequently (60.34) gives the exact first local correction

$$G_1(n, s; c) = \binom{s}{2} \phi_n(c), \quad \phi_n(c) = \frac{(n-1)T_{n-1}(c)/T_n(c) + 2(1+c)}{c(n+1)}. \quad (60.38)$$

For $n = k - s$, $s \leq S_0$, the quantitative ratio lemma implies

$$\phi_n(c) = \frac{w+2+2c}{ck} + O_K \left(\frac{(s+1)w+w^2}{k^2} \right).$$

Here $(n-1)T_{n-1}/T_n = w_n + O_K(w_n^2/n)$, $w_n - w = O_K(s/k)$, and the remaining denominator change is $O(s/k^2)$. Averaging against (60.6), using (60.7), yields

$$\begin{aligned} A_1(k, c) &= \frac{w+2+2c}{2ck} \mathbb{E} S^2 + O_K \left(\frac{w \mathbb{E} S^3 + w^2 \mathbb{E} S^2}{k^2} \right) \\ &= P_1(k, c) + O_K(w^4/k^2). \end{aligned}$$

This proves (60.35). Equation (60.36) follows immediately from (60.32) at $p = 1$.

For the more informative logarithmic form, split A_2 into its two permitted patterns: two $h = 2$ blocks, and one $h = 3$ block. For the first pattern, retain the whole polynomial $P_2(w; c)^2$, rather than only its highest monomial. The fixed-shift ratio estimate in (60.33) gives its partition average as

$$\frac{n^2}{w_n^2} P_2(w_n; c)^2 (1 + O_K(w_n/n)).$$

The term with Stirling index $v = 1$, all falling-factorial corrections, and the errors of the Touchard ratios are included in this bound. Averaging its urn factor $S^4/(2n^4)$, using the exact fourth fresh factorial moment, gives

$$A_{2,\text{pair}}(k, c) = \frac{P_1(k, c)^2}{2} + O_K(w^7/k^3). \quad (60.39)$$

To see that replacing $n = k - S$ by k causes no larger error, differentiate only the displayed leading expression, in which $w_n = W(n/c)$ is an ordinary smooth function. Its relative derivative on $[k - S_0, k]$ is bounded by C_K/k . The already uniform error is kept as a bound, without being differentiated. Now use $\mathbb{E} S^4 S = O_K(w^5)$. The relative error is $O_K(w/k)$.

For the single $h = 3$ pattern, only the leading coefficient of d_3 is needed. At fixed t , the leading power of q in $E_{t,q}^c$ is supplied by $F_{t,t}(c) = T_t(c)$. Hence

$$\lim_{q \rightarrow \infty} R_c(q, t) = c^{-t} T_t(c), \quad [q^3] d_3(q; c) = \frac{1}{6} \mathbb{E}(X/c - 1)^3 = \frac{1}{6c^2}.$$

The final equality is the third centered Poisson moment. Thus the $h = 3$ marked partition average is

$$\frac{nw_n^2}{6c^2} + O_K(nw_n) + O_K(w_n^3),$$

and its urn factor and third fresh factorial moment imply

$$A_{2,\text{single}}(k, c) = \frac{cw^5}{6k^2} + O_K(w^4/k^2). \quad (60.40)$$

For example, the error $O_K(w^6/k^3)$ from changing nearby indices is absorbed in the displayed remainder.

By (60.32) after two exact transforms, the remaining full-moment contribution is $O_K(w^9/k^3)$, which is $o_K(w^4/k^2)$. Equations (60.35), (60.39), and (60.40), together with $\log(1+x) = x - x^2/2 + O(x^3)$, prove

$$\log \frac{m_k(c)}{2H_k(c)} = P_1(k, c) + \frac{cw^5}{6k^2} + O_K(w^4/k^2).$$

This is (60.37). The final normalization comparison uses (60.3) and $e^{cw} = (k/(cw))^c$. $\square$

[write] At $c = 1$, 9 October 2026. At $c = 1$, $w = W(k)$, $H_k(1) = B_{k+1}$, $P_1(k, 1) = W(k)^3/(2k) + 2W(k)^2/k$, and the logarithmic form reads as displayed after (58.11). Part I's (6) records only the leading $W(k)^3/(2k)$ of this scale, and its (7) the coefficient $\frac{1}{8}$, which here is $c^2/8$ and comes from $P_1^2/2$ in the exponential; the write checked that $A_1(k, 1)$ computed from (60.38) equals Part I's T_k/B_{k+1} of (5) exactly for $3 \leq k \leq 60$. The new content at $c = 1$ is therefore the evaluated coefficient 2 of $W(k)^2/k$ and the term $W(k)^5/(6k^2)$ of the logarithm, with the remainder $O(W(k)^4/k^2)$ (dated notes at the end of Section 56 and after Remark 46.1).

61 Sharp Poisson laws for the dominant-vertex deficit

The moment expansion also identifies the geometry of a typical contributing walk. Here “typical” refers to the probability on normalized, time-rooted closed tree walks of length $2k$ that assigns a walk ω probability proportional to $c^{e(\omega)}$. It does not refer to a uniformly sampled vertex or walk in a finite random graph. Write

$$D_k(\omega) = k - \max_v d_\omega(v), \quad \nu_{k,c}(s) = \mathbb{P}(D_k = s),$$

where $d_\omega(v)$ counts departures from v , with repetitions. In particular, the statistic is a departure deficit, rather than the deficit of the number of distinct neighbors. The complement theorem shows that $D_k \leq \lceil \log^2 k \rceil$ with a superpolynomially small exceptional probability.

Throughout this section c belongs to a fixed compact interval $K = [a, b] \subset (0, \infty)$, and all uniformity assertions concern this interval. Set

$$w = W(k/c), \quad \lambda = cw, \quad \epsilon = \frac{w^2}{k}, \quad \theta_0 = \lambda + \lambda\epsilon = cw + \frac{cw^3}{k}. \quad (61.1)$$

Let $\pi_\theta(s) = e^{-\theta}\theta^s/s!$ denote the Poisson mass function, and define the second Charlier polynomial by

$$C_2(s; \lambda) = s^2 - 2\lambda s + \lambda^2 = (s - \lambda)^2 - s. \quad (61.2)$$

Our convention is $d_{\text{TV}}(\rho, \sigma) = \frac{1}{2} \sum_{s \geq 0} |\rho(s) - \sigma(s)|$. The factor $1/2$ in this definition matters for both constants below.

Theorem 61.1 (Signed expansion and two sharp distances). *Uniformly for $c \in K$, as $k \rightarrow \infty$,*

$$\nu_{k,c}(s) = \pi_\lambda(s) \left\{ 1 + \epsilon(s - \lambda) + \frac{\epsilon}{2\lambda} C_2(s; \lambda) \right\} + r_{k,c}(s), \quad (61.3)$$

$$\sum_{s \geq 0} |r_{k,c}(s)| = O_K \left(\frac{w^{3/2}}{k} + \frac{w^{16}}{k^2} \right) = o_K(\epsilon). \quad (61.4)$$

Consequently,

$$d_{\text{TV}}(\nu_{k,c}, \text{Pois}(cw)) \sim_K \sqrt{\frac{c}{2\pi}} \frac{w^{5/2}}{k}, \quad (61.5)$$

$$d_{\text{TV}}(\nu_{k,c}, \text{Pois}(\theta_0)) \sim_K \varphi(1) \frac{w^2}{k}, \quad \varphi(1) = \frac{e^{-1/2}}{\sqrt{2\pi}}. \quad (61.6)$$

Here $\sim_K$ means that the ratio tends to 1 uniformly on K . More precisely, the recentered signed expansion is

$$\nu_{k,c}(s) - \pi_{\theta_0}(s) = \frac{\epsilon}{2\lambda} C_2(s; \lambda) \pi_\lambda(s) + \tilde{r}_{k,c}(s), \quad (61.7)$$

$$\sum_s |\tilde{r}_{k,c}(s)| = O_K \left(\frac{w^{3/2}}{k} + \frac{w^{16}}{k^2} \right). \quad (61.8)$$

The shift cw^3/k removes the leading change in the mean parameter. The remaining error has a quadratic shape, represented by C_2 . This observation yields a stronger assertion: changing the Poisson mean further cannot remove the leading quadratic error.

Theorem 61.2 (Optimal Poisson mean and its error profile). *For each fixed $T < \infty$, uniformly for $c \in K$ and $|t| \leq T$,*

$$\frac{1}{\epsilon} d_{\text{TV}}(\nu_{k,c}, \text{Pois}(\theta_0 + t\sqrt{\lambda}\epsilon)) \longrightarrow F(t) := \frac{1}{2} \mathbb{E} \left| \frac{Z^2 - 1}{2} - tZ \right|, \quad Z \sim N(0, 1). \quad (61.9)$$

The continuous, even, convex function F is strictly minimized at $t = 0$, where $F(0) = \varphi(1)$. An explicit expression is

$$F(t) = \frac{1}{2} \{u_+(t)\varphi(u_-(t)) - u_-(t)\varphi(u_+(t))\}, \quad u_{\pm}(t) = t \pm \sqrt{1+t^2}, \quad (61.10)$$

with $\varphi(x) = (2\pi)^{-1/2} e^{-x^2/2}$. In particular,

$$\inf_{\theta \geq 0} d_{\text{TV}}(\nu_{k,c}, \text{Pois}(\theta)) \sim_K \varphi(1) \frac{w^2}{k}. \quad (61.11)$$

For any sequence $c = c_k \in K$, a sequence of means $\theta_k \geq 0$ is asymptotically optimal, in the sense that its distance divided by the infimum in (61.11) tends to 1, if and only if

$$\theta_k = \theta_0 + o\left(\sqrt{\lambda} \frac{w^2}{k}\right). \quad (61.12)$$

61.1 The fresh deficit law and a quantitative Poisson estimate

The exact probability that appears before collision corrections is

$$p_{k,c}(s) = \frac{\binom{k}{s} c^s T_{k-s}(c)}{H_k(c)}, \quad 0 \leq s \leq k. \quad (61.13)$$

Its Poisson approximation must be more accurate than w^2/k to identify the constant in (61.6). A qualitative Poisson limit would not suffice.

Lemma 61.3 (A Poisson estimate from birth ratios). *Let p be a probability on the nonnegative integers with $(s+1)p(s+1) = a_s p(s)$, and let S have law p . For $\lambda > 0$,*

$$d_{\text{TV}}(p, \text{Pois}(\lambda)) \leq \frac{C}{\sqrt{\lambda}} \mathbb{E}|a_S - \lambda|, \quad (61.14)$$

provided the right-hand side and $\mathbb{E}S$ are finite. The constant C is absolute.

Proof. For an indicator h , the Poisson Stein equation is

$$\lambda f(s+1) - sf(s) = h(s) - \mathbb{E}h(Y), \quad Y \sim \text{Pois}(\lambda).$$

There is a solution with $\sup_s |f(s)| \leq C/\sqrt{\lambda}$. For completeness, this bound follows directly from the immigration and death semigroup

$$P_t h(j) = \mathbb{E}h(B_{j,t} + Y_t), \quad B_{j,t} \sim \text{Bin}(j, e^{-t}), \quad Y_t \sim \text{Pois}(\lambda(1 - e^{-t})),$$

where the variables on the right are independent. The function $g(j) = -\int_0^\infty (P_t h(j) - \mathbb{E}h(Y)) dt$ solves the generator equation, and $f(j) = g(j) - g(j-1)$, $j \geq 1$, with $f(0) = 0$, solves the displayed

Stein equation. Couple $B_{j,t}$ and $B_{j-1,t}$ by one additional Bernoulli variable. Convolution cannot increase total variation, so

$$|f(j)| \leq \int_0^\infty e^{-t} d_{\text{TV}}(Y_t, Y_t + 1) dt \leq \frac{C}{\sqrt{\lambda}} \int_0^\infty \frac{e^{-t}}{\sqrt{1 - e^{-t}}} dt = \frac{2C}{\sqrt{\lambda}}.$$

Here $d_{\text{TV}}(\text{Pois}(u), \text{Pois}(u) + 1)$ equals the largest Poisson mass, by unimodality, and this mass is at most $C/\sqrt{u}$ by Stirling's formula (with the bound 1 used when $u < 1$). The semigroup integral converges by the same coupling with a stationary initial population.

The birth-ratio identity gives $\mathbb{E}[Sf(S)] = \mathbb{E}[a_S f(S + 1)]$. Taking expectations in the Stein equation, applying the supremum bound, and then taking the supremum over indicators proves (61.14). $\square$

Lemma 61.4 (Fresh Poisson approximation). *Uniformly for $c \in K$,*

$$d_{\text{TV}}(p_{k,c}, \text{Pois}(\lambda)) = O_K(w^{3/2}/k). \quad (61.15)$$

For every fixed nonnegative integer r , the tails of both distributions past $L_k = \lceil \log^2 k \rceil$ are superpolynomially small even after multiplication by $(1 + s)^r$. Moreover,

$$\mathbb{E}_p |S^2 - \mathbb{E}_p S^2| = O_K(w^{3/2}). \quad (61.16)$$

Proof. For (61.13), the exact birth rates are

$$a_s = c b_{k-s}(c), \quad b_n(c) = n \frac{T_{n-1}(c)}{T_n(c)}, \quad s < k, \quad a_k = 0.$$

The ratio estimate in Lemma 60.2 gives $b_n(c) = W(n/c) + O_K(W(n/c)^2/n)$. Since $\frac{d}{dn} W(n/c) = W(n/c)/(n(1 + W(n/c))) \leq 1/n$, for $s \leq L_k$ we obtain

$$|a_s - \lambda| \leq C_K \frac{s + w^2}{k} = O_K(w^2/k).$$

For $n \geq 2$, $T_{n-1}(c) \leq T_n(c)$ because these are moments of an integer-valued nonnegative variable; also $cT_0(c)/T_1(c) = 1$. Thus $a_s \leq C_K k$ everywhere. The fresh tail bound in Lemma 60.2 now shows $\mathbb{E}_p |a_S - \lambda| = O_K(w^2/k)$. Lemma 61.3 proves (61.15), since $\lambda \asymp_K w$.

The fresh tail assertion is part of Lemma 60.2. The Poisson assertion follows from the Chernoff bound, because its mean is $O_K(w)$ and $L_k \asymp_K w^2$. Consequently, for any fixed r, D ,

$$\sum_{s \geq 0} (1 + s)^r |p_{k,c}(s) - \pi_\lambda(s)| \leq C_{K,r} w^{2r} \sum_s |p_{k,c}(s) - \pi_\lambda(s)| + o_K(k^{-D}). \quad (61.17)$$

For $Y \sim \text{Pois}(\lambda)$,

$$\mathbb{E} Y^2 = \lambda^2, \quad \text{Var}(Y^2) = 4\lambda^3 + 2\lambda^2.$$

Use (61.17) with $r = 2$, then Cauchy–Schwarz and (61.15), to transfer the resulting $O_K(w^{3/2})$ centered absolute moment from Y to S . The transfer error is $O_K(w^{11/2}/k) = o_K(w^{3/2})$. Changing the centering from λ^2 to $\mathbb{E}_p S^2$ costs at most the same absolute-moment bound. This proves (61.16). $\square$

61.2 From collisions to a signed law

Proof of (61.3)–(61.4). The complement and weighted rotation identities in Theorem 60.4 allow the maximum-departure statistic to be read from the dominant-root row. The exceptional contribution of two dominant vertices is superpolynomially small. The exact cancellation

$$\frac{k}{k-s} \binom{k-1}{s} = \binom{k}{s}$$

then produces precisely the fresh law (61.13). The order-one local expansion in Lemma 60.5, followed by normalization of its total mass, therefore gives

$$\nu_{k,c}(s) = p_{k,c}(s) \{1 + G_1(k-s, s; c) - A_1(k, c)\} + u_{k,c}(s), \quad (61.18)$$

$$\sum_s |u_{k,c}(s)| = O_K(w^{16}/k^2). \quad (61.19)$$

Initially this identity is used on $s \leq L_k$; when it is extended to all $s \geq 0$, its G_1 correction is set to zero outside that cutoff. All omitted tails can be absorbed into its stated ℓ^1 error. To justify normalization at this accuracy, note that $A_1 = O_K(w^3/k)$ and the local correction is $O_K(w^5/k)$ on the cutoff. The quadratic errors in expanding the normalizing factor are smaller than w^{16}/k^2 .

The exact first coefficient is

$$G_1(n, s; c) = s^2 \left\{ \frac{n-1}{2c(n+1)} \frac{T_{n-1}(c)}{T_n(c)} + \frac{1+c}{c(n+1)} \right\}.$$

For $n = k - s$, the ratio estimate yields

$$G_1(k-s, s; c) = \alpha_* s^2 + O_K \left(\frac{s^2((s+1)w + w^2)}{k^2} \right), \quad \alpha_* := \frac{w+2+2c}{2ck}. \quad (61.20)$$

The fresh factorial moments supplied by Lemma 60.2 imply $\mathbb{E}_p S^r = O_{K,r}(w^r)$ for every fixed r . Averaging the error in (61.20) thus costs $O_K(w^4/k^2)$. Subtracting its average in (61.18) gives

$$\nu_{k,c}(s) = p_{k,c}(s) \{1 + \alpha_*(s^2 - m_2)\} + O_{\ell^1, K}(w^{16}/k^2), \quad m_2 = \mathbb{E}_p S^2.$$

Set $\alpha = w/(2ck) = \epsilon/(2\lambda)$. By (61.16), replacing α_* by α costs only $O_K(w^{3/2}/k)$ in ℓ^1 . This centering is essential: an uncentered bound would lose the precision needed for the recentered constant.

Replace $p_{k,c}$ by π_λ in the resulting expression. The constant term costs $O_K(w^{3/2}/k)$ by (61.15). For the polynomial term, (61.17) and $|m_2 - \lambda^2| \leq \sum_s s^2 |p_{k,c}(s) - \pi_\lambda(s)|$ bound the replacement error by $O_K(w^{13/2}/k^2)$, which is absorbed by w^{16}/k^2 . We have proved

$$\nu_{k,c}(s) = \pi_\lambda(s) \{1 + \alpha(s^2 - \lambda^2)\} + O_{\ell^1, K}(w^{3/2}/k + w^{16}/k^2). \quad (61.21)$$

Finally, $s^2 - \lambda^2 = 2\lambda(s - \lambda) + C_2(s; \lambda)$, and $2\alpha\lambda = \epsilon$. This proves the asserted signed expansion. $\square$

61.3 Poisson Taylor expansion and the sharp constants

For $u > 0$, direct differentiation gives

$$\partial_u \pi_u(s) = \frac{s-u}{u} \pi_u(s), \quad \partial_u^2 \pi_u(s) = \frac{C_2(s; u)}{u^2} \pi_u(s). \quad (61.22)$$

The exact identity $\mathbb{E}[C_2(Y; u)^2] = 2u^2$, $Y \sim \text{Pois}(u)$, implies

$$\sum_s |\partial_u^2 \pi_u(s)| \leq \frac{\sqrt{2}}{u}. \quad (61.23)$$

Consequently, whenever $|v - u| \leq u/2$,

$$\sum_s \left| \pi_v(s) - \pi_u(s) - \frac{v - u}{u}(s - u)\pi_u(s) \right| \leq C \frac{(v - u)^2}{u}. \quad (61.24)$$

Proof of the remaining assertions of Theorem 61.1. Take $u = \lambda$, $v = \theta_0$ in (61.24). Since $\theta_0 - \lambda = \lambda\epsilon$, its error is $O_K(\lambda\epsilon^2) = O_K(w^5/k^2) = o_K(\epsilon)$. The linear term cancels exactly with the linear term of (61.3), proving (61.7).

Let $Y \sim \text{Pois}(\lambda)$ and $Z_\lambda = (Y - \lambda)/\sqrt{\lambda}$. The Poisson central limit theorem gives $Z_\lambda \Rightarrow Z \sim N(0, 1)$, and its moment formulas give the uniform integrability needed below. In particular,

$$\frac{1}{\lambda} \mathbb{E}|C_2(Y; \lambda)| = \mathbb{E} \left| Z_\lambda^2 - 1 - \frac{Z_\lambda}{\sqrt{\lambda}} \right| \rightarrow \mathbb{E}|Z^2 - 1| = 4\varphi(1).$$

One may check uniform integrability directly from $\mathbb{E}(C_2/\lambda)^2 = 2$. For the last equality, use $\mathbb{E}(Z^2 - 1) = 0$ and $\int_1^\infty (z^2 - 1)\varphi(z) dz = \varphi(1)$. Taking half the ℓ^1 norm in (61.7) proves (61.6).

For the uncentered approximation, the quadratic term in (61.3) has ℓ^1 norm $O_K(\epsilon)$, whereas the linear term has norm $\epsilon \mathbb{E}|Y - \lambda|$. The identity $\mathbb{E}|Y - \lambda| = 2\lambda\pi_\lambda(\lfloor \lambda \rfloor)$ and Stirling's formula yield $\mathbb{E}|Y - \lambda| \sim \sqrt{2\lambda/\pi}$. Taking half the norm therefore proves (61.5). All limiting statements are uniform on K , because $\inf_{c \in K} \lambda \rightarrow \infty$. $\square$

61.4 Why no other Poisson mean improves the leading error

Lemma 61.5 (Separation of Poisson means). *There are absolute constants $c_0 > 0$ and $u_0 < \infty$ such that, for $u \geq u_0$ and every $v \geq 0$,*

$$d_{\text{TV}}(\text{Pois}(u), \text{Pois}(v)) \geq c_0 \min \left\{ 1, \frac{|v - u|}{\sqrt{u}} \right\}. \quad (61.25)$$

Proof. Put $m = \lfloor u \rfloor$ and $Q_m(v) = \mathbb{P}(\text{Pois}(v) \leq m)$. Differentiating the finite sum shows that $Q'_m(v) = -\pi_v(m)$ for $v > 0$. Stirling's formula, uniformly for $|v - u| \leq \sqrt{u}$, gives $\pi_v(m) \geq c_0/\sqrt{u}$ after increasing u_0 if necessary. Integration proves (61.25) in this interval. Outside the interval, the monotonicity of Q_m retains at least the difference accumulated between u and the nearer endpoint. The event $\{0, \dots, m\}$ therefore proves the bound for every $v \geq 0$. $\square$

Proof of Theorem 61.2. For $\theta = \theta_0 + t\sqrt{\lambda}\epsilon$ with $|t| \leq T$, (61.24) and (61.3) give

$$\nu_{k,c}(s) - \pi_\theta(s) = \epsilon\pi_\lambda(s) \left\{ \frac{C_2(s; \lambda)}{2\lambda} - t \frac{s - \lambda}{\sqrt{\lambda}} \right\} + o_{\ell^1, K, T}(\epsilon). \quad (61.26)$$

Indeed, the Taylor error is $O_{K,T}(\lambda\epsilon^2) = o_K(\epsilon)$. The Poisson central limit theorem and the second-moment bounds used above identify half the limiting norm as $F(t)$. Uniformity for t in compact intervals follows either by the same uniform integrability argument or by the Lipschitz estimate

$$|\mathbb{E}|Q_\lambda - tZ_\lambda| - \mathbb{E}|Q_\lambda - t'Z_\lambda|| \leq |t - t'| \mathbb{E}|Z_\lambda| \leq |t - t'|, \quad Q_\lambda = C_2(Y; \lambda)/(2\lambda).$$

This proves (61.9).

Evenness and convexity of F follow from symmetry and convexity of the absolute value. More explicitly, symmetry of Z gives

$$F(t) = \frac{1}{2} \mathbb{E} \max \left\{ \frac{|Z^2 - 1|}{2}, |tZ| \right\}.$$

Thus $F(t) \geq F(0)$, and the inequality is strict whenever $t \neq 0$: a neighborhood of $Z = 1$ has positive probability and satisfies $|tZ| > |Z^2 - 1|/2$. To obtain (61.10), integrate $z^2 - 1 - 2tz$ between its two roots $u_-(t)$ and $u_+(t)$, using its zero Gaussian expectation. The antiderivatives

$$\int (z^2 - 1)\varphi(z) dz = -z\varphi(z), \quad \int z\varphi(z) dz = -\varphi(z)$$

give the displayed formula.

It remains to justify the infimum over all means, including means outside the scale in (61.9). Theorem 61.1 gives a candidate with distance $O_K(\epsilon)$. If any other mean θ has distance at most $C\epsilon$, the triangle inequality implies

$$d_{\text{TV}}(\text{Pois}(\theta), \text{Pois}(\theta_0)) = O_K(\epsilon).$$

Apply Lemma 61.5 with $u = \theta_0 \sim_K \lambda$. Since $\epsilon \rightarrow 0$, it follows that

$$|\theta - \theta_0| = O_K(\sqrt{\lambda}\epsilon). \quad (61.27)$$

Thus every competitive mean lies on the bounded- t scale already covered by (61.9); unbounded scaled shifts and more distant means cannot improve the error.

Choose means within $o(\epsilon)$ of the infimum. By (61.27), their scaled shifts have convergent subsequences. Every subsequential limit t has limiting normalized distance $F(t) \geq F(0) = \varphi(1)$. The choice $\theta = \theta_0$ attains this lower bound, proving (61.11). The same compactness argument and the unique minimizer of F show that asymptotic optimality forces $t \rightarrow 0$; conversely, $t \rightarrow 0$ implies asymptotic optimality by the uniform profile limit. This proves (61.12) and completes the proof. $\square$

The distributional results use only the first collision correction and an error bound for the remaining collisions. The optimality statement also explains the limit of changing a single Poisson parameter: it can cancel the linear correction, while the quadratic correction has a nonzero, explicitly evaluated total variation norm.

Corollary 61.6 (A positive approximation with jumps of size two). *Let $Y \sim \text{Pois}(\lambda)$ and $J \sim \text{Pois}(\lambda\epsilon/2)$ be independent auxiliary variables, and let $Q_{k,c}$ be the law of $Y + 2J$. Uniformly for $c \in K$,*

$$Q_{k,c}(s) = \pi_\lambda(s) \left\{ 1 + \frac{\epsilon}{2\lambda}(s^2 - \lambda^2) \right\} + O_{\ell^1, K}((\lambda\epsilon)^2), \quad (61.28)$$

$$d_{\text{TV}}(\nu_{k,c}, Q_{k,c}) = O_K \left(\frac{w^{3/2}}{k} + \frac{w^{16}}{k^2} \right) = o_K(\epsilon). \quad (61.29)$$

The approximating variable has mean θ_0 and variance $\lambda + 2\lambda\epsilon$.

Proof. Put $r = \lambda\epsilon/2$ and interpret $\pi_\lambda(s-2)$ as zero for $s < 2$. Conditioning on $J = 0, 1$, and using $\mathbb{P}(J \geq 2) \leq r^2/2$, gives

$$\|Q_{k,c} - (1-r)\pi_\lambda - r\pi_\lambda(\cdot - 2)\|_1 \leq |e^{-r} - 1 + r| + r|e^{-r} - 1| + \mathbb{P}(J \geq 2) \leq 2r^2.$$

Since $\pi_\lambda(s-2) = s^2\pi_\lambda(s)/\lambda^2$, this proves (61.28). Comparing with (61.21) proves (61.29), because $(\lambda\epsilon)^2 = O_K(w^6/k^2)$. Independence of the auxiliary variables gives the stated mean $\lambda + 2r$ and variance $\lambda + 4r$. $\square$

62 Growing mean degree: a Catalan transition

The fixed-parameter equivalent has a nontrivial transition when the mean degree grows. This section proves that transition directly, including the contribution of walks of arbitrary depth. Throughout, $m_k(c)$ is the moment of order $2k$, so that k is the half-length of a walk.

Theorem 62.1 (Critical growing-degree asymptotics). *Fix $0 < a < b < \infty$. For each k , let $c > 0$ and set*

$$w = W(k/c), \quad \lambda = cw, \quad u = \frac{\lambda}{\sqrt{k}}.$$

Uniformly over all c for which $a \leq u \leq b$,

$$\frac{m_k(c)}{2H_k(c)} = \exp\left(\frac{c^2 w^2}{k}\right) \left\{ 1 + O_{a,b}\left(\frac{\log^{13/4} k}{k^{1/4}}\right) \right\}. \quad (62.1)$$

Consequently, if $c_k W(k/c_k)/\sqrt{k} \rightarrow u_0 \in (0, \infty)$, then

$$\frac{m_k(c_k)}{2H_k(c_k)} \rightarrow e^{u_0^2}.$$

In particular, for every fixed $\beta > 0$,

$$c_k = \frac{\beta \sqrt{k}}{\log k} \implies \frac{m_k(c_k)}{2H_k(c_k)} \rightarrow e^{\beta^2/4}. \quad (62.2)$$

Here and below an implicit constant can depend on a, b . The identity $we^w = k/c$ gives the useful exact relations

$$w = \log(k/\lambda) = \frac{1}{2} \log k - \log u, \quad c = \frac{u\sqrt{k}}{w}. \quad (62.3)$$

Thus $w \asymp \log k$ and $c \asymp \sqrt{k}/\log k$ throughout the uniform range. The proof uses

$$S = \left\lceil \sqrt{k} \log^{1/10} k \right\rceil. \quad (62.4)$$

The error in (62.1) is deliberately conservative. Its purpose is to provide a proved vanishing bound in the critical regime; it is not asserted to be the first neglected asymptotic term.

62.1 Uniform Poisson estimates

We record the uniform estimates needed below, including their elementary justification. Write $w_j = W(j/c)$. For $k^{2/3} \leq j \leq k+1$, $w_j \asymp \log k$. Uniformly in this range,

$$\log T_j(c) = j \log(j/w_j) - j + j/w_j - c + O(\log k), \quad (62.5)$$

$$\frac{T_{j+1}(c)}{T_j(c)} = \frac{j}{w_j} \{1 + O(j^{-1/2})\}, \quad \frac{T_{j-1}(c)}{T_j(c)} = \frac{w_j}{j} \{1 + O(j^{-1/2})\}. \quad (62.6)$$

To see this, use the positive Dobinski summands

$$a_r = e^{-c} \frac{c^r r^j}{r!}, \quad \log \frac{a_{r+1}}{a_r} = \log c - \log(r+1) + j \log(1 + 1/r).$$

Put $y = j/w_j$. The logarithmic ratio at $r = y$ is $O(w_j^2/j)$. Its derivative, for real $r > 0$, is

$$-\frac{1}{r+1} - \frac{j}{r(r+1)}.$$

On $[y/2, 2y]$ its magnitude is bounded above and below by positive constant multiples of w_j^2/j . The mode is therefore within $O(1)$ of y , and summing the ratios gives Gaussian bounds on that interval, with scale $\sqrt{j}/w_j$. Outside the interval the monotonicity of the ratio gives geometric tails after an $\exp\{-\Omega(j)\}$ loss. In particular, under the r^j -tilted Poisson law,

$$\mathbb{E}|r - y| = O(\sqrt{j}/w_j), \quad \mathbb{E}r = y\{1 + O(j^{-1/2})\}, \quad \mathbb{E}(1/r) = y^{-1}\{1 + O(j^{-1/2})\}.$$

The inverse is used only on $r \geq 1$, since the tilted mass at zero vanishes. These estimates prove (62.6). For (62.5), a single summand with r an integer nearest to y gives the lower bound by Stirling's formula. The positive-radius coefficient bound for $\exp(c(e^z - 1))$ at $z = w_j$ gives the upper bound. Their logarithms differ by $O(\log k)$.

The same ratio argument applies to the shifted tilt defining $H_k(c)$. It will be convenient to state its consequences as a lemma.

Lemma 62.2 (Critical fresh deficit law). *The masses*

$$p_k(s) = \frac{\binom{k}{s} c^s T_{k-s}(c)}{H_k(c)}, \quad 0 \leq s \leq k, \quad (62.7)$$

sum to one. If Z has this distribution, then

$$\mathbb{E}|Z - \lambda| = O(k^{1/4}). \quad (62.8)$$

There are constants $K, \eta > 0$, depending only on a, b , such that

$$\mathbb{P}(Z > K\sqrt{k}) \leq e^{-\eta\sqrt{k}} \quad (62.9)$$

for all sufficiently large k .

Proof. Tilt $X \sim \text{Pois}(c)$ by $(X + c)^k/H_k(c)$ and, conditionally on X , take $Z \sim \text{Bin}(k, c/(X + c))$. Expanding the binomial gives (62.7) exactly. Put $x_0 = k/w$. The tilted summands have ratio

$$\frac{c}{r+1} \left(1 + \frac{1}{r+c}\right)^k.$$

The derivative of its logarithm is

$$-\frac{1}{r+1} - \frac{k}{(r+c)(r+c+1)}.$$

At $r = x_0$ the logarithm is $O(cw^2/k)$; on $[x_0/2, 2x_0]$ the derivative has magnitude comparable to w^2/k . The mode is within $O(c+1)$ of x_0 . The same summation of ratios as above, with (62.3), gives

$$\mathbb{E}|X - x_0| = O(\sqrt{k}/w), \quad \mathbb{P}(X < x_0/2) \leq e^{-\eta_0 k} \quad (62.10)$$

for some $\eta_0 > 0$. This argument also controls the corresponding first moments on the exceptional tail.

Let $\Lambda_X = kc/(X + c)$, the conditional binomial mean. On $X \geq x_0/2$,

$$|\Lambda_X - \lambda| \leq \frac{Ccw^2}{k}(|X - x_0| + c).$$

Equations (62.3) and (62.10), together with $0 \leq \Lambda_X \leq k$ on the exceptional event, show that $\mathbb{E}|\Lambda_X - \lambda| = O(1)$ and $\mathbb{E}\Lambda_X = O(\sqrt{k})$. Conditional variance and Cauchy–Schwarz now give

$$\mathbb{E}|Z - \lambda| \leq \sqrt{\mathbb{E}\Lambda_X} + \mathbb{E}|\Lambda_X - \lambda| = O(k^{1/4}).$$

On $X \geq x_0/2$, the conditional mean is at most $2b\sqrt{k}$. Choose, for example, $K > 8b + 1$ and apply the binomial Chernoff bound. Combining it with the exceptional probability in (62.10) proves (62.9). $\square$

62.2 Plane forests and an improved allocation bound

Let

$$C(z) = \frac{1 - \sqrt{1 - 4z}}{2z}, \quad f_q(t) = [z^t]C(z)^q = \frac{q}{q + 2t} \binom{q + 2t}{t} \quad (q \geq 1, t \geq 0).$$

The expression at $t = 0$ is one. A segmented child walk with q segments and t interior edge pairs has one of these $f_q(t)$ root-distance paths. Requiring every interior edge to be new gives exactly $c^t f_q(t)$ weighted walks. More generally, at the i th outward step there are at most $i - 1$ previously discovered child choices, and a new child has weight c . Consequently

$$c^t f_q(t) \leq E_{t,q}(c) \leq f_q(t)c^{\bar{t}} = c^t f_q(t)v_t, \quad v_t = \frac{c^{\bar{t}}}{c^t}. \quad (62.11)$$

The weighted choice bound is applied to each prescribed root-distance path. It counts a new edge once, at its first outward traversal; inward steps are forced. The q ordered closed segments are already part of the path, so no additional labeling or cutting factor occurs.

For a partition π of $[n]$, let q_b denote its block sizes and $A = \max_b q_b$. The exact partition decomposition and (62.11) imply that the root count contributed by this partition lies between

$$c^{|\pi|+s} f_n(s) \quad \text{and} \quad c^{|\pi|+s} \sum_{\sum t_b = s} \prod_b f_{q_b}(t_b) v_{t_b}.$$

The first expression follows from $\prod_b C(z)^{q_b} = C(z)^n$. In particular the exact weighted count with all off-root edges simple is

$$F_{n+s,n}^{\text{pl}}(c) = T_n(c) c^s f_n(s). \quad (62.12)$$

The following bound retains the factor $1/c$ in the cost of reusing an interior edge. That factor is essential when c grows.

Lemma 62.3 (Forest allocation bound). *For every partition of $[n]$, every $s \geq 0$, and every $c > 0$,*

$$\frac{\sum_{\sum t_b = s} \prod_b f_{q_b}(t_b) v_{t_b}}{f_n(s)} \leq \exp \left\{ \frac{s(s-1)}{2n} + \frac{As(s-1)}{2cn} + \frac{4s^3}{3cn} \right\}. \quad (62.13)$$

Proof. An exact coefficient ratio is

$$a(q, t) := \frac{(t+1)f_q(t+1)}{f_q(t)} = q + 3t + \frac{t(t-1)}{q+t+1}. \quad (62.14)$$

In particular $q + 3t \leq a(q, t) \leq q + 4t$. Expand $s! \sum_{\sum t_b = s} \prod_b f_{q_b}(t_b) v_{t_b}$ as a sum over all ordered sequences of s selections of blocks. When a block already selected t times is selected again, its multiplicative factor is $a(q, t)(1 + t/c)$. This expansion is exact: summing over the $s! / \prod_b t_b!$ orders produces the desired factorial factors. After j selections the sum of all possible next factors is at most

$$\sum_b (q_b + 4t_b)(1 + t_b/c) \leq n + 4j + \frac{Aj + 4j^2}{c}.$$

On the other hand, $s! f_n(s) = \prod_{j=0}^{s-1} a(n, j) \geq \prod_{j=0}^{s-1} (n + 3j)$. Taking the quotient of these product bounds and using $\log(1+x) \leq x$ proves

$$\log \frac{\sum \prod_b f_{q_b}(t_b) v_{t_b}}{f_n(s)} \leq \sum_{j < s} \frac{j + Aj/c + 4j^2/c}{n + 3j},$$

which implies (62.13). □

Give partitions of $[n]$ probability $c^{|\pi|}/T_n(c)$, and denote expectations by $\mathbb{E}_{n,c}$. For $n \geq k/2$, the adjacent ratio bound (62.6) gives

$$\mathbb{E}_{n,c} \# \{b : |b| = q\} = c \binom{n}{q} \frac{T_{n-q}(c)}{T_n(c)} \leq c \frac{(C_0 \log k)^q}{q!}, \quad q \leq n/2. \quad (62.15)$$

For $q > n/2$, retain the first $\lfloor n/2 \rfloor$ adjacent ratios. The remaining ratios have product at most one: $T_j(c)$ is increasing in j when $c \geq 1$. It follows that, uniformly for $0 \leq \theta \leq 0.01 \log k$,

$$\log \mathbb{E}_{n,c} e^{\theta A} \leq C_1 e^\theta \log k + C_1 \log k. \quad (62.16)$$

Indeed, $e^{\theta A} \leq 1 + \sum_b e^{\theta |b|}$, and the part with $q > n/2$ is at most $c(1 + e^\theta)^n (C_0 \log k/k)^{\lfloor n/2 \rfloor} = \exp\{-\Omega(k \log k)\}$. The same estimates show that, for $B = \lceil \log^2 k \rceil$ and $\theta = o(1)$,

$$\mathbb{E}_{n,c} [e^{\theta A}; A > B] \leq \exp\{-\Omega(\log^2 k \log \log k)\}. \quad (62.17)$$

For the latter estimate use $e^{\theta A} \mathbf{1}_{A > B} \leq \sum_{|b| > B} e^{\theta |b|}$ in (62.15). Thus these are weighted tail estimates, not only estimates of the probability of a large block.

62.3 The entire near-hub window is asymptotically planar

Lemma 62.4 (Uniform near-hub estimate). *For $0 \leq s \leq S$ and $n = k - s$,*

$$F_{k,n}(c) = T_n(c) c^s f_n(s) \left\{ 1 + O\left(k^{-1/4} \log^{13/4} k\right) \right\}, \quad (62.18)$$

uniformly in the critical parameter range.

Proof. The lower bound is (62.12). First fix a partition with $A \leq B = \lceil \log^2 k \rceil$. Put $\alpha = 1 + s/c$. For $t \leq s$, telescoping the product defining v_t gives

$$0 \leq v_t - 1 \leq \frac{t(t-1)}{2c} \alpha^{t-2}. \quad (62.19)$$

For example, one expands the difference of the product and one by changing the factors one at a time, and bounds every unchanged nonunit factor by α . Their linear increments sum to $t(t-1)/(2c)$.

Let

$$g_q(z) = C(z)^q + \sum_{t=2}^s f_q(t)(v_t - 1)z^t.$$

Up to the coefficient of z^s , this majorizes the normalized branch series in (62.11). It also majorizes $C(z)^q$ coefficientwise. For positive z ,

$$0 \leq g_q(z) - C(z)^q \leq \frac{z^2}{2c} \frac{d^2}{dy^2} C(y)^q \Big|_{y=\alpha z}.$$

For $s \geq 1$ use the exact Catalan saddle

$$z_* = \frac{s(n+s)}{(n+2s)^2}, \quad C(z_*) = \frac{n+2s}{n+s}. \quad (62.20)$$

Here $z_* \asymp s/n$, $\alpha z_* = o(1)$, and $B\alpha z_* = o(1)$ uniformly for $s \leq S$. The first two derivatives of $\log C(y)$ are bounded for $0 \leq y \leq 1/8$, so

$$\frac{g_q(z_*)}{C(z_*)^q} \leq 1 + \frac{C_2 q(q+1)z_*^2}{c}, \quad q \leq B.$$

Since $\sum_b q_b(q_b + 1) \leq (B+1)n$, coefficient positivity yields

$$[z^s] \left(\prod_b g_{q_b}(z) - C(z)^n \right) \leq z_*^{-s} C(z_*)^n (e^\Delta - 1), \quad \Delta = O\left(\frac{Bs^2}{nc}\right). \quad (62.21)$$

There is a uniform constant $C_3 > 0$ such that

$$f_n(s) z_*^s C(z_*)^{-n} \geq \frac{C_3}{\sqrt{s+1}}. \quad (62.22)$$

For completeness, the left side is exactly

$$\frac{n}{n+2s} \binom{n+2s}{s} \left(\frac{s}{n+2s} \right)^s \left(\frac{n+s}{n+2s} \right)^{n+s}.$$

This is $n/(n+2s)$ times a binomial probability at its mean, and Stirling's upper and lower factorial bounds prove (62.22). It holds uniformly since $s = o(n)$. The cases $s = 0, 1$ are exact and can be handled separately.

Dividing (62.21) by $f_n(s)$ gives, for every good partition, a relative error at most

$$O\left(\sqrt{s+1} \frac{Bs^2}{nc}\right) = O\left(k^{-1/4} \log^{13/4} k\right).$$

The exponent $13/4 = 2 + 1 + \frac{5}{2} \cdot \frac{1}{10}$ comes from B , the factor $1/c$, and the powers of s .

For $A > B$, apply Lemma 62.3 before averaging over partitions. In this window,

$$\theta = \frac{s(s-1)}{2cn} = O(k^{-1/2} \log^{6/5} k) = o(1), \quad \frac{s(s-1)}{2n} + \frac{4s^3}{3cn} = O(\log^{13/10} k).$$

Equation (62.17) therefore makes the entire bad-partition contribution smaller than every fixed negative power of k , relative to $T_n(c)c^s f_n(s)$. Combining the two parts proves (62.18). $\square$

62.4 Control of the full complement

The near-hub estimate is not by itself a full-moment theorem. We next prove the uniform estimate that permits its use for $m_k(c)$.

Lemma 62.5 (Critical complement). *Let $\mathcal{U}_{k,S}$ be the walks with a vertex having at least $k - S$ departures, and let the weight of a set of walks be the sum of $c^{\#edges}$ over it. For every fixed $D > 0$,*

$$\text{wt}_c(\mathcal{W}_k \setminus \mathcal{U}_{k,S}) = o(k^{-D} H_k(c)), \quad (62.23)$$

uniformly in the critical range.

Proof. Set

$$h = \left\lfloor \frac{k}{64 \log k} \right\rfloor, \quad L = \frac{1}{100} \sqrt{ck \log k}. \quad (62.24)$$

Then $S \ll L \asymp k^{3/4} \ll h$, and $c = o(h)$.

The high-root shoulder. Let $S \leq s \leq L$ and $n = k - s$. The exact quotient

$$Q(k, s) := \frac{f_n(s)}{\binom{k-1}{s}} = \prod_{j=1}^{s-1} \left(1 + \frac{s}{n+j} \right) = \prod_{j=1}^{s-1} \frac{k+j}{k-j} \quad (62.25)$$

has the empty-product value one for $s = 0, 1$, and $\log Q(k, s) \leq s(s-1)/(n+1)$. The partition decomposition, Lemma 62.3, and (62.6) consequently give

$$\log \frac{F_{k,k-s}(c)}{T_k(c)} \leq -s \log \frac{s}{cw} + O(s) + \frac{3s^2}{2n} + \frac{4s^3}{3cn} + \log \mathbb{E}_{n,c} e^{\theta A}, \quad (62.26)$$

$$\theta = \frac{s(s-1)}{2cn}. \quad (62.27)$$

Here the fresh term was bounded using $\binom{k-1}{s} c^s T_{k-s}(c) / T_k(c) \leq (Ccw/s)^s$. Put $J(s) = s \log(s/(cw))$. Uniformly over this shoulder, $O(s) = o(J)$ because $S/(cw) \rightarrow \infty$, and $s^2/n = o(J)$. The function $s^2/\log(s/(cw))$ is increasing once $\log(s/(cw)) > 1/2$. At the upper endpoint,

$$\frac{\frac{4}{3} L^3 / (c(k-L))}{J(L)} \longrightarrow \frac{16}{3 \cdot 100^2} < \frac{1}{100}.$$

Thus the cubic term in (62.26) is at most $J/100$ eventually. Also $\theta \leq 0.001 \log k$ eventually, so (62.16) gives $\log \mathbb{E} e^{\theta A} = O(k^{0.001} \log k) = o(J)$. We obtain

$$\frac{F_{k,k-s}(c)}{H_k(c)} \leq e^{-J(s)/2}, \quad S \leq s \leq L, \quad (62.28)$$

using $H_k(c) \geq T_k(c)$. The sum of these bounds, even after a polynomial factor, is smaller than k^{-D} for every fixed D .

The intermediate return range. Now let $m + s = k$, $m \geq h$, and $s \geq L$. Define

$$\text{Ent}(m, s) = k \log k - m \log m - s \log s.$$

We shall prove

$$\frac{F_{k,m}(c)}{H_k(c)} \leq \exp\{-\text{Ent}(m, s)/2\}. \quad (62.29)$$

The coefficientwise Bauer–Golinelli bound [1], applied to the positive edge weights, gives

$$T_t(c) \leq m_t(c) \leq T_t(4c).$$

The same positive-radius estimate used in (62.5) gives, for $t \geq L$,

$$m_t(c) \leq T_t(c) \exp\{3t/W(t/c) + O(\log k)\}. \quad (62.30)$$

For smaller indices we use the direct path bound

$$m_t(c) \leq \text{Cat}_t c^{\bar{t}} \leq 4^{t-1} c^{\bar{t}} \quad (t \geq 1), \quad m_1(c) = c. \quad (62.31)$$

Finally, the moment identification and Hölder's inequality give $m_t(c) \leq m_s(c)^{t/s}$ for $0 \leq t \leq s$. Each assertion concerns finitely many moments at each selected value of c ; no graph-size limit is interchanged with $k \rightarrow \infty$.

For a fixed target s , truncate each branch to $t \leq s$. The labeled-partition formula and positivity give, for $x, z > 0$,

$$F_{m+s,m}(c) \leq m! x^{-m} z^{-s} \exp\left\{c(e^x - 1) + c \sum_{t=1}^s m_t(c) z^t \mathcal{A}_t(x)\right\}, \quad (62.32)$$

where

$$\mathcal{A}_t(x) = e^x \sum_{j=1}^t \binom{t-1}{j-1} \frac{x^j}{j!} \leq e^x x(1+x)^{t-1}, \quad \mathcal{A}_t(x) \leq e^x e^{2\sqrt{tx}}.$$

These are finite coefficient majorants. In particular, evaluating (62.32) does not assume convergence of the ordinary moment series.

Choose

$$x = W(m/c), \quad \epsilon = \frac{m}{8k}, \quad \eta = 1 - \epsilon, \quad z = \eta m_s(c)^{-1/s}, \quad T = \left\lceil \frac{64 \log k}{\epsilon^2} \right\rceil.$$

Then $T = O(\log^3 k) = o(c)$, $T < s$, and $z \leq CW(s/c)/s \leq C \log k/s$. For $t > T$, Hölder's inequality and the second bound for $\mathcal{A}_t$ imply

$$c \sum_{t>T}^s m_t(c) z^t \mathcal{A}_t(x) \leq c e^x \sum_{t>T} e^{-\epsilon t/2} = O(k^{-10}).$$

For $t \leq T$, (62.31) gives $m_t(c) \leq 4^{t-1} c(c+T)^{t-1}$. Since $c x e^x = m$,

$$c \sum_{t=1}^T m_t(c) z^t \mathcal{A}_t(x) \leq m c z \sum_{t=1}^T \{4(c+T)(1+x)z\}^{t-1} = m c z \{1 + o(1)\}.$$

The last equality is uniform: its geometric ratio is at most $C c \log^2 k/L = O(k^{-1/4} \log k)$. Comparing the x -part of the Cauchy bound with $T_m(c)$ by (62.5) yields

$$\log \frac{F_{k,m}(c)}{T_k(c)} \leq \log \frac{T_m(c) m_s(c)}{T_k(c)} - s \log \eta + O(\log k) + m c z \{1 + o(1)\}. \quad (62.33)$$

For clarity, the required cancellation in the Touchard terms is uniform:

$$\log \frac{T_m(c)T_s(c)}{T_k(c)} = -\{1 + o(1)\} \text{Ent}(m, s). \quad (62.34)$$

Here is a direct verification. Put $p = \min(m, s)$ and $q = k - p$. Use (62.5) for $T_p(c)$ and sum the adjacent logarithmic ratios (62.6) only from q to $k - 1$. This gives

$$\log \frac{T_p(c)T_q(c)}{T_k(c)} = -\text{Ent}(m, s) + p \log \frac{w_k}{w_p} + O(p/w_p + c + \log k) + o(p).$$

The derivative of $\log W(t/c)$ with respect to $\log t$ is $1/(1 + W(t/c))$, so

$$0 \leq p \log(w_k/w_p) \leq \frac{p \log(k/p)}{1 + w_p} \leq \frac{\text{Ent}(m, s)}{1 + w_p}.$$

In this range $p \geq L \asymp k^{3/4}$, $w_p \asymp \log k$, and $c = o(p)$. The remaining errors are $o(\text{Ent})$, proving (62.34).

The envelope loss $3s/W(s/c) + O(\log k)$ in (62.30) is also $o(\text{Ent})$. For $s \leq k/2$ use $\text{Ent} \geq s \log 2$. For $s > k/2$, use $m \geq h$ and $\text{Ent} \geq m \log(k/m) \geq Ck \log \log k / \log k$. Moreover

$$-s \log \eta \leq \frac{ms}{7k} \leq \frac{\text{Ent}(m, s)}{14}.$$

Finally, $mcz = o(\text{Ent})$ uniformly. For $L \leq s \leq k^{7/8}$ use

$$\frac{mcz}{\text{Ent}} \leq \frac{Cck \log k}{s^2 \log(k/s)} \leq \frac{Cck}{L^2} = O(1/\log k).$$

For $k^{7/8} \leq s \leq k/2$, the bound $Cck \log k/s^2 = O(k^{-1/4})$ suffices. For $s > k/2$, use $mcz/\text{Ent} \leq Cc \log k/k = o(1)$. Thus (62.33) is at most $-(13/14 - o(1)) \text{Ent}$, proving (62.29). Since there are at most k such entries, their sum with rotation weights $2k/m \leq 2k/h$ is superpolynomially small relative to $H_k(c)$.

The maximum-departure split. If every vertex of the walk has at most h departures, then at each outward step the total weight of old-child choices and the possible new child is at most $h + c$. There are Cat_k root-distance paths, so the weight of this class is at most $[4(h + c)]^k$. Equations (62.5) and (62.3) show

$$\frac{[4(h + c)]^k}{T_k(c)} \leq \exp\{-(\log 32 - 1 + o(1))k\}.$$

Every remaining walk outside $\mathcal{U}_{k,S}$ has a vertex with departure count between h and $k - S$. The weighted rotation identity covers its weight by

$$\sum_{h < m < k-S} \frac{2k}{m} F_{k,m}(c).$$

The shoulder (62.28) and the intermediate bound (62.29) cover this entire interval. Together with the low-maximum bound they prove (62.23). $\square$

62.5 Completion of the transition theorem

Proof of Theorem 62.1. For large k , $k - S > k/2$. A walk has at most two vertices with at least $k - S$ departures; if two exist, they are adjacent, and their joining edge has multiplicity at least $k - 2S$. Deleting its traversals leaves two segmented child walks of total half-length at most $2S$. The usual root-distance bound and an overcount of the possible origins therefore bound the weight of this double-hub class by

$$2k(2S + 1)^2(4k)^{2S}c^{2S+1} = \exp\{O(S \log k)\}.$$

Here the factor c^{2S+1} is valid because $c \geq 1$ and the entire walk uses at most $2S + 1$ distinct edges. This bound is $o(k^{-D}H_k(c))$ for every fixed D . The weighted rotation identity and Lemma 62.5 imply

$$\frac{m_k(c)}{2H_k(c)} = \frac{1}{H_k(c)} \sum_{s=0}^S \frac{k}{k-s} F_{k,k-s}(c) + o(k^{-D}). \quad (62.35)$$

Rotation does not change the edge weight, so this identity has precisely the same indexed-orbit justification as in the fixed-parameter case.

Insert Lemma 62.4 and use the exact cancellation

$$\frac{k}{k-s} \binom{k-1}{s} = \binom{k}{s}.$$

With p_k from (62.7) and Q from (62.25), this gives

$$\frac{m_k(c)}{2H_k(c)} = \left\{1 + O(k^{-1/4} \log^{13/4} k)\right\} \sum_{s=0}^S p_k(s) Q(k, s) + o(k^{-D}). \quad (62.36)$$

For $s \leq S$, the symmetric final product in (62.25) gives

$$\log Q(k, s) = \frac{s(s-1)}{k} + \frac{s^2(s-1)^2}{6k^3} + O(s^6/k^5). \quad (62.37)$$

Indeed, expand $\log((1+x)/(1-x)) = 2 \sum_{r \geq 0} x^{2r+1}/(2r+1)$ and use the sums of the first and third powers of $1, \dots, s-1$; the remaining series is absolutely bounded by $O(s^6/k^5)$. For $s \leq K\sqrt{k}$, replacing this expression by s^2/k costs $O(k^{-1/2})$, and $x \mapsto \exp(x^2/k)$ has derivative $O(k^{-1/2})$ on this interval. Lemma 62.2 therefore implies

$$\sum_{s \leq K\sqrt{k}} p_k(s) Q(k, s) = \exp(\lambda^2/k) + O(k^{-1/4}).$$

For $K\sqrt{k} < s \leq S$, use $Q(k, s) \leq \exp\{O(\log^{1/5} k)\}$ and (62.9); this part is smaller than every negative power of k . Thus the sum in (62.36) is $\exp(\lambda^2/k) + O(k^{-1/4})$ and is bounded above and below by positive constants. Equation (62.1) follows. The limit (62.2) follows from $W(k/c_k)/\log k \rightarrow 1/2$. $\square$

The proof gives an additional structural consequence. Let $\mathcal{V}_{k,S}$ consist of walks with a vertex having at least $k - S$ departures and with every edge not incident to that vertex traversed exactly once in each direction. The fresh count (62.12), the double-hub estimate, and Lemma 62.4 imply

Corollary 62.6 (Simple off-hub edges dominate). *Uniformly in the regime of Theorem 62.1,*

$$\frac{\text{wt}_c(\mathcal{W}_k \setminus \mathcal{V}_{k,S})}{m_k(c)} = O(k^{-1/4} \log^{13/4} k).$$

In particular the entire critical exponential multiplier is obtained from a class with no repeated off-hub edge.

The source of the multiplier is the finite Catalan quotient (62.25). Its logarithm is $s^2/k + o(1)$, while the fresh deficit is $s = cw + O_{L^1}(k^{1/4})$. Hence this quotient tends to $\exp(c^2 w^2/k)$. At the same scale, the fixed-parameter repeated-edge quantity cw^3/k is $O(\log^2 k/\sqrt{k})$ and tends to zero. The growing-degree transition therefore has a precise combinatorial description: fresh planar interiors alter the full moment by a finite factor even while repeated off-hub edges carry a vanishing fraction of its weight.

63 Exact checks, numerical diagnostics, and reproducibility

The accompanying computations have three distinct purposes. Exact enumeration tests the combinatorial identities at finite sizes. Symbolic algebra checks the low collision polynomials and their normalizations. Floating-point diagnostics illustrate the scales in the theorems and expose slow onset. None of these calculations replaces the analytic estimates proved above.

63.1 Independent exact enumeration

The program `code/verify_exact.py` implements Proposition 59.2 using integers for the rational parameters $c = 1/2, 1, 2$. Separately, it enumerates normalized words directly by depth-first search. At every step it either follows an existing tree edge or creates the next fresh vertex. A word is retained only when it closes at the required length. The direct enumerator does not use the recurrence, a partition decomposition, or the Catalan formula.

The recorded run computes all rooted rows through $k = 128$ and directly enumerates all 12,850 words at half-lengths $1 \leq k \leq 7$. It checks:

- 84 weighted root-departure row identities against the recurrence;
- 84 weighted simple-off-root forest identities at the three rational parameters, and 168 coefficientwise identities for that class, using its directly counted edge numbers;
- 21 weighted change-of-root identities, including all periodic words present in the enumerated range;
- 1,515 exact factorial-moment identities for the fresh law;
- the first 13 terms of A094149 against the published prefix.

All checks pass. The coefficientwise forest check compares the coefficient of c^j in the direct count with that of $T_n(c)c^{k-n}[z^{k-n}]C(z)^n$, so it tests the dependence on c as well as the numerical totals. The change-of-root test uses the property satisfied by every vertex, for which the left-hand side counts each walk with its number of vertices. The general identity for arbitrary intrinsic vertex properties is proved in Section 59.

63.2 Symbolic collision polynomials

The program `code/verify_symbolic.py` computes the low rooted rows with an indeterminate c . It forms the finite Newton differences and checks the seven identities for block orders $h = 0, \dots, 6$. The initial nonzero polynomials are

$$\begin{aligned} d_2(q; c) &= \frac{q(q+1)}{2c} + q, \\ d_3(q; c) &= \frac{q(q+1)(q+2)}{6c^2} + \frac{q(q+2)}{c} + q, \\ P_2(z; c) &= \frac{z(z+2+2c)}{2c}, \\ P_3(z; c) &= \frac{z^3}{6c^2} + \left(\frac{1}{c^2} + \frac{1}{c}\right)z^2 + \left(\frac{1}{c^2} + \frac{3}{c} + 1\right)z. \end{aligned}$$

The script also verifies 132 exact instances of the Catalan quotient product identity. The larger polynomials are included in `data/symbolic_checks.json` and a generated TeX fragment. These finite calculations verify signs and coefficients that enter the proof; they do not establish the all-order tail estimate.

63.3 Exact moments and slow asymptotic onset

Table 9 compares the exact rationally computed ratio with the first two logarithmic approximations. Only the final decimal display and evaluations of W and the exponential are floating-point. The moment and shifted-Poisson normalizer in the ratio are computed exactly before conversion.

c	k	$m_k/(2H_k)$	e^{P_1}	$e^{P_1+cw^5/(6k^2)}$
1/2	32	2.118075	1.549589	1.582971
1/2	128	1.313016	1.267645	1.275392
1	32	3.222888	1.927875	1.961030
1	128	1.549910	1.460536	1.469267
2	32	8.299412	2.888992	2.923511
2	128	2.173148	1.925804	1.936093

Table 9: Finite moment diagnostics. Here $P_1 = cw^2(w+2+2c)/(2k)$ and $w = W(k/c)$. The entries show that asymptotic accuracy can still be modest at $k = 128$, especially at larger fixed c . No finite- k error bound is inferred from this table.

The slow approach is mathematically compatible with the theorems. Terms separated only by a factor $1/w$ are separated only logarithmically in k . The hidden constants and the onset threshold in the compact-parameter O estimates are not numerically optimized. In particular, a small asymptotic remainder does not imply a useful relative error at the modest indices for which exact full moments are inexpensive to compute.

63.4 Distributional and critical-window illustrations

For large k , enumerating the complete departure-deficit law is not feasible with the direct code. Figure 1 therefore uses the explicitly labeled first-collision model

$$\hat{\nu}_{k,c}(s) = \frac{p_{k,c}(s)\{1 + G_1(k-s, s; c)\}}{\sum_t p_{k,c}(t)\{1 + G_1(k-t, t; c)\}}, \quad (63.1)$$

evaluated in a concentrated finite window. It is not an exact computation of $\nu_{k,c}$. The exact local coefficient, including its lower logarithmic terms, is retained. The limiting horizontal lines are the constants proved in Section 61; the visible finite values remain noticeably separated from them. This plot illustrates the relevance of logarithmic suborders and the difficulty of using moderate indices to estimate the limiting constant.

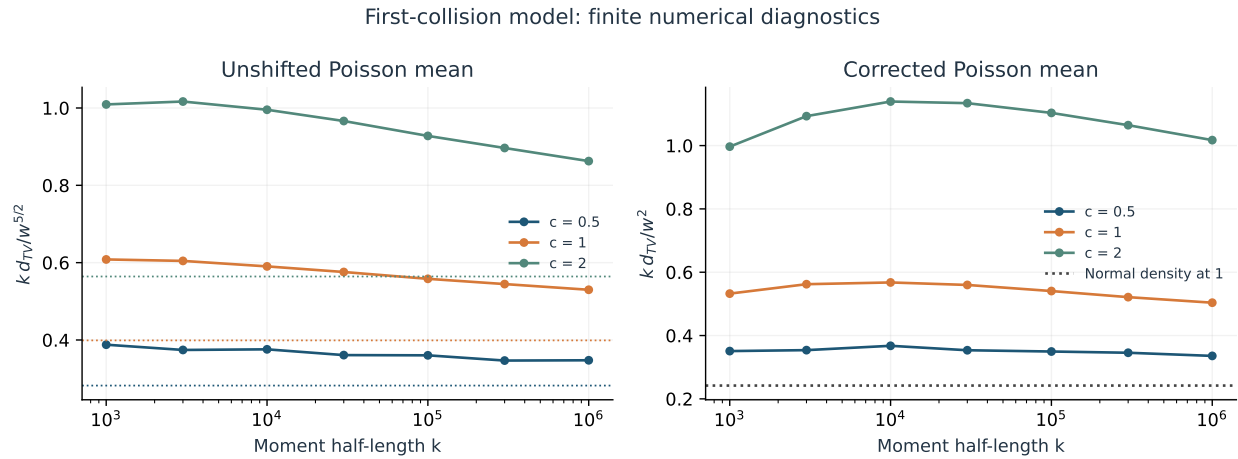


Figure 1: Scaled total variation diagnostics for the normalized first-collision model, at $c = 1/2, 1, 2$. Left: comparison with $\text{Pois}(cw)$, scaled by $k/w^{5/2}$. Right: comparison with $\text{Pois}(cw + cw^3/k)$, scaled by k/w^2 . The full walk law is controlled analytically in the proofs, not simulated by this plot.

For Figure 2, set

$$w = \frac{1}{2} \log(k/\tau), \quad c = \frac{\sqrt{\tau k}}{w}, \quad \tau \in \{1/4, 1/2, 1, 2\}.$$

Then $c^2 w^2 / k = \tau$ holds exactly. The plotted quantity is the fresh-law average of the exact Catalan quotient $Q(k, s)$ in its concentrated window. It is the explicit term identified in the proof of Theorem 62.1, and the horizontal lines are e^τ . The proof, not this numerical average, controls the omitted classes of walks.

The numerical program uses positive Dobiński sums centered at the Touchard saddle, with log-sum-exp arithmetic. Repeating the sums with widths 12 and 15 saddle standard deviations, each with a 20-integer cushion, changes the computed fresh distribution by at most 9.3×10^{-15} in ℓ^1 over the recorded cases. This is a stability check, not an interval-arithmetic certificate of all discarded tails or floating-point errors. The JSON files record all 12 finite-moment rows, 21 deficit-model cases, and 28 critical-window cases, with their scope written into the metadata.

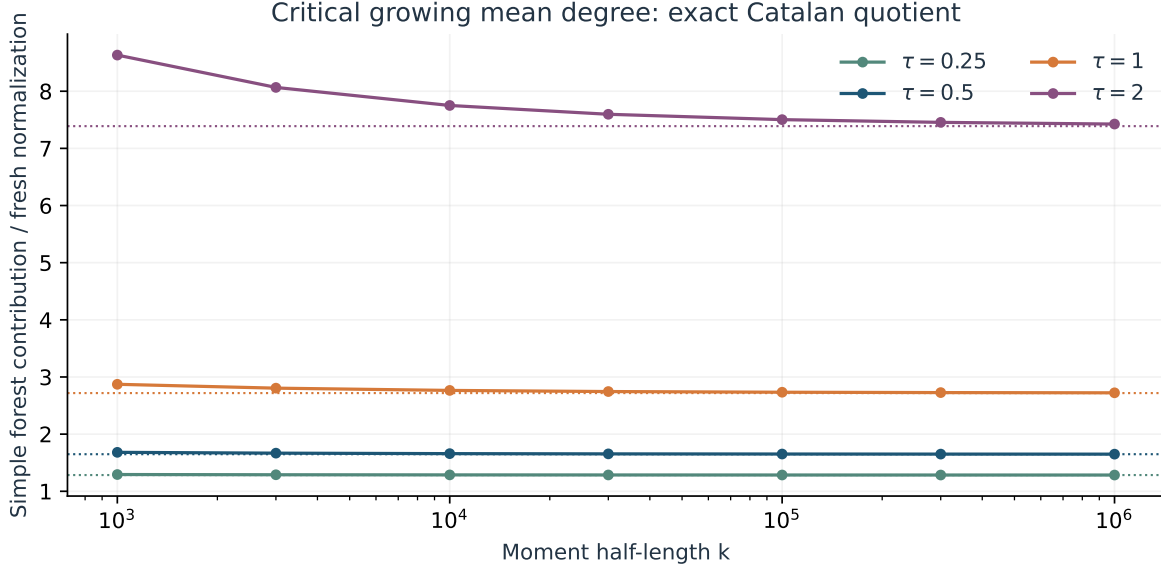


Figure 2: Critical-window diagnostics for the simple-forest contribution. The numerical average of $Q(k, s)$ approaches the predicted factor e^τ . The reported global error bound includes additional contributions that are not evaluated by this figure.

63.5 Reproduction and integration

The ZIP contains the TeX source, the compiled PDF, all input sections, the bibliography, the scripts, raw exact rows, verification summaries, and vector and raster figures. A `README.md` gives the build commands and a suggested report directory in the ProveIt repository. The exact integer checks need only Python’s standard library; symbolic and numerical diagnostics additionally use SymPy, NumPy, SciPy, and Matplotlib. The supplied dependency versions record the environment used for the archived run. No external dataset or network call is needed to rebuild the article or rerun the computations.

The claim ledger distinguishes proved results, inherited ingredients, finite checks, illustrations, and open questions. The provenance file pins the inspected repository revisions, and `source_audit.txt` records the source-access and priority boundaries. Integration can therefore preserve both the mathematical statements and the evidence supporting the scope of the novelty claim. No theorem-prover source is included: the present deliverable is a research article with proofs and reproducibility code.

[write] **The package here, and the write’s recomputation, 9 October 2026.** In this report the programs named above are `code/06-meandeg-verify_exact.py`, `code/06-meandeg-verify_symbolic.py` and `code/06-meandeg-numerical_diagnostics.py`; the data files are `data/06-meandeg-*`, the figures `figures/06-meandeg-*`, the claim ledger, provenance file and verification record `data/06-meandeg-claim_ledger.json`, `data/06-meandeg-provenance.json` and `data/06-meandeg-verification_report.json`, and `source_audit.txt` is `06-meandeg-source_audit.txt`. The ZIP, the compiled PDF, the section sources, the bibliography and the delivered `README.md` are not shipped (Section 57.1). The programs read and write the delivered names (`data/exact_rows.json` and so on), so they run only on a copy with the delivered layout (the `README` gives the recipe). With its own integer-scaled implementation of the first-child recurrence, the write recomputed every scaled moment of `data/06-meandeg-exact_rows.json` at $c = \frac{1}{2}, 1, 2$ for $k \leq 128$ (all equal) and all eighteen entries of Table 9 (all equal to the printed six decimals); with SymPy it recomputed the four displayed polynomials, d_4 and P_4 (equal to the generated fragment

`data/06-meandeg-collision_polynomials.tex`). The two figures plot explicitly labelled models (the first-collision law (63.1) and the fresh-law average of $Q(k, s)$), not the full walk law; they illustrate, and were not recomputed.

64 Further research questions and topics

The results leave several concrete problems. We separate improvements to the present estimates from changes of regime or model; a result in one category need not automatically supply a result in another. The questions below are proposed projects, not conclusions established by the current article.

64.1 Sharper critical asymptotics and a wider parameter range

Research question 64.1 (The first correction in the critical regime). What is the first nonzero asymptotic correction to $\log\{m_k(c)/(2H_k(c))\} - c^2w^2/k$ when $cw/\sqrt{k}$ stays bounded away from zero and infinity?

The present $k^{-1/4} \log^{13/4} k$ bound is produced by a coefficient-extraction penalty and a deliberately broad cutoff. It does not identify the first neglected term. The exact expansion

$$\log Q(k, s) = \frac{s(s-1)}{k} + \frac{s^2(s-1)^2}{6k^3} + O(s^6/k^5)$$

already separates the pure forest correction from repeated off-hub edges. A sharper weighted allocation estimate, averaged over the actual partition and deficit distributions instead of bounded by their largest admissible values, is a natural route. The useful output would be a uniform equivalent with an explicit correction function of $u = cw/\sqrt{k}$.

Research question 64.2 (A uniform passage from bounded to growing mean degree). Can one obtain a single controlled expansion that joins compact c to the critical regime $cw \asymp \sqrt{k}$?

The quantities cw^3/k and c^2w^2/k measure different effects: fixed-parameter collision corrections and the growth of fresh planar interiors. Their relative size changes with c . The present paper proves the two regimes separately; it does not establish a uniform theorem throughout the intervening range. A successful result should state the admissible range of c explicitly and bound the residual dependence on both quantities. Substituting $c = c_k$ into an O_K term is not a valid substitute for that task.

Research question 64.3 (Beyond the first critical window). For $cw/\sqrt{k} \rightarrow \infty$, how long does the Catalan quotient continue to determine the leading logarithm of the full moment?

Higher terms in $\log Q$ become relevant as the deficit increases, and the factor used to penalize repeated off-hub edges changes as well. A larger-window saddle might still isolate a dominant vertex, or a different geometry might become competitive. The main technical target is a new complement estimate with matching logarithmic precision, not simply an extrapolation of $e^{c^2w^2/k}$. The relation to the large-mean, fixed-excess enumerations of [4] and the high-moment diluted-matrix regimes of [15] deserves a systematic comparison.

Research question 64.4 (Vanishing mean degree). What replaces the compact-parameter expansion when $c = c_k \rightarrow 0$?

The lower endpoint of the compact interval enters branch bounds through inverse powers of c . Meanwhile cw may remain bounded or vanish, so the Gaussian limit used to evaluate the sharp Poisson constants no longer applies. A useful classification would distinguish $cw \rightarrow 0$, $cw \rightarrow \lambda_0 \in (0, \infty)$, and $cw \rightarrow \infty$, and identify when walks using very few distinct edges become dominant.

64.2 Probability laws and the geometry of contributing walks

Research question 64.5 (Sharper positive approximations). Can the positive two-jump compound Poisson approximation in Corollary 61.6 be extended to the next asymptotic order, with an optimal error constant in an explicitly specified family?

The signed quadratic correction explains why changing only the Poisson mean has an irreducible error. Allowing a rare jump of size two removes that term. At the next order, additional Charlier coefficients must be identified together with sufficiently accurate fresh-law estimates. A positive approximation imposes constraints that a formal signed expansion does not. Determining the minimal positive family capable of matching those coefficients is a concrete approximation-theoretic problem.

Research question 64.6 (Local limits and moderate deviations at critical degree). What are the local limit law and the moderate-deviation range for the dominant-vertex deficit when $cw \asymp \sqrt{k}$?

The present proof gives concentration and an ℓ^1 comparison after multiplying by $Q(k, s)$, but not a sharp pointwise law or moderate-deviation expansion. A local theorem must track the binomial mixture, the shifted Poisson saddle, and the Catalan tilt on the same scale. Relative error in a shrinking tail probability requires substantially more than the total-mass error used here.

Research question 64.7 (Joint hub statistics). Can one determine the joint law of the departure deficit, the number of distinct neighbors of the dominant vertex, its largest excursion block, and the number and depths of off-hub edges?

Departure count and graph degree are different statistics. A joint generating function with separate marks would quantify which features create the correction terms. In the critical regime, Corollary 62.6 says that repeated off-hub edges are rare in total weight, but it does not give a limiting law for their number or locate the deepest fresh branch. Marked Catalan forests and weighted block counts provide an exact starting point.

Research question 64.8 (Several competitive dominant vertices). Is there a parameter range in which configurations with two or more heavily visited vertices contribute at leading order?

They are negligible in the two regimes proved here. Establishing their onset, or proving a much larger exclusion range, would require an optimization over weighted tree skeletons and excursion allocations. A rigorous answer should compare the full combinatorial weight, not just the adjacency spectral radius of a candidate tree.

64.3 Finite graphs, other models, and verification

Research question 64.9 (A simultaneous finite-graph transfer theorem). For what explicit ranges of $k = k_N$ and $c = c_N$ do these tree-moment equivalents approximate $N^{-1} \mathbb{E} \operatorname{Tr} A_{N,c}^{2k}$?

The fixed- k limit in Proposition 59.1 is elementary, but its error is not uniform in k . A transfer theorem must bound non-tree normalized words, repeated graph-label assignments, and their dependence on c . To reach statements about typical graphs, one would then need concentration or higher trace moments. This is an essential step before using the present formula as a finite-matrix spectral estimate.

Research question 64.10 (Spectral tails from sharp moments). What rigorous upper and lower bounds on the limiting spectral measure can be recovered from the corrected moment growth?

Optimizing Markov’s inequality gives a route to tail upper bounds, but matching lower bounds and refined prefactors require additional information. The sparse limiting measure can have atoms, so a pointwise density equivalent is not automatically meaningful. The single-defect spectral calculations of [23] and the earlier sparse-tail work [24] motivate the question while leaving the required rigorous inversion separate from the moment results.

Research question 64.11 (Weighted edges and other offspring laws). Which parts of the collision and forest analysis survive for random edge weights, configuration models, or local weak limits with a non-Poisson offspring distribution?

Random edge weights attach a moment depending on edge multiplicity, so the single factor c for each distinct edge is no longer sufficient. A non-Poisson offspring law changes the weighted partition structure and may change both the typical hub and its correction scale. A useful theorem would state conditions on the weight moments and offspring tails under which a dominant vertex still controls the high moments, together with a replacement for the exact Touchard normalizer. The weighted-graph moment framework of [19] is a natural point of comparison.

Research question 64.12 (Effective bounds and formal certification). Can the arguments be converted into explicit finite- k inequalities and a machine-checked proof of the principal theorems?

There are two separate milestones. Effective analysis would replace the unspecified constants and onset thresholds in the saddle, collision, and complement estimates by usable values, ideally with interval-certified diagnostics. Formalization could begin with the finite combinatorial core: the weighted rotation identity including periodic words, the first-child recurrence, the partition decomposition, and the Pólya factorial identities. A complete proof-assistant development would also certify the Bauer–Golinelli comparison, the uniform analytic estimates, and the ℓ^1 limit arguments. The exact code in this archive is a regression resource for that effort, not a formal proof.

64.4 Suggested order of attack

The nearest extensions are the first critical correction, the intermediate-parameter uniform theorem, and the next positive probability approximation: all three can reuse the exact decompositions in this article. Finite-graph transfer has a larger combinatorial scope but would make the results more directly useful for random matrices. Spectral inversion and general offspring laws require additional ideas and should be treated as separate research projects. Formal certification can proceed in parallel, beginning with the identities that are already finite and exact.

[write] **Further questions of Part VI, 9 October 2026.** Following Vladimir Reshetnikov’s standing rule of 4 October 2026, every claim of the manuscript that is not proved in full is stated as an open question; these twelve research questions are the manuscript’s own, as delivered, and the write labelled them. No claim of the manuscript was found wrong, and nothing is moved or refuted. Relations to Section 56: Question 64.12 is

item 1 for general c ; Questions 64.2 and 64.4 are the parts of item 5 that Theorem 60.6 leaves open (a uniform passage from compact to growing c , and $c \rightarrow 0$); Question 64.10 is close to item 8 (spectral information from moments). Item 4, a growing collision order, is none of these questions and stays open; the growing-degree transition of Section 62 is a different limit.

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